INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

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Abstract

Users of decentralized finance (DeFi) let smart contracts move their tokens. Before an owner trades through a router or deposits into a vault, the owner signs an ERC-20 approval for that contract and has to judge whether the contract deserves this trust. Reputation scores computed on transaction graphs could support that judgement. An existing method, Adaptive Weighted PageRank (AWP), ranks addresses on token transfers. A transfer records a payment, whereas an allowance records that an owner authorized a spender to move its tokens. This thesis asks what ERC-20 allowance edges add to PageRank as a reputation score for smart contracts on Arbitrum One, compared with the transfer walk and with the raw degrees that both walks smooth.

The study defined EndorseRank, PageRank on the latest non-zero allowances with AWP's edge weights and uniform restarts, which ranks the spender contracts that owners approve. It also defined a coupled PageRank (C-PR) that walks both edge types in one chain. An analysis plan that the study committed on 9 October 2026, before it extracted any log, fixed the cohorts, scores, labels, statistics and the decision rule for C-PR, with amounts in raw base units. The study compared every score and two raw degrees through Kendall's τ_b with paired bootstrap 95% intervals, in the same window and on two holdouts. In W0 the study froze the scores on 31 March 2026 and counted the labels from April to June. In W1 it froze them on 30 June 2026 and counted the labels from July to September, under rules that it registered before extracting those logs. After the results of this registered analysis were known, the study revised the amounts. In raw base units the value weight had been close to one for almost every amount, and the coupled walk had added raw amounts across tokens, such as USDC and WETH, whose base units differ by twelve orders of magnitude, and had let unlimited allowances take whole rows. Any token contract can also emit logs that name owners and senders who took no part. The study therefore valued every amount in US dollars at daily prices, kept only five widely traded tokens with verified prices (WETH, USDC, USDT0, ARB and WBTC), set the

slope of the value weight so that the median amount weighs one half, and capped each allowance in the coupled walk at $20{,}000$ dollars. From public Arbitrum One logs of October 2023 to June 2026, the study kept the $15{,}107$ contracts that at least three distinct owners had approved on these tokens; externally owned accounts entered the graphs only as owners, senders and recipients. The study applied the rules of the plan and the registration unchanged to the revised scores, as a re-analysis and not as registered tests, and the registered analysis, reported in an appendix, reached the same verdicts.

EndorseRank and AWP each agreed most with the checks built from their own edges, as intended and partly by construction: EndorseRank reached $\tau = 0.585$ $[0.578, 0.592]$ on the allowance checks and AWP $\tau = 0.572$ $[0.564, 0.580]$ on the transfer checks. The two scores ordered the contracts differently ($\tau = 0.174$ $[0.163, 0.185]$). Out of window, AWP ranked later new transfer senders better than the raw transfer in-degree in W0 and W1 ($\Delta\tau = +0.043$ $[0.038, 0.048]$ and $+0.022$ $[0.017, 0.027]$). EndorseRank, with the plan's uniform restarts, ranked later new approval pairs below the raw count of approving owners ($\Delta\tau = -0.019$ $[-0.026, -0.011]$ and -0.034 $[-0.038, -0.029]$), although the allowance graph did carry propagation beyond one hop. Neither the plan nor the registration fixed these comparisons with the raw degrees in advance; the study computed them after the labels were known. A variant of EndorseRank with AWP's activity restarts beat the count in both windows ($+0.059$ $[0.050, 0.067]$ and $+0.027$ $[0.021, 0.034]$), but the study chose to look at that variant against the count only after seeing the labels, so the result is a hypothesis for a registered test, not a finding. On the revised scores, C-PR at $\lambda = 0.5$ met neither rule A of the plan nor rule B of the registration: in W1 it beat the transfer walk on new approval pairs ($+0.110$ $[0.098, 0.122]$) but fell short of the margin of 0.01 on new senders (-0.028 $[-0.033, -0.023]$) and, narrowly, on the liquidation-free close rate of GMX V2 contract traders ($+0.000$ $[-0.014, 0.010]$). On that outcome, which GMX closes record and neither graph contains, the comparison that the registration fixed for W1, applied to the revised counts, found no count-level lead of allowance over transfer ($\Delta\tau = +0.084$ $[-0.013, 0.176]$, Bonferroni 97.5% interval $[-0.026, 0.188]$, 70 traders), and the PageRank versions are not resolved. AWP's graph had 2.23 times EndorseRank's edges

(25.0 against 11.2 million, on 7.9 against 6.2 million nodes), and its mean solve time over five timed runs after one warm-up was 2.17 times EndorseRank's (22.4 against 10.3 seconds); the difference in time follows from the number of edges, not from the algorithm.

Compared with existing techniques, EndorseRank gives an ordering of contracts that the transfer-based AWP does not, and it ranks a contract by the approvals that other addresses give it, whereas deployed attestation tools score the holder of an address. It did not beat the approval count that any protocol can compute without PageRank, whereas AWP did beat its own count. The thesis contributes the allowance as a reputation edge for contracts and an out-of-window test of each score against the count it smooths. It also reports that one walk over both edge types failed the rules fixed for it, in the registered analysis and again on the revised scores.

For the people who approve contracts, the number of distinct owners that approve a contract is a public signal that wallet software could show before a user signs, and EndorseRank with uniform restarts did not improve on it out of window. A Sybil model of EndorseRank on the same graph shows how cheaply such signals move: 22 colluding addresses lift a contract past the cohort's 90th percentile, 2,074 past the 99th and about 135,895 to the top. Looking ordinary is cheap and reaching the top is expensive, so no score studied here should decide warnings or rewards on its own, and none says whether a contract's code is safe.

Key terms: decentralized finance; smart contract; on-chain reputation; ERC-20 allowance; PageRank; EndorseRank; Adaptive Weighted PageRank; coupled PageRank; temporal holdout; Arbitrum One; Kendall's tau

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Abbreviations and Notation

The abbreviations and mathematical notation used throughout this research are listed below. Each symbol is also defined where it first appears in the main chapters.

Abbreviations

Table I gives the abbreviations in alphabetical order, each with the meaning it has in this thesis.

Mathematical notation

Table II gives the symbols that recur across the chapters. The full notation of the graphs, scores and statistics is in Table 3.1 of Chapter 3.

Table I: Abbreviations used in this research.

Abbreviation	Meaning
ABI	Application binary interface (the encoding of a contract's calls and event logs)
ARB	Arbitrum governance token (one of the five tokens, Section 3.5.1)
AWP	Adaptive Weighted PageRank
C-PR	Coupled PageRank (per-node mixture of the allowance and transfer layers with weight λ)
CI	Confidence interval (here a percentile interval of the paired bootstrap)
DeFi	Decentralized Finance
DEX	Decentralized Exchange
EIP	Ethereum Improvement Proposal
EIP-7702	Ethereum Improvement Proposal 7702, which lets an EOA delegate to contract code through a delegation designator; such an address counts as an EOA in this research
EOA	Externally owned account (an address controlled by a private key)
ER	EndorseRank (in table headings)
ERC-20	Ethereum Request for Comments 20 (fungible token standard)
EVM	Ethereum Virtual Machine
JSON	JavaScript Object Notation (format of the machine-readable summaries)
KYC	Know your customer (identity checks of a regulated service)
L2	Layer 2 (a network that settles its transactions on Ethereum)
NFT	Non-fungible token
PR	PageRank
RPC	Remote procedure call (the node interface through which contract code is read)
RQ	Research Question
S-PR	Endorsement-Seeded PageRank (transfer layer walked with restarts drawn from the allowance walk)
SD	Standard deviation
SHA-256	Secure Hash Algorithm with a 256-bit digest
SQL	Structured Query Language
USD	United States dollar, the unit of every amount in the revised analysis
USDC	USD Coin, issued natively on Arbitrum One (one of the five tokens)
USDT0	Tether USD on Arbitrum One, backed one-to-one by USDT (one of the five tokens)
UTC	Coordinated Universal Time
W0	Holdout window: scores frozen on 31 March 2026, labels counted from April to June 2026
W1	Registered window: scores frozen on 30 June 2026, labels counted from July to September 2026 (its rules were registered before its logs were extracted)
WBTC	Wrapped Bitcoin (one of the five tokens)
WETH	Wrapped Ether, the form in which ether enters the graphs (one of the five tokens)

Table II: Mathematical notation used in this research.

Symbol	Meaning
G_{approve}	Directed endorsement graph from the latest allowances of the five tokens, valued in US dollars (owner $\to$ spender)
G_{transfer}	Directed transfer graph of the five tokens, amounts in US dollars (sender $\to$ receiver; AWP input)
$\lvert V\rvert$, $\lvert E\rvert$	Numbers of nodes and of directed edges in a reputation graph
w_{ij}	Weight of the directed edge from node i to node j
o_i	Out-strength of node i, $o_i = \sum_j w_{ij}$ (written $o(u)$, or $o_A(u)$ and $o_T(u)$ per layer, in Chapter 3)
P	Transition matrix, $P_{ij} = w_{ij}/o_i$ for $o_i > 0$; a dangling row is uniform in Equation (2.1) and zero in Equation (3.2), where its mass is redistributed separately
G_{PR}	Google matrix $dP^\top + (1-d)\,\mathbf{1}\mathbf{1}^\top/\lvert V\rvert$ (Equation (2.2))
$\mathbf{r}$, r_i	PageRank vector and its entry for node i; $\mathbf{r}^{(t)}$ is the iterate after t power-iteration steps
$\boldsymbol{\pi}$	Teleportation (restart) distribution: uniform for EndorseRank and C-PR, proportional to activeness X_u for AWP and for the variant of EndorseRank with activity restarts, and the normalized allowance-walk score for S-PR
d	PageRank damping factor (default 0.85); the teleport probability is $1-d$
ε	Convergence tolerance of power iteration (10^{-8}, on the $\lVert\cdot\rVert_1$ change between iterates)
$I_{\max}$	Iteration cap of power iteration (300)
λ	Allowance-layer weight of the coupled operator C-PR ($\lambda = 1$ and $\lambda = 0$: one layer walked alone)
$P^{(\lambda)}$	Coupled transition matrix of C-PR with allowance-layer weight λ (Equation (3.9))
$\alpha_A(u)$, $\alpha_T(u)$	Weights of the allowance row and the transfer row in the coupled transition row of node u
U	Cap on an allowance in the allowance layer of C-PR and in the in-approve value: an allowance above U counts U; U = 20,000 dollars, the 99th percentile of the US-dollar values of single transfers up to t_1, rounded to two significant digits
k, t_0	Steepness and midpoint (days) of the logistic time decay
Δt	Age in days of a transfer or approval
$\sigma(\Delta t)$	Logistic time weight of a transfer (AWP) or latest approval (EndorseRank)
$V(z)$, b	Bounded value weight of AWP and EndorseRank on a US-dollar amount z, $2/(1+e^{-bz}) - 1$ with slope $b_T = \ln 3/m_T$ for transfers and $b_A = \ln 3/m_A$ for allowances, so that an amount of m_T or m_A gets $V = 1/2$
m_T, m_A	Medians, rounded to two significant digits, of the US-dollar amounts of the transfers up to t_1 (200 dollars) and of the finite latest allowances at t_1, valued at t_1 prices (62 dollars) (Section 3.5.1)
X_u	Activeness of address u, AWP's restart weight
n	Number of contracts in a cohort or a benchmark stage (Section 4.1.1)
ρ	Spearman's rank correlation coefficient
τ	Kendall's rank correlation coefficient; every reported value is τ_b
$\Delta\tau$	Paired difference of two Kendall coefficients on the same resamples
T_{obs}	End of the observation window and score freeze of W1 (30 June 2026)
t_1, t_2	Score freeze of W0 (31 March 2026) and end of its label window (30 June 2026)
$\mathcal{W}$	Matched cohort: contracts that received non-zero approvals of the five tokens from at least three distinct owners, counted over the five tokens together, in the observation window; the registered analysis used the plan's cohort, approved on any token (Appendix 8.6.10)
$\mathcal{C}$	Contracts evaluated in one analysis: $\mathcal{W}$, or a spender or trader cohort of a holdout
$\mathbf{s}_M$	Reputation score vector of method M on the contracts evaluated
$\mathbf{p}_q$	Value vector of same-window proxy q on the matched cohort
B	Number of paired bootstrap resamples
δ	Non-inferiority margin of rule B

1 Introduction and Research Problem

Decentralized Finance (DeFi) has changed how capital is allocated on public blockchains: lending, borrowing and leveraged trading now run through smart contracts without traditional intermediaries (Schär, 2021). To use most of these services, an owner first authorizes a contract to move its tokens. The ERC-20 standard records each such authorization as an allowance (Vogelsteller & Buterin, 2015), and users, the software they sign with and other protocols all have to decide which spender contracts should receive one. Under pseudonymity that decision is hard to make well, and the central issue of this thesis is what the public record of earlier authorizations can say about a contract.

This thesis asks what ERC-20 allowance edges add to PageRank as a reputation score for smart contracts. The allowance edges come from the `Approval` events that `approve` and `permit` emit, and EndorseRank is PageRank computed on them with the edge weights of the transfer-based Adaptive Weighted PageRank (AWP) of T. Do et al. (2023), applied to the US-dollar amounts of five widely traded tokens. Both are run on the same Arbitrum One contracts, beside a coupled walk over both edge types and the raw degrees that the walks smooth, and checked by the rank-correlation method used with AWP (T. Do et al., 2023), extended with paired contrasts, raw-degree baselines and two temporal holdouts, the second of them in a window registered before its logs were extracted. The amounts of the main analysis were revised after the results of the registered analysis were known (Section 1.4), and the registered analysis is kept in an appendix (Appendix 8.6.10). The scores use on-chain events and, to value amounts, public daily token prices; Section 1.3 defines an on-chain reputation score together with the other main terms.

1.1 Background: DeFi, collateralization, and pseudonymity

Public blockchains make programmable settlement possible without a central authority (Nakamoto, 2008). A smart contract can encode the terms of a deal (Szabo, 1997), and Ethereum lets such a contract hold and move tokens under the rules of its code (Buterin, 2014). DeFi platforms built from these contracts clear trades on the blockchain instead of routing them through a custodian, and protocols stack them into decentralized markets for lending, automated market making and perpetual futures (Werner et al., 2022). On Arbitrum One and other L2 networks, fees are low enough that ordinary users transact often, and every transfer, approval and protocol event can be inspected address by address.

Even with this visibility, DeFi credit markets remain over-collateralized. Most protocols make a borrower or trader post more collateral than the position is worth, because the counterparty is a pseudonymous account whose owner has not been verified (Harvey et al., 2021). Pseudonymity is a basic property of public ledgers. Anyone can create addresses at almost no cost, and one entity can control many of them (Douceur, 2002). Conventional credit ratings depend on personal histories that translate poorly to this setting (Packin & Lev-Aretz, 2024), so protocols control risk through collateral reserves, liquidation mechanisms and protocol rules instead of personalized reputation. Researchers have turned to on-chain signals as a substitute for off-chain credit histories: transfer-graph centrality (H.-D. Do & Do, 2023), network risk propagation (D. Lin et al., 2025), credit scoring based on prospect theory (Hassija et al., 2020), and machine learning pipelines over lending protocols (Kotb, 2023). All of these score borrowers or the accounts behind them.

Collateral protects a lending protocol against its borrowers, but it does nothing for an owner who must decide whether a contract may spend its tokens. In that decision the pseudonymous counterparty is the contract. Before a router, vault, aggregator or lending pool can move an owner's tokens, the owner grants it an allowance, which stays in force until it is spent, replaced or revoked (Vogelsteller & Buterin, 2015). Most approvals on Ethereum are for unlimited amounts (Wang et al., 2022), so an allowance granted to a malicious or later compromised contract exposes the owner's

whole balance of that token. Phishing sites exploit this by leading users to sign transactions that hand their assets to attackers (He et al., 2023), and the toolkits that do so are now sold as a service (He et al., 2025). Protocols face the same decision when their own contracts approve other contracts, as a vault does when it lets a lending pool pull the tokens it supplies. A contract's code is public, but who deployed it, who can change it and how it will use an allowance often are not.

In practice, trust in a spender contract forms in a few ways. An owner usually decides at the moment its wallet software presents the approval for signature, and weaknesses in how wallets present and validate token approvals make that moment easy to exploit (Adamczyk, 2025). Code analysis judges a contract by its functions and can flag, for example, functions written for a rug pull (Z. Lin et al., 2024). Deployed reputation tools work at the level of persons. Human Passport collects weighted credentials, on-chain and off-chain, to decide whether an address is controlled by a distinct human, and it serves as a gate against Sybil attacks, not as a graded ordering (Human Passport, n.d.). That is a question about the holder of a key. It has no direct counterpart for a contract, whose behavior is fixed by its code and by whoever may upgrade it. None of these routes uses the public record of what other owners have decided: which contracts many owners have authorized, and how strongly those owners are endorsed in turn.

Graph ranking algorithms appear well suited to such a record. PageRank and its variants rank nodes by recursive endorsement, so that a node's score depends on the endorsements it receives from nodes that are endorsed in turn (Page et al., 1999). Web trust algorithms such as EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) apply similar ideas to peer-to-peer trust and to web spam. H.-D. Do and Do (2023) use both PageRank and HodgeRank on a graph of transfers to judge how trustworthy Ethereum users are, and call the result a form of social credit. AWP (T. Do et al., 2023) is another transfer-graph ranking algorithm. It weights each transfer by a bounded function of its value and a logistic function of its age, and it restarts the walk at addresses that sent recently; its authors check it by showing that it correlates with inbound transfer count and value, taken as proxies for economic importance. These studies make transfer-graph ranking the baseline against which a score built on another edge type has to be compared. They rank addresses

by the payments they receive, however, and leave out explicit spending authorization, a separate on-chain action recorded as ERC-20 allowances (Vogelsteller & Buterin, 2015).

1.2 Motivation for on-chain reputation

Four factors drive the need for a reputation score for spender contracts.

- Authorization decisions: owners, the software they sign with and the protocols that integrate other contracts all decide which spenders to approve. An ordinal signal of how other owners have treated a contract could inform these decisions, for instance as one input to the warning shown before an approval is signed, or to the list of contracts a protocol allows to integrate with it.
- Auditability: curated lists and off-chain scores are opaque, but graph-based scores computed from public event data and public token prices can be recomputed and audited by anyone who holds the same inputs, which answers part of the concern about black-box credit scoring in decentralized finance (Packin & Lev-Aretz, 2024).
- Operational cost: a score that is refreshed often, before each approval prompt for instance, should be cheap to recompute.
- Standardization: other markets use screening ratios to tell counterparties apart, but DeFi has no standard ordinal language for doing so at the level of the contract. Chapter 5 discusses where the scores studied here could serve that role and where they cannot.

Transfer-based AWP is public and auditable, but its graph grows with every payment (T. Do et al., 2023). Allowance graphs are a structurally different alternative. Whenever an owner calls `approve`, or signs a `permit` (Lundfall, 2020), the token contract records how much of that token a spender may transfer (Vogelsteller & Buterin, 2015), and each new approval supersedes the older ones, so the latest approval of each owner–spender pair and token stands for the history of that pair. Approvals are also rarer than transfers. On the same contracts the endorsement graph G_{approve} therefore has fewer than half as many edges as the transfer graph G_{transfer}, and the PageRank solve,

whose cost per iteration grows linearly with $|E|$ (Leskovec et al., 2020), takes correspondingly less time (Section 4.2). The same solver runs in both cases, so the difference comes from the sparser graph and is not an algorithmic improvement; Claim C1 treats it as a property of the representation. Efficiency aside, allowances also mean something different. A transfer might reflect routing, market making, or a passive receipt, whereas an allowance is an active grant of permission to spend funds. Treating allowances as endorsement edges carries PageRank's citation reading (Page et al., 1999) over to a trust relationship that owners and protocols already depend on, and the endorsed node is the spender contract, the object that owners have to decide about. Whether this produces different rankings, and whether those rankings agree differently with authorization and transfer checks and with later activity, is examined on a matched cohort of $n = 15{,}107$ contracts on Arbitrum One.[1]

1.3 Research problem

AWP infers trust from transfer history through weighted sums and logistic time decay (T. Do et al., 2023). It gives each transfer an economic weight, but it (i) must aggregate the whole transfer history, (ii) models a high-value address as one that receives many inbound transfers of large value, and (iii) assigns no role to spending authorization. EndorseRank, the score studied here, keeps AWP's edge weights and solver but builds the directed graph from the latest allowance of each owner–spender pair and token, valued in US dollars. The resulting graph is sparser, encodes a different signal and ranks a different side of each relationship. An allowance points from the owner to the spender, so EndorseRank scores the contracts that receive approvals, such as routers, vaults and lending pools, and each owner, an EOA or another contract, passes its endorsement to the spenders it approves.

The main research question is: what do ERC-20 allowance edges add to PageRank as a reputation score for smart contracts on Arbitrum One, compared with the transfer walk and with the raw degrees that both walks smooth? Research questions RQ1–RQ5 (Section 1.6) break it down. Because both edge types live on the same addresses, a single walk can also mix them node by node with a

[1] Matched means that every method scores the same contracts, a list drawn from the approval logs of five tokens alone, without reference to any transfer; Section 3.2 gives the rule.

fixed weight λ, which defines a Coupled PageRank (C-PR); at $\lambda = 1$ and $\lambda = 0$ it walks one layer alone. The mixture is worth testing because the two edge types record different acts on the same contracts. An operator that already runs a transfer walk could add allowance edges to it instead of running a second score, and the open question is whether the walk then carries the allowance signal without losing the transfer signal.

Agreement with statistics of a score's own edges, or prediction of more of the same edges, does not yet show that a score measures reputation in a substantive sense. A further comparison, which adds no research question, therefore also uses an outcome built from neither edge type and observed after the freeze: the later liquidation avoidance of the GMX V2 accounts that are contracts and trade in the label window (Section 3.7.3). Its rules were registered with the second holdout and are applied to the revised scores as a re-analysis. Few such contracts traded in these months, so this comparison rests on small samples.

The scores appear in Table 1.1. Table 1.2 lists hypotheses (H1–H3), research questions (RQ1–RQ5), and empirical claims (C1–C6) from Chapter 4.

Table 1.1: The scores evaluated in this thesis (matched contract cohort, $n = 15{,}107$).

Score	Input graph	Role
EndorseRank	Latest ERC-20 allowances of the five tokens (G_{approve}), in US dollars, weighted as in AWP; uniform restarts	Allowance delegation; the object of study in this research. A variant with AWP's restarts in proportion to activity is reported beside it
AWP	Transfers of the five tokens (G_{transfer}), in US dollars	Transfer flow; the published transfer-graph method of T. Do et al. (2023), with the slope of its value weight set from the median amount (Section 3.6.1)
C-PR	Both layers, US-dollar amounts with allowances capped, mixed per node with weight λ	Coupled operator used to test whether the two edge types combine (RQ5); at $\lambda = 1$ and $\lambda = 0$ it walks one layer alone. Its ablation is S-PR, a transfer walk restarted from the allowance walk
Raw degrees	G_{approve} and G_{transfer}, counted per node	In-approve degree and transfer in-degree, the local counts that the PageRanks smooth: proxies in the same window and baselines out of window

The main terms are defined as follows:

- *DeFi (Decentralized Finance):* Financial services such as lending, trading and borrowing, implemented as smart contracts on public blockchains and generally run without a central operator in charge of custody or settlement (Schär, 2021).
- *Smart contract:* Self-executing program code that runs on a blockchain and carries out an agreement without a trusted intermediary (Szabo, 1997). In this thesis the allowance and transfer logs are emitted by the contracts of five tokens, WETH, USDC, USDT0, ARB and WBTC, and the addresses that are scored are contracts as well.
- *Address, contract account and EOA:* An address identifies an account. A contract account is governed by the code it stores, whereas an externally owned account (EOA) is controlled by a private key (Antonopoulos & Wood, 2018). An address counts as a contract here when `eth_getCode` at one pinned block returns bytecode; no bytecode, or an EIP-7702 delegation designator (Buterin et al., 2024), makes it an EOA (Section 3.2). EOAs enter the graphs as owners, senders and recipients and are never scored. No off-chain identity is assumed or recovered.
- *On-chain reputation score:* A number computed from on-chain data that expresses an address's relative position among the addresses it interacts with (H.-D. Do & Do, 2023). In this thesis the scored addresses are contracts, and the only off-chain inputs are the list of five tokens and the daily US-dollar prices that value their amounts (Section 3.5.1). Scores are ordinal and meant for ranking; their absolute scale has no meaning.
- *PageRank:* A graph ranking algorithm that spreads influence through a directed graph with a damping factor (Page et al., 1999). Adaptive Weighted PageRank (AWP) is PageRank on the transfer graph, with edges weighted by the value and age of each transfer and restarts at recently active senders (T. Do et al., 2023). Here values are in US dollars, and the value weight is one half at the median amount and approaches one for larger amounts. EndorseRank applies AWP's edge weights to the latest-allowance graph and restarts uniformly (Section 3.6).
- *Coupled PageRank (C-PR):* One random walk over both the allowance and the transfer layer,

which at each node mixes the two rows with weights λ and $1 - \lambda$ over the layers that node occupies (Section 3.6.3). Endorsement-seeded PageRank (S-PR) keeps the transfer walk but restarts it from the allowance walk, as TrustRank restarts from a seed set (Gyöngyi et al., 2004).

- *ERC-20 allowance / Approve / Permit:* Contract methods that record how many of the owner's tokens a spender may spend on the owner's behalf (Vogelsteller & Buterin, 2015). `Approve` is an ordinary on-chain call; EIP-2612 `permit` makes the same state change through a signature (Lundfall, 2020). For each owner–spender pair and each of the five tokens, EndorseRank uses the latest allowance, valued at the token's price on the day of the anchor date, and keeps it only when it is positive.
- *Endorsement graph:* A directed graph whose edges denote an explicit delegation of spending authority, as opposed to an actual transfer of assets. An edge $u \to v$ means that the owner u, an EOA or a contract, has approved v to spend tokens belonging to u (Vogelsteller & Buterin, 2015). The spenders v that are scored are contracts.
- *Domain alignment:* The degree to which a reputation ranking agrees with an observable proxy ranking, measured with Spearman's ρ (Spearman, 1904) and Kendall's τ (Kendall, 1938). Agreement with proxies built from a score's own edges is read as a check of the intended construct in the sense of Cronbach and Meehl (1955). A method can agree more with the graph of its own construct than with a different check.

1.4 Temporal holdout

The same-window allowance coefficients (Table 4.3) are an intended-construct check: PageRank on G_{approve} should agree with the in-approve degree of each spender, because it smooths that degree globally. To test whether a score conveys information beyond the same-window graph, the scores are frozen at t_1 (31 March 2026, 23:59:59 UTC) and compared with two labels counted from 1 April to 30 June 2026: the new owner–spender approval pairs that a spender contract gains (the endorsement

construct) and its new transfer senders (the flow construct). This first window, W0, is evaluated on the spender cohort of every contract that holds at least one positive latest allowance of one of the five tokens at t_1 in the extracted approval data ($n = 17{,}050$). Each PageRank is also compared with the raw degree it smooths at t_1, so the gain of the propagated score over the local count can be seen directly; these degree contrasts belong to no rule fixed in advance and were computed after the labels were known. W0 carries the primary part of rule A, the rule for the coupled operator fixed in the analysis plan, and an exploratory run on the GMX V2 contract traders of the same months (Section 3.7).

A second window, W1, was registered at 16:49 UTC on 9 October 2026, after the W0 results were known and before any log dated after 30 June 2026 was extracted. The extraction script refuses to run unless the registration and the configuration are committed, and it records the hashes of both files. W1 freezes the scores at the end of the observation window (30 June 2026, 23:59:59 UTC), so they come from the same graphs as the same-window evaluation, and counts its labels from 1 July to 30 September 2026. It repeats the two spender labels on the contract spenders with a positive latest allowance at that freeze (in the revised analysis, of one of the five tokens: $n = 18{,}152$) and adds one for trader contracts: the share of closes that did not end in liquidation, for the GMX V2 accounts that are contracts, appear in the freeze-date graph and close at least three positions in the window ($n = 70$). The registration fixes rule B for the coupled operator, with its margin, three further contrasts reported outside the decision and two sensitivity analyses, and a comparison of allowance-side with transfer-side scores on that trader label, with the wording of each possible conclusion (Sections 3.7.2 and 3.7.3). That comparison uses W0 as its secondary window, and the liquidation labels of the W0 traders had not been computed for any score when the registration was committed.

On 10 October 2026, after the results of this registered analysis in the same window, W0 and W1 were known, the amounts were revised (Section 3.5.1). The plan had used raw base units of every ERC-20 token and a value weight that is close to one for almost every amount, so in the registered analysis the amount played almost no part in EndorseRank or AWP. The revised analysis values

every amount in US dollars at the daily price of its token, keeps only five widely traded tokens with reliable prices, sets the slope of the value weight from the median amount and rebuilds the cohort, the graphs, the proxies and the spender labels on those tokens. The revision was committed before any revised score was computed. The revised analysis is the main analysis of this thesis. Rules A and B and the comparison on trader outcomes are applied to it with their thresholds and wording, but because its scores were defined after the registered results were known, its verdicts are a re-analysis and not registered tests. The registered analysis reached the same verdicts; Appendix 8.6.10 keeps its results, and Table 8.13 sets them beside the revised ones.

Table 1.2: Summary of hypotheses, research questions, and empirical claims.

Item	Addresses	Reported in
H1 / RQ1	Do EndorseRank and AWP order the same contracts differently?	§4.4; inter-method τ with 95% CI; Table 4.5; Claim C3
H2 / RQ2	How closely does each score agree with the allowance checks and with the transfer checks?	Tables 4.3 and 4.5; Claim C2
H3 / RQ3	What does each score cost to compute at equal n?	§4.2; Claim C1
RQ4	Do scores frozen at t_1 predict the later new approval pairs and new senders of spender contracts, and does any PageRank do better than the raw degree it smooths?	§4.6; Tables 4.10, 4.11, 4.12, 4.14 and 4.16; Claim C4
RQ5	Does the coupled operator C-PR improve on a walk over the transfers alone, and does it rank contracts better than a walk over either layer alone?	§4.3.1, §4.6 and §4.6.2; Tables 4.3, 4.4, 4.11, 4.14 and 4.15; Claim C5
C1	Solve time follows the edge count of each graph: EndorseRank's call takes less time than AWP's on its sparser graph, while C-PR and S-PR, which walk the transfer layer, take longer than AWP	§4.2
C2	Same-window agreement: each score agrees most with the checks of the edge type it walks on, an intended-construct check that is partly built in; AWP follows the summed value of inbound transfers almost as closely as their number; EndorseRank leaves few ties, the isolated-node rule changes no coefficient, and the C-PR mixtures lie between their two layers	Tables 4.3 and 4.5
C3	EndorseRank and AWP order the cohort differently	Table 4.5; Figure 4.3

Table 1.2: (continued)

Item	Addresses	Reported in
C4	Out of window, in contrasts computed after the labels were known, AWP ranks new transfer senders above the raw transfer in-degree in both windows, while EndorseRank with uniform restarts ranks new approval pairs below the raw in-approve degree in both; its variant with activity restarts ranks them above the degree, a hypothesis to be tested on new data and not a finding	Tables 4.10, 4.11, 4.12 and 4.16
C5	On the revised scores, in a re-analysis under the rules fixed for the registered analysis, one walk over both edge types does not beat its layers walked alone: rule A fails in both parts, and rule B is not met, because the gain over the transfer walk on new approval pairs replicates but non-inferiority fails on new senders and, narrowly, on the traders' liquidation-free close rate; against AWP it fails clearly; in the S-PR ablation, allowance edges help only when they carry the walk; the registered analysis reached the same verdicts	Tables 4.3, 4.4, 4.11, 4.14, 4.15 and 8.13; §4.6.2
C6	Comparison on an outcome built from neither edge type, with the registered rules applied to the revised scores: no count-level lead of allowance over transfer on later liquidation avoidance in the primary window, the PageRank versions not resolved, and the approval count below the transfer count on the profitable and non-loss shares; small trader samples	Tables 4.18, 4.19 and 8.13

1.5 Research objectives

The research has five objectives, one for each of RQ1–RQ5. Each is stated with the measurement used to operationalize it and with the criterion that Chapter 5 applies when judging whether the objective was met.

O1. Establish whether EndorseRank and AWP order the contracts of the same cohort differently (RQ1). Measurement: Kendall τ between the two score vectors on the matched cohort with

a 95% bootstrap interval. Criterion: the interval contains neither 0 (unrelated orderings) nor 1 (identical orderings).

O2. Assess how closely each score agrees with each construct in the same window (RQ2). Measurement: family-level Kendall τ with 95% intervals for every score and construct on the matched cohort. Criterion: a score agrees with a construct when its interval on that family excludes 0; agreement with the construct of the score's own edges is read as an intended-construct check, because those proxies come from the same events as the graph.

O3. Measure what each score costs to compute on the same contracts, and how runtime responds to the number of sampled contracts (RQ3). Measurement: solve time (mean and standard deviation of five timed runs after one warm-up run), peak memory, iterations, $|V|$ and $|E|$, all under one protocol. Criterion: the time ratio is reported only together with the ratio of $|E|$, so that any difference can be traced to the size of the graph.

O4. Test whether scores frozen before a label window predict the new approval pairs and new transfer senders that spender contracts gain in it, and whether any PageRank does better than the raw degree it smooths (RQ4). Measurement: Kendall τ with 95% intervals between every score, the two raw degrees included, and both labels, on the spender contracts of two windows: W0, with scores frozen on 31 March 2026 and labels counted from April to June 2026, and W1, with scores frozen on 30 June 2026 and labels counted from July to September 2026 (Section 1.4); in each, paired contrasts of each PageRank against its own degree. Criterion: a score predicts a label when its interval excludes 0, and a PageRank improves on its degree only when the interval of the paired difference lies above 0. Neither the degree contrasts nor this criterion was fixed before the W0 labels were known.

O5. Test whether one walk across both edge types improves on a walk over the transfers alone, and whether it beats a walk over either layer alone (RQ5). Measurement: C-PR at $\lambda = 0.5$ (with 0.25 and 0.75 as sensitivity values) and S-PR, compared through paired intervals on $\Delta\tau$ with the two layers walked alone and with EndorseRank and AWP, on the holdout and same-

window labels. Criteria, each fixed for the registered analysis before the data it was judged on, and applied to the revised scores with the same thresholds after the registered results were known, as a re-analysis and not as registered tests: (a) rule A, fixed in the analysis plan of 9 October 2026 before any log was extracted, under which C-PR does no worse than the allowance layer walked alone on new approval pairs and better than the transfer layer walked alone on new senders in W0, and in the same window lies within the interval of the better layer on the transfer and allowance families and above both layers on Sybil stability; (b) rule B, registered later on 9 October 2026, after the W0 results were known and before the W1 logs were extracted, under which C-PR beats the transfer layer walked alone on new approval pairs in W1 and is not worse than it by more than a margin of 0.01, set from the half-width of the corresponding W0 interval of the registered analysis and rounded up, on new senders and on the trader contracts' liquidation-free close rate. A sensitivity analysis registered with rule B repeats (b) with AWP in place of the transfer layer; it does not change the rule. Every other contrast of C-PR and S-PR is exploratory: it was computed after the scores and labels it compares were known.

1.6 Research questions

Three hypotheses go with the first three research questions. H1: on the same set of contracts, EndorseRank and AWP produce different orderings. H2: each score agrees with the checks built from the edges it walks on. H3: at the same value of n, a score's computation time follows the number of edges in its graph, so EndorseRank, on the sparser graph, is faster to compute than AWP. H1–H3 hold largely by construction and act as checks of the instruments. The falsifiable tests are RQ4, whose degree contrasts belong to no registered rule and, on the revised scores, were computed after the labels of both windows were known, and RQ5, answered by rules fixed in advance for the registered analysis (Section 3.7) and applied to the revised scores as a re-analysis. Table 1.2 summarizes the questions.

RQ1. On the same set of Arbitrum One contracts, do EndorseRank and AWP produce different orderings?

RQ2. How closely does each score agree with current allowance checks and with current transfer checks (Spearman's ρ and Kendall's τ)?

RQ3. What does each score cost to compute on the same set of contracts, as measured by time, memory, and iterations?

RQ4. Given scores computed at a freeze date t_1, do the scores predict the new owner–spender approval pairs and the new transfer senders that spender contracts gain after t_1, and does any PageRank predict them better than the raw degree it smooths at t_1?

RQ5. Does a coupled PageRank running on both the allowance and the transfer layer improve on a walk over the transfers alone, such as AWP, and does it rank contracts better than a walk over either layer alone, out of window as well as in the same window?

RQ4 uses later labels of the same two constructs because the contracts ranked here do not borrow, so no default or repayment label exists for them (Section 2.8). RQ5 moves from comparing the two edge types to combining them. Rule A, which asks whether C-PR beats a walk over either layer alone, was fixed with λ in the analysis plan and is judged on W0 and in the same window. The rule for improving on the transfer walk alone was written after the W0 results were known, so it is judged only on W1, whose logs were extracted after it was registered (Sections 3.7 and 3.7.2). Both rules are applied to the revised scores after the registered results were known (Section 3.5.1), and Appendix 8.6.10 gives their registered verdicts.

1.7 Summary of the literature review

Chapter 2 reviews four areas. In graph ranking, PageRank (Page et al., 1999) ranks nodes by recursively propagated endorsement at a per-iteration cost linear in the number of edges (Langville & Meyer, 2006), and EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) adapt it to peer-to-peer trust and web spam with normalization and seed sets. In on-chain reputation, H.-D.

Do and Do (2023) apply PageRank and HodgeRank to Ethereum transactions, and AWP (T. Do et al., 2023) weights transaction edges by value and age and checks its ranking against inbound transfer count and value, statistics drawn from the same data as its graph. D. Lin et al. (2025) propagate account risk through the transaction network, and surveys of blockchain trust and reputation systems (Bellini et al., 2020) and of smart-contract reputation designs (Almasoud et al., 2020) find explicit ratings and transfer volume to be the usual trust signals. Deployed DeFi tooling such as Human Passport instead aggregates on-chain and off-chain credentials to decide whether an address belongs to a distinct person (Human Passport, n.d.), and code analysis judges a contract by the functions it contains (Z. Lin et al., 2024). None of this work takes the ERC-20 allowance (Vogelsteller & Buterin, 2015) as a ranking edge or ranks contracts by the approvals they receive. Address clustering uses approvals only to link addresses of one entity (Victor, 2020), and an analysis of an approval-based fraud scheme shows that an approval can be obtained by deception (Adamczyk, 2025). In DeFi risk and measurement, systematizations classify risk by protocol layer (Werner et al., 2022) and by the attacks seen in practice (L. Zhou et al., 2023), and wash trading (Victor & Weintraud, 2021), extractable value (Qin et al., 2022), scam tokens (Xia et al., 2021) and airdrop farming (Luo et al., 2025) show that much transfer activity is artificial. In measurement theory, construct validity (Cronbach & Meehl, 1955) separates what a score measures from external criteria, and the bootstrap (Efron, 1979) gives distribution-free intervals for rank statistics. The gap (Section 2.21) is an allowance-edge ranking of contracts evaluated on one cohort beside the transfer method and the raw degrees, with confidence intervals, paired differences, a fixed timing protocol and temporal holdouts, one of them in a window registered before its data were extracted.

1.8 Summary of the research methodology

Chapter 3 applies a quantitative, comparative design to public ledger data. An analysis plan, committed on 9 October 2026 before any log of the study was extracted, fixes the cohorts, scores, labels, statistics and rule A. ERC-20 `Approval` and `Transfer` logs on Arbitrum One from 1 October 2023

to 30 September 2026 are extracted from the public Google BigQuery dataset for the ego network of the plan's cohort, whose rule and account-type check Section 3.2 defines. The revision of 10 October 2026, made after the registered results were known, rebuilds the cohort on five tokens, keeps the subset of the extracted logs of those tokens that involve the revised cohort, values their amounts with the daily prices of the DefiLlama service (DefiLlama, n.d.) and sets the slope of the value weight from the median amount (Section 3.5.1); the main analysis is the revised one, and the registered one is kept in Appendix 8.6.10. One power-iteration PageRank implementation scores the latest-allowance graph and the transfer graph, and the coupled operator runs on the same solver. Each score is compared with Spearman's ρ and Kendall's τ against same-window transfer, allowance and Sybil-stability proxies and, out of window, against new approval pairs and new transfer senders on the spender contracts of W0 and W1, with the raw degrees at each freeze as baselines. Every coefficient has a 95% interval from 400 paired contract resamples (seed 42), and method comparisons are paired. The registration of W1 adds rule B and a comparison of allowance-side with transfer-side scores on the later liquidation avoidance of GMX V2 contract traders, a label built from neither edge type, in both windows (2,000 paired resamples; Bonferroni 97.5% intervals for its two decision contrasts); on the revised scores both are applied as a re-analysis. Runtime, memory, iterations and graph size are recorded under one timing protocol. The code, configuration and machine-readable summaries behind every table are in a public repository (Section 3.13), and the study uses only public data (Section 3.16).

1.9 Contributions

The thesis makes five contributions to the study of on-chain reputation and DeFi risk. EndorseRank and AWP both apply weighted PageRank (Page et al., 1999) to a single edge type, each with its own restart distribution; only the coupled operator changes the transition rule. What is new is therefore an edge type and the objects it ranks, an evaluation design, an analytic account of what the allowance walk computes, a test of whether the two edge types combine, and a comparison on an outcome

that neither edge type contains, under rules registered before the logs of its primary window were extracted.

- Allowance edges as a ranking edge for contracts (EndorseRank): an endorsement graph in which an edge $u \to v$ stands for the latest in-window ERC-20 allowances of the five tokens, set by `approve` or `permit`, that the spender v holds from the owner u, valued in US dollars and weighted as AWP weights a transfer. The edge records a delegation of spending rights, not a history of payments, and it points to the spender, so the score ranks the contracts that owners authorize. Approvals have been used to cluster the addresses of one entity (Victor, 2020), but no reputation method reviewed ranks contracts on the approvals they receive (Section 2.21).
- Evaluation of every score against the counts it smooths, in the same window and in two hold-out windows: every score is computed on one set of contracts and checked against identical transfer, allowance and Sybil-stability proxies under the protocol used with AWP (T. Do et al., 2023), with 95% bootstrap intervals for every coefficient and paired intervals for every contrast (Efron, 1979). Scores frozen before a label window are then tested against the new approval pairs and new transfer senders that spender contracts gain in it, beside the raw degrees that PageRank smooths, in W0 and again in W1, whose cohort definitions and labels were registered before its logs were extracted and are applied in the revised analysis to the five tokens. The design separates same-window agreement from prediction and the propagated score from its local count. The degree contrasts belong to no fixed rule; on the revised scores they were computed after the labels of both windows were known. On these data the two walks differ: the transfer walk ranked later senders above the raw transfer count in both windows, and the allowance walk with uniform restarts did not rank later approvals above the raw approval count in either (Claim C4).
- Closed form of EndorseRank at depth one, and evidence of propagation: with uniform restarts and uniformly redistributed dangling mass, a spender whose owners receive no approvals themselves scores $c\,(1 + d\,\delta(v))$, where c is the common score of nodes without in-edges, d

the damping factor and $\delta(v)$ the owner-normalized in-strength, which divides each owner's endorsement over all the spenders it approves (Section 5.3.2). On the cohort's allowance graph the formula holds within 1% for every depth-one spender but for only a minority of the spenders with a deeper in-neighbourhood, so multi-hop propagation is present in this graph. It did not, however, make EndorseRank rank later approvals better than the raw count.

- Coupled operator and its tests: C-PR blends an allowance layer and a transfer layer, both weighted by US-dollar amounts without the bounded transform and with each allowance capped, node by node with a fixed weight λ, and S-PR is a seeded ablation. Rule A, fixed with λ in the analysis plan before any log was extracted, asked C-PR to do no worse than the allowance layer and better than the transfer layer, each walked alone; rule B, registered before the logs of W1 were extracted, asked it to improve on the transfer layer walked alone. Applied to the revised scores after the registered results were known, both rules fail, as they did in the registered analysis; these verdicts are a re-analysis, not registered tests. The gain over the transfer walk on new approval pairs nevertheless replicated in W1 (Claim C5). The ablation, in contrasts that were not fixed in advance, shows that the allowance signal survives when allowance edges carry the walk and is lost when they only decide where it restarts.
- Comparison on an outcome built from neither edge type, under registered rules: allowance-side and transfer-side scores are set against each other on the later liquidation avoidance of GMX V2 contract traders, a label that enters neither graph, with the decision contrasts, their intervals and the wording of each conclusion registered before the July–September logs were extracted. On the revised scores, in the primary window and as a re-analysis, the registered wording gives the conclusions that there is no count-level lead of allowance over transfer on later liquidation avoidance and that the PageRank versions are not resolved; on the profitable and non-loss shares the approval count falls below the transfer count (Claim C6). The registered analysis reached the same conclusions (Appendix 8.6.10). The trader samples are small.

1.10 Scope and limitations

The study covers Arbitrum One. Its observation window runs for 33 months, from 1 October 2023 to 30 June 2026, and the registered window W1 adds the quarter from 1 July to 30 September 2026. Its main sample is the $n = 15{,}107$ contracts that received a non-zero approval of WETH, USDC, USDT0, ARB or WBTC from at least three distinct owners in the observation window, so the results apply to contracts of this kind on Arbitrum One in these months, and not to contracts approved by fewer owners or only on other tokens. The holdout samples are the spender contracts with a positive latest allowance of one of these tokens at each freeze ($n = 17{,}050$ in W0 and $n = 18{,}152$ in W1). Account type comes from `eth_getCode` at blocks pinned within three hours on 9 October 2026 (Section 3.2), so an address is classified by its code at the block at which it was read, not by its code at the time of each event. Only events with a cohort contract at one end are extracted, so each graph is the ego network of the cohort, and a path longer than one edge is seen only where every edge on it touches a cohort contract; multi-hop propagation is therefore measured on a truncated graph (Section 5.3.2). Only the events of the five tokens enter the graphs, and native ether only as WETH; Table 8.4 counts the rows of other tokens that are left out (Appendix 8.4.2). The comparison on outcomes built from neither edge type uses the GMX V2 accounts that are contracts, and few of them traded in these months ($n = 88$ in W0 and $n = 70$ in W1), so every trader result has wide intervals (Section 4.6.3). Runtime scaling is measured on subsamples of the cohort and does not test graphs larger than the cohort's ego network.

Apart from the daily token prices and the token list drawn from Chainlink's directory of price feeds (Chainlink, n.d.), the scores use no off-chain information: no audit reports, no curated lists of contracts and no identity credentials of the kind Human Passport collects. No score is tested as a detector of exploits, rug pulls or phishing, or against loan defaults (Section 2.8), and no claim is made that the results generalize beyond Arbitrum One or the months studied. Amounts are in US dollars at the daily price of their token, a choice made after the registered results were known (Section 3.5.1). EndorseRank and AWP pass them through a bounded function that gives the median

amount half weight, so a larger approval or transfer counts for more up to a ceiling, and an unlimited allowance counts as much as any large one; the layers of C-PR use the amounts themselves, with each allowance capped so that unlimited approvals do not take over the rows of the owners that hold them (Chapter 3). Adversarial manipulation of allowances or transfers is a real threat, because addresses are cheap to create (Douceur, 2002). Chapter 5 shows that a small farm of Sybil owner addresses can lift a contract's EndorseRank above most of the cohort at little cost, while reaching the top of the cohort is expensive (Section 5.6); keeping five tokens does not raise this cost, because an approval needs no balance of the token. The Sybil-adjusted proxies describe some manipulation patterns without protecting the score against them.

1.11 Research roadmap

Chapter 2 reviews graph-based on-chain reputation, ERC-20 allowance semantics and the reputation tooling deployed in DeFi, and develops the theoretical framework. Chapter 3 defines the cohorts, the tokens, prices and revised amounts, the graphs and scores, the construct checks, the temporal holdouts W0 and W1, the timing protocol, and the standards for reproducibility and ethics. Chapter 4 reports the revised analysis: the benchmark (Claim C1), the same-window alignment of every score (C2), rank divergence (C3), checks of damping, value slope, sample definition and restarts, the temporal holdouts with the registered window (C4 and C5) and the comparison on outcomes built from neither edge type (C6), with the rules fixed for the registered analysis applied as a re-analysis; Appendix 8.5 collects the direct comparisons between EndorseRank and AWP, and Appendix 8.6.10 sets the registered analysis beside the revised one. Chapter 5 interprets the findings by research question and objective, works out their implications, models the cost of Sybil attacks on allowance graphs, and weighs threats to validity and limitations. Chapter 6 answers RQ1–RQ5, states the contribution to knowledge and sets out future work.

2 Literature Review and Theoretical Framework

The research problem set out in Chapter 1 is that transfer-graph reputation scores give no role to spending authorization, an explicit act of trust that every token following the ERC-20 standard records, and that their graphs grow with every payment. The objectives O1–O5 and research questions RQ1–RQ5 follow from it. The addresses ranked are smart contracts: the routers, vaults, aggregators, lending pools and other contracts that owners approve to spend their tokens. To compare EndorseRank with Adaptive Weighted PageRank (AWP) on these contracts, we run each method's graph through PageRank and check the resulting rankings with rank correlations. The framework combines hyperlink–citation propagation, domain alignment and computational complexity. The gap that remains is taken up in Chapters 3 and 4.

2.1 Blockchain and DeFi foundations

A public blockchain records transactions in hashed blocks, and each block is secured by the ones added after it (Nakamoto, 2008). Once a majority of the participating nodes has accepted a block, rewriting it is costly. That cost allows parties to settle on the ledger without each of them needing a known legal identity (*trust-minimized* settlement).

Bitcoin built a peer-to-peer electronic cash network on this kind of ledger (Nakamoto, 2008). Ethereum keeps the append-only ledger but lets an account hold a program alongside its balance. A transaction can then call one of the program's functions and change the contract's state (Buterin, 2014), and these functions run on the Ethereum Virtual Machine (EVM) (Wood, 2025). Programs stored on the ledger in this way are called smart contracts, a term that Szabo (1997) used for contract terms embedded in computer protocols. Execution on the EVM is deterministic, so any honest node that replays a function call gets the same output (Wood, 2025). As a result,

collateralization ratios, liquidation conditions and fee schedules can be enforced by the contract instead of by a trusted third party (Schär, 2021). The contracts that users let act on their tokens are the objects this thesis ranks.

Collectively, these contracts make up Decentralized Finance (DeFi). Between them they do the work of exchanges, lending platforms, derivatives markets and asset managers; Werner et al. (2022) define the protocol classes, and Schär (2021) describes how protocols are layered on one another. Because one protocol can interact with another, new financial instruments can be composed from existing ones (Kitzler et al., 2023). In such compositions contracts call other contracts, and a contract can be the owner of an allowance as well as its spender, since the owner is simply the account that grants it (Vogelsteller & Buterin, 2015). The same property lets a failure spread across several protocols. Successful attacks have included flash-loan exploits (Qin et al., 2021) and price-oracle manipulation (Angeris & Chitra, 2020). Transaction histories on these chains are public, yet the addresses in them are pseudonymous, which makes the identity checks of conventional finance hard to carry out (D. Lin et al., 2025).

2.1.1 Pseudonymity and the credit problem

Conventional credit scoring ties a credit history to a legal entity. On a public blockchain the basic unit of analysis is instead an address. It may be an externally owned account controlled by a private key or a contract governed by its code, and it may stand for a legal entity, a trust, an algorithmic agent or a protocol. One entity can own several addresses (Douceur, 2002), and address-clustering heuristics can recover ownership only in part (Victor, 2020). On-chain lending therefore requires borrowers to post collateral worth more than the amount they borrow; if the collateral value falls below a set threshold, a liquidator may repay part of the debt and receive the corresponding collateral at a discount (Bartoletti et al., 2021). The arrangement is secure, but it ties up capital. An on-chain credit score tries to quantify from on-chain behavior alone how likely an actor is to repay. Such scores draw on prospect theory (Hassija et al., 2020) and on machine learning applied to lending data (Kotb, 2023), and a related line of work rates the fraud risk of Ethereum accounts by propagat-

ing it through the transaction network (D. Lin et al., 2025). More recent work predicts liquidations from DeFi transaction features gathered on several chains (Palaiokrassas et al., 2024), and a credit score for DeFi lending combines account-level financial features with relational features from a weighted transaction graph (M et al., 2026).

All of these scores rank borrowers. A contract does not borrow, but the same lack of identity arises when an owner decides whether to let a contract spend its tokens. The contract's code is public, while who deployed it, who can change it and how it will use the allowance may not be. A reputation score for contracts asks what the on-chain record of other owners' decisions says about a given contract.

2.2 Ethereum, Layer-2 rollups, and Arbitrum One

Most DeFi applications were first deployed on the Ethereum mainnet. While Ethereum ran on proof of work, it could not raise its block size or shorten its block interval far without weakening its security (Gervais et al., 2016), so growing use of the mainnet raised gas costs. Layer-2 (L2) protocols try to keep Ethereum's security guarantees while making transactions cheaper. Some execute transactions off-chain, while others batch them and post a succinct summary of the execution to the Ethereum mainnet (Gudgeon, Moreno-Sanchez, et al., 2020). Thibault et al. (2022) survey the protocols of the second kind, called rollups, and compare their implementations. Optimistic L2s such as Arbitrum treat each submitted batch of transactions as valid unless evidence against it is presented within a dispute period (Kalodner et al., 2018).

Arbitrum One is an optimistic L2 of this kind and can execute EVM-based smart contracts. It has become a popular platform for DeFi applications, among them GMX (n.d.), an exchange for perpetual swaps. Contracts on Arbitrum do not simply behave as they would on Ethereum mainnet: a study of contracts migrated from Ethereum to Arbitrum traces their security risks to differences in cross-chain messaging, block properties, contract address aliasing and gas fees (Tang & Shi, 2024). We use Arbitrum One because the public BigQuery dataset includes both its ERC-20 `Approval`

and `Transfer` events, and because its fees are low enough that the observation window contains dense activity around the contracts studied. We make no claim that our results generalize to other L2s or to Ethereum mainnet without further study. The framework (approval and transfer graphs, construct checks and rank correlation) can, however, be applied to any chain that emits the same logs.

2.3 ERC-20 tokens and the allowance lifecycle

The ERC-20 standard defines a common interface for fungible tokens, made up of `transfer`, `approve`, `transferFrom`, `allowance` and the associated events (Vogelsteller & Buterin, 2015). After the owner calls `approve(spender,amount)`, the token contract records that the spender may move up to `amount` out of the owner's balance. The spender can then call `transferFrom` to carry out a token transfer. With EIP-2612 `permit`, an allowance can be updated through an off-chain signature that settles in a single transaction (Lundfall, 2020). The user experience improves, and the underlying semantics of authorization are unchanged. The owner is whichever account grants the allowance, an EOA or a contract, and the spender can be any address; in this thesis the spenders that are scored are contracts.

Figure 2.1 traces an ERC-20 allowance through its lifecycle. An allowance is an authorization and involves no actual transfer. Token owners can grant allowances to routers, vaults, aggregators, lending pools or trading contracts that they trust to act within set limits. Revoking (approving zero) and updating (setting a new non-zero allowance) both change the allowance currently in force. EndorseRank accounts for this by building each owner–spender edge from the most recent allowance on each of the five tokens it uses (Section 3.5.1), so a newer allowance replaces the previous one outright; old values are not averaged in or left to decay gradually.

Allowances are sparse compared with transfers. A single approval can cover many transfers, and many accounts never grant an approval to any other address. The sparsity lowers the cost of a PageRank solve, as a consequence of the edge count (Section 2.5.2), and it also marks a semantic

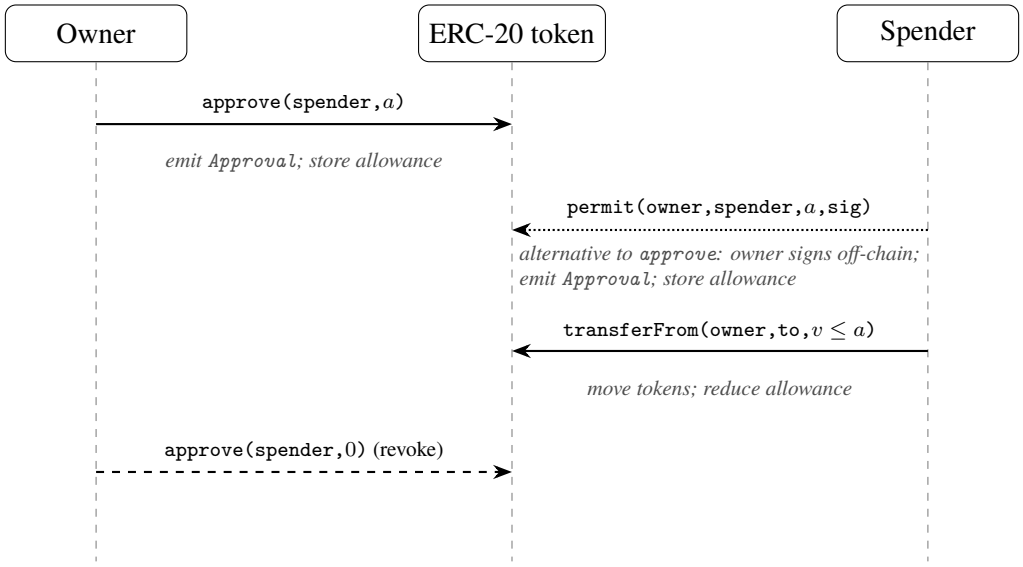


Figure 2.1: ERC-20 allowance lifecycle: `approve`, or its signature-based alternative `permit` (dotted), an optional `transferFrom`, and revoke. An EndorseRank edge sums, over the five tokens of the study, one term for the most recent `Approval` event of its owner–spender pair in the observation window, and the term weights that approval by its age and its US-dollar amount as AWP weights a transfer. A later approval replaces an earlier one.

difference, since EndorseRank quantifies authorization to spend whereas a transfer graph records payment flows.

2.4 Graph-based on-chain reputation

PageRank and its variants rank nodes by the influence that flows to them along directed edges (Page et al., 1999). Two recent studies apply this approach to blockchain transaction graphs, and the two should be distinguished. H.-D. Do and Do (2023) run PageRank and HodgeRank on Ethereum transactions as a measure of social credit, meant for people who lack a credit history at financial institutions. T. Do et al. (2023) propose Adaptive Weighted PageRank (AWP), the transfer-graph method that this thesis evaluates beside EndorseRank on the same contracts (Section 2.18). D. Lin et al. (2025) treat the risk rating of Ethereum accounts as a node-ranking problem and propagate risk through the transaction network, so that accounts without labels are rated as well, and Wu et al. (2022) detect phishing addresses from a network embedding of the transaction graph that takes the

amounts and times of transactions into account. Random walks have also been used to detect scam behavior on Ethereum from a graph of interactions between addresses (Yan et al., 2023), and a graph convolutional network labels the business role of Ethereum accounts from their features and their position in the transaction network (D. Lin et al., 2024). Bellini et al. (2020) survey blockchain-based trust and reputation management, and Almasoud et al. (2020) review reputation systems built on smart contracts, in which users rate how trustworthy the providers of online services or products are. Data mining on cryptocurrency transactions likewise works from transaction histories, for address linking and for the study of user behavior (X. F. Liu et al., 2021). Packin and Lev-Aretz (2024) warn that decentralized credit scores can turn biased decisions into automated processes that outsiders cannot inspect.

Outside the research literature, trust in deployed DeFi and web3 systems is usually established another way. An address builds standing through activity that anyone can inspect on the ledger, and tooling bound to the address summarizes it. One such tool is Human Passport, formerly Gitcoin Passport (Human Passport, n.d.). It collects verifiable credentials, called stamps, for checks such as government identity, biometrics, a web of trust and earlier web2 and web3 activity. Each stamp carries a weight that the documentation describes as the strength of its human signal, and an address passes when the sum of its weights reaches a threshold. According to the same documentation, Passport also classifies EVM addresses on several networks, Arbitrum among them, as human or Sybil from their on-chain activity. Its purpose is Sybil resistance, so it combines on-chain with off-chain evidence and works as a gate and not as a graded ordering. A review of proof-of-personhood protocols describes how they resist Sybil attacks with subjective inputs such as vouching and voting in place of attestations from central issuers (Siddarth et al., 2020), and a recent review of Sybil attacks on blockchains traces airdrop farming on layer-2 networks such as Arbitrum and assesses proof-of-personhood systems among the defenses (Bordeianu & Popescu, 2026).

Such tools answer whether an address is controlled by a distinct person. That is a question about the holder of a key, and it has no direct counterpart for a contract, whose behavior is fixed by its code and by whoever may upgrade it. Trust in a contract is formed in other ways. An owner usually

decides whether to trust a spender contract at the moment its wallet software presents the approval for signature, and approval phishing exploits that moment (Section 2.17); code analysis judges the contract by its functions (Section 2.14). EndorseRank answers a different question at the level of the contract: which contracts other addresses authorize to spend their tokens, and how strongly those addresses are endorsed in turn. It orders the contracts from on-chain records, with amounts valued at public daily token prices, and certifies nothing about the people behind them (Table 2.1). Graph-based methods need only public information, their scores are continuous and so can order every address, and their cost can be stated in terms of the edge count (Section 2.5.2). They are, however, vulnerable to Sybil attacks (Douceur, 2002) and wash trading, and graph centrality is neither the safety of a contract nor the default risk of an account; PageRank scores are not credit scores in the regulatory sense, and they are not security ratings (Section 2.8).

A propagated score must also beat a simpler one. Where degree correlations are weak, as on the Web graph, PageRank is on average proportional to in-degree and can be approximated from it (Fortunato et al., 2008). New links concentrate on nodes that already have many, as preferential attachment assumes (Barabási & Albert, 1999) and measured network growth shows (Newman, 2001), so link prediction uses degree-based predictors as standard baselines (Liben-Nowell & Kleinberg, 2007). On Ethereum, the frequency and the amount of past transactions both bear on which new transactions appear (D. Lin et al., 2022).

2.5 PageRank formalism

Consider a directed graph $G = (V, E)$ with $|V|$ nodes, and write $w_{ij} > 0$ for the weight of the edge from i to j. That weight stands for influence from i to j, in either the endorsement or the transfer sense introduced above. Let W be the corresponding weighted adjacency matrix, and let

$o_i = \sum_j w_{ij}$ be the out-strength of node i. The row-stochastic transition matrix P is then

$$P_{ij} = \begin{cases} w_{ij}/o_i & \text{if } o_i > 0, \\ 1/|V| & \text{if } o_i = 0 \end{cases} \tag{2.1}$$

(dangling nodes spread their mass evenly). Given a damping parameter $d \in (0, 1)$, the Google matrix is

$$G_{\text{PR}} = d\, P^\top + \frac{1-d}{|V|}\, \mathbf{1}\mathbf{1}^\top, \tag{2.2}$$

and the PageRank vector $\mathbf{r}$ satisfies $\mathbf{r} = G_{\text{PR}}\mathbf{r}$ with $\mathbf{r} \geq 0$ and $\sum_i r_i = 1$ (Langville & Meyer, 2006). Element by element, with the dangling rows of P written as a separate middle term, this becomes

$$r_j = d \sum_{i:(i\to j)\in E} r_i \frac{w_{ij}}{o_i} + \frac{d}{|V|} \sum_{i:\, o_i = 0} r_i + \frac{1-d}{|V|}. \tag{2.3}$$

2.5.1 Random-surfer interpretation

Equation (2.3) can be read probabilistically. At each step a surfer at a node with out-edges follows one of them with probability d, choosing it in proportion to its weight, or teleports to a uniformly chosen node with probability $1 - d$ (Figure 2.2). At a node without out-edges the surfer always jumps to a uniformly chosen node; this is the dangling rule of Equation (2.1). The stationary distribution then ranks nodes by how often they are visited in the long run. Damping keeps rank sinks (groups of nodes with no edge leaving them) from absorbing all the mass. With $d < 1$ and uniform teleportation, every entry of G_{PR} is positive, so G_{PR} is primitive and its Perron vector is unique (Langville & Meyer, 2006). Appendix 8.3.1 gives the argument.

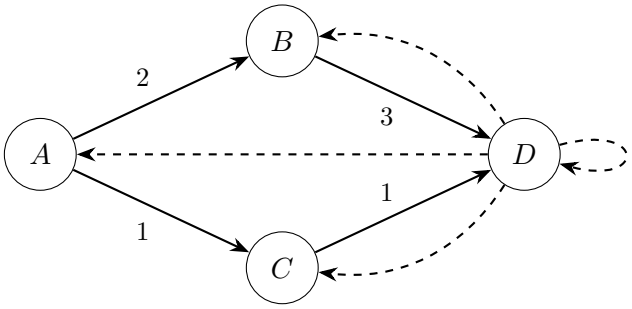


Figure 2.2: Random-surfer intuition for PageRank on the four-node graph of Section 2.13; edge labels are the allowance weights. Solid arrows are graph transitions: with total probability d the surfer follows one of them, chosen in proportion to its weight. Dashed arrows show uniform teleportation, which reaches every node (the current one included) with probability $(1-d)/|V|$; the dashed fan is drawn only from D, although every node has one. D has no outgoing edge, so under the dangling rule of Equation (2.1) a surfer at D always jumps to a uniformly chosen node, each with probability $1/|V| = 0.25$. That rule, and not damping, keeps D from absorbing the chain.

2.5.2 Power iteration and complexity

In practice, $\mathbf{r}$ is usually obtained by power iteration from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/|V|$,

$$\mathbf{r}^{(t+1)} = G_{\text{PR}}\mathbf{r}^{(t)}, \tag{2.4}$$

until $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or until the iteration cap $I_{\max}$ is reached. On a sparse graph, each iteration costs $O(|E|)$ arithmetic operations (Leskovec et al., 2020). A method that generates fewer edges, for instance by using latest-allowance snapshots instead of full transfer histories, therefore lowers the cost per iteration and sometimes the memory required. Hypothesis H3 is motivated by this complexity fact.

By default this work sets damping to the conventional $d = 0.85$ (Page et al., 1999). The convergence tolerance ε and the iteration cap $I_{\max}$ are study-specific solver settings, given with their values and the reason for them in Section 3.1.[1] The damping sensitivity checks of Section 4.5 vary d.

[1] They are not put forward as part of the classical PageRank formulation.

2.6 Reputation systems lineage

PageRank belongs to a wider family of spectral and propagation-based reputation mechanisms. The HITS algorithm of Kleinberg (1999) assigns each page two scores, one as a hub and one as an authority. EigenTrust extends the idea to peer-to-peer networks and normalizes local trust, so a malicious node cannot raise its own score at will (Kamvar et al., 2003). To suppress web spam, TrustRank starts from a seed set of trusted nodes and propagates their trust outward (Gyöngyi et al., 2004). Reputation is relational in all of these methods. Where trust flows only from a vetted seed set, as in TrustRank, a newcomer cannot raise its score without endorsements from nodes that already hold reputation. With uniform teleportation, which EndorseRank uses (Section 3.6.2), that protection is weaker. Gyöngyi and Garcia-Molina (2005) derive the link farms and farm alliances that most raise a target's PageRank, and Cheng and Friedman (2005) show that symmetric reputation functions such as PageRank can be manipulated with Sybils, whereas some asymmetric functions based on flows or paths from a trusted node resist them. The Sybil cost model of Section 5.6 prices such a farm on an allowance graph.

On chain, the Sybil attack (Douceur, 2002) is the main concern for a graph score. New addresses cost little to create, whereas addresses that receive allowances or transfers from distinct counterparties cost more. For a spender contract the cheap route is plain: whoever deploys it can create owner addresses and have each of them approve it, paying only gas. Coordinated multi-address behavior does leave traces in the transaction graph. Graph learning on NFT transactions identifies airdrop hunters who amass many addresses (C. Zhou et al., 2024), and features of each address's transaction subgraph, among them the timing of its first transaction and of its first gas funding, separate Sybil addresses in token airdrops (Q. Liu et al., 2025). The Sybil-adjusted stability proxies used in this work (inbound counterparty ratio, tenure, active months) partly capture that diversity of interaction, but they do not prevent a malicious actor from manipulating them, and no Sybil detector is applied here.

Machine-learning credit models fitted to DeFi lending data (Kotb, 2023) learn from labeled credit

outcomes, and an ensemble of fitted models is harder to audit than a fixed rule. EndorseRank and AWP apply fixed propagation rules that anyone can recompute from the public logs and a public series of daily token prices (Sections 2.4 and 3.5.1).

2.7 ERC-20 allowances as endorsement

Through the ERC-20 `approve` function (Vogelsteller & Buterin, 2015) and EIP-2612 `permit` (Lundfall, 2020), owners state explicitly how much a spender may move on their behalf. EndorseRank takes the latest allowance from an owner to a spender on each of the five tokens of the study (Section 3.5.1) as one endorsement, weights its US-dollar value as AWP weights a transfer, and runs PageRank on the resulting directed graph. Newer approvals supersede older ones, so each pair and token contributes a single event. The endorsed node is the spender contract, and the endorsing node is the owner, which may be an EOA or another contract.

Concretely, for each owner–spender pair the latest in-window approval on each of these tokens contributes its time weight times the bounded value weight of its US-dollar amount, as AWP weights a transfer; a token on which the pair's latest value is zero (a revocation) contributes no term, an unlimited allowance takes its full time weight, and the terms are summed over the tokens (Equation (3.7) in Section 3.6.2). The resulting graph G_{approve} is sparser than the transfer graph over the same contracts and window (Chapter 4 gives both edge counts), and PageRank on it yields the EndorseRank scores $\mathbf{s}_{\text{ER}}$ of the cohort contracts.

Treating allowances as endorsements is a deliberate choice, and an imprecise one. Owners approve routers because it is convenient, not necessarily to make a social statement about whether the router is trustworthy. Contract addresses and EOAs coexist on the owner side as well. A contract that approves another contract does so because its code says so, and the approval records a decision its developers made in advance, not a judgement made when the event was emitted. Even so, an allowance signals approval more explicitly than a received transfer, which could come from market making, wash trading (Cong et al., 2023), airdrop farming (Luo et al., 2025) or routing without any

intent to endorse. This holds only for a token contract that follows the standard: a spam or address-poisoning token can emit an `Approval` log that names an owner who never approved anything, which is one reason the study keeps only five widely traded tokens (Section 3.5.1).

Approvals have been read as a relation between addresses before: the self-authorization heuristic of Victor (2020) uses ERC-20 approvals to cluster addresses that it attributes to one entity. That reading links addresses and ranks nothing. It is also a caution, since an approval between two addresses of one entity is a self-endorsement, the edge that the Sybil cost model of Section 5.6 prices at gas alone.

2.8 What this research does not claim

This thesis ranks contracts by how other addresses relate to them on chain, and its claims stop at that ordering. No score is offered as a predictor of credit default or of any borrower's repayment, and no score is tested against loan defaults. No score is a security audit: a contract can contain malicious functions (Z. Lin et al., 2024) whatever its standing among owners, and the scores read neither code nor upgrade permissions. No score measures trading success. Where GMX V2 trader contracts enter the evaluation (Sections 3.7.1 and 3.7.2), their outcomes serve as out-of-window labels, not as a target that any score was built to predict. Agreement of a score with checks built from its own edges is an intended-construct check (Cronbach & Meehl, 1955), so part of it is built in, and the holdout coefficients are associations over time. None of the results supports a causal claim, for instance that approvals make a contract safer or that a high score brings a contract later approvals.

2.9 Rank correlation theory

Reputation scores and the proxies they are checked against are both continuous variables, and neither has a natural boundary for classification. Following the evaluation design of AWP (T. Do et al., 2023) and classical psychometrics (Cronbach & Meehl, 1955), this research measures two kinds

of rank agreement, monotonic and pairwise. Bootstrap confidence intervals quantify the sampling uncertainty of every coefficient of the main analysis (Efron, 1979).

Spearman's rank correlation (Spearman, 1904) is the Pearson correlation of the ranks. For paired ranks (R_i, S_i) of n items without ties, it reduces to

$$\rho = 1 - \frac{6 \sum_{i=1}^{n} (R_i - S_i)^2}{n(n^2 - 1)}. \tag{2.5}$$

This no-ties form does not hold when ties occur. Spearman's ρ is then the Pearson correlation of average ranks (mid-ranks), in which tied items share the mean of the positions they occupy, and every reported ρ is computed this way. In both forms ρ measures monotonic association.

Kendall's τ (Kendall, 1938) rests on counts of concordant and discordant pairs. Of the $n_0 = n(n-1)/2$ pairs (i, j) with $i < j$, n_c are concordant (both rankings order them the same way) and n_d are discordant (the rankings order them oppositely); a pair tied in either ranking is neither. Let $n_1 = \sum_k t_k(t_k - 1)/2$, where t_k is the size of the k-th group of tied values in the first ranking, and define n_2 in the same way for the second ranking. The τ-b variant is then

$$\tau_b = \frac{n_c - n_d}{\sqrt{(n_0 - n_1)(n_0 - n_2)}}, \tag{2.6}$$

and without the tie corrections it becomes $\tau_a = (n_c - n_d)/n_0$. Every τ reported in this thesis is τ_b, computed on the raw scores and proxies. The difference matters for heavily tied scores, such as the raw degrees, which are integer counts, and any PageRank that gives many contracts the same value (Section 4.4). For agreement about pairwise orderings, Kendall's τ is usually the easier coefficient to interpret, and Chapter 4 uses it as the main family-level summary. Its values are compared with those of other methods on the same cohort; no uncited absolute benchmark is used.

2.10 Theoretical framework

This study compares EndorseRank and AWP on reputation structure (H1), construct alignment (H2) and computational cost (H3), and then on prediction out of window (RQ4) and on coupling the two layers (RQ5). Three bodies of theory frame the first part: hyperlink–citation propagation, construct (domain) alignment, and computational complexity. Two further arguments bear on the second part.

- Hyperlink–citation semantics: In the original PageRank algorithm (Page et al., 1999), an inbound hyperlink counts as a citation. H.-D. Do and Do (2023) apply this reasoning to Ethereum transfers, and AWP (T. Do et al., 2023) weights each transfer edge by the age and the value of its transfers, through a logistic decay in age and a bounded transform of value (Section 3.6.1); this study sets the slope b of that transform from the median US-dollar amount (Section 3.5.1). EndorseRank extends citation-based propagation to ERC-20 approve and permit edges, in which the cited node is a spender contract and the citing nodes are the owners that approve it. An allowance is an explicit, on-chain delegation of spending authority that remains in force until it is revoked or replaced (Vogelsteller & Buterin, 2015), whereas a transfer may reflect trading or routing without any intent to trust. Risk propagation on Ethereum rates accounts that address clustering and illicit-account classification leave unassessed (D. Lin et al., 2025), so network structure can carry information that per-account methods miss, which argues for a directed endorsement graph alongside transfer volume.
- Domain alignment framework: Measurement theory uses construct alignment to ask how closely an indicator follows the construct it is meant to track (Cronbach & Meehl, 1955). Messick (1995) widens the idea into one unified judgment about what a score means and what using it implies. With rank-based reputation scores, Spearman's ρ and Kendall's τ measure monotonic and pairwise agreement in ordering and need no classification threshold, as in the evaluation of AWP (T. Do et al., 2023). Graph coherence is checked with contemporaneous transfer and allowance statistics of each contract. The out-of-window checks are temporal

holdouts: new approval pairs and new transfer senders for spender contracts, and outcome labels built from neither edge type for the contracts that trade on GMX V2, among them the liquidation-free close rate, whose definition the registration of W1 fixed (Sections 3.7.1 and 3.7.2).

- Computational complexity: Each PageRank iteration costs $O(|E|)$ (Section 2.5.2). Allowance graphs are sparser than transfer graphs, with fewer edges $|E|$, so an iteration needs less time and memory. The number of iterations is a separate factor, whose rate of convergence is governed by the damping factor (Langville & Meyer, 2006), so a sparser graph is not expected to need fewer iterations. This argument underlies H3. Both methods use the same solver, so differences in cost come from the graphs and from the restart vectors, which can change the number of iterations (Chapter 4 reports the edges and iterations of each solve); a speed that comes from a sparser graph does not count as an algorithmic contribution.
- Local counts as a baseline: Where contracts that already have many counterparties gain the most new ones (Section 2.4), the raw degree can already rank future arrivals, and a propagated score can beat it only when who the endorsers are matters more than how many. RQ4 therefore sets each PageRank against its own raw degree at the freeze date.
- Layer coupling: Multiplex PageRank (Halu et al., 2013) and multilayer ranking (De Domenico et al., 2015) treat several relations as layers over one node set. Mixing the layers node by node, as C-PR does, leaves a question open for layers of very different density: a node with no edge in the sparse layer follows the dense one, so the signal of the sparse layer may or may not survive. RQ5 therefore asks two things: whether the mixture improves on the transfer walk, and whether it beats both layers.

The conceptual model in Figure 2.3 links the scores of the thesis to H1–H3 and RQ1–RQ4. RQ5 is the dashed box on the right. It takes input from H2/RQ2 and from RQ4 and asks whether the two constructs combine.

Only EndorseRank and AWP enter H1. H2/RQ2 and H3/RQ3 concern every score, and RQ4 is

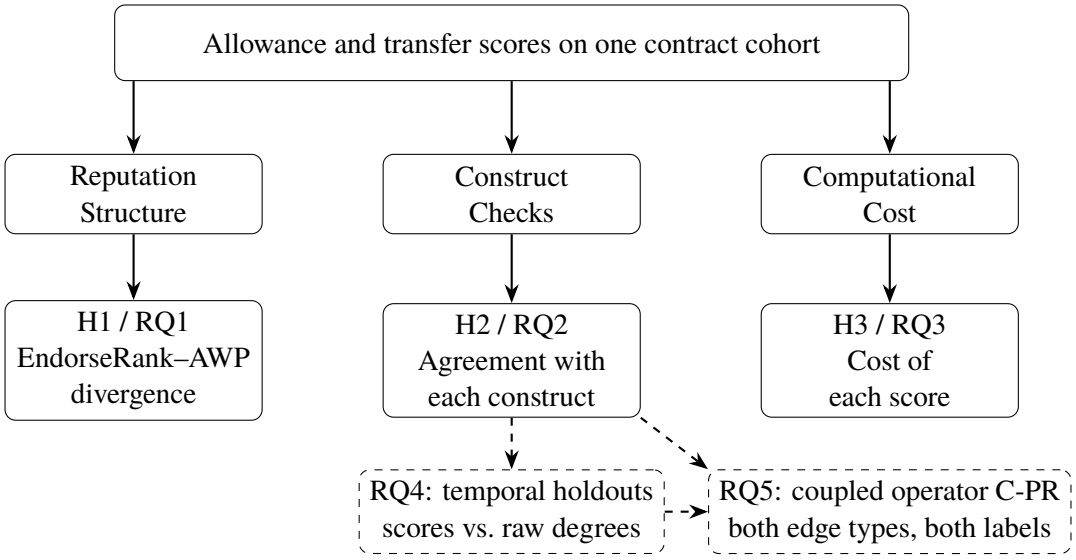


Figure 2.3: Conceptual model linking the allowance and transfer scores of one contract cohort to reputation structure (H1/RQ1), construct checks (H2/RQ2), computational cost (H3/RQ3) and the temporal holdout, in which each PageRank is set against the raw degree it smooths (RQ4). The dashed box on the right is RQ5, the coupled operator C-PR.

a time-split test on the spender contracts, with the raw degrees at the freeze as its baselines. The contracts that trade on GMX V2 are evaluated on outcomes built from neither edge type: profit labels in an exploratory W0 run, and the liquidation-free close rate in a comparison whose rules were registered before W1 was extracted and are applied to the revised scores after the registered results were known (Sections 3.7.1, 3.7.2 and 3.5.1).

2.11 Terminology

The formal constructs are defined in Chapter 3 (Sections 3.1 and 3.6). The terms below are used when results are interpreted:

- Single-layer methods: EndorseRank (allowance graph) and AWP (transfer graph), the two PageRank scores that walk on one edge type. The coupled walks of Section 3.6.3 and the raw degrees are evaluated beside them.
- Primary graph construct: the graph that a method is *designed* to rank, for example G_{approve} for

EndorseRank, as distinct from the checks applied after scoring.

- Contemporaneous check: a local transfer or allowance statistic taken from the same window as the score (Cronbach & Meehl, 1955).
- Temporal holdout: scores are frozen at t_1, and new owner–spender approval pairs and new transfer senders of the five tokens are counted from t_1 up to t_2 (Section 3.7); a second window, W1, registered before its labels were extracted, repeats the design with a later freeze and labels counted after the observation window ends (Section 3.7.2). Appendix 8.6.10 sets the decision contrasts of the registered analysis beside those of the revised one.
- Matched contract cohort, spender holdout cohort, trader cohort and benchmark subsamples: the address sets of Section 4.1.1. Every scored and evaluated address in them is a contract; EOAs enter the graphs only as owners, senders and recipients.

2.12 Domain alignment metrics

Construct alignment is measured by Spearman's ρ and Kendall's τ between scores and check rankings (Section 2.10), on contemporaneous transfer and allowance checks and on the holdout labels; Chapter 3 gives the operational definitions. Each coefficient of the main analysis is reported with a 95% bootstrap interval, and two scores are compared through the paired difference of their coefficients on the same resamples (Efron, 1979); the value-slope check and the sample-definition stages of Section 4.5 are reported without intervals.

2.13 Worked example: PageRank on a four-node endorsement graph

Take four addresses $\{A, B, C, D\}$ joined by the latest-allowance edges

$$A \to B\ (2), \quad A \to C\ (1), \quad B \to D\ (3), \quad C \to D\ (1),$$

with damping $d = 0.85$. One reading is that A is an owner's EOA, B and C are two vault contracts that A has approved, and D is a lending pool that both vaults approve in turn so that it can pull the tokens they supply. The out-strengths are $o_A = 3$, $o_B = 3$, $o_C = 1$ and $o_D = 0$ (the last is dangling). The edges $B \to D$ and $C \to D$ are the only out-edges of their nodes, so each is followed with probability one and its weight has no effect on the scores. Teleportation adds $(1 - d)/4 = 0.0375$ to every node at every step, the first one included. One power-iteration step from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/4$ gives $\mathbf{r}^{(1)} = (0.0906,\ 0.2323,\ 0.1615,\ 0.5156)$ for (A, B, C, D). The mass already tilts toward D, since both B and C endorse D, while A splits its mass between B and C. The iteration converges to the stationary vector $\mathbf{r} = (0.1375,\ 0.2154,\ 0.1765,\ 0.4706)$, so D comes first, B and C follow, and A is last. No address endorses A. Apart from teleportation, its whole score comes from the dangling rule, which spreads D's mass evenly over the four nodes ($d\, r_D/4 = 0.1000$). The example shows that a node gains score by *receiving* endorsements from nodes that are endorsed in turn, which is the citation logic of PageRank (Page et al., 1999).

Suppose the same addresses were linked by transfers instead of allowances. The transfers would in general join other pairs, or the same pairs with other per-pair totals. That gives another transition matrix P, and possibly a different ranking of exactly the same nodes. Hypothesis H1 is this situation in miniature: EndorseRank and AWP need not agree.

2.14 DeFi risk taxonomy and where reputation fits

Werner et al. (2022) separate the technical security of DeFi protocols from their economic security, and L. Zhou et al. (2023) place the attacks and accidents seen in practice in one reference frame. Bugs in contract code are smart-contract risks, one of the several levels at which DeFi vulnerabilities arise (W. Li et al., 2022). Token contracts add risks of their own: customizations that break the ERC-20 standard put the DeFi applications that handle those tokens at financial risk, and a large-scale study of Ethereum token contracts finds such behaviors in most of the contracts it examines (Z. Li et al., 2026). Some contracts are hostile by design. In a contract-related rug pull the developers

abandon a token project and take investors' funds through malicious functions written into the contract, and code analysis can flag such functions (Z. Lin et al., 2024). On rollups, risk also arises between layers: a systematization of rollup security separates architectural risks, such as sequencing and data availability, from risks in how applications integrate with the rollup, such as cross-domain messaging (Efimov et al., 2026). A stale price, or a feed under someone's control, is an oracle risk (Angeris & Chitra, 2020). Market risks (Perez et al., 2021) cover volatility and liquidation cascades, including stress in stablecoin and funding markets (Klages-Mundt et al., 2020), and the market crash of March 2020 showed how these two layers can interact (Gudgeon, Perez, et al., 2020). Token-weighted voting belongs to governance risk: in the on-chain governance of Compound, Uniswap and ENS, a small number of addresses hold most of the voting power (Fritsch et al., 2024). We study a narrower layer, *counterparty* or *behavioral* risk, at the level of the contract: whether the approvals and transfers a contract has received mark it as a counterparty that owners trust to spend their tokens.

On-chain reputation scores cover only part of the behavioral risk layer and are no substitute for collateral, auditing, or oracle design. Their purpose is to compress a system's relational structure (who authorizes whom, who pays whom) into an ordinal measure that others can audit and reproduce from public data. Within behavioral risk, EndorseRank looks at authorization and AWP at payment flows. Neither tries to predict exploits, rug pulls, phishing or governance attacks.

Finally, fake activity makes behavioral risk harder to measure. Wash trading inflates volume on decentralized exchanges (Victor & Weintraud, 2021) and, on a larger scale, on unregulated centralized exchanges (Cong et al., 2023). Front-running and related forms of extractable value create transfers that follow ordering strategies instead of relationships (Daian et al., 2020); Qin et al. (2022) estimate how much of that value is captured on Ethereum. Scam tokens on Uniswap (Xia et al., 2021) and cross-exchange arbitrage trades (Makarov & Schoar, 2020) likewise add transactions without forming sustained relationships. The transfer graph is particularly sensitive to fake volume, while the allowance graph is vulnerable to approval farming and to approvals obtained by phishing (Section 2.17). Time-based splits, Sybil detection and a list of widely traded tokens, which keeps out

the logs of spam and address-poisoning tokens (Section 3.5.1), each solve the problem only in part.

2.15 Related measurement traditions

Outside blockchains, reputation systems have been studied in e-commerce (Resnick & Zeckhauser, 2002), in peer-to-peer networks (Kamvar et al., 2003), and in work on web spam (Gyöngyi et al., 2004). Four insights from this work apply here: (i) reputation is a property of relations; (ii) when identities are easy to create, Sybil attacks become possible (Douceur, 2002); (iii) seeding and normalization limit the effect of inflation; and (iv) evaluation needs external benchmarks, because internal coherence alone is not enough (Cronbach & Meehl, 1955). EndorseRank takes up (i) directly and treats (ii) as a threat. It does not take up (iii), since it restarts uniformly (Section 2.6). On (iv) it goes part of the way: the contemporaneous checks, measured by Spearman's ρ (Spearman, 1904) and Kendall's τ (Kendall, 1938), test coherence, while the temporal holdouts test prediction out of window, on the same constructs for spender contracts and on outcomes built from neither edge type for trader contracts.

Personalized PageRank (Jeh & Widom, 2003) and topic-sensitive PageRank (Haveliwala, 2003) bias the teleportation vector toward a chosen group of nodes, the mechanism that TrustRank uses to turn a small seed set into a global ranking (Gyöngyi et al., 2004); AWP's restarts in proportion to sending activity are another instance (T. Do et al., 2023). Multiplex PageRank lets a node's centrality on one layer shape the walk on another (Halu et al., 2013), and the multilayer framework of De Domenico et al. (2015) lets one walker move within and between the layers of an interconnected network. Both treat layers as different relations over one node set, which matches allowances and transfers among the same addresses. C-PR takes the simplest form of this idea, a per-node weighted sum of two transition matrices with one weight, and S-PR borrows the personalized-teleportation idea, with the allowance score as the teleportation vector (Section 3.6.3).

2.16 Accounts, gas, and why logs are the measurement surface

Ethereum has two kinds of account: externally owned accounts (EOAs) and contract accounts. A private key controls an EOA, whereas a contract account is governed by the code it stores (Antonopoulos & Wood, 2018). The difference decides what a score can attach to. An EOA acts when its key holder signs, while a contract acts only as its code allows when another account calls it, so the reputation of a contract belongs to its code and to whoever controls that code, not to a person. In the ERC-20 transfer network the two kinds of account also behave differently: transfers that involve contracts, especially transfers between two contracts, show heavier-tailed and less stable statistical patterns than transfers between EOAs (Mukhia et al., 2026). This thesis scores contracts only, and EOAs enter its graphs as owners, senders and recipients (Section 4.1.1).

Every state-changing transaction consumes gas. Logs come from a contract's `LOG` instructions, which run when its code emits an event (the Solidity `emit` statement); writing to storage produces no log. A reputation system that means to be auditable must take its graph from logs and not from off-chain databases, because logs are a permanent part of the blockchain history that any full node or index provider can recompute independently (Narayanan et al., 2016). Both graphs of this study come from logs; valuing their amounts in US dollars adds a public series of daily token prices kept off the chain, which Appendix 8.4.2 compares with Chainlink's on-chain price feeds.

Gas prices also shape the graphs. Cheaper gas on L2 networks densifies G_{transfer} more than G_{approve}, because transfers recur routinely while an approval is granted once so that later transfers can go through; Chapter 4 reports the resulting graph sizes.

2.17 Approvals as security-relevant events

Unlimited approvals (`type(uint256).max`) save the owner from approving again, at a cost in safety. If the approved contract is compromised, or was malicious from the start, the owner's whole balance of that token can be taken until the owner resets the approval. An allowance is therefore a capability and not a record: whoever holds it can move the owner's tokens until it is spent or

revoked (Vogelsteller & Buterin, 2015). EndorseRank interprets a token approval as endorsing one spender's capability to move those tokens. This is why an allowance does not mean the same thing as a transfer, even though both involve the movement of tokens. A contract could receive many transfers and never be granted permission to move funds on anyone else's behalf.

Wang et al. (2022) find that unlimited approvals make up the majority of ERC-20 approvals on Ethereum and quantify the risk they carry. Unlimited approvals appear as amounts of $2^{256} - 1 \approx 1.16 \times 10^{77}$ base units. EndorseRank's value weight, applied to US-dollar amounts, gives them the full weight of one, which a finite allowance nears only at a few times the median amount. The allowance layer of C-PR counts any allowance up to a cap in US dollars, so an unlimited approval weighs no more than a large finite one; in the registered analysis, whose layer summed raw base units, it took almost all of its owner's row whenever that owner also held finite ones (Section 3.5.1; Appendix 8.6.10). Appendix 8.4.2 reports how many owners hold both kinds.

The capability is what attackers seek. In transaction-based phishing, fake websites lead users to sign transactions that let scammers take their assets, and the first empirical study of the attack on Ethereum detected and reported more than 26,000 such websites in about eight months (He et al., 2023). A later study of payload-based phishing, in which malicious payloads manipulate the contract calls a user signs, sorts the tactics into four main categories and finds more than 130,000 phishing transactions on Ethereum over 300 days (Chen et al., 2025). Phishing of this kind is now sold as a service: operators build the toolkits, called wallet drainers, affiliates deploy the phishing websites, and the proceeds are split through profit-sharing contracts (He et al., 2025). Approval phishing also drives high-yield investment frauds that run across several blockchains, and an analysis of eighteen popular wallets finds that weaknesses in how wallet software presents and validates token approvals make these frauds easier (Adamczyk, 2025).

These attacks bear on a contract-level score in two ways. An approval obtained by deception emits the same `Approval` event as an informed one, so a spender that collects approvals by phishing gains endorsement edges just as an honest router does. EndorseRank cannot tell the two apart from the event; only who the approving owners are, and how strongly they are endorsed in turn, can separate

them. In the other direction, an owner asked to sign an approval has little on-chain evidence about the contract in front of it, and a score that summarizes which other owners have approved the same contract is one possible input to that decision. This thesis tests neither score as a phishing or drainer detector (Section 2.8).

2.18 AWP in depth: what the transfer-graph baseline measures

Adaptive Weighted PageRank (AWP), proposed by T. Do et al. (2023), ranks the addresses of a transaction graph. Each transaction adds to its edge a logistic function of its time multiplied by a bounded function of its value, so that no transaction adds more than one to the edge weight. The walk restarts at each address in proportion to its activeness, the sum over the addresses it has sent to of the time weight of its latest transaction to each. The paper then checks the rankings against inbound degree and inbound value, taken as proxies for the economic attention an address attracts. The form of the decay (Equation (3.3)) and the in-degree/in-value proxies in Chapters 3 and 4 are taken from the AWP paper. The paper gives no parameter values; this study fixes the decay parameters in its plan and sets the slope b of the value transform from the median US-dollar amount (Sections 3.6.1 and 3.5.1), and its in-value proxy sums inbound transfers in US dollars. As published, the approach is entirely on-chain and rank-based, and it needs no off-chain identity of any kind; applied to several tokens, as here, it also needs their prices, so that amounts in different tokens can be added. In this thesis AWP is applied to the same cohort contracts as EndorseRank, through the transfers of the five tokens that they send and receive.

EndorseRank keeps the same template, AWP's edge weights under PageRank plus a rank-correlation check, and changes what the edges mean. In Chapter 4 we test, on the same contracts, whether allowance edges give an ordering that differs from AWP's and agrees with the allowance checks, whether the cost of the solve follows the smaller edge set, and whether scores taken at t_1 rank the new approval pairs gained by t_2 better than the raw in-approve degree does.

2.19 Comparison of graph reputation methods

Table 2.1 compares EndorseRank with the other approaches covered in this chapter by edge semantics, temporal model and evaluation design.

As Table 2.1 indicates, EndorseRank's contribution is the use of the latest allowance as the edge of a reputation ranking of contracts, evaluated out of window; it is neither the ranking algorithm nor the reading of approvals as a link between addresses (Section 2.7). The two rows without edges judge different things: Human Passport judges the person behind an address, and CRPWarner judges the code of a contract. Neither ranks a contract by the relations that other addresses have formed with it. On the operator side, our contribution is to couple the two relations into one walk, with a seeded ablation to measure how much this coupling contributes. The coupling is tested out of window against both layers walked alone, under two rules that were fixed before the data they are judged on were extracted, rule A in the plan and rule B in the registration of W1, and are applied to the revised scores after the registered results were known (Sections 3.7, 3.7.2 and 3.5.1).

2.20 Summary of the reviewed literature

Table 2.2 collects in one place the sources underlying this study. Each row records a source's methodology, the strengths this thesis draws on, the limitations it tries to address, and how the source relates to this work. The research gap stated in Section 2.21 is grounded in the evidence collected in this table.

Table 2.1: Comparison of selected graph reputation methods (conceptual).

Method	Edge semantics	Temporal model	Typical evaluation
PageRank (web) (Page et al., 1999)	Hyperlinks	None / static snapshot	Retrieval quality
EigenTrust (Kamvar et al., 2003)	Local trust ratings	Iterative aggregation	P2P file authenticity
TrustRank (Gyöngyi et al., 2004)	Links from seed goods	Static / seeded	Spam demotion
PageRank/ HodgeRank on Ethereum (H.-D. Do & Do, 2023)	Token transfers	Static window	Not reproduced in this thesis
AWP (T. Do et al., 2023)	Transactions, bounded value weights	Logistic decay of edges and restarts	In-degree / in-value
RiskProp (D. Lin et al., 2025)	Transactions, with a de-anonymization score	Propagation of risk	Labeled illicit accounts
Multiplex (Halu et al., 2013) / multilayer PageRank (De Domenico et al., 2015)	Several relations over one node set	Static layers	Comparison with single-layer rankings; versatility of top-ranked nodes
Human Passport (n.d.)	No edges; weighted credentials about the address holder, on-chain and off-chain	Current credential set	Admission threshold; known human and Sybil addresses
CRPWarner (Z. Lin et al., 2024)	No edges; functions in a contract's code	Code at the time of analysis	Contracts tied to known rug pulls
EndorseRank (this study)	Latest allowances of five tokens in US dollars, AWP's weights	Logistic decay of each latest approval	Allowance checks; t_2 new approvals
C-PR / S-PR (this study)	Allowances mixed with, or seeding, transfers; US-dollar amounts, allowances capped	Allowance snapshot and logistic decay of transfers	Holdout labels; registered rules applied to the revised scores

Table 2.2: Summary of the reviewed literature: methodology, strengths, limitations, and relation to this study.

Source	Methodology	Strengths	Limitations	Relation to this study
Page et al. (1999); Langville and Meyer (2006)	Eigenvector ranking of a hyperlink graph, solved by power iteration	Interpretable propagation; $O(\|E\|)$ per iteration; damping yields a unique Perron vector	Edges can only mean hyperlinks; no notion of time	Same solver and complexity argument, used unchanged for both methods compared
Kamvar et al. (2003); Gyöngyi et al. (2004)	Aggregation of trust in P2P networks; propagation from seeds to demote spam	Normalization and seed sets curb inflation; the adversary is modeled explicitly	Depends on explicit ratings or curated seeds; not applied on-chain	Informs the Sybil-cost discussion and the uniform-teleportation baseline
H.-D. Do and Do (2023)	PageRank and HodgeRank applied to Ethereum transaction graphs as a measure of social credit	Treats transfer connectivity as relevant to credit; data are fully public	Transfers conflate payment with trust; allowance semantics absent	Prior transfer-graph study, kept separate from AWP in this thesis
T. Do et al. (2023), AWP	Transaction PageRank with bounded value weights, logistic recency and restarts weighted by sending activity; rank correlation with in-degree/in-value, presented as validation	Evaluates on ranks, so no default labels are needed; recency modeled explicitly	Proxies are computed from the same transfer data that forms the graph (mechanical alignment); dense graph; no parameter values reported	Baseline, used as published on the same contracts (Section 3.6.1); the decay parameters are set here and b from the median US-dollar amount, and EndorseRank takes the same edge weights with the median allowance

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Table 2.2 (continued)

Source	Methodology	Strengths	Limitations	Relation to this study
D. Lin et al. (2025), RiskProp	Network propagation of account risk, paired with a de-anonymization score	Rates every account, including those that clustering and classification leave unassessed; checked against labeled accounts	Checked against illicit-account labels; aimed at fraud, not trust or authorization	Shows that propagation over transactions adds to per-account methods; not reproduced here, since this study uses no fraud labels
Wu et al. (2022)	Phishing detection through network embedding of Ethereum transaction graphs	Embedding that uses the amounts and times of transactions; more effective than earlier embedding methods	Needs labeled phishing addresses; its labels mark phishing, not creditworthiness	Backs the premise that graph position is informative; not a competing method
D. Lin et al. (2024); Yan et al. (2023); C. Zhou et al. (2024); Q. Liu et al. (2025)	Account labelling with a graph convolutional network; scam detection with biased random walks; detection of airdrop hunters and Sybil addresses from transaction graphs	Show that account roles, scams and coordinated multi-address behavior leave traces in graph structure	Need labels; classify addresses and do not rank them	Back the premise that graph position is informative, and frame the Sybil threat to an allowance ranking
Hassija et al. (2020); Kotb (2023)	Blockchain-data credit scoring using prospect theory and machine learning	Aimed directly at lending decisions	Need labels or off-chain features; only partly interpretable	A contrasting class; EndorseRank remains within auditable graph propagation

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Table 2.2 (continued)

Source	Methodology	Strengths	Limitations	Relation to this study
Palaiokrassas et al. (2024); M et al. (2026)	Machine-learning prediction of liquidations from multichain DeFi features; a DeFi credit score that adds features from a weighted transaction graph	Show that on-chain behavior and graph position carry credit-relevant signal	Need liquidation or default labels; score borrowers	Support the use of graph position; this study ranks spender contracts, which do not borrow
Bellini et al. (2020); Almasoud et al. (2020)	Surveys of blockchain-based trust and reputation systems	Taxonomies of the design space; a review of thirty studies of reputation systems built on smart contracts	Reputation taken from users' ratings of services; authorization edges not considered; nothing measured on L2	None of the surveyed designs uses latest-allowance edges
Victor (2020); Victor and Weintraud (2021); Cong et al. (2023)	Heuristics for clustering addresses, one of which links addresses through ERC-20 approvals; detection of wash trading on DEXs and exchanges	Measure the share of on-chain activity that is artificial or multi-address	No reputation score proposed	Closest prior use of approvals, as a clustering link and not a ranking edge; show why transfer-based signals are open to inflation
He et al. (2023); Chen et al. (2025); He et al. (2025); Adamczyk (2025)	Measurement and detection of transaction-based, payload-based and approval phishing and of drainer-as-a-service operations	Large datasets of phishing websites, transactions and accounts; analysis of how wallets present approvals	Aimed at detecting attacks, not at ranking the contracts that hold approvals	Show why an approval is security-relevant and that an approval edge can be obtained by deception (Section 2.17)

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Table 2.2 (continued)

Source	Methodology	Strengths	Limitations	Relation to this study
Z. Lin et al. (2024); Z. Li et al. (2026); Tang and Shi (2024); Efimov et al. (2026)	Code analysis of rug-pull functions and of token contracts that break ERC-20; security analysis of contracts on Arbitrum and of rollup ecosystems	Judge a contract by its code and by the chain it runs on	Say nothing about how owners treat a contract	Contract-level risk work that a relational score complements and does not replace
Cronbach and Meehl (1955); Messick (1995)	Psychometric construct validity	Distinguish what a score is designed to measure from external criteria	Not specific to blockchains	Basis for treating the contemporaneous checks as construct checks and the temporal holdouts as out-of-window evidence
Human Passport (n.d.); Siddarth et al. (2020); Bordeianu and Popescu (2026)	Weighted credentials about an address holder, on-chain and off-chain, against a passing threshold; reviews of proof-of-personhood defenses	In deployed use; adds off-chain evidence against Sybil attacks	A gate and not a graded ordering; relations between addresses play no part; Sybil strategies keep changing	Shows how trust is established in practice, and what a contract-level score does not offer: no personhood guarantee
Efron (1979); DiCiccio and Efron (1996)	Bootstrap resampling and bootstrap-based confidence intervals	Uncertainty for rank statistics with no distributional assumption; paired contrasts	Units must be exchangeable	Source of the 95% intervals and $\Delta\tau$ contrasts reported in Chapter 4

2.21 Research gap

Five gaps emerge from Table 2.2 and Section 2.15. First, every on-chain reputation method discussed here (PageRank and HodgeRank (H.-D. Do & Do, 2023), AWP (T. Do et al., 2023), RiskProp (D. Lin et al., 2025), and the designs surveyed by Bellini et al. (2020)) builds its graph from transfers or explicit ratings, and the attestation tooling deployed in DeFi scores the holder of an address without any graph of relations between addresses (Human Passport, n.d.). None uses the ERC-20 allowance, the on-chain record of a deliberate delegation of spending power that every token following the ERC-20 standard carries. Approvals have served address clustering and security analysis (Sections 2.7 and 2.17), and contract risk has been assessed from code (Z. Lin et al., 2024), but in the work reviewed here approvals have not served as the edges of a reputation ranking, and contracts have not been ranked by the approvals they receive. It remains unknown whether a latest-allowance endorsement graph orders spender contracts differently from a transfer graph (O1 and RQ1). A second gap lies in evaluation. The transfer-graph baseline checks its scores against in-degree and in-value computed from the same transfer data used to build the graph, so some agreement is unavoidable; no report applies this procedure to an allowance construct and its transfer counterpart with confidence intervals and paired difference tests on the same set of contracts (O2 and RQ2). Third, a runtime says little about a method unless it is reported with the edge count, the memory use and the number of iterations under a fixed protocol; otherwise a difference in speed cannot be traced to sparsity as opposed to implementation (O3 and RQ3). The fourth gap is methodological and has to do with time. Agreement with graph statistics from the same window, the check used to evaluate AWP, cannot show that a score predicts later behavior; the out-of-window check used here is instead a temporal holdout, with a label for each construct counted after the freeze and the raw degree at the freeze as the baseline (O4 and RQ4). Finally, multiplex PageRank (Halu et al., 2013) and multilayer ranking (De Domenico et al., 2015) have been applied to social networks and to other networks with several kinds of relation, but not to two on-chain relations over one set of addresses. No study reports whether combining an authorization layer with a payment layer im-

proves on the payment layer alone, or beats both layers, under decision rules written down before the outcome is seen (O5 and RQ5). This study fixed such rules in its plan and in the registration of W1, and applies them to the revised scores as a re-analysis after the registered results were known (Section 3.5.1).

2.22 Open problems motivating this research

The literature leaves several problems in on-chain reputation unresolved, and these motivate the present study:

1. Edge semantics: When identities are pseudonymous (D. Lin et al., 2024), which on-chain relations capture trust or authorization best?
2. Evaluation without defaults: On what basis can methods be compared when credit records are missing (H.-D. Do & Do, 2023), and when the ranked contracts do not borrow at all?
3. Computational cost: As graphs grow, can reputation scores still be refreshed at interactive latencies (Langville & Meyer, 2006)?
4. Time-split checks: What keeps an evaluation from mistaking same-window graph statistics for out-of-window evidence (Cronbach & Meehl, 1955)?
5. Adversarial robustness: How much does it cost to forge reputation when owner addresses and contracts are cheap to create (Douceur, 2002)?

This study takes on (1)–(4) through allowance edges, contemporaneous checks, runtime benchmarks and temporal holdouts. Problem (5) is analyzed at the model level in Chapter 5; a full empirical adversarial evaluation is left for future work.

Chapter 3 turns this gap statement into an operational design: the data window and cohorts, the construction of both graphs and their solution with one shared PageRank implementation, the proxy families, the temporal holdout, and the timing and bootstrap protocols behind the results of Chapter 4.

3 Research Methodology

Chapter 2 ended with five gaps. This chapter gives each of them a measurable procedure and sets out how EndorseRank, Adaptive Weighted PageRank (AWP) and the coupled walks built from their two layers are evaluated as reputation scores for smart contracts on Arbitrum One. EndorseRank runs on the ERC-20 allowance graph, and AWP (T. Do et al., 2023) on the transfer graph. Both weight their edges as AWP's authors specify and are solved with weighted PageRank (Page et al., 1999); AWP restarts the walk in proportion to sending activity, and EndorseRank restarts it uniformly. Every scored and evaluated address is a contract. Externally owned accounts (EOAs) enter the graphs as owners, senders and recipients and are never scored. Both graphs keep the approvals and transfers of five widely traded tokens and value every amount in US dollars (Section 3.5.1).

The design follows construct alignment. Reputation scores are continuous rankings, and the checks rest on observable contract-level statistics instead of binary default labels. Each ranking method is treated as one indicator of a reputation construct that cannot be observed directly, in the sense of construct validity (Cronbach & Meehl, 1955). The claim is comparative, every score set against the checks of each construct and against the raw degrees it smooths; it is not a prediction of trading profit, of credit default or of the safety of a contract's code.

Every procedure runs on public blockchain logs and public daily token prices inside fixed windows. Seeds are deterministic and the evaluation code is public, so anyone who reruns the extraction, or holds the downloaded pair tables, can regenerate the reported tables.

Four principles guide each operational choice.

- Same-seed comparability: every score is computed on the graphs of one seed set, the matched contract cohort, and evaluated on identical sets of contracts, so no difference between scores can come from sampling.
- Fixing choices in advance: the plan wrote down every choice that decides a result before the

data it is judged on, and the revision of the amounts was written down before any revised score was computed, though after the registered results were known. The events took place in this order.[1]

1. At 13:41 UTC on 9 October 2026 the analysis plan and the versioned configuration were committed to the public repository (commit `43d8045`) before any log of the study was extracted. They fix the windows, the cohort rule, the account-type rule, every score with its damping, decay, value transform and restart rule, the values of λ, the proxies, the labels of the holdout W0, the bootstrap resamples and seed, the benchmark protocol, the robustness checks and rule A for the coupled operator (Section 3.7).
2. The same day, scan A extracted the observation window. The cohort was fixed, and the ego events of the cohort, the pair tables, the proxies and the W0 labels were built.
3. The same-window and W0 results were computed, and rule A was judged.
4. At 16:49 UTC on the same day the registration of the window W1 was committed (commit `0d3a057`): rule B with its margin, three further results reported in a fixed order, two sensitivity analyses, and the comparison on outcomes built from neither edge type with the wording of its conclusions (Sections 3.7.2 and 3.7.3). The script that runs scan B refuses to run without this commit and records the hashes of the registered files.
5. Scan B extracted July to September 2026, the W1 labels were counted, and the registered evaluation was run.
6. After the W1 labels were known came the supplementary contrasts (EndorseRank with activity restarts against the in-approve degree, and EndorseRank against AWP in W1), the robustness checks, the Sybil model and the benchmark.
7. On 10 October 2026, after these results were known, the amounts were revised (Section 3.5.1). Every amount is valued in US dollars, only five tokens with reliable prices are kept, the value transform takes its slope from the median amount, and the cohort,

[1] All dates and times are given in UTC. The commits record the local time of the machine on which they were made (UTC+9), in which the registration commit carries the date 10 October 2026.

the graphs, the proxies and the spender labels are rebuilt on those tokens. The revision was committed between 07:54 and 11:24 UTC, before any revised score was computed, and the revised analysis was then run with the protocols, rules and thresholds of the registered one. Chapter 4 reports the revised analysis.

In the registered analysis, only rule A (its two contrasts in W0 and its same-window part) and the results registered for W1 (rule B, the three further results, the two sensitivity analyses and the comparison on outcomes built from neither edge type) were fixed before the data they are judged on. In the revised analysis none was: the same rules and thresholds are applied, but to scores defined after the registered results were known. Every other contrast, the contrast of each PageRank with its own raw degree among them, was computed after the data it is judged on were known; it is reported as such, with its interval.

- Construction and timing: the pair tables are aggregated once for each anchor date and loaded before any timer starts, and the benchmark times the solver call.
- Scope: H1 and Claim C3 compare the orderings of EndorseRank and AWP; H2, H3 and Claims C1, C2 and C4 concern every score, C5 the coupled operator and its ablation, and C6 the approval and transfer counts and their PageRanks on trader outcomes. RQ4 and RQ5 draw on the spender cohorts of W0 and W1; rule B and the comparison on outcomes built from neither edge type add the trader cohort.

3.1 Notation

The notation used in this chapter and in Chapters 4 and 5 is collected in Table 3.1 below, and a condensed list appears in Table II. Vectors are set in bold and sets in calligraphic type. Addresses are treated as opaque identifiers; we do not map them to any external identity.

Table 3.1: Notation for graphs, scores, proxies, and parameters.

Symbol	Meaning
$\mathcal{W}$	Matched contract cohort ($\lvert\mathcal{W}\rvert = n = 15{,}107$), the seed set of both graphs

Table 3.1: (continued)

Symbol	Meaning
$\mathcal{C}$	Contracts evaluated in one analysis: $\mathcal{W}$, or a spender or trader cohort of a holdout
$G = (V, E)$	Graph passed to the PageRank solver: V holds the end points of the retained edges E
$G_{\text{approve}} = (V_A, E_A, w_A)$	EndorseRank allowance graph
$G_{\text{transfer}} = (V_T, E_T, w_T)$	AWP transfer graph
$a^\star(u, v, \text{token})$	Latest allowance of owner u to spender v on one token up to the anchor (US dollars at the anchor-day price)
$\Delta t^\star(u, v, \text{token})$	Age of the approval that set $a^\star$
$w_A(u, v)$	EndorseRank edge weight, owner $u \to$ spender v: $\sigma(\Delta t^\star)V(a^\star)$ summed over tokens
$w_T(u, v)$	AWP edge weight, sender $u \to$ receiver v: $\sigma(\Delta t_e)V(x_e)$ summed over transfers
x_e	Amount of transfer e in US dollars at the price of its day
$V(z)$, b	Bounded value transform of AWP, $V(z) = 2/(1 + e^{-bz}) - 1$, with $b = \ln 3/m$ on US-dollar amounts (b_T for transfers, b_A for allowances)
m_T, m_A	Medians, rounded to two significant digits, of the US-dollar amounts of the transfers up to t_1 and of the finite latest allowances at t_1 at t_1 prices
U	Cap on an allowance in the allowance layer of C-PR and in the in-approve value: an allowance above U counts U
X_u	Activeness of address u, the restart weight of AWP
$\tilde{w}_A(u, v)$, $\tilde{w}_T(u, v)$	Weights of the two layers of C-PR, in US dollars
$o(u)$; $o_A(u)$, $o_T(u)$	Out-strength $\sum_v w(u, v)$ of node u; the same in the allowance and transfer layers
P	Transition matrix, $P_{u,v} = w(u, v)/o(u)$; rows of dangling nodes are zero
$\mathbf{1}[\cdot]$	Indicator: 1 if the condition holds, 0 otherwise
$\mathbf{1}_{\text{dang}}$	Indicator vector of the dangling nodes, entries $\mathbf{1}[o(u) = 0]$
$\mathbf{r}$, $\mathbf{r}^{(t)}$	PageRank vector and its iterate after t steps
$\boldsymbol{\pi}$	Restart (teleportation) distribution, also used for dangling mass: uniform for EndorseRank and C-PR, proportional to X_u for AWP and for EndorseRank with activity restarts, and for S-PR the allowance layer's score restricted to V_T and renormalized
d	PageRank damping factor (default 0.85)
ε	PageRank convergence tolerance (10^{-8})
$I_{\max}$	Maximum PageRank iterations (300)
λ	Allowance-layer weight of the coupled operator C-PR ($\lambda = 0.5$ primary; 0.25, 0.75 sensitivity; at $\lambda = 1$ and $\lambda = 0$ one layer is walked alone)
$\alpha_A(u)$, $\alpha_T(u)$	Layer weights in node u's C-PR row; $D_\lambda(u)$ is their common denominator
$P^{(\lambda)}$	Coupled transition matrix of C-PR
$\mathbf{s}_M$	Reputation score vector for method M
$\mathbf{p}_q$	Proxy value vector for proxy q

Table 3.1: (continued)

Symbol	Meaning
$\sigma(\Delta t)$	Logistic time-decay on the age of an approval or transfer
k, t_0	Decay steepness (0.01) and midpoint (180 days)
Δt	Age of an event in days before the decay anchor (T_{obs}; t_1 in W0)
T_{obs}	End of the observation window and freeze of W1 (2026-06-30 23:59:59 UTC)
t_1, t_2	Score freeze of W0 (2026-03-31 23:59:59 UTC) and end of its label window (2026-06-30 23:59:59 UTC)
W0, W1	Holdout windows: labels from April to June 2026 with scores at t_1, and from July to September 2026 with scores at T_{obs}
δ	Non-inferiority margin of rule B (0.01)
B	Paired bootstrap resamples (400; 2,000 in Section 3.7.3)
$\Delta\tau$	Paired difference of two Kendall coefficients on the same resamples
ρ	Spearman's rank correlation
R_i, S_i	Mid-ranks of contract i's score and proxy value (tied values share their average position)
τ, τ_b	Kendall's rank correlation; every reported value is the tie-corrected τ_b
n_c, n_d	Numbers of concordant and discordant contract pairs
n_0; n_1, n_2	All pairs, $n(n-1)/2$; pairs tied in the first and in the second ranking

Except where stated otherwise, PageRank means the weighted variant of the algorithm of Page et al. (1999), whose update at iteration t is

$$\mathbf{r}^{(t+1)} = (1-d)\,\boldsymbol{\pi} + d\,P^{\top}\mathbf{r}^{(t)} + d\left(\mathbf{r}^{(t)\top}\mathbf{1}_{\text{dang}}\right)\boldsymbol{\pi}, \tag{3.1}$$

where P is the transition matrix of Equation (3.2) below, $\boldsymbol{\pi}$ is the restart distribution over the nodes, uniform ($\pi_u = 1/|V|$) unless a method specifies another (AWP in Section 3.6.1, S-PR in Section 3.6.3), and $\mathbf{1}_{\text{dang}}$ is the indicator vector of the dangling nodes, with entries $\mathbf{1}[o(u) = 0]$. The third term therefore collects the mass held by nodes without out-edges. With $w(u,v)$ denoting the weight of the directed edge from u to v and $o(u) = \sum_v w(u,v)$ the out-strength, the transition matrix is

$$P_{u,v} = \begin{cases} w(u,v)/o(u) & \text{if } o(u) > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{3.2}$$

The rows of dangling nodes are zero, so P is substochastic; the mass lost at those nodes returns through the third term of Equation (3.1) and is spread according to $\boldsymbol{\pi}$. Filling each zero row with $1/|V|$ gives the row-stochastic matrix of Equation (2.1), and with uniform $\boldsymbol{\pi}$ Equation (3.1) is then the Google-matrix form of Equation (2.2). The iteration starts from $\mathbf{r}^{(0)} = \mathbf{1}/|V|$ and stops once $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or after $I_{\max}$ iterations. V is the set of end points of the retained edges, and the final vector is normalized to sum to one over V. Contracts can also be listed in descending order of score, ties broken by address, but such ordinal ranks serve only for listing; the correlations of Section 3.10 are computed on the raw scores, where tied scores stay tied.

The damping factor d interpolates between pure propagation ($d \to 1$) and uniform scores ($d \to 0$). Its default of 0.85 is the classical web setting (Page et al., 1999), and the robustness sweep varies it with edge weights held fixed (Section 3.12). The strict $\varepsilon = 10^{-8}$ and $I_{\max} = 300$ keep the measured runtimes tied to the structure of the graph and not to early termination.

3.2 Data sources and sampling

This section fixes the chain, the time interval, the public tables, the events and the contracts of the main sample; no later step brings in events from outside it. Figure 3.1 places every kind of on-chain record used in the thesis on one calendar, with the months each analysis scores, the months it holds out as labels, and what each record is used for.

- Primary chain: All events come from Arbitrum One (chain ID 42161), an optimistic rollup on Ethereum (Kalodner et al., 2018) whose logs are in the public BigQuery export. One chain avoids cross-chain aliases, where the addresses of one entity on several chains would appear as distinct nodes, and every contract in the sample runs under the same fees and latencies.
- Observation window: Thirty-three months, from 2023-10-01 00:00:00 UTC to 2026-06-30 23:59:59 UTC; the queries take 2026-07-01 as an exclusive end. With the registered label window W1, from July to September 2026, the data cover three years. Both graphs use exactly the same dates. The window is long enough for a contract to collect approvals from several

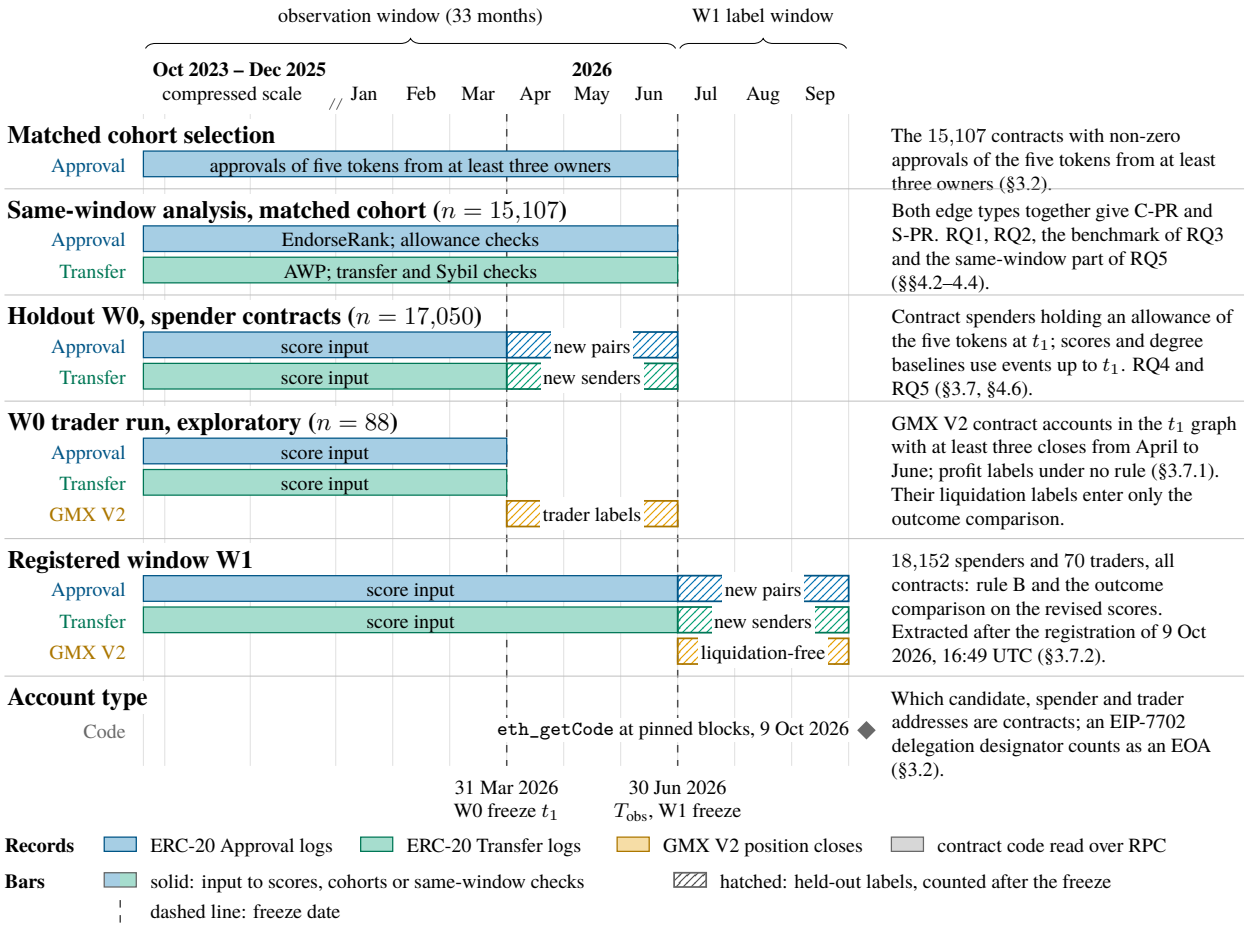


Figure 3.1: Blockchain data used in this thesis, all from Arbitrum One. Each lane is one analysis, and each bar is one kind of on-chain record placed on the months it covers; the 27 months before January 2026 are drawn at a compressed scale. Solid bars are inputs to scores, to cohorts or to same-window checks. Hatched bars are labels counted after a freeze date and kept out of the scores; the July–September logs were extracted only after the registration of W1 was committed at 16:49 UTC on 9 October 2026. The text in a bar says what the records compute, and the right-hand column names the cohort, the research questions and the sections that report the results. The approval logs of five tokens select the matched cohort, and every approval and transfer query is then filtered to it and to those tokens.

owners and for the tenure and active-month proxies to vary. The time decay is anchored at $T_{\text{obs}} =$ 2026-06-30 23:59:59 UTC, so an event on the final day has $0 \leq \Delta t < 1$ and one on the first day is about $1{,}004$ days old.

- BigQuery source: Logs are read from the public BigQuery table

  `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`.

  One record is one event emitted by a contract, with its block timestamp, block number, log index, emitting address, topics and data. Scan A reads this table once for the observation window and copies the events the study needs into a table of the study's own Google Cloud project, and scan B does the same for July to September 2026 (Section 3.3). Every later query reads the project tables. Anyone with a Google Cloud account can rerun the same queries against the same public table; Appendix 8.2 shows the query patterns.

- Reputation events: EndorseRank uses ERC-20 `Approval` logs (Vogelsteller & Buterin, 2015), which also cover approvals made through EIP-2612 `permit` (Lundfall, 2020), and AWP uses `Transfer` logs. Both are kept only when the log has exactly three topics, which leaves out ERC-721 tokens (Section 3.5), and only for the five tokens of Section 3.5.1. GMX V2 position closes supply the trader labels. The plan also listed the borrow, repay and liquidation events of the Aave V3 pool, and scan A copied them; no analysis in this thesis uses them.

- Account type: Whether an address is a contract is read with the `eth_getCode` call of a public Arbitrum One RPC endpoint at one pinned block. For the cohort candidates this is block $513{,}216{,}506$, read on 9 October 2026. An address whose code is non-empty is a contract. An address without code is an EOA, and so is an address whose code is an EIP-7702 delegation designator, the prefix `0xef0100` followed by a 20-byte address: under EIP-7702 an EOA keeps its private key and points to code that runs on its behalf (Buterin et al., 2024). Holdout spenders and GMX accounts that were not among the cohort candidates were read by the same call at later blocks, within three hours of the first read; each read is stored with its block number.

- Matched cohort: The cohort $\mathcal{W}$ is every contract that received non-zero `Approval` events of the five tokens from at least three distinct owners inside the observation window, the owners counted over the five tokens together. The plan's rule counted approvals of every token: the spender addresses that met it are the candidates, and those with contract code formed the plan's cohort. The revised cohort, $n = |\mathcal{W}| = 15{,}107$, is the part of it that meets the rule on the five tokens; Section 4.1.1 gives the account types of all candidates and the size of both cohorts. The plan's cohort was fixed from the approval logs and the code check alone, before any transfer was aggregated, and the revised cohort is fixed from the approvals of the five tokens alone. Because the rule reads approvals, the cohort is not independent of the allowance graph: every cohort contract held allowances at some time in the window, while the rule asks nothing about transfers. A cohort contract need not appear in either graph at T_{obs}. It is outside the allowance graph when every allowance it had received was later set to zero and it held no positive allowance as an owner, and outside the transfer graph when it sent and received no non-zero transfer in the window. The method concerned then scores it 0 (Section 3.6); Section 4.1.1 counts these contracts. EndorseRank and AWP also score 0 a cohort contract whose only pairs carry amounts so small that the value weight is zero in double precision, since such pairs are dropped as non-positive edges; Section 4.4 counts them.
- Ego network: Only events that involve a cohort contract are extracted: an approval in which a cohort contract is the owner or the spender, and a non-zero transfer in which it is the sender or the recipient. Each graph is thus the one-hop ego network of the cohort, since an event between two addresses outside it is never extracted. The plan extracted the events of every token that involve a contract of its cohort; the revised graphs keep those of the five tokens that involve a contract of the revised cohort, a subset, so no log had to be read again. Appendix 8.4.1 counts the extracted events, and Section 4.1 gives the nodes and edges of both graphs at T_{obs}. Owners that are themselves cohort contracts, or that a cohort contract has approved, receive approvals of their own, so the allowance walk can carry endorsement over more than one hop; Section 5.3.2 examines how far it does.

- Exclusions: Mempool transactions, off-chain order books, centralized exchange accounts and bridges lie outside the data, and no attempt is made to merge addresses controlled by one operator into one entity. Tokens other than the five are left out, and native ether, which moves without `Transfer` logs, enters only as WETH. No EOA is scored or evaluated.

3.3 Two-phase collection strategy

Data collection and evaluation run as two separate phases, so the tables, benchmarks and robustness runs can be rebuilt as often as needed without cloud costs, by us or by anyone who holds the downloaded tables.

- Phase 1 (BigQuery): Scan A copies the approval, transfer and GMX events of the observation window from the public logs into a table of the study's project, partitioned by month, and scan B copies those of July to September 2026 into a second table once the registration of W1 is committed (Section 3.7.2). One query lists the spenders approved by at least three owners, the code check of Section 3.2 fixes the cohort, and a further query extracts the ego events of the cohort. At each anchor date, T_{obs} and t_1, the ego events are then aggregated into pair tables. For the allowance graph the latest allowance of each (token, owner, spender) triple is taken in (block number, log index) order, kept when it is positive, and summed into one row per owner–spender pair that carries the EndorseRank weight and the capped US-dollar amount. For the transfer graph each sender–recipient pair gets one row with its AWP weight, its US-dollar weight, its largest time weight and its number of transfers. In the revised analysis these queries run on the ego events of the five tokens, valued with a table of daily prices loaded into the project. The same phase computes the same-window proxies and the holdout labels and maps every address to an integer node identifier.
- Phase 2 (local evaluation): The pair tables, the node table, the proxies, the labels and the GMX closes are downloaded once. From them a local machine decodes the GMX closes and builds the graphs, and it computes the scores, the checks, the alignment statistics, the holdout

evaluations, the benchmark and the robustness runs. No step of Phase 2 queries BigQuery.

- Cost guardrails: Every query is first run as a dry run, and its estimated cost is checked against the budget of USD 50 that the configuration sets for the whole study. A query whose estimate would take the spending past the budget is refused unless the operator confirms it with an extra flag.[2] Every job is written to a cost ledger with the bytes it billed and its cost, so the cost of acquisition can be audited (Appendix 8.1).
- Storage: Every file written under the data folder of the repository is split into parts below 90 MB, so that the extracts and summaries can be kept in Git under GitHub's file-size limit.

3.4 Pipeline architecture

Figure 3.2 traces the whole pipeline, from public logs to the research tables. All remote steps write tables in the study's BigQuery project, and the downloaded parts are not changed afterwards. Given its inputs and the versioned configuration file, every local step is deterministic.

The Phase 1 queries write the pair tables, the proxies and the labels, which the Phase 2 drivers read to build the graphs, compute the scores and write the machine-readable summaries behind the tables of Chapter 4. Unit tests run the scoring and statistics code on small synthetic fixtures.

3.5 Event decoding

A log emitted by a smart contract has topics and a data payload but no field names. Decoding turns each row into a typed record for graph construction and the checks, and an error here would corrupt both graphs silently. The topic hashes and field mappings are listed below.

- Log anatomy: Each log records the emitting contract, a sequence of topics and an ABI-encoded data blob. The first topic is the keccak-256 hash of the event signature; indexed parameters follow as 32-byte words (addresses left-padded), and non-indexed parameters go into the data field in ABI order (Vogelsteller & Buterin, 2015). ERC-721 tokens emit `Tran`

[2]The budget is enforced by the study's own pipeline; BigQuery does not impose it.

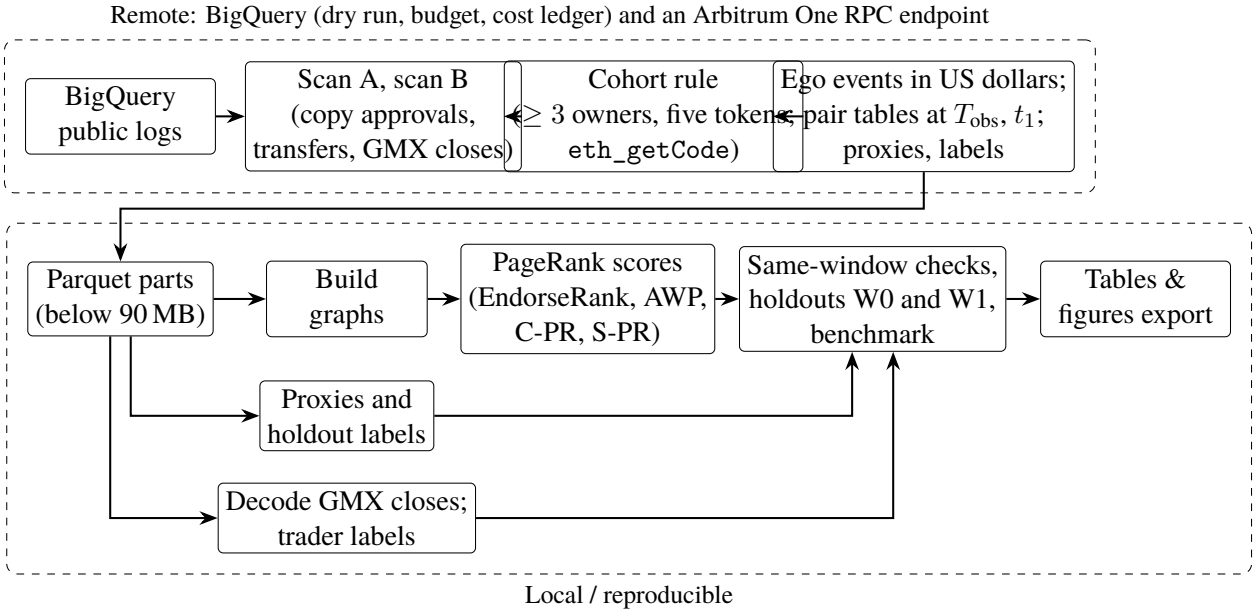


Figure 3.2: Pipeline overview. In BigQuery, scan A (observation window) and scan B (W1, run only after its registration was committed) copy the study's events from the public logs into a project table; the cohort is fixed from the approvals of five tokens and the account code; the ego events of the cohort are valued in US dollars with daily token prices and aggregated into pair tables at each anchor date, together with the proxies and the holdout labels. Every query is dry-run against the study's budget and written to a cost ledger. Locally, the downloaded parts become the graphs from which EndorseRank, AWP, C-PR and S-PR are computed, and the GMX closes are decoded into trader labels. The same-window checks, the holdouts and the benchmark follow, and the tables and figures are exported last.

sfer and Approval under the same signatures but index the token ID as a fourth topic, so the condition of exactly three topics keeps them out. The ERC-20 events are decoded in SQL during scan A and the ego extraction; the GMX events are decoded locally.

- ERC-20 Approval: Its topic hash is

  0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925,

  the keccak-256 hash of Approval(address,address,uint256). Owner and spender occupy Topics 1 and 2, and the allowance is decoded from the data field in the token's base units and valued in US dollars (Section 3.5.1). An unlimited allowance (type(uint256).max) keeps its value, and an allowance of zero is kept, since it sets the allowance in force to zero. A permit emits the same Approval event, so an allowance counts whether it came from an on-chain approve() call or from an off-chain signature (Lundfall, 2020).

- ERC-20 Transfer: Its topic hash is

  0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef,

  the keccak-256 hash of Transfer(address,address,uint256). Sender and recipient are indexed, and the value is decoded in base units and valued in US dollars at the price of its day. Zero-value transfers are dropped, and so are transfers made before their token's first price, which are counted (Appendix 8.4.2). Mint and burn logs, which use the zero address, are kept whenever a cohort contract is involved. Self-transfers stay in the events but are left out of the inbound-counterparty computations (Section 3.9), since they reveal no counterparty.

- GMX V2 position closes: A close is one EventLog1 log of the GMX V2 event emitter whose event name is PositionDecrease, so partial closes and liquidations count. The trading account is an indexed topic, so the accounts can be listed in SQL; the data field is decoded locally with the emitter's ABI into the account, the base profit or loss in USD, the order type and the size of the close. A close is a liquidation when its order type is the liquidation type or when the liquidation handler emitted it. The decoder read every close of a contract account in the observation window and in W1 without a decoding error (Section 4.1.1 gives the counts).

- Address normalization and units: Addresses are compared as 20-byte values and written in lower-case hexadecimal. Amounts are the `uint256` value as a double-precision float, divided by ten to the power of the token's decimals and multiplied by its US-dollar price (Section 3.5.1). An unlimited allowance, $2^{256} - 1 \approx 1.16 \times 10^{77}$ base units, is valued in the same way; it has $V = 1$ in EndorseRank and counts U in the allowance layer of C-PR (Sections 3.6.2 and 3.6.3). A data field that cannot be read as a number gives the amount zero, so such a transfer is dropped and such an approval counts as setting the allowance to zero.

3.5.1 Tokens, prices and the revision of the amounts

The analysis plan fixed amounts in raw base units, every ERC-20 token, and the value transform of Equation (3.4) with $b = 1$. Three defects of that setting came to light after the registered results of the same window, W0 and W1 had been computed. First, $V(z)$ is within 10^{-4} of 1 for any amount of 10 base units or more, and almost every transfer and latest allowance reaches that level (Appendix 8.4.2), so the amount played no part in EndorseRank or AWP, although AWP's value weight exists so that larger transactions count for more (T. Do et al., 2023). Second, the layers of C-PR added raw base units across tokens: one USDC is 10^6 base units and one WETH 10^{18}, and an unlimited allowance, $2^{256} - 1$, took almost the whole row of every owner that held one. Third, every ERC-20 contract counted. A token contract can emit `Approval` and `Transfer` logs that name any address as owner or sender without that address taking part, as spam and address-poisoning tokens do, so a log of such a token is no evidence that the named owner endorsed or paid anyone.

The amounts were therefore revised on 10 October 2026, and the revised analysis is the one that Chapter 4 reports. The revision is a deviation from the plan and from the registration of W1, decided after their results were known, so none of its results is a registered result. It was written down and committed to the public repository before any revised score, proxy or label was computed: the revision document at 07:54 UTC (commit `a897475`), the choice of five tokens at 08:12 UTC (commit `09dfc32`) and an amendment of the allowance cap described below at 11:24 UTC (commit `2705b28`). The registered analysis is kept as it was obtained; its decision contrasts and verdicts are

set beside the revised ones in Appendix 8.6.10, and its complete results are published unchanged (Appendix 8.1).

- Token list: A token qualified when Chainlink's directory of data feeds on Arbitrum One, read on 10 October 2026, has a market-price feed for it, quoted in US dollars or in ether, whose market-risk category is low or medium, when that feed prices the token itself, the asset that the token wraps one-to-one (WETH for ether), or the Ethereum token of which the token is the copy minted by Arbitrum's canonical bridge, and when the token occurs in the extracted transfers or latest allowances of the observation window. Chainlink assigns the category from the depth, spread and concentration of an asset's markets, and leaves exchange-rate and other custom feeds, which do not follow the market, outside it (Chainlink, n.d.). Of the 133 Arbitrum One tokens that the directory maps, 64 qualified and 46 of those have a daily price. The analysis keeps the five of these with the most rows in the data of the observation window: WETH, USDC, USDT0, ARB and WBTC. All five are priced by feeds rated low risk. Appendix 8.4.2 gives their addresses and the rows of the other tokens that are left out.
- Prices: The US-dollar price of a token on a UTC day is its closing price in the daily series of the DefiLlama coins API (DefiLlama, n.d.), the point nearest to 00:00 UTC of the following day. Where the series under the token's Arbitrum address starts later than the token's use (for WETH and WBTC in August 2024, for USDT0 in January 2025), the days are filled from DefiLlama's series of the same asset under its CoinGecko identifier, provided the two series differ by less than 1% in median on the days they share. A daily point more than twice or less than half of both neighbouring days, while those two lie within a factor of two of each other, is treated as a data error and replaced by the previous day's price; a day without a point takes the previous day's price. For a sample of dates, the prices are compared with the Chainlink feeds read on the chain (Appendix 8.4.2).
- Amounts: A transfer of v base units of a token with q decimals, on a day with price p, is worth $x_e = v\,10^{-q}p$ US dollars. A latest allowance is valued at the price of the anchor's day,

30 June 2026 at T_{obs} and 31 March 2026 at t_1, since the edge stands for the permission in force at the anchor. Native ether moves without `Transfer` logs and was never in the data; it enters the graphs through WETH only.

- Value slopes: The slope of V is set so that the median amount weighs one half, $b = \ln 3/m$, with m the median amount rounded to two significant digits. For transfers, m_T is the median US-dollar value of the ego transfers up to t_1, $m_T = 200$ dollars; for allowances, m_A is the median value at t_1 prices of the finite positive latest allowances at t_1, $m_A = 62$ dollars. Both use events up to the earlier freeze only, so no label window enters them.
- Allowance cap of C-PR: In the allowance layer of C-PR an allowance counts at most $U = 20{,}000$ dollars, the 99th percentile of the US-dollar values of single transfers up to t_1, rounded to two significant digits. The revision document first defined U as the 99th percentile of the finite latest allowances at t_1. On the five tokens that percentile is about 10^{47} dollars, because the allowances below $2^{256} - 1$ include many that are unlimited in effect, such as 2^{255} or 2^{128}, and a cap at that level would have left unlimited allowances in charge of every owner's row. The definition was amended after the medians and percentiles had been computed and before any revised score existed; the medians were unchanged.
- Scope: Every quantity of the revised analysis uses the five tokens only. The matched cohort is the set of contracts that received a non-zero `Approval` of one of the five tokens from at least three distinct owners in the observation window. The owners are counted over the five tokens together, as the plan's rule counted them over every token, so such a contract meets the plan's rule and belongs to the cohort of the plan, whose account types were already known; the graphs, the proxies, the spender cohorts and the spender labels are drawn from the ego events of the plan's cohort, filtered to the five tokens and to the revised cohort, and no public log was scanned again. The GMX trader labels do not depend on tokens.

On the revised scores, rules A and B and the comparison on outcomes built from neither edge type are applied with the thresholds and the wording that were fixed for them, $\delta = 0.01$ included

(Sections 3.7, 3.7.2 and 3.7.3). Because the scores they judge were defined after the registered results were known, these verdicts are a re-analysis and not registered tests; Chapter 4 says so wherever it reports them.

3.6 Graph construction

EndorseRank and AWP weight their edges the same way and run the same PageRank solver, Equation (3.1); they differ in the events that make the edges and in where the walk restarts. Both graphs hold the ego network of the matched cohort $\mathcal{W}$ (Section 3.2), which brings in the cohort's one-hop neighbors as nodes. A method reports a score for every contract in the evaluated set $\mathcal{C}$: the matched cohort in the same window, and the spender and trader cohorts in the holdouts. A contract missing from the graph scores 0.

Edges in EndorseRank stand for endorsements and edges in AWP for transfers (Section 2.7). Both methods weight an event by its age, so that recent events count for more, and pass its US-dollar amount through a bounded function, so that a larger amount counts for more but one event counts at most once. AWP is described first, since EndorseRank takes its weights from it.

3.6.1 AWP transfer graph

AWP was proposed by T. Do et al. (2023) at IEEE RIVF 2023. A transfer e worth x_e US dollars, made Δt_e days before the anchor (T_{obs}, or t_1 in W0), enters the edge from its sender to its receiver with the weight $\sigma(\Delta t_e)\, V(x_e)$. The time weight is the logistic decay

$$\sigma(\Delta t) = \frac{1}{1 + \exp\big(k(\Delta t - t_0)\big)}, \qquad k = 0.01, \quad t_0 = 180 \text{ days}, \tag{3.3}$$

and the value weight is

$$V(z) = \frac{2}{1 + e^{-bz}} - 1, \qquad b = \frac{\ln 3}{m}, \tag{3.4}$$

which is 0 at $z = 0$, $1/2$ at $z = m$ and rises towards 1 as the amount grows. For transfers $m = m_T = 200$ dollars and for allowances $m = m_A = 62$ dollars, the medians of Section 3.5.1, so $b_T = \ln 3/200$ and $b_A = \ln 3/62$ per dollar. The edge weights are sums over transfers:

$$w_T(u, v) = \sum_{e:\, u \to v} \sigma(\Delta t_e)\, V(x_e). \tag{3.5}$$

The sum runs over every transfer of the five tokens from u to v up to the anchor. A self-transfer ($u = v$) is kept and becomes a self-loop on u. The walk restarts at an address in proportion to its activeness,

$$X_u = \sum_{v \neq u} \max_{e:\, u \to v} \sigma(\Delta t_e), \tag{3.6}$$

which adds up, over the addresses that u has sent to, the time weight of its latest transfer to each. The restart distribution of Equation (3.1) is $\pi_u = X_u / \sum_{v \in V_T} X_v$, and it also takes the mass of the dangling nodes. An address that never sends to another has $X_u = 0$ and receives no restart mass, so its score comes from its in-edges alone. The damping factor is $d = 0.85$.

The paper scores Ethereum transactions and gives no parameter values; its time weight is the logistic function of Equation (3.3) written in calendar time instead of age. The values of k and t_0 are this study's choices, fixed in the plan; the plan fixed $b = 1$ on raw base units, and the revision replaced it (Section 3.5.1). With $b_T = \ln 3/m_T$ a transfer of the median amount weighs one half, one of twice the median 0.8 and one of half the median 0.27, so an edge weight grows with the amounts transferred as well as with their number. The activeness of an address still counts the addresses it has sent to, each by the time weight of its latest transfer.

Over the 33 months of the window the decay discounts old events strongly (Figure 3.3). The weight falls from $\sigma(0) = 0.858$ through $\sigma(90) = 0.711$, $\sigma(182) = 0.495$ and $\sigma(365) = 0.136$ to $\sigma(730) = 0.004$, and an event from the first day of the window, $1{,}004$ days before T_{obs}, weighs $\sigma(1{,}004) = 0.0003$. Events older than about a year still create edges but add little weight. Because each row of P is divided by the out-strength, the decay changes the walk only where one address has out-

edges of different ages: an owner whose approvals are all old passes on as much mass as one whose approvals are recent, and only the split of that mass among its edges follows their ages. In AWP the decay also enters the restart weights, so senders that were active only long ago receive little restart mass.

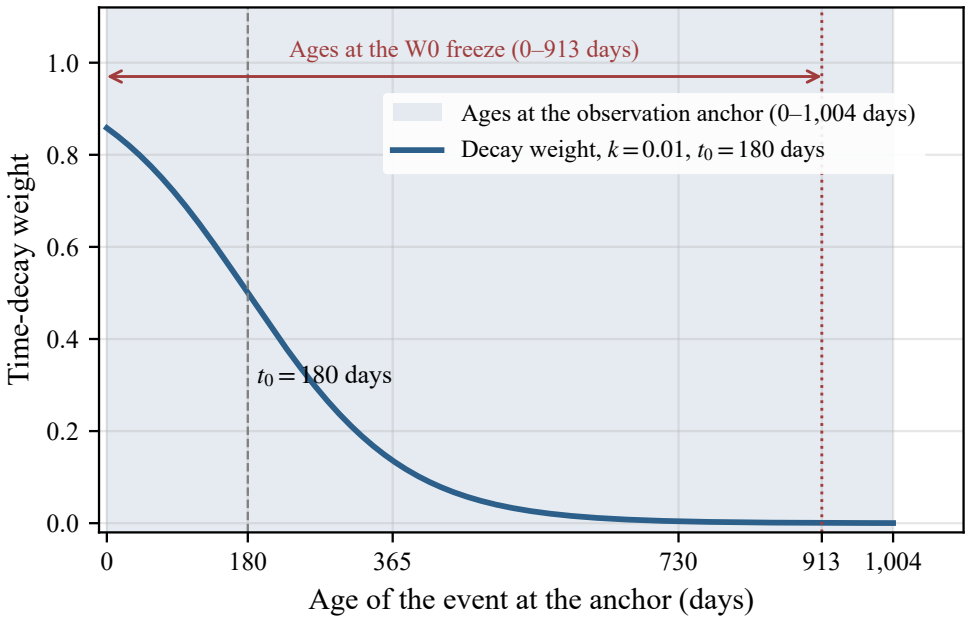


Figure 3.3: Logistic time-decay $\sigma(\Delta t)$ of AWP and EndorseRank ($k = 0.01$, $t_0 = 180$ days). The horizontal axis shows the age of an event in days before the anchor, $T_{\text{obs}} =$ 30 June 2026 or, in W0, $t_1 =$ 31 March 2026, and the vertical axis shows the time weight of a transfer or an approval. The shaded band marks the ages that occur at T_{obs} (0 to 1,004 days) and the arrow those at the W0 freeze (0 to 913 days). Over the band the weight falls from 0.86 to below 0.001.

Algorithm 1 states the construction and scoring steps formally.

3.6.2 EndorseRank allowance graph

EndorseRank puts AWP's edge weights on the allowance graph. Let $\mathcal{A}$ denote the set of latest allowances after preprocessing. For each (token, owner, spender) triple only the most recent `Approval` event up to the anchor is kept, in (block number, log index) order, and a zero allowance deletes the edge. The surviving allowance $a^{\star}(u, v, \text{token})$, valued in US dollars at the anchor's price, was set $\Delta t^{\star}(u, v, \text{token})$ days before the anchor, and it enters the edge as one event of that age and

Algorithm 1 AWP graph construction and scoring

Require: Transfers $\mathcal{T}$ with US-dollar amounts, seed contracts $\mathcal{W}$, evaluated contracts $\mathcal{C}$, anchor T, parameters $k, t_0, b_T, d, \varepsilon, I_{\max}$
Ensure: Scores $\mathbf{s}_{\text{AWP}}$ on $\mathcal{C}$

$E \leftarrow \emptyset$; $X_u \leftarrow 0$ for every address u
for each transfer $e = (u, v, x_e, t_e)$ in $\mathcal{T}$ with $t_e \le T$ **do**
    $\Delta t \leftarrow (T - t_e)$ in days
    accumulate $\sigma(\Delta t)\, V(x_e)$ on edge $(u \to v)$ ▷ Equations (3.3)–(3.5); $u = v$ gives a self-loop
end for
for each sender u **do**
    $X_u \leftarrow \sum_{v \neq u} \max_{e:\, u \to v} \sigma(\Delta t_e)$ ▷ Equation (3.6)
end for
remove non-positive edges; retain edges incident to $\mathcal{W}$
$\pi_u \leftarrow X_u / \sum_v X_v$ for the nodes u of the retained edges
$\mathbf{s} \leftarrow \text{WeightedPageRank}(E, \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equations (3.1)–(3.2)
return $\mathbf{s}_{\text{AWP}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{C}$ (else 0)

amount:

$$w_A(u, v) = \sum_{\text{token}} \sigma\big(\Delta t^\star(u, v, \text{token})\big)\, V\big(a^\star(u, v, \text{token})\big), \tag{3.7}$$

with σ, V, k and t_0 as for AWP and the slope $b_A = \ln 3 / m_A$. A directed edge $u \to v$ means that owner u endorses spender v. An allowance of the median amount enters with half its time weight and an unlimited allowance with its full time weight, so $w_A(u, v)$ grows with the amounts that u has approved to v, but no single allowance counts for more than one. The damping factor is $d = 0.85$. The walk restarts uniformly. AWP's restart rule, carried over to allowances, would restart the walk at owners in proportion to the number of spenders they approved recently. That rewards granting endorsements, while EndorseRank is meant to rank the contracts that receive them. The analysis plan fixed uniform restarts before any data were extracted and named the variant with AWP's restarts as one to report beside EndorseRank (Table 4.9).

The weight keeps the amount and the time of the latest approval; it is not the allowance in force at the anchor. Within a pair, on each token, the newest approval replaces the older ones (Vogelsteller & Buterin, 2015). Approvals granted before 1 October 2023 are not observed, a finite allowance

that `transferFrom` has since reduced keeps its approved value, and an owner who approves an allowance-manager contract such as Uniswap's Permit2 appears only with an edge to that contract, not to the application that later spends through it.

The steps for building the graph and running PageRank are listed in Algorithm 2.

Algorithm 2 EndorseRank graph construction and scoring

Require: Latest allowances $\mathcal{A}$ up to the anchor in US dollars, seed contracts $\mathcal{W}$, evaluated contracts $\mathcal{C}$, anchor T, parameters $k, t_0, b_A, d, \varepsilon, I_{\max}$
Ensure: Scores $\mathbf{s}_{\text{ER}}$ on $\mathcal{C}$

$E \leftarrow \emptyset$
for each latest allowance $(u, v, \text{token}, a^\star, t^\star)$ in $\mathcal{A}$ **do**
    $\Delta t^\star \leftarrow (T - t^\star)$ in days
    accumulate $\sigma(\Delta t^\star)\, V(a^\star)$ on edge $(u \to v)$ ▷ Equation (3.7)
end for
remove non-positive edges; retain edges incident to $\mathcal{W}$
$\mathbf{s} \leftarrow \text{WeightedPageRank}(E, \text{uniform } \boldsymbol{\pi}, d, \varepsilon, I_{\max})$ ▷ Equations (3.1)–(3.2)
return $\mathbf{s}_{\text{ER}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{C}$ (else 0)

Approvals are rarer than transfers, so at T_{obs} the allowance graph of the cohort has fewer edges than its transfer graph (Section 4.1 gives both counts). Since one PageRank iteration costs $O(|E|)$ (Leskovec et al., 2020), this sparsity is the structural basis of H3, and any difference in runtime that follows from it is a property of the graph and not of the algorithm.

For both methods, edge lists are stored as (from, to, weight) triples; duplicate edges are merged by summing their weights, and non-positive weights are dropped before the solve. The solver builds the transposed transition matrix in compressed sparse-row form and the restart vector with vectorised array operations, iterates Equation (3.1) and returns a score for each node. The scores of the evaluated contracts are read from that vector, with zeros for contracts outside the graph.

3.6.3 Coupled operator over both layers (C-PR) and the seeded ablation (S-PR)

EndorseRank and AWP each walk on a single edge type, and RQ5 asks whether one walk over both edge types ranks contracts better than a walk over one. The analysis plan fixed the coupled operator on 9 October 2026, before any log was extracted, on two layers that weight each event by its amount;

the revision values the amounts in US dollars and caps the allowances (Section 3.5.1):

$$\tilde{w}_A(u,v) = \sum_{\text{token}} \min\big(a^\star(u,v,\text{token}), U\big), \qquad \tilde{w}_T(u,v) = \sum_{e:\,u\to v} x_e\,\sigma(\Delta t_e). \tag{3.8}$$

The allowance layer has the edges of EndorseRank and the transfer layer those of AWP; only the weights differ. Amounts enter in US dollars without the bounded transform, an allowance counts at most $U = 20{,}000$ dollars (Section 3.5.1), and the allowance layer carries no time decay. Rule A was fixed in the plan and rule B in the registration of W1, both on the same layers in raw base units; here they are applied to the layers in US dollars. The two layers form one graph on the union node set $V = V_A \cup V_T$, with transition matrices P_A and P_T from Equation (3.2). For node u the coupled row is a convex mixture of the two layer rows, renormalized over the layers in which u has out-edges:

$$\begin{aligned} P^{(\lambda)}_{u,\cdot} &= \alpha_A(u)\,P_{A,u,\cdot} + \alpha_T(u)\,P_{T,u,\cdot}, \\ \alpha_A(u) &= \frac{\lambda\,\mathbf{1}[o_A(u)>0]}{D_\lambda(u)}, \qquad \alpha_T(u) = \frac{(1-\lambda)\,\mathbf{1}[o_T(u)>0]}{D_\lambda(u)}, \end{aligned} \tag{3.9}$$

where $D_\lambda(u) = \lambda\,\mathbf{1}[o_A(u)>0] + (1-\lambda)\,\mathbf{1}[o_T(u)>0]$, and $o_A(u)$ and $o_T(u)$ are the out-strengths of u in the two layers. A node with out-edges in only one layer follows that layer with weight one. A layer of weight 0 is dropped with its edges and nodes, and a node with $D_\lambda(u) = 0$ is dangling, its mass handled by the third term of Equation (3.1). Teleportation is uniform, $d = 0.85$, and $\mathbf{s}_{\text{CPR}}$ is the PageRank vector of Equation (3.1) with $P = P^{(\lambda)}$, under the solver settings of the single-layer methods.

At $\lambda = 1$ the operator walks the allowance layer alone and at $\lambda = 0$ the transfer layer alone; we write these walks C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$). They are the comparators of both rules for RQ5 (Sections 3.7 and 3.7.2). Unit tests check on small fixtures that each end point gives the scores of its layer walked alone with uniform restarts. The family is not continuous at the end points. For $0 < \lambda < 1$ the node set is $V_A \cup V_T$ and a node with out-edges in only one layer follows it whatever λ is, whereas at $\lambda = 1$ a node whose out-edges are all transfers is dangling or absent; as λ approaches

1, C-PR therefore tends to a walk that still follows the transfer edges of such nodes, and the same holds at the other end. The values $\lambda = 0.5$ (primary) and $\lambda \in \{0.25, 0.75\}$ (sensitivity) were fixed in the configuration of 9 October 2026.

In raw base units the plan's layers were very uneven: an owner that held an unlimited allowance passed almost all of its mass through it, and in the transfer layer a token with many decimals outweighed one with few (Section 3.5.1). In US dollars the transfer layer adds comparable amounts, and the cap U lets an unlimited allowance weigh as much as an allowance of the size that 99% of transfers do not exceed, so in the row of an owner that holds unlimited and finite allowances side by side an unlimited allowance weighs as much as a finite allowance of U dollars or more, and the smaller allowances keep a share of the row in proportion to their amounts (Appendix 8.4.2 gives the counts). EndorseRank and AWP bound every amount through V instead.

In the S-PR ablation the transfer layer is walked, but the uniform teleportation vector $\boldsymbol{\pi}$ of Equation (3.1) is replaced by the score of C-PR ($\lambda = 1$), restricted to V_T and renormalized over it. Let $\mathbf{r}_A$ be the PageRank vector of the allowance layer walked alone on V_A, extended by $r_{A,u} = 0$ for $u \notin V_A$. Then

$$\pi_u = \frac{r_{A,u}}{\sum_{v \in V_T} r_{A,v}} \qquad (u \in V_T),$$

so a transfer-graph node without an allowance score receives zero restart mass. If no node of V_T lies in V_A, the denominator is zero and $\boldsymbol{\pi}$ falls back to the uniform distribution on V_T. The same vector redistributes dangling mass. The result is the TrustRank construction (Gyöngyi et al., 2004), but with allowance-derived seeds instead of hand-labeled ones. Allowances decide where the walk restarts; transfers decide how it spreads. A comparison of S-PR with C-PR separates what the allowance edge contributes as a propagation medium from what it contributes as a seed.

The edge set determines what the coupled operator costs. Because C-PR walks on $E_A \cup E_T$, one of its iterations costs as much as an iteration on each layer, and its runtime is expected to be at or above AWP's. S-PR needs one solve on each layer. Claim C1 therefore concerns the allowance graph that EndorseRank walks; the coupled walks carry the edge count of the transfer layer with them. Both

operators are timed under the benchmark protocol of Section 3.11.

3.7 Temporal holdout protocol

In the holdout W0, scores use every extracted event from the start of the data, 1 October 2023, up to t_1 = 31 March 2026, 23:59:59 UTC. Labels are counted on events from 1 April 2026, 00:00:00 UTC, up to t_2 = 30 June 2026, 23:59:59 UTC. The spender cohort consists of the contracts that hold at least one positive latest allowance of one of the five tokens at t_1 in the extracted approval data; spenders without code or with an EIP-7702 designator are left out (Section 4.1.1 gives the counts). It includes spenders outside the matched cohort, which entered the extract because a cohort contract approved them. Every score (EndorseRank with both restart rules, AWP, C-PR at three values of λ and at its two end points, S-PR and two raw-degree baselines) is computed from events up to t_1 alone. The time decay is anchored at t_1 as well: Δt in Equation (3.3) is measured back from t_1, so ages run up to 913 days and the weights from 0.858 down to below 0.001.

We count two labels per spender, one for each construct, so that in the future window each single-layer method is tested on its own construct and the coupled walks on both:

1. New approval pairs: the number of owners that give the spender a positive approval in the label window although their owner–spender pair held no positive allowance at t_1. This label measures the endorsement construct.
2. New transfer senders: the number of distinct addresses that send the spender a transfer in the label window without having sent it one between the data start (1 October 2023) and t_1, self-transfers excluded. It measures the flow construct and is the transfer-side counterpart of the first label.

Both labels count approvals and transfers of the five tokens only. For baselines we take each spender's in-approve degree (the number of distinct owners with a positive latest allowance) and its transfer in-degree (the number of distinct senders other than the spender itself), both at t_1. Each

single-layer PageRank smooths one of these local statistics. Because their rows appear in the table, the increment of each PageRank over its own raw degree can be read off directly. That increment answers RQ4; the PageRank coefficient on its own does not.

Uncertainty comes from the paired bootstrap over spender contracts of Section 3.10 ($B = 400$, one shared index matrix), so every difference between two coefficients on the same label is a paired difference. The two contrasts fixed in the plan before any log was extracted are those of rule A below, applied here to the revised scores after the registered results were known: C-PR minus C-PR ($\lambda = 1$) on new approval pairs, and C-PR minus C-PR ($\lambda = 0$) on new transfer senders. Every other contrast in this window was computed after the labels were known: each single-layer PageRank against its own degree, C-PR ($\lambda = 1$) against the in-approve degree, the two contrasts of S-PR with the layers walked alone, C-PR against C-PR ($\lambda = 0$) on new approval pairs, and C-PR against EndorseRank and AWP. The differences between EndorseRank and AWP are in Appendix 8.5. AWP scores zero a spender without a transfer edge at t_1, so the degree contrasts are also computed on the spenders that had received a transfer before t_1 (Table 4.12); this restriction was not fixed in advance either.

Rule A, fixed in the plan, judges C-PR against its own layers walked alone and has two parts. The primary part uses W0: C-PR at $\lambda = 0.5$ should do no worse than C-PR ($\lambda = 1$) on new approval pairs (the interval of $\Delta\tau = \tau_{\text{CPR}} - \tau_{\lambda=1}$ contains zero or lies above it) and should be better than C-PR ($\lambda = 0$) on new transfer senders (the interval of $\tau_{\text{CPR}} - \tau_{\lambda=0}$ lies above zero). The secondary part uses the same window and the matched cohort. Its condition is that C-PR's point estimate lies within the 95% interval of the better of the two layers on the transfer family and on the allowance family, and that C-PR exceeds both layers on the Sybil-stability family. The paired contrasts of Table 4.4 are reported beside this rule as a stricter check; they do not decide it. If its primary part fails, the second half of RQ5, whether C-PR beats a walk on either layer alone, receives a negative answer, and the result is reported with its intervals. The first half, whether C-PR improves on a walk over the transfers alone, is not covered by rule A; rule B tests it in W1 (Section 3.7.2). Once labels have been observed, λ is not tuned. When rule A was fixed, no log of the study had been extracted. On the revised scores it is applied with the same conditions, but after the registered results were

known, so its verdict there is not a registered test.

The pipeline also counts two further quantities over the label window: revocations, the owners of a positive allowance at t_1 that set it to zero, and a drain heuristic, the owners whose new approval to the spender is followed by a transfer of the same token out of the owner's account, within six hours and of at least half the approved amount, or within one hour when the approval is of 10^{30} base units or more. Both count events of the five tokens only. Neither is offered as evidence. Both rise with the number of owners a spender has, so they track use and cannot separate trust from it (Section 4.6).

3.7.1 Trader outcomes in W0

The spender cohort says nothing about contracts that trade. An exploratory run applied the freeze t_1 to the GMX V2 accounts that are contracts, appear in the freeze-date graph in either layer and close at least three positions between 1 April and 30 June 2026 (Section 4.1.1 gives their number). Its labels are the number of profitable closes, the realized gain, and the shares of closes that were profitable and that were not losses. A close is profitable when it is not a liquidation and its base profit is positive, and a loss when it is not a liquidation and its base profit is negative; the realized gain sums the positive base profits of a contract's closes in USD. Every score of the spender holdout was set against these labels with 400 paired resamples, under no rule, and Chapter 4 reports the run as exploratory. No liquidation label was computed in this run; the liquidation labels of both windows were first computed under the registration of Section 3.7.2. With so few contract traders, every interval of this run is wide.

3.7.2 Registered window W1: July to September 2026

In the registered analysis the W0 holdout also showed something that no rule had asked about. Against its transfer layer walked alone, C-PR ranked the spenders that later gained new approval pairs higher, with an interval above zero, while it was worse on new transfer senders (Appendix 8.6.10). This was seen only after the labels were in, so it counts as a hypothesis and

not as a finding, and it is tested on a window that had not been extracted when the test was fixed. The test was written into a registration document after the W0 results were known. That document and the configuration, in which the status of the window was set to registered, were committed to the public repository at 16:49:32 UTC on 9 October 2026 (commit `0d3a057`), together with the extraction and evaluation code. Scan B, the only query that reads logs dated after 30 June 2026, ran after that commit. The extraction script refuses to run unless the registration and the configuration are committed without local changes and the status is registered. It writes the commit and the SHA-256 hashes of both files into an extraction manifest, which records that the extraction started one second after the registration commit (Appendix 8.1).

The method is the one fixed in the plan; in the revised analysis the amounts, the tokens and the cohort are those of Section 3.5.1, and everything else is unchanged. C-PR is used at $\lambda = 0.5$, together with the two sensitivity values and S-PR. The solver settings, the SQL, the extraction filter of the matched cohort, the GMX decoder and the paired bootstrap (400 resamples, seed 42) are those of W0. Scores are frozen at $T_{\text{obs}} =$ 30 June 2026, 23:59:59 UTC, so they are the scores of the same-window evaluation, built from every event since 1 October 2023 with ages up to $1{,}004$ days. Labels are counted from 1 July to 30 September 2026.

Two cohorts are evaluated, and Section 4.1.1 gives their sizes. The spender cohort is defined as in W0, as every contract spender holding at least one positive latest allowance at the freeze. The trader cohort is the GMX V2 accounts that are contracts, are nodes of the freeze-date graph in either layer and close at least three positions in the label window. The spenders keep their two labels. The traders get the liquidation-free close rate, one minus the share of a contract's July–September closes that ended in liquidation. It is the nearest thing to a default that these data contain, it is built from neither graph, and because it is a rate it does not reward a contract merely for trading more. When it was chosen, no score had been set against a liquidation label in either window, and the W0 trader run had used only the profit labels of Section 3.7.1. What was known about liquidations came from the decoding step, which had shown how common they are among the closes of contract accounts in the observation window (Section 4.1.1).

Rule B compares C-PR with its transfer layer walked alone, C-PR ($\lambda = 0$), through the paired difference $\Delta\tau = \tau_{\text{CPR}} - \tau_{\lambda=0}$:

- F1, spenders, new approval pairs: the 95% interval of $\Delta\tau$ lies above zero;
- F2, spenders, new transfer senders: the lower end of the interval lies above $-\delta$;
- F3, traders, liquidation-free close rate: the lower end of the interval lies above $-\delta$.

C-PR counts as improving on a walk over the transfers alone only if all three hold, so no correction for multiple testing is needed. The margin $\delta = 0.01$ comes from W0: it is the half-width of the paired interval of the corresponding contrast on new transfer senders, C-PR minus C-PR ($\lambda = 0$) in the registered analysis (Appendix 8.6.10), rounded up to two decimals, since a smaller difference could not be resolved at these sample sizes. One weakness of F3 was written down with the rule. If neither score predicts liquidation, F3 can pass with both coefficients near zero, so the two coefficients are reported next to the verdict. Three further results are reported in a fixed order and do not enter the decision: C-PR against C-PR ($\lambda = 0$) on the liquidation-free close rate as a superiority test (F4), C-PR against C-PR ($\lambda = 1$) on the same label (F5), and the primary part of rule A, rerun on the new window (F6).

Two sensitivity analyses were registered in the same commit. Both are reported next to the decision, and neither enters it. The first repeats F1–F4 with AWP (Section 3.6.1) in place of C-PR ($\lambda = 0$). The second repeats F1–F6 with the cohort contracts that a graph leaves out kept as isolated nodes instead of scored zero (Section 4.4). Only the two layers walked alone change under the second analysis, since every spender is in the allowance graph and every trader in the union graph that C-PR walks. The run that judged rule B also set EndorseRank against the in-approve degree, AWP against the transfer in-degree and C-PR against EndorseRank. These contrasts were not registered; they were computed after the W1 labels were known and are reported apart from the decision, as are the supplementary contrasts computed later.

3.7.3 Outcomes built from neither edge type

A comparison of the two edge types on the GMX trader labels, which are built from neither, was registered in the same commit as rule B, before scan B. As in a preregistration (Nosek et al., 2018), its contrasts, decision rules and the wording of each conclusion were written down before the data that decide them were extracted. It reuses the trader cohorts and the label code of Sections 3.7.1 and 3.7.2 in two windows: W1 (primary; freeze on 30 June 2026, labels from July to September) and W0 (secondary; freeze on 31 March 2026, labels from April to June, the cohort defined in the same way). When the comparison was registered, the W0 liquidation labels had not been computed for any score; the W0 profit labels and their single-score coefficients were known from the exploratory run.

Four pairs set allowance against transfer: in-approve degree and transfer in-degree; EndorseRank with AWP's activity restarts and AWP, the like-for-like PageRank pair; EndorseRank and AWP; and the amount-weighted layers C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$). C-PR ($\lambda = 0$) minus AWP isolates AWP's weights and restarts. All six trader labels are reported: the liquidation-free close rate (primary), a no-liquidation flag, profitable closes, realized gain, and the profitable and non-loss shares of closes. Two contrasts on the liquidation-free close rate in W1 decide: P, in-approve degree minus transfer in-degree, and Q, EndorseRank with activity restarts minus AWP, each with a Bonferroni 97.5% interval from $B = 2{,}000$ paired resamples (seed 42). Because the chance of at least one liquidation rises with the number of closes, τ_b is also computed within strata of 3–4, 5–9, 10–24 and 25 or more closes in the label window and weighted by each stratum's contract pairs. Recipients, the traders with an in-approve degree above zero at the freeze, are compared with the other traders on every label, and an influence check leaves out the ten recipients of highest in-approve degree.

R1, a count-level lead on liquidation avoidance, needs the 97.5% interval of P and the 95% interval of the stratified P above zero in W1, and a positive P in W0. R2, a PageRank-level lead, needs the 97.5% interval of Q above zero in W1 and a positive Q in W0. R3 applies if P on the profitable or

the non-loss share in W1 is below zero with a 95% interval excluding zero; the allowance signal is then read as specific to liquidation avoidance. The wording of the conclusion for each outcome was fixed in the same commit. If R1 holds, the approval count at the freeze ranked contract traders above the transfer count on later liquidation avoidance, and if it fails there is no count-level lead of allowance over transfer on that outcome. R2 is worded in the same way for the PageRank pair, and its failure reads as the PageRank versions not being resolved. If R3 applies, the signal concerns forced-liquidation avoidance and not trading success. On the revised scores the same contrasts, rules and wording are applied after the registered results were known.

3.8 Construct-check design

Every score enters the same-window evaluation on the matched cohort: EndorseRank with both restart rules, AWP, C-PR at its five values of λ and S-PR. For each score M and check q the pipeline reports Spearman's ρ and Kendall's τ between $\mathbf{s}_M$ and $\mathbf{p}_q$ with 95% bootstrap intervals (Section 3.10), each check oriented so that a higher value means higher expected reputation, as in the evaluation of AWP by its authors (T. Do et al., 2023).

The same-window evaluation is organized around two contemporaneous families:

1. Transfer (the proxies against which T. Do et al. (2023) evaluated AWP): the contract's inbound transfer activity.
2. Allowance: inbound approval activity, where the contract is the spender.

Both families are graph-adjacent: their inputs are the event classes that make up G_{transfer} and G_{approve}, although each check is a local statistic and not a global score. Agreement of AWP with the transfer checks, or of EndorseRank with the allowance checks, is an intended-construct check: it shows coherence within a construct, part of it is built in, and it is not evidence of independent predictive power. A third family, the Sybil-adjusted activity summaries of Section 3.9.3, enters the secondary part of rule A, but no hypothesis rests on it.

The temporal holdouts of Section 3.7 provide the out-of-window checks. Packin and Lev-Aretz

(2024) warn that decentralized credit scores can become opaque; publishing explicit formulas (Section 3.9) and keeping same-window checks apart from the time splits reduce that opacity.

3.9 Operational definitions of proxies

In what follows, $\mathcal{W}$ denotes the matched cohort, and the proxies are computed over the observation window up to T_{obs}. The notation is that of the proxy queries and the evaluation code in the open pipeline.

3.9.1 Transfer family

Write $\mathcal{T}_{\text{in}}(w)$ for the set of transfers received by contract w in the window and u_e for the sender of transfer e. Self-transfers belong to $\mathcal{T}_{\text{in}}(w)$, so a contract that sends tokens to itself counts itself as one sender, and the amounts x_e are summed in US dollars over the five tokens. Define

$$\text{in_degree}(w) = \big|\{\, u_e : e \in \mathcal{T}_{\text{in}}(w) \,\}\big|, \tag{3.10}$$

$$\text{in_value}(w) = \sum_{e \in \mathcal{T}_{\text{in}}(w)} x_e. \tag{3.11}$$

Both are baselines defined in the AWP paper (T. Do et al., 2023), on the premise that economically important addresses receive transfers from more distinct counterparties and in greater total value. Unlike PageRank they are local and ignore the senders' reputations, so comparing AWP with them shows how far propagation, recency weighting and restarts at active senders move its ordering away from the raw inflow.

3.9.2 Allowance family

On the allowance side, w is the spender (owner $\rightarrow$ w). With $a^\star$ the latest allowances of Equation (3.7), define

$$\text{in_approve_degree}(w) = \big|\{\, u : a^\star(u, w, \cdot) > 0 \,\}\big|, \tag{3.12}$$

$$\text{in_approve_value}(w) = \sum_{u,\text{token}} \min\big(a^\star(u, w, \text{token}), U\big). \tag{3.13}$$

Like the transfer baselines, they count and sum incoming approvals without following multi-hop endorsement chains; the in-approve value uses the capped amounts of the allowance layer of C-PR.

3.9.3 Sybil-adjusted stability family

The inbound counters leave out self-transfers ($u = w$). Let $N_{\text{tx}}(w)$ be the number of transfers that w receives from other addresses and $N_{\text{uniq}}(w)$ the number of distinct inbound counterparties. Tenure and active months use a second set of rows. Every transfer that involves a cohort contract is attributed to its sender if the sender is in the cohort, and otherwise to its recipient; a transfer between two cohort contracts thus counts only for the sender. We take $t_{\min}(w)$ and $t_{\max}(w)$ to be the earliest and latest timestamps among the rows attributed to w. $N_{\text{mon}}(w)$ is the larger of two counts of distinct calendar months: the months in which w received a transfer from another address, and the months of the rows attributed to w. Define

$$\text{inbound_counterparty_ratio}(w) = \frac{N_{\text{uniq}}(w)}{\max(N_{\text{tx}}(w), 1)}, \tag{3.14}$$

$$\text{transfer_tenure_days}(w) = \frac{t_{\max}(w) - t_{\min}(w)}{86400} \quad \text{(seconds to days)}, \tag{3.15}$$

$$\text{active_months}(w) = N_{\text{mon}}(w). \tag{3.16}$$

The counterparty ratio penalizes contracts that receive many transfers from few addresses, a simple wash pattern. It does not penalize the usual Sybil pattern, in which many fresh addresses send one

transfer each; a contract fed that way scores close to the maximum of one. Tenure and active months track how persistent on-chain activity is over the observation window. These proxies are therefore weak Sybil checks, and colluding adversaries can satisfy all three. They rest on an assumption of this study: activity that lasts and involves many counterparties says more about reputation than short bursts of transfers from few addresses.

3.10 Statistical tests

Neither the reputation scores nor the proxies are binary default indicators; both are continuous and are handled as ranks. As in the evaluation of AWP (T. Do et al., 2023), concordance is measured with rank correlations, which are invariant under any monotonic re-scaling of either variable. Each coefficient is reported with a bootstrap confidence interval.

- Spearman's ρ: the Pearson correlation of the mid-ranks R_i and S_i of the raw score and the raw proxy value of contract i, tied values sharing the average of their positions (Section 2.9). It measures monotonic agreement: if higher reputation consistently goes with higher proxy values, ρ is positive whatever the nonlinearity in the raw scores.
- Kendall's τ: every reported value is the tie-corrected τ_b of Equation (2.6), computed on the raw scores and proxies. Without ties, $(1+\tau)/2$ is the probability that a randomly chosen pair is ordered the same way by both rankings. With ties that reading belongs to $\tau_a = (n_c - n_d)/n_0$ and not to τ_b. Ties are rare among the PageRank scores of the matched cohort (Section 4.4), but they are common in the raw degrees, which are integer counts, in the holdout labels, which are zero for most spenders, and in the trader labels, whose rates are often one.
- Why both: the AWP protocol (T. Do et al., 2023) reports both, and reporting both keeps the conclusions from hinging on one coefficient. The score distributions are heavy-tailed, with most contracts near zero and a few hubs holding much of the mass, which is why rank coefficients are used instead of a Pearson correlation on raw scores, which the hubs would dominate.

- Family aggregation: A family is summarized by the plain average over its proxies. The transfer-family mean τ, for example, is $\frac{1}{2}(\tau_{\text{in_degree}} + \tau_{\text{in_value}})$. On a fixed cohort and fixed proxies, a comparison between scores is only a relative statement about those scores and implies no universal property of reputation. Our conclusions are drawn from the confidence intervals, and no absolute τ cutoff is used as a decision criterion.
- Uncertainty: Sampling uncertainty is estimated with a paired bootstrap over contracts (Efron, 1979). The evaluated cohort (the matched cohort in the same window, the spender or trader cohort in a holdout) is resampled with replacement $B = 400$ times (seed 42, fixed in the configuration), and one index matrix is reused for every method and proxy or label, so the resampled statistics are paired across methods. For each τ, family mean and inter-method difference, the 2.5th and 97.5th percentiles of the bootstrap distribution form a 95% percentile interval (DiCiccio & Efron, 1996). Because τ_b is a smooth function of U-statistics, the percentile interval is first-order accurate at the size of the matched cohort and of the spender cohorts; for the trader cohorts, which are small (Section 4.1.1), the intervals are wide and their coverage is less certain. The small number of resamples keeps the $O(B\,n \log n)$ cost of the tie-corrected τ tractable across all methods and proxies. Its price is a Monte Carlo error in each interval end, which for a normal-shaped bootstrap distribution is about 0.13 times the bootstrap standard error. Only the analysis of Section 3.7.3 departs from this, with $B = 2{,}000$.
- Difference tests: Every comparison is a paired contrast on the same resamples. Resample by resample we compute $\Delta\tau = \tau_A - \tau_B$ and report the 95% percentile interval of its bootstrap distribution; a contrast counts as a difference when its interval excludes zero. Because resampling is paired, the contrast keeps the covariance of the two estimates, $\text{Var}(\hat{\tau}_A - \hat{\tau}_B) = \text{Var}\,\hat{\tau}_A + \text{Var}\,\hat{\tau}_B - 2\,\text{Cov}(\hat{\tau}_A, \hat{\tau}_B)$, and with positive covariance its interval is narrower than the two separate intervals suggest. The contrasts that decide a rule were fixed before the data they are judged on: those of rule A (Section 3.7) before any log was extracted, and those of rule B and the decision contrasts P and Q (Sections 3.7.2 and 3.7.3) before the W1 logs were extracted, together with F4–F6 and the two sensitivity analyses, which are reported beside

the decision. These dates hold for the registered analysis; on the revised scores the same contrasts are computed after the registered results were known (Section 3.5.1). C-PR against its transfer layer walked alone on new approval pairs in W0 was not among them; it is marked as exploratory wherever it appears. In the same window, eight contrasts set C-PR and S-PR against whichever layer, walked alone, leads each family (the transfer walk on transfer, the allowance walk on allowance, both on Sybil stability), and four set C-PR against AWP on transfer, against EndorseRank on allowance and against both on Sybil stability. None of these twelve was fixed in the plan; they were computed with the same-window results and are reported beside the secondary part of rule A. Five same-window contrasts that involve EndorseRank and AWP, none fixed in advance either, are reported in Appendix 8.5.

- Directionality and imputation: Each proxy is oriented so that a higher value means a better reputation. After re-alignment to the evaluated cohort, a contract absent from a given proxy source is imputed as zero. With sparse transactions this is a conservative choice, and both scoring approaches apply it in the same way. After this step every evaluated contract has a value in every score and proxy vector, so each correlation uses the whole cohort.
- Non-inferiority: Two of the three conditions of rule B ask only that C-PR not fall behind its transfer layer walked alone by more than a margin fixed in advance, $\delta = 0.01$. Such a contrast passes when the lower end of its 95% interval lies above $-\delta$. The test is weaker than a difference test and says nothing about superiority on that label. It is used where the claim under test is that C-PR gives up nothing that matters.
- What is excluded: The study attempts no causal inference, trains no supervised models on separate training and testing sets, and reports no p-values; it asks whether the interval of a contrast excludes the null value.

3.11 Computational benchmark (Claim C1)

Claim C1 concerns the computational cost of each score. The benchmark times the solver call that turns an edge list into scores. It leaves out the one-time cost of aggregating and loading the pair tables, which a scoring service that refreshes its scores periodically would amortize.

- Timed component: The edge arrays and AWP's restart weights are built before the timer starts. The timed call indexes the nodes, assembles the sparse transposed transition matrix, forms the restart vector and runs power iteration with $d = 0.85$, $\varepsilon = 10^{-8}$ and $I_{\max} = 300$. Matrix assembly is vectorised: node identifiers are mapped through a lookup array, the out-strengths come from one weighted count over the source nodes, and the matrix is built in one sparse conversion. Assembly and each iteration therefore both take time linear in $|E|$, and a deployment would repeat both at every update. The calls timed are EndorseRank, AWP, C-PR at $\lambda = 0.5$ and S-PR, the last including the solve of the allowance layer that seeds it. The same graphs give the alignment scores.
- Measurement protocol: Every configuration is measured in one session on an otherwise idle machine. An untimed warm-up run, on which alone peak memory is traced, is followed by five timed runs; we report the mean and standard deviation of the five run times, the number of iterations, and the nodes $|V|$ and edges $|E|$ of the solver graph. Peak memory is the largest allocation traced by Python's `tracemalloc` module during the warm-up call. The same graph under the same settings occurs in the main benchmark, in the $d = 0.85$ rows of the damping sweep and in the scaling stage with $n = 15{,}107$, which is the full cohort graph. Those rows reuse the one measurement, and the tables say so; they are not independent measurements of one quantity. Every runtime table lists node and edge counts, since edges drive the cost of an algorithm with $O(|E|)$ work per iteration (Leskovec et al., 2020), and the runtimes are written to the machine-readable output (Section 3.13).
- Cohorts: The main benchmark runs on the graphs of the matched cohort at T_{obs}, the graphs of the same-window evaluation. For the growth of runtime with graph size we draw deterministic

subsamples of the cohort (Section 3.12).

- Fairness: Every algorithm runs with the same solver, tolerance, iteration cap, damping (outside the damping sweep) and graph construction, so differences in cost come from the graphs and from the restart vectors, which can change the number of iterations, and not from the implementation. The benchmark summary records the platform, the processor and the Python version, Appendix 8.1.1 names the machine and the library versions, and the cost of two scores is compared through time ratios, each reported beside the corresponding edge ratio.

3.12 Scaling and robustness extensions

We also examine how runtime grows with the size of the graph, and how far the family-level pattern moves when the damping, the slope of the value weight or the evaluation sample changes. These analyses qualify Claim C1 and test the reach of the family-level conclusions; none of them supplies an alternative primary sample. Unless stated otherwise, alignment claims in Chapter 4 refer to the matched cohort.

3.12.1 Cohort subsamples for runtime scaling

The scaling stages are deterministic subsamples of the matched cohort. Contracts are sorted by

$$\texttt{SHA256(}\textit{seed}\texttt{:}\textit{address}\texttt{)} \text{ with } \textit{seed} = \texttt{contract-benchmark-v1},$$

that is, by the hash of one string made of the seed, a colon and the lower-cased address, and the stage of size n takes the first n contracts in that order. The seed string is a constant fixed in the configuration of 9 October 2026. The stages are $n = 1{,}000$, $2{,}000$, $5{,}000$, $10{,}000$ and the full cohort, $n = 15{,}107$. Since the hashing is deterministic, the subsamples cannot drift between library versions the way pseudorandom draws do, and each stage contains the one before it.

At every stage the graph keeps the edges of the cohort graph at T_{obs} that have at least one end point among the sampled contracts, and the solver's node set consists of the end points of those edges. The edge count therefore grows with the stage, up to the full cohort graph, but not in proportion

to n: a contract with many counterparties brings many edges, and an edge between two sampled contracts is counted once. The stage graphs are ego networks of subsets of one cohort, not samples of a larger chain, and no stage goes beyond the full cohort graph. The experiment therefore shows how runtime grows with $|E|$ up to that size and says nothing about larger graphs. Every scaling table reports per-stage $|V|$ and $|E|$ for both methods, and the scaling statement in Chapter 4 goes no further than those columns allow.

3.12.2 Damping sensitivity

On the matched cohort we vary the damping over $d \in \{0.75, 0.85, 0.95\}$, the values fixed in the plan, with the graphs and proxies held fixed. A smaller d raises teleportation; a larger one gives multi-hop paths more weight and may slow convergence. The sweep asks whether alignment on the allowance and transfer families depends on the teleportation probability.

3.12.3 Value-slope sensitivity

The plan fixed a check on subgraphs of twenty tokens, chosen by summed raw amount and by number of rows. Once every graph keeps five tokens that check loses its point, and it is replaced by a check of the parameter that the revision introduced (Section 3.5.1). EndorseRank and AWP are recomputed at T_{obs} with the slope b of V divided and multiplied by ten, which moves the point of half weight to ten times and to a tenth of the median amount. The check asks whether the family-level results depend on how steeply the amount weight rises. The registered analysis keeps its token check, published with its results.

3.12.4 Sample-definition sensitivity

At each scaling stage of Section 3.12.1 the family means of EndorseRank and AWP are recomputed on the stage's contracts, scored on the stage's graph. A stage differs from the full cohort in size and in its graph: a contract's score changes because the edges of the contracts left out of the stage are missing. The per-stage coefficients are therefore a sensitivity check on the sample definition and

not evidence of stability, and no intervals were computed for them; the question is how far each coefficient moves when the sample changes. This analysis was not in the plan; it was run with the robustness checks after the W1 labels were known.

3.13 Reproducibility

No table in Chapter 4 was transcribed by hand. Once the Phase 1 tables are on disk, the evaluation drivers recompute every check, every alignment statistic with its interval, every holdout result and the benchmark, and write them to machine-readable summaries. An export script writes the tables and figures of Chapter 4 and Appendix 8.1 from those summaries, and a publishing script copies the summaries with their SHA-256 checksums and a digest that names, for every number, the file and the key it comes from. One versioned configuration file holds the windows, the cohort threshold, the event topics, the damping, the decay parameters, the values of λ, the bootstrap count and seed, the benchmark protocol and seed, the cost budget and the part size. The revised analysis adds the token list, the daily price table and the constants of Section 3.5.1 as files in the repository, and one switch selects which of the two analyses the scripts compute.

The design was fixed in commits to the public repository at https://github.com/abitocodes/phd_works: the analysis plan and the configuration on 9 October 2026 (commit `43d8045`), before any log was extracted; the registration of W1 together with the extraction and evaluation code at 16:49 UTC on the same day (commit `0d3a057`), before scan B; and the revision of the amounts on 10 October 2026 (commits `a897475`, `09dfc32` and `2705b28`), before any revised score was computed. The reproducible elements are these: deterministic subsampling of the cohort by SHA256 hashes; a fixed PageRank tolerance and iteration cap; one bootstrap seed and one resample-index matrix for each cohort; the proxy formulas of Section 3.9; the benchmarking procedure of Section 3.11, with the time of every run kept in the output; the coupling weight λ and the holdout logic of Section 3.7, both fixed in the configuration before any log was extracted; and the extraction manifest of scan B with the hashes of the registered files. From the downloaded tables the alignment, holdout and

benchmark results can be rebuilt without any new BigQuery query. The machine-readable summaries behind all the tables, together with the configuration, the analysis plan and the registration, are committed to the repository under `experiments/data/3-published-results-for-thesis/`, those of the revised analysis in its subfolder `revised-usd/`, in the same project folder as the source code. The numbers in this thesis can thus be checked against those summaries without rerunning the pipeline. Each summary records the code revision that wrote it, and the digest lists any script changed since that revision. Appendix 8.1 lists the repository URL, the commit hashes, the configuration file and the summary files.

3.14 Sampling considerations and missingness

The matched cohort is a constructed sample, not a random draw from all Arbitrum contracts: eligibility needs non-zero approvals of one of the five tokens from at least three distinct owners in the window, and transfer activity plays no part in it. Same-window findings therefore apply to contracts on Arbitrum One that at least three owners approved on WETH, USDC, USDT0, ARB or WBTC in these months, and holdout findings to the contract spenders in the cohort's approval data. Contracts that fewer than three owners approved, and contracts that no owner approved at all, are excluded by design, and cohort contracts outside a graph score 0 under that method (Section 3.2). Because the cohort is defined from approvals, it favors the allowance side in coverage: every cohort contract received approvals at some time in the window, while AWP scores zero the cohort contracts that sent and received no transfer in the window (Table 4.5).

The spender cohorts of the holdouts include spenders outside the matched cohort, which entered the extract because a cohort contract approved them. For such a spender both the scores and the labels see only the approvals of the five tokens that cohort contracts granted it and the transfers of those tokens it exchanged with cohort contracts, so its counts understate its activity on the chain. The trader cohorts are small (Section 4.1.1) because few contract accounts closed GMX V2 positions in these months, and every trader result has wide intervals.

Data can be missing in a few ways. An amount that cannot be read gives zero (Section 3.5). EndorseRank does not see approvals that emit no standard ERC-20 event, nor allowances set before 1 October 2023 and not changed since, and AWP does not see transfers made off-chain or inside custodial accounts. Both are thus biased toward on-chain, standard-token behavior, which is acceptable for an on-chain reputation study but matters once results are extrapolated to all contracts of a protocol. The revision leaves out the tokens other than the five by design and counts their rows. Its price rules would also leave out a transfer made before its token's first daily price and give a day without a price point the previous day's price, but neither case occurs for the five tokens, and no point of theirs was replaced as a data error; the only days without a point under a token's Arbitrum address are those filled from the series under its CoinGecko identifier (Appendix 8.4.2).

3.15 Implementation notes

The pair tables, the proxies and the labels are computed in BigQuery SQL. The PageRank code is written in Python, with a SciPy sparse transition matrix, NumPy power iteration and vectorised matrix assembly, so that graphs with tens of millions of edges fit one solver call. Each configuration builds its edge arrays once. The timing benchmarks leave out I/O and the aggregation of the pair tables and cover operator assembly and iteration only (Section 3.11). The benchmark ran on one machine, an Intel Core Ultra 9 185H processor with 64 GB of memory under Windows 11, with Python 3.11.9, NumPy 2.3.2 and SciPy 1.16.1 (Appendix 8.1.1). Randomness enters only through the bootstrap, whose seed is fixed; given the downloaded tables and the configuration, evaluation is deterministic. Small floating-point differences between platforms are to be expected, but they would be negligible against the three-decimal-place precision of the τ metric.

Figures are generated from the same evaluation summaries as the tables, so the plotted values agree with the tabulated numbers.

3.16 Ethics

Only publicly available blockchain data are used in this research, and it involves no human subjects. The study was reviewed by the institutional ethics committee (Unisa SoC/CAES). Appendix 8.7.1 contains the ethics clearance certificate (reference [Unisa SoC/CAES ethics clearance ref: TBD], dated [TBD]).

- Privacy and anonymity: A blockchain address is a pseudonym and does not by itself reveal the real-world identity behind it (Packin & Lev-Aretz, 2024). Pseudonymity is not the same as anonymity, however, and in theory a determined adversary could tie an address to a person through exchange deposits, online profiles or heuristics. This study makes no such links. The scored addresses are contracts; EOAs, which may belong to individuals, appear only as owners, senders and recipients, and no result is reported for any of them. Our data collection is limited to on-chain `Approval`, `Transfer` and GMX V2 position events and to the code stored at an address, and we collected no off-chain personally identifiable information (such as IP addresses, exchange KYC records, social media handles or deanonymization graphs). We did not cluster addresses into real-world entities, publish leaderboards of addresses likely to be personally identifiable, or try to join an address's transaction history with other databases to infer identity. The published results are aggregate measures only (correlation coefficients, runtimes, family means) and contain no individualized rankings. Where the main text quotes examples, they name broad categories (for example, 'spenders with high allowances') and never specific addresses.
- Data minimization: Scan A keeps every ERC-20 `Approval` and `Transfer` event of the window in a table of the study's own cloud project, because the cohort rule needs every approval. All later queries and every downloaded table are restricted to the ego network of the cohort and to the fields that graph construction and the checks require, namely the relevant addresses, token amounts and timestamps. The extracted tables are kept so that the research pipeline can be audited. They will not be released for the purpose of deanonymizing users.

- Data usage and compliance: Every query ran against public data hosted in the `bigquery-public-data.goog_blockchain_arbitrum_one_us` dataset and complied with the Google Cloud terms of service and rate limits. A dry run before every query and a budget for the whole study prevented unnecessary scans. The account-type check sent read-only `eth_getCode` calls to public RPC endpoints, and the token decimals and the Chainlink feed values were read with read-only `eth_call` requests; daily token prices came from the public DefiLlama API. We sent no transactions and accessed no privileged or non-public indexing endpoints, and no smart contract was invoked in any way that might interfere with the normal operation of the network.
- Fairness and transparency: Decentralized credit scores raise concerns about fairness as well as opacity (Packin & Lev-Aretz, 2024). Graph-based contract scores can undervalue new contracts, rarely used contracts and contracts that receive no public approvals, and both EndorseRank and AWP carry structural biases of this kind. In the allowance graph, for instance, a contract that has received no spending authority is at a disadvantage relative to contracts that have; transfer graphs, in turn, favor addresses embedded in value-flow networks. Chapter 5 discusses the limitations, among them cold-start effects and the Sybil susceptibility of approval graphs. The code, the parameters and the evaluation metrics are in the public repository (Appendix 8.1), so that others can review the work and repeat it.
- Responsible disclosure and impact: EndorseRank may inform risk assessment in DeFi, but it is not financial advice, credit advice, a security rating or a consumer reporting product. Lending, trading and compliance decisions still need human oversight. Chapter 5 discusses the potential for adversarial manipulation, for example Sybil attacks on G_{approve} or wash transfers on G_{transfer}. The study reports aggregated, cohort-level findings and names no contract as trustworthy or untrustworthy. Any future operational deployment would require additional governance, appeal mechanisms and fairness monitoring, all of which lie beyond the scope of this research.
- Summary of methodological commitments: The analysis plan fixed the chain, the windows,

the cohort and account-type rules, the graph formulas, the proxy definitions, the spender labels, the bootstrap and timing protocols, the statistical tests and rule A before any log was extracted. The registration fixed the trader labels, rule B and the comparison on trader outcomes after the W0 results were known and before the W1 logs were extracted. The revision of 10 October 2026 changed the amounts, the token set and with them the cohort, after the registered results were known and before any revised score was computed (Section 3.5.1). The primary claims concern all the scores together, on the matched contract cohort and the spender holdouts, with each PageRank set against the raw degree it smooths. The cohort subsamples show how runtime grows with $|E|$, and the sensitivity sweeps show how far results depend on parameters and on the sample definition.

Chapter 4 reports what this protocol yields for Arbitrum One: the dataset and cohorts, the runtime benchmark, the same-window alignment of every score, the rank divergence, the sensitivity checks, the temporal holdouts W0 and W1 with the rules fixed for them, the outcomes built from neither edge type, and finally Claims C1–C6.

4 Implementation and Empirical Results

This chapter runs the design of Chapter 3 on Arbitrum One as a single case study: one chain and one observation window, one matched cohort of contracts for every same-window comparison, spender and trader cohorts for the two holdout windows W0 and W1, and subsamples of the matched cohort for runtime against size. It reports the revised analysis of Section 3.5.1, in which every amount is valued in US dollars and the cohort, the graphs, the proxies and the spender labels keep the five tokens WETH, USDC, USDT0, ARB and WBTC. The revision was decided after the registered results of the same window, W0 and W1 were known, so no result in this chapter is a registered result. Rule A was fixed in the analysis plan of 9 October 2026 before any log was extracted; rule B and the comparison on outcomes built from neither edge type were registered after the W0 results of the registered analysis were known and before the logs of W1 were extracted. This chapter applies these rules, with their thresholds and wording, to the revised scores: rule A on W0 and in the same window, rule B on W1, and the comparison on W1 with W0 as its secondary window (Sections 4.6.2 and 4.6.3). Their verdicts here are a re-analysis, not registered tests; the registered analysis reached the same verdicts (Appendix 8.6.10, Table 8.13). Every table is generated from the evaluation summaries of Section 3.13, each Kendall τ comes with its 95% confidence interval except in the value-slope and sample-definition checks, which were computed without intervals, and each PageRank is read against the raw degree it smooths. Each section states which of its comparisons the plan or the registration named.

4.1 Case study setting: Arbitrum One

Arbitrum One was chosen because it is an Ethereum rollup on which many DeFi protocols run (Section 2.2), because its logs are in the public BigQuery export that the pipeline reads, and because

one of its margin-trading protocols, GMX V2, records the position closes behind the trader labels. The observation window runs from 1 October 2023 to 30 June 2026 (33 months), and the time decay is anchored at its end, T_{obs}. An event at the anchor has time weight 0.858; the weight falls to 0.5 at 180 days and is close to zero for events more than two years old (Figure 3.3). Events from the early months still create edges, but the last year carries most of the edge weight in AWP and EndorseRank.

4.1.1 Cohorts

Four sets of contracts are scored or evaluated. Every address in them holds contract code; externally owned accounts (EOAs) enter the graphs as owners, senders and recipients and are never scored.

- Matched cohort ($n = 15{,}107$), defined in Section 3.2: the contracts that received a non-zero `Approval` of one of the five tokens from at least three distinct owners. They are part of the plan's cohort of 27,844 contracts, which applied the same rule to the approvals of every token; its other 12,737 contracts do not meet the rule on the five tokens. Of the plan's 29,022 candidate spenders, `eth_getCode` at block 513,216,506 (9 October 2026) returned contract code for 27,844, no code for 1,121 and an EIP-7702 delegation designator (Buterin et al., 2024) for 57; the last two groups are EOAs and were left out. Membership does not depend on either graph at T_{obs}. Of the cohort contracts, 523 are outside the allowance graph, because by the end of the window the latest `Approval` of every owner and token that had approved them carried the value zero and they held no positive allowance of the five tokens as owners. The transfer graph leaves out 2,670, because no non-zero transfer of the five tokens in the window names them as sender or recipient (a spender that moves tokens with `transferFrom` from an owner directly to a third address does not appear in the resulting `Transfer` event), and 80 contracts are outside both graphs. A contract absent from a graph scores zero under that graph's method. Every same-window alignment, difference test and benchmark is computed on this cohort.
- Spender cohorts of the two holdout windows. In W0 the scores are frozen at t_1 (31 March 2026,

23:59:59 UTC) and the labels are counted from 1 April to 30 June 2026. The cohort is every contract spender that holds at least one positive latest allowance of one of the five tokens at t_1 in the extracted approval data. Of the 17,167 such spenders, 17,050 hold contract code, 104 have no code and 13 carry an EIP-7702 designator; the 117 EOAs are left out. In W1 the scores are frozen at T_{obs} (30 June 2026, 23:59:59 UTC), the labels are counted from 1 July to 30 September 2026, and the same rule gives 18,152 contract spenders. The spender cohorts are not subsets of the matched cohort. The extracted approvals include those that cohort contracts grant as owners, so the contracts they approve enter as spenders, and a spender needs only one owner with a positive allowance.

- Trader cohorts. The GMX V2 accounts that hold contract code, are nodes of the freeze-date graph (either layer) and close at least three positions in the label window: 88 in W0, 40 of which had received an approval by the freeze, and 70 in W1, 37 of which had. Contract accounts closed 133,685 GMX V2 positions in the observation window, 6,473 of them by liquidation, with 88,002 closes in 2025 against 9,809 in the first half of 2026, and 6,805 from July to September 2026, 182 of them by liquidation. Every trader result therefore rests on few contracts and has wide intervals.
- Subsamples of the matched cohort. Deterministic SHA256 subsamples (seed `contract-benchmark-v1`) with $n = 1{,}000$, $2{,}000$, $5{,}000$ and $10{,}000$ contracts, followed by the full cohort of 15,107 (Section 3.12.1). Each stage keeps the extracted edges incident to its sampled contracts, so its edge count grows with n, from 424,888 allowance and 1,692,513 transfer edges at $n = 1{,}000$ to the full graphs. The stages serve the runtime against size (Section 4.2.1) and the sensitivity to the sample definition (Section 4.5.3).

The five tokens hold most of the extracted rows. Of the ego events of the plan's cohort in the observation window, the revised analysis keeps 224,954,202 approvals (57.7%) and 1,149,168,538 transfers (74.7%). It leaves out 164,202,859 approvals and 364,468,917 transfers of other tokens, and 709,821 approvals and 24,766,612 transfers of the five tokens in which no contract of the revised

cohort takes part; every kept transfer has a price. At T_{obs}, 5,763,757 of the 12,590,613 positive latest allowances (45.8%) are unlimited, and 6,776,289 (53.8%) reach the cap U of the allowance layer of C-PR. On 38 sampled dates the DefiLlama prices differ from the Chainlink feeds read on the chain by a median of 0.01% to 0.08% per token and by at most 0.72%. Table 8.4 in Appendix 8.4.2 gives these figures per token, with the addresses and the feeds.

At T_{obs} the allowance graph has 6,221,665 nodes and 11,210,114 edges, and the transfer graph 7,884,602 nodes and 25,001,216 edges; at t_1 the two pair tables hold 10,650,258 allowance and 23,680,692 transfer pairs. The edge counts at T_{obs} are those of pairs with a positive EndorseRank or AWP weight. The node counts also include 2,602 and 2,246 addresses whose only pairs carry amounts so small that the value weight is zero in floating-point arithmetic; the solver graphs of EndorseRank and AWP leave them out (Table 4.1). On the same contracts the transfer graph thus has 2.23 times as many edges as the allowance graph, which has 44.8% of the transfer graph's edge count. An owner–spender pair is an allowance edge only while its latest allowance of one of the five tokens is positive, whereas every pair of addresses that exchanged a non-zero transfer of these tokens in the window is a transfer edge. Of the allowance graph's nodes, 6,208,769 are owners with at least one positive allowance, so that graph consists almost entirely of owners pointing at a much smaller set of spenders. Both graphs come from the same extraction, the ego network of the revised cohort, so the differences between the scores trace back to their edges and restarts.

4.1.2 Implementation

Every algorithm runs on one Python implementation of PageRank ($d = 0.85$, $\varepsilon = 10^{-8}$ in the L_1 norm, $I_{\max} = 300$). EndorseRank and AWP differ in their edge lists and restart vectors, and C-PR mixes its two edge lists node by node (Equation (3.9)). Each evaluation run writes its scores, per-proxy statistics and bootstrap intervals to one summary file, and the benchmark writes its timings to a file of its own. The registered analysis computed its same-window and W0 results after the observation window had been extracted and before W1 was registered, and its W1 results after the July–September logs had been extracted. Every result of the revised analysis was computed on

10 October 2026, from 11:36 UTC, after the revision had been committed and after the registered results of all three windows were known (Section 3.5.1). At $d = 0.85$ every solve converged well inside the iteration cap: at T_{obs} EndorseRank needed 69 iterations, AWP 92, C-PR at $\lambda = 0.5$ 79 and C-PR ($\lambda = 0$) 92, and the longest single solve, the seeded transfer walk of S-PR, 95 (Section 4.2). Appendix 8.1 lists the runs, the repository revision, the configuration file, the summary files and the machine on which the benchmark was timed.

4.2 Computational benchmark (Claim C1)

Claim C1 concerns what each score costs to compute on the same input. Following the timing procedure of Section 3.11, Table 4.1 reports for each algorithm, on the full graphs of the matched cohort at T_{obs}, the mean solve time and its standard deviation over five timed runs after one untimed warm-up, the peak memory of the warm-up solve, the number of iterations, and the nodes and edges of the solver graph. Each graph and setting was timed once. The same measurement reappears as the full-cohort row of Table 4.2 and as the $d = 0.85$ row of Table 4.6, and Table 8.2 in Appendix 8.1 lists every timed run.

Table 4.1: Computational benchmark on the full graphs of the matched cohort ($n = 15{,}107$ contracts; AWP/EndorseRank runtime ratio 2.2× at an edge ratio of 2.2×). Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up, on which alone peak memory is traced; SD over the same 5 runs. A timed call turns an edge list into scores: node indexing, sparse-matrix assembly, row normalisation, restart vector and power iteration. $|V|$ and $|E|$ are the nodes and distinct edges of each method's solver graph. C-PR walks on the union of the allowance layer and the transfer layer, so its $|E|$ counts the distinct pairs of both; S-PR runs on the transfer layer, and its runtime and iterations include the solve of the allowance layer that supplies its restart vector. Table 8.2 lists every run.

Method	Runtime (s)	SD (s)	Peak mem. (MB)	Iters	$\lvert V\rvert$	$\lvert E\rvert$
EndorseRank	10.294	0.601	879.7	69	6,219,063	11,210,114
AWP	22.367	1.825	1,712.6	92	7,882,356	25,001,216
C-PR ($\lambda = 0.5$)	36.162	2.112	1,768.1	79	9,151,010	31,552,359
S-PR	34.422	3.833	1,235.8	158	7,884,602	25,018,892

EndorseRank solves the allowance graph (6,219,063 nodes, 11,210,114 edges) in 10.29 s on average

and AWP solves the transfer graph (7,882,356 nodes, 25,001,216 edges) in 22.37 s. AWP thus takes 2.17 times as long as EndorseRank, at an edge ratio of 2.23. Assembling the sparse matrix and running each power iteration are both linear in $|E|$ (Appendix 8.3.3), and the time ratio stays close to the edge ratio, slightly below it, although AWP needs more iterations (92 against 69, as in the evaluation runs at T_{obs}). Peak memory is 879.7 MB for EndorseRank and 1,712.6 MB for AWP, and the standard deviations over the five timed runs are 0.60 s and 1.83 s. The speed is a consequence of $|E|$: on the same contracts the allowance graph has 44.8% of the transfer graph's edges, and both scores use the same solver.

The two hybrid rows show what the speed result covers and where it stops. C-PR walks on the union of both edge sets (9,151,010 nodes, 31,552,359 edges) and needs 36.16 s, 1.62 times as long as AWP on 1.26 times its edges. S-PR runs on the transfer layer (7,884,602 nodes, 25,018,892 edges) and takes 34.42 s, 1.54 times as long as AWP. Both its time and its iteration count include the solve of the allowance layer that supplies its restart vector; in the evaluation run at T_{obs} its 158 iterations are 63 for that solve and 95 for the seeded transfer walk. The lower cost is thus confined to the score that walks the allowance graph alone and comes from that graph's size: any algorithm that walks the transfer layer, on its own or combined with allowance information, pays for the edges of that layer.

Two caveats attach to this result. The time covers only the solver call on edge lists built in advance; a full refresh would add parsing, decoding and edge-list construction, which all algorithms share wholly or in part. And a faster method is not necessarily a better one, so the runtime has to be read against the alignment and holdout results that follow.

4.2.1 Runtime on subsamples of the matched cohort

Table 4.2 reports the runtime of EndorseRank and AWP at each stage of Section 4.1.1, together with the ratio of the two runtimes, the ratio of the two edge counts and the nodes and edges of both solver graphs at that stage. The stages change the edge ratio as well as the size. The transfer graph has 3.98 times as many edges as the allowance graph at $n = 1{,}000$ (1,692,513 against 424,888),

2.49 times at $n = 2{,}000$, 2.60 times at $n = 5{,}000$, 2.57 times at $n = 10{,}000$ and 2.23 times on the full cohort. If the time of a solve follows $|E|$, the time ratio should follow these edge ratios; it is 2.89, 4.55, 3.60, 2.70 and 2.17 at the five stages. From $n = 10{,}000$ on, the time ratio stays within 0.13 of the edge ratio, and on the full cohort it lies just below it. At $n = 5{,}000$ it exceeds the edge ratio by 1.00, and at the two smallest stages, where EndorseRank's solve takes less than a second, it is 0.73 and 1.83 times the edge ratio. The time ratio thus follows the edge ratio on the two largest graphs but not on the three smaller ones. Figure 4.1 plots both runtimes against the stage size, with the edge count of each graph beside its point. The stages are subsamples of one cohort graph, so they show how runtime grows with the edges up to the full graph and not beyond it.

Table 4.2: Runtime of EndorseRank (ER) and AWP against the size of the evaluation set: deterministic SHA256 subsamples of the matched cohort (seed `contract-benchmark-v1`) up to the full cohort of $n = 15{,}107$ contracts. Each stage keeps the extracted edges incident to its sampled contracts, so $|V|$ and $|E|$ of both solver graphs are reported per stage. AWP/ER time and AWP/ER $|E|$ are the ratios of the two runtimes and of the two edge counts in the same row. Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up. [a]The full-cohort row is the measurement of Table 4.1, reused, not timed again.

n	ER (s)	AWP (s)	AWP/ER time	AWP/ER $\|E\|$	ER $\|V\|$	ER $\|E\|$	AWP $\|V\|$	AWP $\|E\|$
1,000	0.580	1.677	2.9×	4.0×	404,563	424,888	1,124,621	1,692,513
2,000	0.939	4.272	4.6×	2.5×	1,112,404	1,263,471	1,917,597	3,148,826
5,000	2.496	8.995	3.6×	2.6×	2,104,606	2,988,450	3,539,528	7,769,268
10,000	6.220	16.821	2.7×	2.6×	4,291,078	6,673,167	6,247,634	17,180,184
15,107[a]	10.294	22.367	2.2×	2.2×	6,219,063	11,210,114	7,882,356	25,001,216

4.3 Same-window alignment (Claim C2)

Claim C2 asks what each score agrees with in the window it is computed from. Table 4.3 reports the family-mean Kendall τ of every score on the transfer, allowance and Sybil-stability families, each with a 95% paired-bootstrap interval. The raw degrees have no row because in this window they are proxies: in-degree belongs to the transfer family and in-approve degree to the allowance family. The secondary part of rule A (Section 3.7) is the only same-window comparison that the plan fixed; here it is applied to the revised scores, after the registered results were known, so its

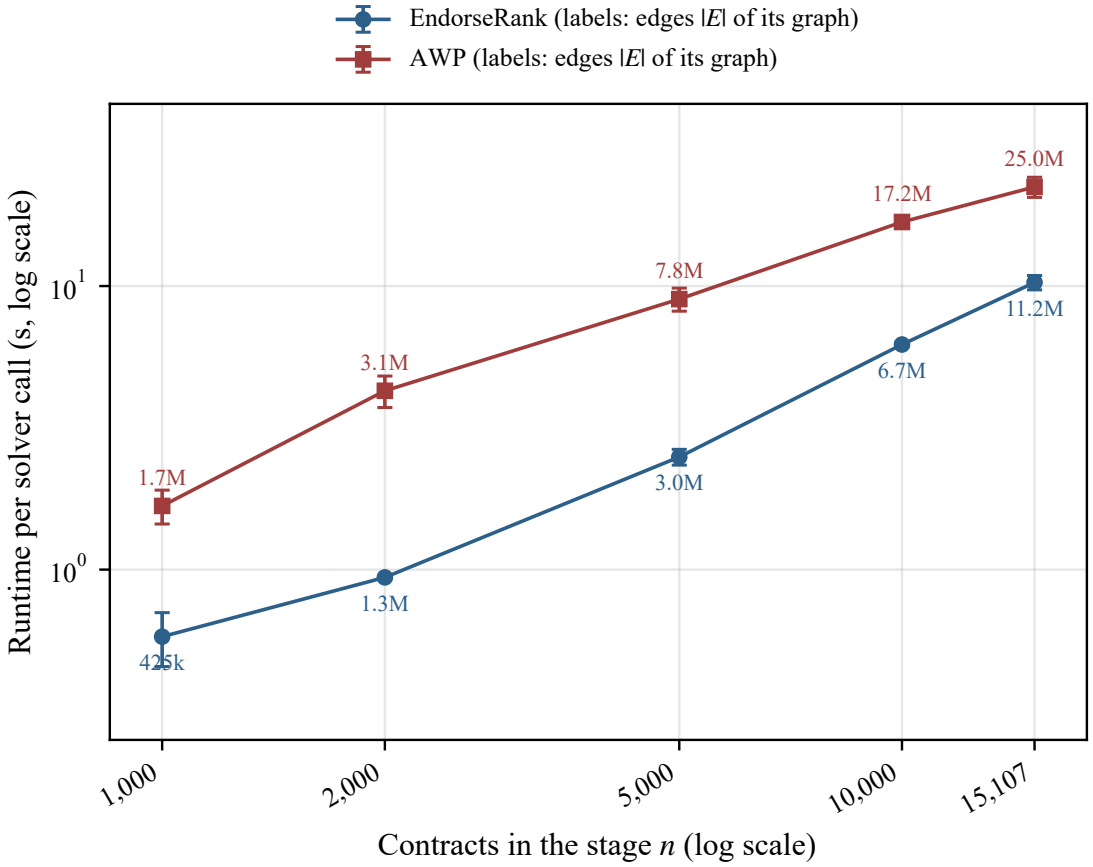


Figure 4.1: Wall-clock time of one solver call of EndorseRank and AWP against the number of contracts in the stage, both axes on a log scale. The stages are the deterministic SHA256 subsamples of the matched cohort (Table 4.2); each keeps the extracted edges incident to its contracts, and the label beside each point gives the edge count $|E|$ of that graph. The rightmost point is the full cohort of 15,107 contracts. Error bars are the standard deviations over five timed runs.

verdict is a re-analysis. The paired contrasts below were not named in the plan and were computed together with the same-window results.

Table 4.3: Same-window family-mean Kendall τ with bootstrap intervals for every score of this study (matched cohort, $n = 15{,}107$ contracts). The scores are EndorseRank and AWP, the coupled operator C-PR at five values of λ and the endorsement-seeded walk S-PR; $\lambda = 1$ is the allowance layer walked alone (latest allowances in USD, each capped) and $\lambda = 0$ the transfer layer walked alone (USD amounts discounted by the decay of AWP); both restart uniformly. The raw degrees have no row because they are proxies of the transfer and allowance families; EndorseRank with activity restarts is in Table 4.9. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Method	Transfer	Allowance	Sybil
EndorseRank	0.158 [0.147, 0.169]	0.585 [0.578, 0.592]	0.007 [-0.001, 0.015]
AWP	0.572 [0.564, 0.580]	0.193 [0.182, 0.203]	0.395 [0.387, 0.403]
C-PR ($\lambda = 1$)	0.162 [0.151, 0.173]	0.597 [0.590, 0.604]	0.011 [0.002, 0.019]
C-PR ($\lambda = 0$)	0.659 [0.653, 0.666]	0.225 [0.214, 0.235]	0.404 [0.396, 0.413]
C-PR ($\lambda = 0.25$)	0.385 [0.374, 0.395]	0.457 [0.449, 0.465]	0.155 [0.147, 0.164]
C-PR ($\lambda = 0.5$)	0.365 [0.355, 0.375]	0.477 [0.469, 0.485]	0.142 [0.134, 0.151]
C-PR ($\lambda = 0.75$)	0.344 [0.333, 0.354]	0.492 [0.484, 0.500]	0.129 [0.121, 0.138]
S-PR	0.629 [0.621, 0.636]	0.323 [0.313, 0.333]	0.398 [0.390, 0.406]

The scores divide by the edges they walk on. EndorseRank reaches 0.585 $[0.578, 0.592]$ on the allowance family and 0.158 $[0.147, 0.169]$ on the transfer family; AWP reaches 0.572 $[0.564, 0.580]$ on the transfer family and 0.193 $[0.182, 0.203]$ on the allowance family. Each score is ahead of the other on the family of its own edges, EndorseRank by $+0.392$ $[0.380, 0.405]$ on the allowance family and AWP by $+0.414$ $[0.401, 0.427]$ on the transfer family (Table 8.7 in Appendix 8.5). The two layers of C-PR walked alone repeat the split: C-PR ($\lambda = 1$) reaches 0.597 $[0.590, 0.604]$ on the allowance family and C-PR ($\lambda = 0$) 0.659 $[0.653, 0.666]$ on the transfer family. On the Sybil-stability family the scores that walk on transfers lead, AWP with 0.395 $[0.387, 0.403]$ against 0.007 $[-0.001, 0.015]$ for EndorseRank. Part of this agreement is built in. The main proxies of each family come from the same events as one of the graphs, in-degree and US-dollar in-value from the transfers of the five tokens and in-approve degree and capped value from their latest allowances, and a PageRank smooths the in-degree of its own graph. Agreement with the proxies of a score's

own edge type is therefore an intended-construct check and cannot count as an external test. The holdouts of Section 4.6 ask the same question out of window.

Appendix 8.5.2 breaks the family means of EndorseRank and AWP into their proxies. AWP follows the summed US-dollar value of the inbound transfers (0.566 $[0.558, 0.575]$) almost as closely as their number of senders, the in-degree (0.578 $[0.569, 0.587]$). That is what its weights lead one to expect. With the slope set at the median transfer (Section 3.6.1), a transfer of 200 dollars weighs one half, one of 400 dollars 0.8 and one of 100 dollars 0.27, so an edge weight grows with the amounts transferred as well as with their number. In the registered analysis, with $b = 1$ on raw base units, V was close to one for nearly every transfer whatever its amount, and AWP followed the in-degree (0.541 $[0.534, 0.547]$) much more closely than the in-value (0.353 $[0.345, 0.361]$; registered results digest, Appendix 8.6.10). On the Sybil-stability family AWP follows transfer tenure (0.508 $[0.499, 0.517]$) and active months (0.549 $[0.540, 0.558]$) and is positive on the inbound-counterparty ratio (0.127 $[0.114, 0.141]$).

EndorseRank follows the in-approve degree (0.655 $[0.648, 0.662]$) more closely than the in-approve value (0.515 $[0.506, 0.524]$). Its value weight is 0.8 at twice the median allowance of 62 dollars and close to one at a few hundred dollars, so such an allowance and an unlimited one weigh nearly alike, whereas the in-approve value counts each allowance up to the cap of 20,000 dollars. Its transfer coefficients are 0.182 $[0.169, 0.194]$ with in-degree and 0.134 $[0.123, 0.145]$ with in-value. On the Sybil-stability family it is negative on the inbound-counterparty ratio (-0.156 $[-0.166, -0.147]$) and reaches 0.080 $[0.069, 0.091]$ on tenure and 0.098 $[0.087, 0.109]$ on active months, so how many owners approve a contract says little about how long it has received transfers. C-PR ($\lambda = 1$) walks the same allowance graph with capped US-dollar amounts and also restarts uniformly; it stays close to EndorseRank on every family, and its point estimate on the in-approve value, which sums the same capped amounts as its own layer does, is higher (0.540 $[0.531, 0.549]$ against 0.515 $[0.506, 0.524]$; no paired contrast was computed).

Because every contract in the cohort received approvals of the five tokens from at least three owners, EndorseRank leaves few ties here. It gives 13,625 distinct scores to the 15,107 contracts, and only

0.17% of contract pairs are tied, most of them among the 535 contracts that score zero (Section 4.4). Its coefficients therefore rest on an ordering of nearly every contract, and keeping the contracts outside a graph as isolated nodes changes none of them (Table 4.5).

Claim C2 is supported as an intended-construct check, with agreement that is partly built in. Every score agrees most with the proxies of the edge type it walks on, with intervals far from zero, and the mixtures of C-PR lie between their two layers on every family (Section 4.3.1).

4.3.1 The coupled operator (RQ5)

Two hybrids were defined in Section 3.6.3. The coupled operator C-PR mixes the US-dollar allowance and transfer layers in each node's transition row with weight λ, and the endorsement-seeded walk S-PR keeps the transfer layer but restarts from the allowance walk. Table 4.4 gives their paired contrasts in this window: first eight against the two layers walked alone, then four against EndorseRank and AWP. RQ5 is decided out of window (Sections 4.6 and 4.6.2). Here we only locate the hybrids on the checks that the single-layer scores were built to pass.

C-PR lies between its two layers on every family. At $\lambda = 0.5$ it reaches 0.365 $[0.355, 0.375]$ on the transfer family, 0.477 $[0.469, 0.485]$ on the allowance family and 0.142 $[0.134, 0.151]$ on the Sybil-stability family, and as λ rises from 0.25 to 0.75 it moves toward the allowance layer on all three (Table 4.3). It stays nearer to its allowance layer than to its transfer layer: on the allowance family it falls 0.120 below C-PR ($\lambda = 1$) ($[-0.127, -0.113]$), and on the transfer family 0.294 below C-PR ($\lambda = 0$) ($[-0.305, -0.284]$). The mixing rule of Equation (3.9) acts at every node with out-edges in both layers. Almost every node of the allowance layer is an owner with out-edges, so the allowance rows enter the walk wherever an owner also sent a transfer that the extraction kept. Against the single-layer scores C-PR is below AWP on the transfer family (-0.207, $[-0.220, -0.194]$) and below EndorseRank on the allowance family (-0.108, $[-0.115, -0.101]$). On the Sybil-stability family it lies above the allowance walk ($+0.131$, $[0.126, 0.137]$) and EndorseRank ($+0.135$, $[0.129, 0.141]$) and below the transfer walk (-0.262, $[-0.273, -0.253]$) and AWP (-0.253, $[-0.263, -0.242]$).

On every family S-PR falls below the single layer that leads that family. Restarting the transfer

Table 4.4: Same-window paired contrasts for the coupled operator (C-PR, $\lambda = 0.5$) and the endorsement-seeded walk (S-PR) against the two layers they are built from, the allowance layer (C-PR ($\lambda = 1$)) and the transfer layer (C-PR ($\lambda = 0$)), and of C-PR against EndorseRank and AWP (matched cohort, $n = 15{,}107$ contracts). Family-mean τ; the same contract resamples are used for a and b. The contrasts with the layers are a stricter check beside the secondary part of rule A, which compares C-PR's point estimate with the better layer's interval in Table 4.3. The secondary part of rule A, fixed in the analysis plan and applied here to the revised scores, is not met: C-PR's point estimate lies outside the interval of C-PR ($\lambda = 0$) on the transfer family and outside the interval of C-PR ($\lambda = 1$) on the allowance family, and C-PR does not exceed both layers on the Sybil-stability family. An interval below 0 means a is below b beyond sampling variation; one above 0 means it exceeds it. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Against the two layers walked alone					
C-PR transfer minus C-PR ($\lambda = 0$) transfer	0.365	0.659	-0.294	[-0.305, -0.284]	0.000
C-PR allowance minus C-PR ($\lambda = 1$) allowance	0.477	0.597	-0.120	[-0.127, -0.113]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.142	0.404	-0.262	[-0.273, -0.253]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.142	0.011	+0.131	[0.126, 0.137]	1.000
S-PR transfer minus C-PR ($\lambda = 0$) transfer	0.629	0.659	-0.031	[-0.035, -0.028]	0.000
S-PR allowance minus C-PR ($\lambda = 1$) allowance	0.323	0.597	-0.274	[-0.285, -0.263]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.398	0.404	-0.007	[-0.009, -0.004]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.398	0.011	+0.387	[0.376, 0.399]	1.000
Against EndorseRank and AWP					
C-PR transfer minus AWP transfer	0.365	0.572	-0.207	[-0.220, -0.194]	0.000
C-PR allowance minus EndorseRank allowance	0.477	0.585	-0.108	[-0.115, -0.101]	0.000
C-PR sybil-stability minus AWP sybil-stability	0.142	0.395	-0.253	[-0.263, -0.242]	0.000
C-PR sybil-stability minus EndorseRank sybil-stability	0.142	0.007	+0.135	[0.129, 0.141]	1.000

walk from the allowance walk lowers the transfer coefficient to 0.629 $[0.621, 0.636]$ ($\Delta\tau = -0.031$, $[-0.035, -0.028]$, against C-PR ($\lambda = 0$)) and the Sybil-stability coefficient to 0.398 $[0.390, 0.406]$ (-0.007, $[-0.009, -0.004]$). It raises the allowance coefficient to 0.323 $[0.313, 0.333]$, against 0.225 $[0.214, 0.235]$ for the transfer walk with uniform restarts (no paired contrast was computed for this pair), yet leaves it 0.274 below the allowance walk ($[-0.285, -0.263]$). A restart vector changes where the walk lands after each teleport. It carries part of the allowance signal into the transfer walk, but less than walking the allowance edges does.

As fixed in the analysis plan, the secondary part of rule A asks C-PR's point estimate at $\lambda = 0.5$ to lie within the 95% interval of the better layer on the transfer and allowance families, and C-PR to exceed both layers on the Sybil-stability family. Applied to the revised scores, all three parts fail, as they did in the registered analysis (Table 8.13). The transfer coefficient (0.365) lies below the interval of C-PR ($\lambda = 0$) ($[0.653, 0.666]$), the allowance coefficient (0.477) lies below that of C-PR ($\lambda = 1$) ($[0.590, 0.604]$), and on the Sybil-stability family C-PR exceeds only the allowance layer. The paired contrasts of Table 4.4, a stricter check that is not part of the rule, point the same way. In the same window the mixture gives up part of each layer's agreement with its own family and exceeds the better layer on no family. The holdout takes up the remaining question of whether it has predictive content that the single layers lack.

4.4 Rank divergence (H1 / RQ1)

Hypothesis H1 and RQ1 concern whether EndorseRank and AWP put the same contracts in different orders. The inter-method Kendall τ_b between the two score vectors is 0.174, with interval $[0.163, 0.185]$ (last row of Table 4.5), so the interval excludes both zero and one. The two rankings are related, yet far from interchangeable. Of the 114,103,171 contract pairs, 96.7% are ordered by both scores, and 41.2% of these are ordered in opposite directions.

Ties play almost no part in this result. The marginal score distributions are in Figure 4.2. EndorseRank gives 13,625 distinct scores to the 15,107 contracts and ties 0.17% of the pairs; AWP

gives 12,435 distinct scores and ties 3.13%. Under each score the largest tie block is formed by the contracts that score zero: 535 under EndorseRank and 2,672 under AWP. They are the 523 and 2,670 contracts outside each graph, 80 of them outside both, and 12 and 2 contracts whose only pairs carry amounts too small for the value weight to differ from zero. Under EndorseRank 166 further contracts share one score, $c = 7.64 \times 10^{-8}$, the score of a node without in-edges; they are in the allowance graph only as owners. Every other cohort contract has received at least one positive allowance and scores above c. A marginal distribution cannot show how two scores order the same contracts, so Figure 4.3 plots the two rankings against each other for a random sample of 3,000 contracts, with tied contracts at their mid-rank as in τ_b. The zero blocks appear as a column at the right edge (EndorseRank) and a row at the top (AWP). Contracts that both scores rank near the top, or both near the bottom, lie close to the identity line; in between, the points spread over most of the square.

The zero blocks come from one rule of the pipeline: a contract missing from a graph scores zero (Section 3.6). The analysis plan fixed the alternative, keeping missing contracts as isolated nodes, as a sensitivity analysis, and Table 4.5 gives its result on the revised scores. No coefficient changes at the third decimal, and the agreement between the two scores stays at 0.174 $[0.163, 0.185]$. The reason is where the missing contracts land. Under EndorseRank's uniform restarts an isolated node receives c, the lowest score that any node of the graph can have, so the 523 contracts outside the allowance graph stay near the bottom of the order and join the 166 contracts at c. The 12 contracts whose only pairs carry zero weight are nodes of the allowance graph, so the rule does not add them, and the solver, which keeps only edges of positive weight, leaves them out; they keep the score zero. No pair changes direction: the pairs between the 523 and the 166 become ties, and the pairs between the 523 and the 12, tied at zero under the main rule, become ordered. Under AWP's restarts an isolated contract that sent nothing receives no restart mass and scores zero under both rules. Because the cohort was defined by approvals received, nearly every cohort contract is a spender in the allowance graph, and the rule for missing contracts decides almost nothing.

H1 is supported. The two scores order the cohort differently, and the difference does not come from

Table 4.5: Same-window family-mean Kendall τ with contracts missing from a graph scored zero or kept as isolated nodes (matched cohort, $n = 15{,}107$ contracts). The main analysis scores a contract that is missing from a method's graph zero; keeping it as an isolated node is the sensitivity analysis fixed in the analysis plan. The rule concerns 523 contracts for EndorseRank and 2,670 for AWP (80 are missing from both graphs). Under EndorseRank's uniform restarts an isolated node gets the score of a node without in-edges. AWP restarts only at addresses that sent a transfer, so an isolated contract keeps the score zero and AWP's coefficients are the same under both rules. The last row is the agreement between the two scores. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Missing contracts scored zero		Missing contracts as isolated nodes	
Family	EndorseRank	AWP	EndorseRank	AWP
Transfer	0.158 [0.147, 0.169]	0.572 [0.564, 0.580]	0.158 [0.146, 0.169]	0.572 [0.564, 0.580]
Allowance	0.585 [0.578, 0.592]	0.193 [0.182, 0.203]	0.585 [0.579, 0.592]	0.193 [0.182, 0.203]
Sybil stability	0.007 [-0.001, 0.015]	0.395 [0.387, 0.403]	0.007 [-0.002, 0.015]	0.395 [0.387, 0.403]
EndorseRank vs. AWP	0.174 [0.163, 0.185]		0.174 [0.163, 0.185]	

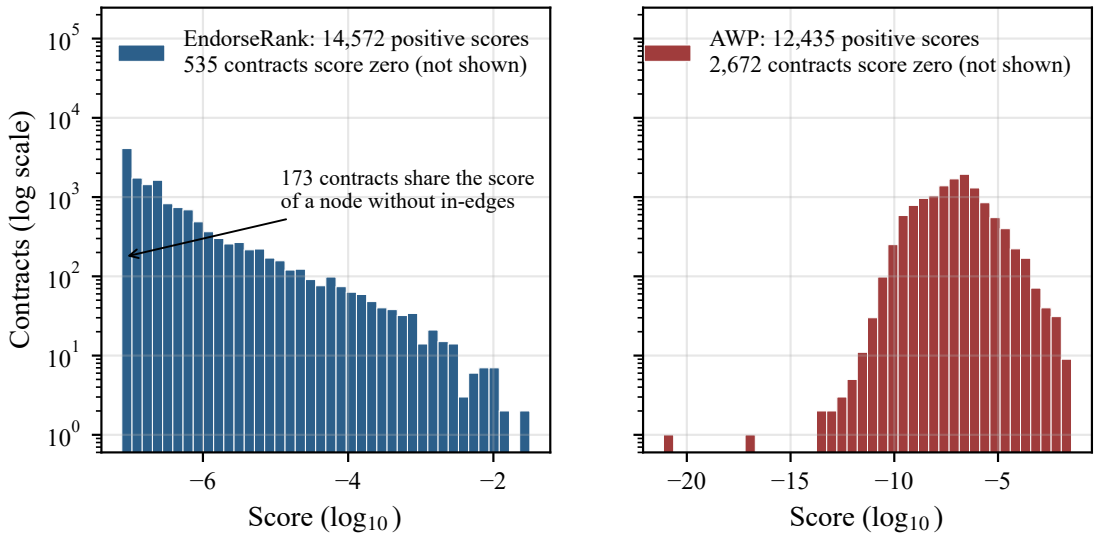


Figure 4.2: Marginal EndorseRank and AWP score distributions on the matched cohort ($n = 15{,}107$ contracts), drawn on a $\log_{10}$ scale for contracts with a positive score; counts are on a log scale. Zero-score contracts, which no edge of positive weight in the respective graph touches, are left out: 535 for EndorseRank and 2,672 for AWP. For EndorseRank, the 166 contracts that are in the allowance graph only as owners share the score $c = 7.64 \times 10^{-8}$ of a node without in-edges, the lowest positive score.

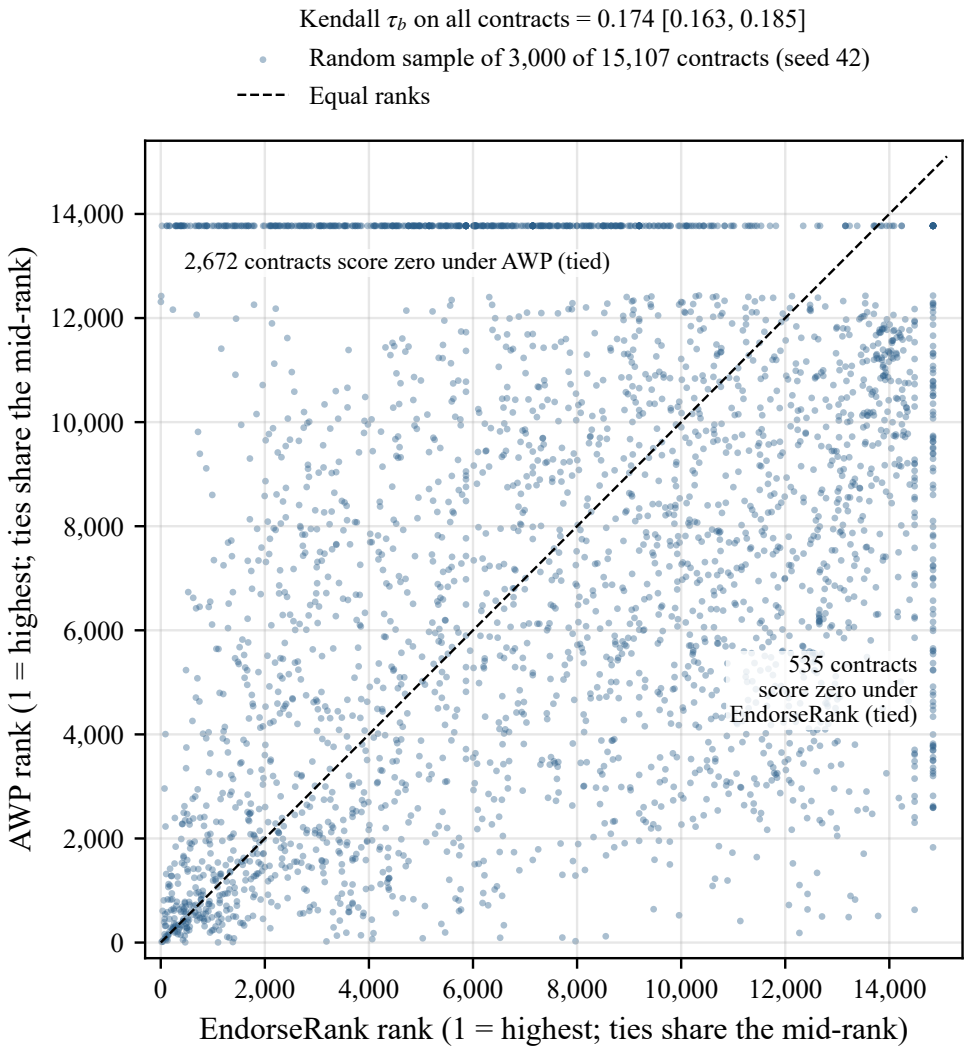


Figure 4.3: Rank–rank scatter of EndorseRank against AWP on the matched cohort (random sample of 3,000 of the 15,107 contracts, seed 42). Tied scores share their average rank, which is the tie treatment of Kendall's τ_b. The column at the right edge holds the sampled contracts among the 535 that score zero under EndorseRank and the row at the top those among the 2,672 that score zero under AWP. A point on the dashed identity line has the same rank in both orderings; the legend gives τ_b on all contracts with its 95% interval.

ties or from the rule for missing contracts. A protocol that relies on transfer-based scores cannot assume that its ordering of contracts survives a switch to allowance-based scores or a blend with them.

4.5 Robustness and sensitivity checks

Three checks, run for EndorseRank and AWP, vary the damping factor, the slope of the value weight and the sample definition. The analysis plan fixed the damping values; the value-slope check takes the place of the plan's token check, which lost its point once every graph keeps five tokens (Section 3.12.3), and the sample-definition check was not in the plan. All three were run on the revised scores after the registered results were known. The first two checks hold the sample fixed and ask whether the results hinge on one setting. The third changes the sample itself, so its coefficients are expected to move, and it is not evidence of robustness. A fourth check asks how much EndorseRank depends on its restart rule.

4.5.1 Damping

Table 4.6 recomputes the family means at $d \in \{0.75, 0.85, 0.95\}$ on the same graphs.

Table 4.6: Robustness: family-mean Kendall τ of EndorseRank (ER) and AWP under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on the matched cohort ($n = 15{,}107$ contracts); all. = allowance, tr. = transfer, Syb. = Sybil stability. Runtimes are means of five timed solver calls on the full graphs; the $d = 0.85$ runtimes are the measurements of Table 4.1, reused, not timed again (Table 8.2 lists every run). CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER Syb.	AWP Syb.	ER (s)	AWP (s)
0.75	0.586 [0.580, 0.594]	0.193 [0.182, 0.203]	0.158 [0.146, 0.168]	0.570 [0.562, 0.578]	0.007 [-0.002, 0.015]	0.397 [0.389, 0.406]	5.984	15.236
0.85	0.585 [0.578, 0.592]	0.193 [0.182, 0.203]	0.158 [0.147, 0.169]	0.572 [0.564, 0.580]	0.007 [-0.001, 0.015]	0.395 [0.387, 0.403]	10.294	22.367
0.95	0.583 [0.577, 0.591]	0.192 [0.181, 0.202]	0.158 [0.147, 0.169]	0.577 [0.569, 0.585]	0.008 [-0.001, 0.016]	0.391 [0.384, 0.400]	31.233	60.962

For EndorseRank no family mean moves by more than 0.003 across the three values; its allowance coefficient is 0.586 $[0.580, 0.594]$ at $d = 0.75$ and 0.583 $[0.577, 0.591]$ at $d = 0.95$. AWP moves

most on the transfer family, from 0.570 $[0.562, 0.578]$ to 0.577 $[0.569, 0.585]$, and its Sybil-stability coefficient falls from 0.397 $[0.389, 0.406]$ to 0.391 $[0.384, 0.400]$. Neither score's pattern is an artifact of the conventional $d = 0.85$. EndorseRank's stability is largely structural. For a spender whose owners receive no approvals, EndorseRank equals $c\,(1 + d\,\delta(v))$, with $\delta(v)$ the owner-normalized in-strength (Section 5.3.2), and the order that this formula gives among such spenders does not depend on d; 12,039 of the 14,406 cohort contracts with in-edges are spenders of this kind. The number of iterations grows with d, from 41 to 69 and 217 for EndorseRank and from 55 to 92 and 261 for AWP. The runtimes are 5.98, 10.29 and 31.23 s for EndorseRank and 15.24, 22.37 and 60.96 s for AWP, time ratios of 2.55, 2.17 and 1.95 at an edge ratio of 2.23, on the full graphs of Table 4.1 (6,219,063 nodes and 11,210,114 edges for EndorseRank, 7,882,356 and 25,001,216 for AWP); each graph and damping pair was timed once, and the $d = 0.85$ values are the main measurement, reused. Figure 4.4 plots the family means with their intervals.

4.5.2 Value-slope sensitivity

Table 4.7 recomputes EndorseRank and AWP at T_{obs} with the slope b of the value weight divided and multiplied by ten (Section 3.12.3), on the same graphs and proxies.

Table 4.7: Robustness: family-mean Kendall τ on the matched cohort ($n = 15{,}107$ contracts) when the slope b of the value weight V is divided or multiplied by ten, next to the main analysis. The main slopes put $V = 1/2$ at the median amount, \$200 for transfers and \$62 for allowances; $b/10$ moves that point to ten times the median and $10b$ to a tenth of it. The rank agreement τ of each variant with the main score of the same method (EndorseRank and AWP) is 0.973 and 0.973 at $b/10$, and 0.968 and 0.971 at $10b$. No intervals were computed for the variants.

Method	Transfer			Allowance			Sybil		
	$b/10$	b	$10b$	$b/10$	b	$10b$	$b/10$	b	$10b$
EndorseRank	0.157	0.158	0.159	0.588	0.585	0.580	0.006	0.007	0.009
AWP	0.577	0.572	0.567	0.193	0.193	0.191	0.397	0.395	0.393

No family mean moves by more than 0.0054, and the pattern stays: EndorseRank leads on the allowance family, and AWP on the transfer and Sybil-stability families. The steeper slope $10b$ puts the half weight at a tenth of the median amount, so that most amounts weigh close to one and an

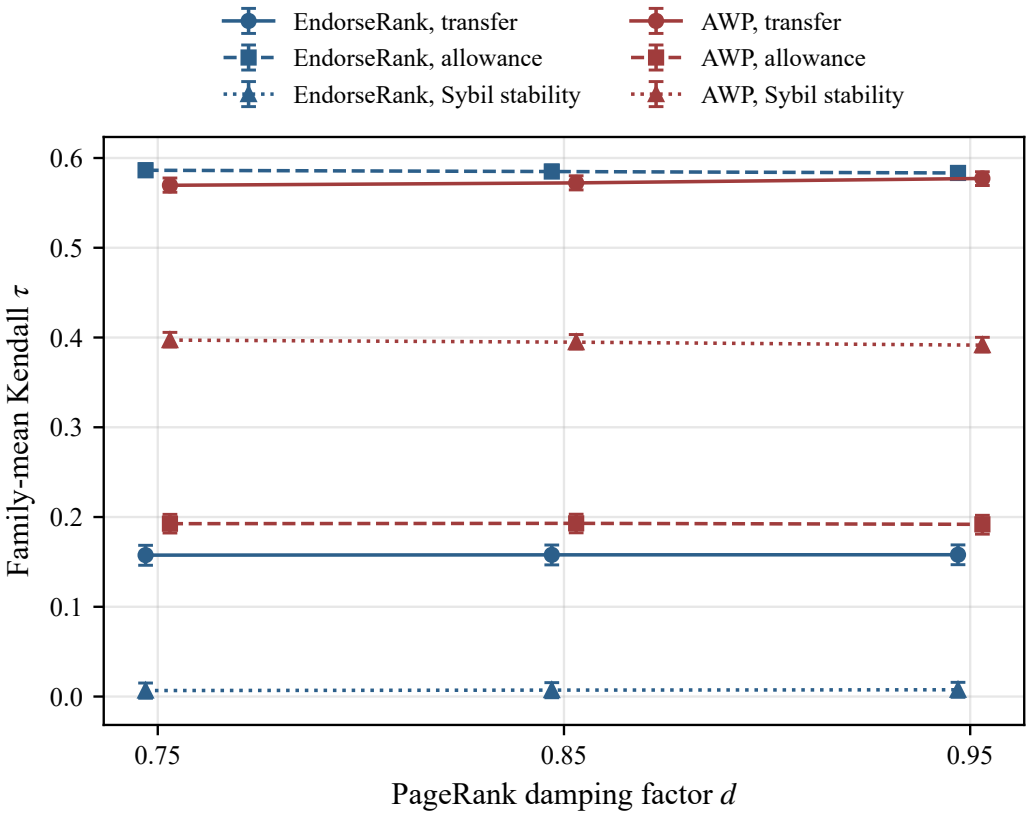


Figure 4.4: Damping sensitivity of the family-mean Kendall τ of EndorseRank and AWP on the transfer, allowance and Sybil-stability families across $d \in \{0.75, 0.85, 0.95\}$, with 95% bootstrap intervals (values from Table 4.6; matched cohort, $n = 15{,}107$ contracts).

edge weight comes nearer to a count of events. It lowers EndorseRank's allowance coefficient from 0.585 to 0.580 and AWP's transfer coefficient from 0.572 to 0.567; the flatter slope $b/10$ raises them to 0.588 and 0.577. Each variant orders the contracts almost as the main score of the same method does, at $\tau = 0.973$ for both scores at $b/10$, and at 0.968 for EndorseRank and 0.971 for AWP at $10b$. No intervals were computed for the variants. The family-level results therefore do not hinge on the amount at which the value weight reaches one half.

4.5.3 Sample-definition sensitivity

Table 4.8 recomputes the family means on the subsamples of Section 4.1.1, each scored on the edges incident to its contracts.

Table 4.8: Sensitivity of the family means to the evaluation set: family-mean Kendall τ of EndorseRank (ER) and AWP on deterministic SHA256 subsamples of the matched cohort (seed `contract-benchmark-v1`, full cohort $n = 15{,}107$ contracts), each stage scored on the edges incident to its contracts; all. = allowance, tr. = transfer, Syb. = Sybil stability. Rows are evaluated on different contract sets, so their τ values are not comparable with Table 4.3; they show how much the statistic moves with the evaluation population. No intervals were computed for the stages. Runtimes are the measurements of Table 4.2.

n	ER all.	AWP all.	ER tr.	AWP tr.	ER Syb.	AWP Syb.	ER (s)	AWP (s)	AWP/ER time
1,000	0.715	0.196	0.205	0.586	0.038	0.408	0.580	1.677	2.9×
2,000	0.686	0.199	0.196	0.588	0.033	0.411	0.939	4.272	4.6×
5,000	0.634	0.211	0.195	0.571	0.034	0.391	2.496	8.995	3.6×
10,000	0.607	0.196	0.167	0.571	0.013	0.394	6.220	16.821	2.7×
15,107	0.585	0.193	0.158	0.572	0.007	0.395	10.294	22.367	2.2×

The coefficients move with the sample. EndorseRank moves most: its allowance coefficient is 0.715 at $n = 1{,}000$ and falls with every stage to 0.585 on the full cohort, and its transfer coefficient falls from 0.205 to 0.158. AWP stays within 0.018 of its full-cohort values on every family. No intervals were computed for the stages. None of these shifts is evidence about the scores. A stage keeps only the edges incident to its sampled contracts, so the paths that pass through unsampled cohort contracts are cut, and each score comes closer to a count of the sampled contract's own edges, which is what the proxies count. We therefore report the table as a sensitivity analysis of the

sample definition, and its values must not be compared with those of Table 4.3; the matched cohort is the primary sample.

4.5.4 Restarts of EndorseRank

EndorseRank restarts the walk uniformly, while AWP restarts it at addresses in proportion to their recent sending activity (Section 3.6.2). Carried over to allowances, AWP's rule restarts the walk at owners in proportion to the spenders they approved, each weighted by the age of the latest approval. The analysis plan fixed uniform restarts for EndorseRank and named the variant with AWP's restarts as a score to report beside it, without fixing any comparison between the two. Table 4.9 scores both versions, with AWP beside them.

Table 4.9: EndorseRank with uniform restarts and with restarts weighted by approving activity, AWP's rule applied to the allowance graph, next to AWP: family-mean Kendall τ on the matched cohort ($n = 15{,}107$ contracts) and τ with the two spender labels in W0 ($n = 17{,}050$ spender contracts, scores frozen at 31 March 2026) and W1 ($n = 18{,}152$, frozen at 30 June 2026). Both EndorseRank rows weight each latest allowance by $\sigma(\Delta t)V(z)$. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

		Same window, matched cohort			W0, spenders		W1, spenders	
Method	Restarts	Transfer	Allowance	Sybil	New approval pairs	New senders	New approval pairs	New senders
EndorseRank	uniform	0.158 [0.147, 0.169]	0.585 [0.578, 0.592]	0.007 [-0.001, 0.015]	0.231 [0.220, 0.242]	0.200 [0.189, 0.210]	0.234 [0.222, 0.244]	0.194 [0.184, 0.205]
EndorseRank	by approving activity	0.164 [0.153, 0.173]	0.426 [0.418, 0.435]	0.049 [0.039, 0.057]	0.308 [0.299, 0.319]	0.269 [0.260, 0.278]	0.295 [0.285, 0.304]	0.242 [0.232, 0.253]
AWP	by sending activity	0.572 [0.564, 0.580]	0.193 [0.182, 0.203]	0.395 [0.387, 0.403]	0.196 [0.181, 0.210]	0.346 [0.336, 0.356]	0.169 [0.157, 0.183]	0.330 [0.321, 0.340]

With activity restarts, EndorseRank's same-window allowance coefficient falls from 0.585 $[0.578, 0.592]$ to 0.426 $[0.418, 0.435]$, and its transfer coefficient changes little (0.164 $[0.153, 0.173]$ against 0.158 $[0.147, 0.169]$). The restart mass then goes to the owners that approve most, so the score weights a spender by how active its owners are as well as by how many there are. Out of window the variant ranks both spender labels higher in both windows. In W0 it reaches 0.308 $[0.299, 0.319]$ on new approval pairs and 0.269 $[0.260, 0.278]$ on new senders, $+0.077$ $[0.071, 0.084]$ and $+0.070$ $[0.064, 0.076]$ above the uniform version. In W1 it reaches 0.295 $[0.285, 0.304]$ and 0.242 $[0.232, 0.253]$, $+0.061$ $[0.054, 0.067]$ and $+0.048$ $[0.042, 0.053]$ above it. These contrasts

were computed after the labels of each window were known and lie outside every rule. The uniform rule keeps the allowance construct in the same window, and the activity rule ranks later arrivals better: in W0 the variant ranks new approval pairs above the in-approve degree ($+0.059$, $[0.050, 0.067]$), where the uniform version falls below it (Section 4.6).

4.6 Temporal holdout (RQ4, RQ5)

A same-window check cannot separate a score from its own construct. The holdout instead tests whether centrality measured up to a cut-off predicts behavior after it. Following Section 3.7, scores on the W0 spender cohort are frozen at t_1, and two labels are then counted for each contract between 1 April and 30 June 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). Table 4.10 sets every score computed at t_1, the two raw-degree baselines included, against both labels. Table 4.11 gives the paired contrasts: in its first block the two of rule A, fixed in the analysis plan before any score existed and applied here to the revised scores, and in its second block those computed after the labels were known.

The first finding answers the second part of RQ4, and the answer differs between the two walks. On the transfer side, AWP ranks new transfer senders at $\tau = 0.346$ $[0.336, 0.356]$, above the transfer in-degree at t_1 (0.303 $[0.292, 0.315]$), with a paired difference of $+0.043$ $[0.038, 0.048]$. Propagation with AWP's weights and activity restarts thus added information to the count on this label. The transfer layer walked with US-dollar amounts and uniform restarts does not clearly do so: C-PR ($\lambda = 0$) reaches 0.296 $[0.285, 0.307]$, slightly below the in-degree (no paired contrast was computed). On the allowance side the count wins. The in-approve degree at t_1 ranks new approval pairs at 0.250 $[0.237, 0.262]$, EndorseRank at 0.231 $[0.220, 0.242]$ (paired difference -0.019, $[-0.026, -0.011]$) and C-PR ($\lambda = 1$) at 0.223 $[0.213, 0.233]$ (-0.027, $[-0.034, -0.019]$). A spender gains more new owners the more it already has, as preferential attachment would predict (Barabási & Albert, 1999), and the count captures this directly. EndorseRank divides each owner's endorsement among all the spenders that owner approves (Section 5.3.2), so a spender whose owners approve many other

Table 4.10: Temporal holdout W0 on the spender cohort ($n = 17{,}050$ contracts): Kendall τ between scores frozen at 31 March 2026 and two labels counted on the contract between 1 April 2026 and 30 June 2026: the number of new owner–spender approval pairs and the number of addresses that sent a transfer to the contract for the first time. The cohort keeps the 17,050 contracts among the 17,167 spenders with a positive latest allowance at the freeze; 104 EOAs without code and 13 EOAs with an EIP-7702 delegation are left out. 2,070 contracts received at least one new pair and 1,695 at least one new sender. Baseline rows are the raw degrees at t_1 that each single-layer PageRank smooths. C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$) are the allowance and transfer layers of the coupled operator walked alone. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at t_1 (baseline)	0.250 [0.237, 0.262]	0.180 [0.169, 0.192]
Transfer in-degree at t_1 (baseline)	0.135 [0.121, 0.150]	0.303 [0.292, 0.315]
EndorseRank	0.231 [0.220, 0.242]	0.200 [0.189, 0.210]
AWP	0.196 [0.181, 0.210]	0.346 [0.336, 0.356]
C-PR ($\lambda = 1$)	0.223 [0.213, 0.233]	0.188 [0.177, 0.199]
C-PR ($\lambda = 0$)	0.160 [0.144, 0.174]	0.296 [0.285, 0.307]
C-PR ($\lambda = 0.25$)	0.253 [0.243, 0.264]	0.269 [0.257, 0.280]
C-PR ($\lambda = 0.5$)	0.252 [0.241, 0.262]	0.262 [0.250, 0.273]
C-PR ($\lambda = 0.75$)	0.249 [0.238, 0.260]	0.253 [0.242, 0.265]
S-PR	0.172 [0.157, 0.185]	0.295 [0.284, 0.305]

Table 4.11: Paired bootstrap differences on the spender holdout W0 ($n = 17{,}050$ contracts; 400 resamples shared by a and b, seed 42). The first block holds the two contrasts of the primary part of rule A, fixed in the analysis plan committed on 9 October 2026 before any data and applied here to the revised scores (C-PR against C-PR ($\lambda = 1$) on new approval pairs and against C-PR ($\lambda = 0$) on new transfer senders). The second block was computed after the labels were known: the increment of C-PR ($\lambda = 1$), the allowance layer walked alone, over its raw degree at t_1, the same comparisons for S-PR, the increments of EndorseRank and AWP over their raw degrees, C-PR against C-PR ($\lambda = 0$) on new approval pairs, and C-PR against EndorseRank and AWP. Primary part of rule A (C-PR no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new transfer senders): no worse holds, better fails, so the rule is not met. The differences between EndorseRank and AWP are in Table 8.11.

Contrast (a minus b)	Label	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Rule A, primary part, fixed in the plan						
C-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.252	0.223	+0.029	[0.022, 0.036]	1.000
C-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.262	0.296	-0.034	[-0.040, -0.029]	0.000
Computed after the labels were known						
C-PR ($\lambda = 1$) minus t_1 in-approve degree	new approval pairs	0.223	0.250	-0.027	[-0.034, -0.019]	0.000
S-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.172	0.223	-0.051	[-0.066, -0.038]	0.000
S-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.295	0.296	-0.001	[-0.004, 0.002]	0.265
EndorseRank minus t_1 in-approve degree	new approval pairs	0.231	0.250	-0.019	[-0.026, -0.011]	0.000
AWP minus t_1 in-degree	new transfer senders	0.346	0.303	+0.043	[0.038, 0.048]	1.000
C-PR minus C-PR ($\lambda = 0$)	new approval pairs	0.252	0.160	+0.092	[0.080, 0.105]	1.000
C-PR minus EndorseRank	new approval pairs	0.252	0.231	+0.021	[0.013, 0.028]	1.000
C-PR minus AWP	new transfer senders	0.262	0.346	-0.085	[-0.092, -0.078]	0.000
C-PR minus AWP	new approval pairs	0.252	0.196	+0.056	[0.042, 0.070]	1.000

spenders ranks lower than its count alone would place it; on this label that division costs a little. None of these degree contrasts was named in the plan; all were computed after the labels were known.

Read across the table, each label is ranked best by scores that use its own edge type. On new approval pairs the C-PR mixtures, the in-approve degree and EndorseRank lead, while AWP (0.196 $[0.181, 0.210]$), S-PR (0.172 $[0.157, 0.185]$), C-PR ($\lambda = 0$) (0.160 $[0.144, 0.174]$) and the transfer in-degree (0.135 $[0.121, 0.150]$) stay lower. On new transfer senders AWP and the transfer in-degree lead, and the allowance-side scores stay lower and close together: EndorseRank reaches 0.200 $[0.189, 0.210]$, C-PR ($\lambda = 1$) 0.188 $[0.177, 0.199]$ and the in-approve degree 0.180 $[0.169, 0.192]$, with overlapping intervals and no paired contrast among them. EndorseRank leaves almost no ties here: it gives 16,208 distinct scores to the 17,050 spenders, its largest tie block holds 156 of them, and only 0.027% of pairs are tied, so its coefficients rest on an ordering of nearly every spender.

Coverage does not explain the degree comparisons. Of the 17,050 spenders, 2,891 have no transfer edge up to the freeze and score zero under AWP. They gained new approval pairs slightly more often than the rest (13.3% against 11.9%) and new senders almost never (0.6% against 11.9%). Table 4.12 restricts the holdout to the 14,094 spenders that had received a transfer from another address before the freeze; the restriction was chosen after the results were known. There AWP still ranks new senders above the transfer in-degree ($+0.056$, $[0.049, 0.064]$), and EndorseRank still ranks new approval pairs below the in-approve degree (-0.013, $[-0.020, -0.006]$). AWP's coefficient on new approval pairs rises from 0.196 $[0.181, 0.210]$ to 0.298 $[0.287, 0.308]$ on this subset, since its zeros had placed at the bottom many spenders that later gained approvals; there its point estimate lies above those of the in-approve degree (0.249 $[0.237, 0.264]$) and EndorseRank (0.236 $[0.224, 0.249]$), although no paired contrast was computed for these pairs.

The second finding is that the coupled operator fails rule A on the revised scores, although in an exploratory comparison it improves on its transfer layer walked alone. Rule A, fixed in the analysis plan before any score existed, asked C-PR at $\lambda = 0.5$ to do no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new transfer senders. Here it is applied to

Table 4.12: Spender holdout W0 restricted to the 14,094 of the 17,050 spender contracts that had received a transfer from another address before the freeze ($n = 14{,}094$). The other 2,956 have no inbound transfer edge from another address, and 2,891 spenders score zero under AWP. The restriction was chosen after the results were known. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
EndorseRank	0.236 [0.224, 0.249]	0.231 [0.219, 0.244]
AWP	0.298 [0.287, 0.308]	0.357 [0.347, 0.367]
In-approve degree at t_1	0.249 [0.237, 0.264]	0.205 [0.193, 0.218]
Transfer in-degree at t_1	0.211 [0.196, 0.224]	0.301 [0.289, 0.313]
EndorseRank minus t_1 in-approve degree (new approval pairs)	$\Delta\tau = -0.013$ [-0.020, -0.006]	
AWP minus t_1 in-degree (new transfer senders)	$\Delta\tau = +0.056$ [0.049, 0.064]	

scores defined after the registered results were known, so its verdict is a re-analysis and not a registered test. The first condition holds: C-PR lies above the allowance walk ($+0.029$, $[0.022, 0.036]$). The second fails: C-PR lies below the transfer walk on new senders (-0.034, $[-0.040, -0.029]$). The registered analysis reached the same verdict (Table 8.13). With the primary part failed, the second half of RQ5, whether C-PR beats a walk over either layer alone, receives a negative answer, as the plan specified. Against the transfer walk alone on new approval pairs, a comparison that rule A did not include, C-PR does better: it reaches 0.252 $[0.241, 0.262]$ against 0.160 $[0.144, 0.174]$, a paired difference of $+0.092$ $[0.080, 0.105]$ on the same resamples. In the registered analysis the same contrast, seen after the labels were known, became the hypothesis F1 of the registration of W1; Section 4.6.2 applies F1 to the revised scores. Against EndorseRank and AWP, also computed after the labels were known, C-PR lies above EndorseRank on new approval pairs ($+0.021$, $[0.013, 0.028]$) and above AWP ($+0.056$, $[0.042, 0.070]$), and it falls well behind AWP on new senders (-0.085, $[-0.092, -0.078]$). Across the three values of λ, C-PR is flat on new approval pairs (0.253 $[0.243, 0.264]$ at $\lambda = 0.25$, 0.249 $[0.238, 0.260]$ at $\lambda = 0.75$), and its point estimate on new transfer senders falls as λ rises, from 0.269 $[0.257, 0.280]$ to 0.253 $[0.242, 0.265]$. No paired contrast between values of λ was computed, and λ was not tuned.

The seeded ablation shows where the allowance signal has to sit. S-PR is level with the transfer walk on new senders (-0.001, $[-0.004, 0.002]$) and falls below the allowance walk on new approval

pairs (-0.051, $[-0.066, -0.038]$). Its restarts lift its coefficient on new approval pairs only a little above that of the transfer walk (0.172 $[0.157, 0.185]$ against 0.160 $[0.144, 0.174]$; no paired contrast), so when transfer edges decide where the walk goes, little of the allowance signal survives. On the spender holdout, allowance edges help when they carry the walk, fully in the allowance walk and mixed in C-PR, and help little when they only decide where the walk restarts. The S-PR contrasts and the increment of C-PR ($\lambda = 1$) over its degree were computed after the labels were known; only the two contrasts of rule A were named in the plan.

Two caveats apply to the whole table. Each label is the same construction carried into the future (new approvals, new senders), so none of these spender results bears on credit or loss; the trader labels below are built from neither edge type but are not credit labels either. Both labels are also sparse. Of the 17,050 spenders, 2,070 (12.1%) gained a new approval pair and 1,695 (9.9%) a new sender, and every coefficient rests largely on how these subsets are ordered.

Two other candidate labels, which the plan did not name, were computed on the holdout, and they are reported here without being offered as support. Revocation counts rise with every score, most with the raw in-approve degree (0.388 $[0.376, 0.398]$), then with EndorseRank (0.337 $[0.327, 0.347]$), and least with the walks over transfers, AWP (0.191 $[0.175, 0.205]$) and C-PR ($\lambda = 0$) (0.200 $[0.185, 0.214]$). The drain heuristic rises with every score as well, from 0.112 $[0.096, 0.126]$ for the transfer in-degree to 0.328 $[0.318, 0.338]$ for EndorseRank with activity restarts, with EndorseRank at 0.234 $[0.224, 0.245]$ and the in-approve degree at 0.214 $[0.200, 0.227]$. For an endorsement reading the sign is wrong, since a trusted spender should lose approvals less often, not more. The likelier reading is volume. A spender with many owners sees more of everything owners do, revocations and drains included, so these labels cannot separate trust from use.

4.6.1 Trader outcomes (exploratory)

The trader run of Section 3.7.1 applies the W0 freeze to the GMX V2 contract traders and is reported as exploratory; its liquidation labels were left for the comparison of Section 4.6.3. Table 4.13 gives every score on the four profit labels of the 88 contract traders with at least three closes from April

to June, 40 of which had received an approval by the freeze. With 88 traders every interval is wide.

Table 4.13: W0 trader run: Kendall τ between scores frozen at 31 March 2026 and GMX V2 labels counted from 1 April 2026 to 30 June 2026, on the $n = 88$ GMX V2 contract accounts in the freeze-date graph with at least three closes in that window, 40 of which had received an approval by the freeze. The run is exploratory; the liquidation labels of W0 were left for the comparison (Table 4.18). Profitable closes and realized gain grow with the number of closes; the two shares do not. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at t_1 (baseline)	0.083 [-0.103, 0.238]	0.024 [-0.154, 0.202]	-0.146 [-0.317, 0.029]	-0.278 [-0.431, -0.119]
Transfer in-degree at t_1 (baseline)	0.008 [-0.135, 0.169]	0.008 [-0.149, 0.175]	-0.191 [-0.329, -0.041]	-0.197 [-0.350, -0.035]
EndorseRank	0.150 [-0.019, 0.295]	0.078 [-0.094, 0.226]	-0.055 [-0.196, 0.084]	-0.178 [-0.298, -0.042]
AWP	0.155 [0.005, 0.297]	0.196 [0.048, 0.351]	-0.124 [-0.265, 0.015]	-0.254 [-0.404, -0.104]
C-PR ($\lambda = 1$)	0.083 [-0.072, 0.223]	0.040 [-0.133, 0.190]	-0.074 [-0.231, 0.068]	-0.188 [-0.322, -0.047]
C-PR ($\lambda = 0$)	0.150 [0.011, 0.272]	0.288 [0.128, 0.438]	-0.101 [-0.243, 0.045]	-0.225 [-0.359, -0.083]
C-PR ($\lambda = 0.25$)	0.152 [0.015, 0.274]	0.294 [0.143, 0.424]	-0.084 [-0.233, 0.069]	-0.210 [-0.352, -0.061]
C-PR ($\lambda = 0.5$)	0.160 [0.030, 0.284]	0.283 [0.142, 0.410]	-0.084 [-0.227, 0.058]	-0.208 [-0.346, -0.061]
C-PR ($\lambda = 0.75$)	0.155 [0.025, 0.279]	0.270 [0.127, 0.398]	-0.082 [-0.221, 0.063]	-0.203 [-0.341, -0.059]
S-PR	0.166 [0.023, 0.290]	0.301 [0.147, 0.436]	-0.075 [-0.221, 0.072]	-0.221 [-0.365, -0.078]

The two count labels separate the walks over transfers from the other scores. On profitable closes and realized gain, which grow with how much a contract trades, AWP, C-PR ($\lambda = 0$), the three mixtures and S-PR reach coefficients with intervals above zero, from 0.150 $[0.011, 0.272]$ for C-PR ($\lambda = 0$) on profitable closes to 0.301 $[0.147, 0.436]$ for S-PR on realized gain, while the intervals of EndorseRank, C-PR ($\lambda = 1$) and both raw degrees contain zero. The transfer in-degree is near zero on both labels (0.008 $[-0.135, 0.169]$ and 0.008 $[-0.149, 0.175]$), so the number of senders alone does not rank them. On the two shares, which do not grow with activity, every coefficient is

negative. On the non-loss share every interval lies below zero, from -0.178 $[-0.298, -0.042]$ for EndorseRank to -0.278 $[-0.431, -0.119]$ for the in-approve degree; on the profitable share only that of the transfer in-degree does (-0.191, $[-0.329, -0.041]$). A contract that ranks high on these scores closed a smaller share of its positions without a loss, and part of that is built into the non-loss share (Section 4.6.3). None of the scores ranks how well a contract trades. In W1, rule B sets C-PR against its transfer layer on a trader label of its own, the liquidation-free close rate (F3).

4.6.2 Window W1: July to September 2026

The registration of Section 3.7.2 was written after the W0 results of the registered analysis were known and committed to the public repository as commit `0d3a057` at 16:49:32 UTC on 9 October 2026 (01:49 on 10 October in the commit's time zone, UTC+9). It was committed before the July–September logs were extracted, as a preregistration in the sense of Nosek et al. (2018), and fixed rule B and its margin, three further results reported beside the decision (F4–F6), two sensitivity analyses and the comparison on outcomes built from neither edge type. The margin $\delta = 0.01$ is the half-width of the W0 interval of the F2 contrast in the registered analysis (0.0040; -0.030, $[-0.034, -0.026]$), rounded up to two decimals. The extraction script refuses to run without the committed registration (Appendix 8.1). Its manifest records the registration commit, the SHA-256 hashes of the registration and of the configuration, no use of the testing override, and a start one second after the commit, at 16:49:33 UTC. The July–September logs held 6,805 GMX V2 closes by contract accounts, all of which decoded. The results below are those of the revised scores, computed after the registered evaluation of W1 was known, with the same 400 paired resamples and seed 42. Rule B, F4–F6 and the two sensitivity analyses are applied to them with the registered thresholds, as a re-analysis and not as registered tests; the registered verdicts are in Appendix 8.6.10.

At the freeze on 30 June 2026, 18,152 contract spenders held a positive latest allowance of one of the five tokens; 1,616 of them gained a new approval pair from July to September and 1,533 a new sender. Seventy GMX V2 contract traders in the freeze-date graph closed at least three positions in the label window; 9 of them had at least one liquidation, and 1 had every close liquidated. With

so few liquidations the trader label varies little, and every trader interval is wide. Table 4.14 gives every score of rule B on its three labels, and Table 4.15 the contrasts with their verdicts.

Table 4.14: Window W1: Kendall τ between scores frozen at 30 June 2026 and the three labels of rule B, counted from 1 July 2026 to 30 September 2026. The spenders are the $n = 18{,}152$ contract spenders holding a positive latest allowance at the freeze; 1,616 of them gained a new approval pair and 1,533 a new sender. The traders are the $n = 70$ GMX V2 contract accounts in the freeze-date graph with at least three closes in the label window; 9 of them had at least one liquidation and 1 had only liquidations. C-PR ($\lambda = 0$) is the comparator of rule B and AWP that of its first sensitivity analysis (Table 4.15); EndorseRank is in Table 4.16. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Spenders ($n = 18{,}152$)		Traders ($n = 70$)
Score at 30 June 2026	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.268 [0.257, 0.279]	0.205 [0.193, 0.215]	0.394 [0.261, 0.524]
Transfer in-degree at the freeze (baseline)	0.131 [0.119, 0.143]	0.308 [0.299, 0.319]	0.310 [0.185, 0.416]
AWP	0.169 [0.157, 0.183]	0.330 [0.321, 0.340]	0.329 [0.215, 0.439]
C-PR ($\lambda = 1$)	0.228 [0.217, 0.238]	0.188 [0.178, 0.198]	0.301 [0.177, 0.425]
C-PR ($\lambda = 0$)	0.141 [0.129, 0.154]	0.291 [0.282, 0.301]	0.350 [0.236, 0.461]
C-PR ($\lambda = 0.25$)	0.251 [0.241, 0.260]	0.270 [0.260, 0.279]	0.349 [0.234, 0.459]
C-PR ($\lambda = 0.5$)	0.250 [0.240, 0.260]	0.263 [0.254, 0.273]	0.350 [0.236, 0.461]
C-PR ($\lambda = 0.75$)	0.249 [0.238, 0.259]	0.256 [0.246, 0.266]	0.347 [0.231, 0.462]
S-PR	0.155 [0.142, 0.168]	0.291 [0.281, 0.301]	0.374 [0.255, 0.498]

Rule B, applied to the revised scores, is not met. F1 holds: C-PR ranks the spenders that later gained new approval pairs at $\tau = 0.250$ $[0.240, 0.260]$ against 0.141 $[0.129, 0.154]$ for its transfer layer walked alone, a difference of $+0.110$ $[0.098, 0.122]$, so the W0 gain of $+0.092$ $[0.080, 0.105]$ recurs in W1. F2 fails. On new transfer senders C-PR reaches 0.263 $[0.254, 0.273]$ against 0.291 $[0.282, 0.301]$; the difference is -0.028 $[-0.033, -0.023]$, and the whole interval lies below the margin of -0.01, so C-PR ranks later inflow worse than the transfer walk by more than the registration allowed. F3 fails narrowly. On the liquidation-free close rate of the traders C-PR and the transfer walk both reach 0.350 $[0.236, 0.461]$, and the difference is $+0.000$ $[-0.014, 0.010]$, whose lower end lies below -0.01. The rule needed all three hypotheses, so, as the registration said for this case, the thesis makes no claim that C-PR improves on a walk over the transfers alone. The registered analysis reached the same verdict (Table 8.13).

Table 4.15: Window W1: the paired contrasts F1–F6 of rule B with their verdicts, and the two sensitivity analyses (rule B and its sensitivity analyses were registered for the raw-unit analysis; here they are applied to the revised scores). $\Delta\tau = \tau_a - \tau_b$, with C-PR ($\lambda = 0.5$) as a. Superiority holds when the interval lies above zero; non-inferiority holds when its lower end lies above $-\delta = -0.01$; F6a, the first part of rule A, holds when the interval contains zero or lies above it. F1, F2, F6a and F6b are on the spenders ($n = 18{,}152$ contracts), F3–F5 on the traders ($n = 70$), as in Table 4.14. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Contrast	Label	Test	τ_a	τ_b	$\Delta\tau$ [95% CI]	Verdict
Rule B: C-PR ($\lambda = 0.5$) against its transfer layer walked alone, C-PR ($\lambda = 0$)							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.250	0.141	+0.110 [0.098, 0.122]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.263	0.291	-0.028 [-0.033, -0.023]	fails
F3	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	non-inferiority, -0.01	0.350	0.350	+0.000 [-0.014, 0.010]	fails
Decision: F1, F2 and F3 must all hold							not met
Reported, not part of the decision							
F4	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	superiority	0.350	0.350	+0.000 [-0.014, 0.010]	fails
F5	C-PR vs. C-PR ($\lambda = 1$)	liquidation-free close rate	superiority	0.350	0.301	+0.049 [-0.055, 0.168]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.250	0.228	+0.022 [0.015, 0.029]	holds
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.263	0.291	-0.028 [-0.033, -0.023]	fails
Sensitivity analysis 1: AWP as the comparator							
F1	C-PR vs. AWP	new approval pairs	superiority	0.250	0.169	+0.081 [0.069, 0.095]	holds
F2	C-PR vs. AWP	new transfer senders	non-inferiority, -0.01	0.263	0.330	-0.067 [-0.072, -0.061]	fails
F3	C-PR vs. AWP	liquidation-free close rate	non-inferiority, -0.01	0.350	0.329	+0.021 [-0.019, 0.060]	fails
F4	C-PR vs. AWP	liquidation-free close rate	superiority	0.350	0.329	+0.021 [-0.019, 0.060]	fails
Sensitivity analysis 2: contracts outside a graph kept as isolated nodes							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.250	0.141	+0.109 [0.098, 0.121]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.263	0.291	-0.028 [-0.033, -0.023]	fails
F3	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	non-inferiority, -0.01	0.350	0.350	+0.000 [-0.014, 0.010]	fails
F4	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	superiority	0.350	0.350	+0.000 [-0.014, 0.010]	fails
F5	C-PR vs. C-PR ($\lambda = 1$)	liquidation-free close rate	superiority	0.350	0.322	+0.028 [-0.047, 0.094]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.250	0.228	+0.022 [0.015, 0.029]	holds
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.263	0.291	-0.028 [-0.033, -0.023]	fails

The registration warned that F3 could pass with both coefficients near zero. Here neither coefficient is near zero, and F3 fails because the paired interval reaches below the margin although the point estimates are equal. Every score ranks this label with an interval above zero: the in-approve degree at the freeze reaches 0.394 $[0.261, 0.524]$, S-PR 0.374 $[0.255, 0.498]$, the three mixtures of C-PR between 0.347 $[0.231, 0.462]$ and 0.350 $[0.236, 0.461]$, AWP 0.329 $[0.215, 0.439]$, the transfer in-degree 0.310 $[0.185, 0.416]$ and C-PR ($\lambda = 1$) 0.301 $[0.177, 0.425]$ (Table 4.14).

The results reported beside the decision follow the registered order. F4 fails: on the liquidation-free close rate C-PR is not better than the transfer walk, and the interval of the difference contains zero. F5 fails, since C-PR and the allowance walk cannot be told apart on that label ($+0.049$, $[-0.055, 0.168]$). F6, the primary part of rule A rerun on W1, fails as in W0: C-PR is no worse than the allowance walk on new approval pairs ($+0.022$, $[0.015, 0.029]$) but lies below the transfer walk on new senders (-0.028, $[-0.033, -0.023]$). As in W0, the point estimates of C-PR are flat across λ on new approval pairs and fall on new senders as λ rises (Table 4.14). S-PR ranks new approval pairs at 0.155 $[0.142, 0.168]$, below the allowance walk (0.228 $[0.217, 0.238]$), and new senders at 0.291 $[0.281, 0.301]$, level with the transfer walk (0.291 $[0.282, 0.301]$); no paired contrast of S-PR was registered for W1.

Neither sensitivity analysis changes the verdict (Table 4.15). With AWP as the comparator, F1 holds ($+0.081$, $[0.069, 0.095]$) and F2 fails clearly. AWP ranks new senders at 0.330 $[0.321, 0.340]$, and C-PR's difference to it is -0.067 $[-0.072, -0.061]$; F3 and F4 fail as well ($+0.021$, $[-0.019, 0.060]$). Against AWP, then, C-PR gives up a measurable part of the ranking of later inflow in exchange for its gain on later approvals. With the contracts outside a graph kept as isolated nodes, every verdict stays as it was, and the point estimates of F1 to F4 and F6 change by at most 0.001. F5 stays unresolved. The rule lifts the traders outside the allowance graph from zero to the score of a node without in-edges, the allowance walk's coefficient on the liquidation-free close rate rises, and C-PR's lead over it shrinks to $+0.028$ $[-0.047, 0.094]$.

Table 4.16 adds EndorseRank and its variant with activity restarts. Neither is a score of rule B, and none of their rows enters the decision. None of the contrasts in this table was named in the registra-

tion; all were computed after the July–September labels were known, those of the variant afterwards from the saved scores. The pattern of W0 holds for both single-layer walks. The in-approve degree at the freeze again ranks new approval pairs above EndorseRank (0.268 $[0.257, 0.279]$ against 0.234 $[0.222, 0.244]$; -0.034, $[-0.038, -0.029]$), and AWP again ranks new senders above the transfer in-degree (0.330 $[0.321, 0.340]$ against 0.308 $[0.299, 0.319]$; $+0.022$, $[0.017, 0.027]$). With activity restarts EndorseRank ranks new approval pairs above the degree ($+0.027$, $[0.021, 0.034]$), as it did in W0. C-PR ranks new approval pairs above EndorseRank ($+0.016$, $[0.010, 0.024]$), as it did in W0, and the two cannot be told apart on the liquidation-free close rate ($+0.053$, $[-0.051, 0.171]$).

Table 4.16: EndorseRank and its variant with activity restarts on the cohorts and labels of the window W1 (spenders $n = 18{,}152$ contracts, traders $n = 70$). Neither these rows nor the contrasts below them are part of rule B, and the contrasts were not named in the registration. The contrasts of EndorseRank with activity restarts were computed afterwards from the saved W1 scores. The AWP, C-PR and degree rows repeat Table 4.14, and every contrast uses the same contract resamples; the differences between EndorseRank and AWP are in Table 8.11. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Spenders ($n = 18{,}152$)		Traders ($n = 70$)
Score at 30 June 2026	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.268 [0.257, 0.279]	0.205 [0.193, 0.215]	0.394 [0.261, 0.524]
Transfer in-degree at the freeze (baseline)	0.131 [0.119, 0.143]	0.308 [0.299, 0.319]	0.310 [0.185, 0.416]
EndorseRank	0.234 [0.222, 0.244]	0.194 [0.184, 0.205]	0.297 [0.174, 0.418]
EndorseRank, restarts by approving activity	0.295 [0.285, 0.304]	0.242 [0.232, 0.253]	0.288 [0.161, 0.405]
AWP	0.169 [0.157, 0.183]	0.330 [0.321, 0.340]	0.329 [0.215, 0.439]
C-PR ($\lambda = 0.5$)	0.250 [0.240, 0.260]	0.263 [0.254, 0.273]	0.350 [0.236, 0.461]
EndorseRank minus in-approve degree, new approval pairs	$\Delta\tau = -0.034$ [-0.038, -0.029]		
AWP minus transfer in-degree, new transfer senders	$\Delta\tau = +0.022$ [0.017, 0.027]		
C-PR minus EndorseRank, new approval pairs	$\Delta\tau = +0.016$ [0.010, 0.024]		
C-PR minus EndorseRank, liquidation-free close rate	$\Delta\tau = +0.053$ [-0.051, 0.171]		
EndorseRank (activity restarts) minus in-approve degree, new approval pairs	$\Delta\tau = +0.027$ [0.021, 0.034]		
EndorseRank (activity restarts) minus EndorseRank, new approval pairs	$\Delta\tau = +0.061$ [0.054, 0.067]		
EndorseRank (activity restarts) minus EndorseRank, new transfer senders	$\Delta\tau = +0.048$ [0.042, 0.053]		

Table 4.17 gives the trader labels of W1 that do not enter the decision, with the 2,000 resamples of Section 4.6.3. The scores built on transfers rank the two count labels. On profitable closes AWP reaches 0.288 $[0.125, 0.442]$ and C-PR ($\lambda = 0$) 0.199 $[0.043, 0.355]$, and on realized gain AWP reaches 0.346 $[0.189, 0.490]$ and C-PR ($\lambda = 0$) 0.338 $[0.178, 0.485]$, while the allowance-side scores stay near zero on both. On the no-liquidation flag every score is positive, the in-

approve degree highest at 0.407 $[0.268, 0.541]$ and EndorseRank with activity restarts lowest at 0.294 $[0.158, 0.417]$. On the two shares the allowance-side scores are negative, and AWP and C-PR ($\lambda = 0$) are positive with intervals that contain zero. On the non-loss share the intervals of the in-approve degree (-0.294, $[-0.468, -0.106]$), C-PR ($\lambda = 1$) (-0.287, $[-0.443, -0.121]$), EndorseRank (-0.264, $[-0.408, -0.106]$) and its variant with activity restarts (-0.269, $[-0.412, -0.118]$) lie below zero, and on the profitable share those of the same four (-0.221, $[-0.406, -0.034]$; -0.208, $[-0.370, -0.030]$; -0.180, $[-0.340, -0.012]$; -0.199, $[-0.346, -0.039]$).

Table 4.17: Window W1, traders: Kendall τ between scores frozen at 30 June 2026 and the other GMX V2 labels counted from 1 July 2026 to 30 September 2026, on the $n = 70$ contract accounts with at least three closes in that window. No liquidation is 1 when none of the account's closes was a liquidation. Profitable closes and realized gain grow with the number of closes; the two shares do not. These labels are reported and do not enter rule B. The intervals use the resamples of Table 4.18. CI = 95% percentile bootstrap over contracts (2,000 paired resamples, seed 42).

Score at 30 June 2026	No liquidation	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at the freeze (baseline)	0.407 [0.268, 0.541]	0.042 [-0.158, 0.243]	0.033 [-0.164, 0.230]	-0.221 [-0.406, -0.034]	-0.294 [-0.468, -0.106]
Transfer in-degree at the freeze (baseline)	0.329 [0.189, 0.451]	0.185 [0.014, 0.341]	0.168 [0.028, 0.291]	-0.025 [-0.194, 0.131]	-0.042 [-0.203, 0.114]
EndorseRank	0.303 [0.172, 0.422]	0.073 [-0.115, 0.260]	0.000 [-0.198, 0.200]	-0.180 [-0.340, -0.012]	-0.264 [-0.408, -0.106]
EndorseRank, restarts by approving activity	0.294 [0.158, 0.417]	0.057 [-0.123, 0.239]	-0.004 [-0.179, 0.181]	-0.199 [-0.346, -0.039]	-0.269 [-0.412, -0.118]
AWP	0.333 [0.207, 0.447]	0.288 [0.125, 0.442]	0.346 [0.189, 0.490]	0.136 [-0.039, 0.300]	0.082 [-0.077, 0.247]
C-PR ($\lambda = 1$)	0.307 [0.174, 0.428]	0.010 [-0.167, 0.192]	-0.066 [-0.249, 0.121]	-0.208 [-0.370, -0.030]	-0.287 [-0.443, -0.121]
C-PR ($\lambda = 0$)	0.358 [0.228, 0.475]	0.199 [0.043, 0.355]	0.338 [0.178, 0.485]	0.136 [-0.034, 0.290]	0.038 [-0.127, 0.201]
C-PR ($\lambda = 0.25$)	0.357 [0.227, 0.474]	0.155 [0.002, 0.305]	0.284 [0.122, 0.435]	0.090 [-0.075, 0.242]	-0.011 [-0.173, 0.148]
C-PR ($\lambda = 0.5$)	0.358 [0.229, 0.475]	0.114 [-0.044, 0.267]	0.240 [0.075, 0.397]	0.053 [-0.115, 0.212]	-0.053 [-0.217, 0.103]
C-PR ($\lambda = 0.75$)	0.353 [0.226, 0.472]	0.081 [-0.078, 0.244]	0.201 [0.023, 0.368]	0.016 [-0.153, 0.182]	-0.090 [-0.256, 0.072]
S-PR	0.381 [0.248, 0.495]	0.167 [0.018, 0.312]	0.286 [0.122, 0.441]	0.100 [-0.048, 0.244]	-0.011 [-0.164, 0.151]

4.6.3 Outcomes built from neither edge type

The GMX trader labels are the only outcomes in these data built from neither edge type. The comparison of Section 3.7.3 was registered with rule B, before the July–September logs were extracted and before the liquidation labels of W0 had been computed for any score. Here its contrasts, rules and wording are applied to the revised scores, after the registered results were known; the verdicts below are a re-analysis, and the registered analysis reached the same ones (Appendix 8.6.10). W1 is the primary window and W0 the secondary one. Intervals come from 2,000 paired resamples and are 95% unless marked 97.5%, the Bonferroni interval of the decision contrasts P and Q. Table 4.18 gives the contrasts on the liquidation-free close rate, Table 4.19 the traders that receive approvals, and Table 8.12 (Appendix 8.5.4) all six trader labels. The samples are small, 70 contract traders in W1 and 88 in W0, and every interval below is wide.

R1 does not hold. In W1 the in-approve degree at the freeze reaches $\tau = 0.394$ $[0.265, 0.518]$ on the liquidation-free close rate and the transfer in-degree 0.310 $[0.184, 0.425]$, so P, their difference, is $+0.084$ $[-0.013, 0.176]$, with a 97.5% interval of $[-0.026, 0.188]$ that includes zero. Stratified by the number of closes, P is $+0.101$ $[-0.024, 0.231]$. In W0, P is $+0.182$ (97.5% interval $[0.032, 0.338]$), positive as R1 required, but the rule also needed both W1 intervals above zero. In the registered wording, there is no count-level lead of allowance over transfer on later liquidation avoidance. R2 does not hold either. Q, EndorseRank with AWP's activity restarts minus AWP (the like-for-like PageRank pair), is -0.041 in W1 (97.5% interval $[-0.148, 0.064]$) and -0.014 in W0 (97.5% interval $[-0.161, 0.151]$), so, in the registered wording, the PageRank versions are not resolved. Neither are the other pairs. C-PR's two layers differ by -0.049 $[-0.169, 0.060]$ in W1 and -0.041 $[-0.187, 0.105]$ in W0, EndorseRank and AWP by -0.032 $[-0.133, 0.066]$ and -0.017 $[-0.156, 0.128]$, and C-PR ($\lambda = 0$) and AWP by $+0.021$ $[-0.016, 0.065]$ and $+0.025$ $[-0.022, 0.077]$. In a contrast that the registration did not name and that was computed after the labels were known, EndorseRank falls below the approval count in both windows, by -0.120 $[-0.203, -0.033]$ in W0 and -0.097 $[-0.174, -0.009]$ in W1.

Table 4.18: Allowance-side against transfer-side scores on the liquidation-free close rate of the GMX V2 contract traders with at least three closes in the label window, a label built from neither edge type ($n = 70$ traders in W1, $n = 88$ in W0). The pairs, the decision contrasts P and Q with their Bonferroni 97.5% intervals and the stratified rows were registered before the July–September logs were extracted, for the raw-unit analysis, and are applied here to the revised scores; W1 is the primary window and W0 the secondary one. Paired contract bootstrap with 2,000 resamples (seed 42). The stratified rows combine Kendall's τ_b within strata of the number of label-window closes (3–4, 5–9, 10–24, 25 or more), weighted by the number of contract pairs. Registered rules applied to the revised scores: R1 (count-level lead) not met; R2 (PageRank-level lead) not met; R3 (specificity) applies.

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI	97.5% CI
Window W1, July–September 2026 ($n = 70$ traders)					
In-approve degree minus transfer in-degree	0.394	0.310	0.084	[-0.013, 0.176]	[-0.026, 0.188]
EndorseRank (activity restarts) minus AWP	0.288	0.329	-0.041	[-0.135, 0.050]	[-0.148, 0.064]
C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$)	0.301	0.350	-0.049	[-0.169, 0.060]	–
EndorseRank minus AWP	0.297	0.329	-0.032	[-0.133, 0.066]	–
C-PR ($\lambda = 0$) minus AWP	0.350	0.329	0.021	[-0.016, 0.065]	–
EndorseRank minus in-approve degree	0.297	0.394	-0.097	[-0.174, -0.009]	–
In-approve degree minus transfer in-degree, stratified by closes	0.381	0.280	0.101	[-0.024, 0.231]	–
EndorseRank (activity restarts) minus AWP, stratified by closes	0.273	0.301	-0.028	[-0.207, 0.098]	–
Window W0, April–June 2026 ($n = 88$ traders)					
In-approve degree minus transfer in-degree	0.344	0.161	0.182	[0.052, 0.314]	[0.032, 0.338]
EndorseRank (activity restarts) minus AWP	0.227	0.240	-0.014	[-0.147, 0.130]	[-0.161, 0.151]
C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$)	0.225	0.265	-0.041	[-0.187, 0.105]	–
EndorseRank minus AWP	0.224	0.240	-0.017	[-0.156, 0.128]	–
C-PR ($\lambda = 0$) minus AWP	0.265	0.240	0.025	[-0.022, 0.077]	–
EndorseRank minus in-approve degree	0.224	0.344	-0.120	[-0.203, -0.033]	–
In-approve degree minus transfer in-degree, stratified by closes	0.338	0.209	0.129	[0.020, 0.293]	–
EndorseRank (activity restarts) minus AWP, stratified by closes	0.215	0.252	-0.037	[-0.173, 0.125]	–

R3 applies. In W1, P lies below zero on the profitable share (-0.196, $[-0.364, -0.046]$) and on the non-loss share (-0.252, $[-0.391, -0.120]$), so, in the registered wording, the signal concerns forced-liquidation avoidance, not trading success. Part of the non-loss gap is built in. The pipeline counts a liquidated close as a close that was not a loss, so a trader with liquidations gains non-loss share, whereas the profitable share counts a liquidated close as not profitable; R3 therefore rests mainly on the profitable share. The other labels do not enter the decision (Table 8.12). On the no-liquidation flag, P is $+0.078$ $[-0.012, 0.166]$ in W1 and $+0.197$ $[0.059, 0.333]$ in W0. On profitable closes and realized gain, which grow with activity, the PageRank pairs favor the transfer side in W1, with intervals below zero for Q (-0.231, $[-0.429, -0.027]$, and -0.350, $[-0.536, -0.157]$), for EndorseRank against AWP (-0.214, $[-0.423, -0.012]$, and -0.346, $[-0.547, -0.144]$) and for C-PR's layers on realized gain (-0.404, $[-0.624, -0.182]$); in W1 the same three pairs also favor the transfer side on both shares (Table 8.12). P points the same way on the two count labels without being resolved, and in W0, among the pairs of an allowance-side and a transfer-side score, only C-PR's layers on realized gain are resolved (-0.248, $[-0.433, -0.067]$).

The count-level comparison rests on few traders. In W1 the highest in-approve degree among the traders is 1, so P in effect compares the 37 recipients with the 33 other traders; in W0 the maximum is 3, and 40 of the 88 traders are recipients. Every recipient avoided liquidation in both windows (37 of 37 and 40 of 40), against 72.7% of the other traders in W1 and 75.0% in W0. Recipients also closed more positions (median 24 against 11 in W1, 31 against 10 in W0) and had lower profitable and non-loss shares (mean profitable share 0.325 against 0.479 in W1), the pattern behind R3. Without the ten recipients of highest in-approve degree, which in W1 are taken from tied values in node order, P is $+0.073$ $[-0.039, 0.186]$ in W1 and $+0.188$ $[0.046, 0.331]$ in W0. Every trader in both windows holds contract code, so account type does not separate the recipients from the others. The transfer count also ranks the liquidation-free close rate in W1, and with 70 and 88 traders the lead of the approval count over it is not resolved in the window that the registration made primary.

Table 4.19: GMX V2 contract traders with a positive in-approve degree at the freeze (recipients) against all other labelled traders, in both windows ($n = 70$ traders in W1, $n = 88$ in W0). The comparison of recipients with the other traders and the influence check were registered with the contrasts of Table 4.18 and are applied here to the revised scores. The highest in-approve degree among the traders is 1 in W1 and 3 in W0, so the ten recipients left out by the influence check are taken from tied values in node order. The influence check is a paired contract bootstrap on the remaining traders (2,000 resamples, seed 42).

	Recipients	Other traders
Window W1, July–September 2026 ($n = 70$ traders)		
Traders	37	33
Share with no liquidation	1.000	0.727
Mean liquidation-free close rate	1.000	0.878
Mean profitable share	0.325	0.479
Mean non-loss share	0.433	0.644
Median closes in the label window	24.0	11.0
P without the ten recipients of highest in-approve degree: $\Delta\tau = 0.073$ [-0.039, 0.186]		
Window W0, April–June 2026 ($n = 88$ traders)		
Traders	40	48
Share with no liquidation	1.000	0.750
Mean liquidation-free close rate	1.000	0.919
Mean profitable share	0.362	0.474
Mean non-loss share	0.402	0.607
Median closes in the label window	31.0	10.0
P without the ten recipients of highest in-approve degree: $\Delta\tau = 0.188$ [0.046, 0.331]		

4.7 Claims (C1–C6)

The chapter's results support the claims below, cross-referenced to the hypothesis and research question columns of Table 1.2. All of them rest on the revised analysis. The verdicts of rules A and B in C5 and of R1–R3 in C6 apply the registered rules to the revised scores, after the registered results were known, so they are a re-analysis; the registered analysis reached the same verdicts (Appendix 8.6.10).

C1. Cost. On the full graphs of the same contracts the solver's time follows the edge count of its graph, and on the subsamples it does so from the stage of 10,000 contracts on. The allowance graph has 44.8% of the transfer graph's edges (11,210,114 against 25,001,216 at T_{obs}), and EndorseRank's call takes correspondingly less time than AWP's: AWP takes 2.17 times as long (22.37 s on 7,882,356 nodes against 10.29 s on 6,219,063 nodes), at an edge ratio of 2.23 (Table 4.1). C-PR (36.16 s on the union of 9,151,010 nodes and 31,552,359 edges) and S-PR (34.42 s on 7,884,602 nodes and 25,018,892 edges) walk the transfer layer and take longer than AWP. The speed belongs to the sparser graph, not to the algorithm.

C2. Same-window agreement. Each score agrees most with the proxies of the edge type it walks on, EndorseRank with the allowance family (0.585, $[0.578, 0.592]$) and AWP with the transfer family (0.572, $[0.564, 0.580]$), with intervals far from zero (Table 4.3). This is an intended-construct check, and the agreement is partly built in. With US-dollar amounts AWP follows the summed value of the inbound transfers almost as closely as their number of senders (in-value 0.566, $[0.558, 0.575]$; in-degree 0.578, $[0.569, 0.587]$). Every contract in the cohort received approvals of the five tokens, so EndorseRank leaves few ties (13,625 distinct scores, 0.17% of pairs tied), and the isolated-node rule changes no coefficient (Table 4.5). The C-PR mixtures lie between their two layers on every family.

C3. Divergence. EndorseRank and AWP order the contracts differently, with $\tau_b = 0.174$ $[0.163, 0.185]$; 96.7% of the pairs are ordered by both scores, and 41.2% of those are discordant (Table 4.5, Figure 4.3).

C4. Degree baselines out of window. The two walks behave differently against the counts they smooth. AWP ranks later new transfer senders better than the raw transfer in-degree in both windows (W0 $+0.043$, $[0.038, 0.048]$; W1 $+0.022$, $[0.017, 0.027]$), so on the transfer side the walk, with its time and value weights and its activity restarts, added information to the count. EndorseRank, with the uniform restarts fixed in the plan, ranks new approval pairs below the raw in-approve degree in both windows (W0 -0.019, $[-0.026, -0.011]$; W1 -0.034, $[-0.038, -0.029]$). The variant with AWP's activity restarts ranks them above the degree in both windows (W0 $+0.059$, $[0.050, 0.067]$; W1 $+0.027$, $[0.021, 0.034]$), but that comparison was made after the labels were known and is a hypothesis, not a finding (Section 4.5.4, Table 4.16). All degree contrasts were computed after the labels were known; none was named in the plan or the registration. The increment of C-PR ($\lambda = 1$) over the in-approve degree in W0 (-0.027, $[-0.034, -0.019]$) and the two S-PR contrasts were also computed after the labels. Restricting W0 to the 14,094 spenders that had received a transfer leaves both degree comparisons in place (EndorseRank -0.013, $[-0.020, -0.006]$; AWP $+0.056$, $[0.049, 0.064]$; Table 4.12).

C5. Coupling. On the revised scores, one walk over both edge types does not beat its layers walked alone. Rule A fails: C-PR is no worse than the allowance walk on new approval pairs ($+0.029$, $[0.022, 0.036]$) but worse than the transfer walk on new senders (-0.034, $[-0.040, -0.029]$), and its same-window part fails as well (Tables 4.11 and 4.4). Rule B is not met (Table 4.15). F1 holds ($+0.110$, $[0.098, 0.122]$, over the transfer walk on new approval pairs), so the W0 gain of $+0.092$ ($[0.080, 0.105]$), computed after the labels were known, recurs; F2 fails (-0.028, $[-0.033, -0.023]$, beyond the margin of 0.01); and F3 fails narrowly ($+0.000$, $[-0.014, 0.010]$, on the traders' liquidation-free close rate, with equal point estimates of 0.350 but a lower end below -0.01). With AWP as the comparator, in the first sensitivity analysis, F2 fails clearly (-0.067, $[-0.072, -0.061]$). In two W0 contrasts computed after the labels were known, S-PR is level with the transfer walk on new senders (-0.001, $[-0.004, 0.002]$) and below the allowance walk on new approval pairs (-0.051,

$[-0.066, -0.038]$): allowance edges help only when they carry the walk. Section 4.3.1 reports where the hybrids stand within the window.

C6. Outcomes built from neither edge type. On the liquidation-free close rate of GMX V2 contract traders, the count-level lead of the approval count over the transfer count (R1) is not supported in the primary window W1 (P $= +0.084$, Bonferroni 97.5% interval $[-0.026, 0.188]$; stratified $+0.101$, $[-0.024, 0.231]$), although P is positive in W0 ($+0.182$, 97.5% interval $[0.032, 0.338]$). The PageRank lead (R2) is not supported (Q $= -0.041$, 97.5% interval $[-0.148, 0.064]$; in W0 -0.014, 97.5% interval $[-0.161, 0.151]$). R3 applies: the approval count ranks traders lower than the transfer count on the profitable share (-0.196, $[-0.364, -0.046]$) and the non-loss share (-0.252, $[-0.391, -0.120]$). Every trader that had received an approval avoided liquidation in both windows (37 of 37 and 40 of 40), against 72.7% and 75.0% of the others, but the samples are small (70 and 88 contract traders), and the transfer count also ranks this label (0.310, $[0.184, 0.425]$, in W1) (Tables 4.18 and 4.19).

The rank divergence answers RQ1, the same-window tables RQ2, the benchmark tables RQ3, and the holdouts RQ4 and RQ5; C6 adds no research question. The direct comparisons between EndorseRank and AWP are in Appendix 8.5 and Table 4.18, and none of them carries a claim of the thesis. Chapter 5 reads these findings against the main research question, what ERC-20 allowance edges add to PageRank as a reputation score for smart contracts on Arbitrum One compared with the transfer walk and with the raw degrees that both walks smooth, and against the objectives of Chapter 1.

5 Discussion

Chapter 4 reported the measurements taken on Arbitrum One. Here each research question and each objective of Chapter 1 is held to the criterion stated there, and every interpretation points to the table of Chapter 4 it rests on. The measurements are those of the revised analysis, which values every amount in US dollars and keeps five tokens. The amounts were revised after the registered results were known (Section 3.5.1), so none of the revised results discussed here is a registered result, and a registered result, where one is mentioned, is labelled as such; the registered analysis, in raw base units over every token, reached the same verdicts (Appendix 8.6.10, Table 8.13). Unless noted otherwise, same-window statements concern the matched cohort of contracts and holdout statements the spender cohort.

5.1 Findings by research question

5.1.1 RQ1: Reputation structure and rank divergence

RQ1 asks whether EndorseRank and AWP order the same contracts in different ways. They do: the interval of their agreement excludes both zero and one (Table 4.5), and of the contract pairs that both scores order, 41.2% are ordered in opposite directions (Section 4.4). Ties play almost no part in this result. Every cohort contract received approvals of one of the five tokens from at least three owners, so EndorseRank leaves few pairs of contracts tied, and keeping the contracts that are missing from a graph as isolated nodes, instead of scoring them zero, leaves the agreement unchanged.

The divergence follows from how the two graphs are built. The allowance graph records the latest approval of each owner, spender and token, whereas the transfer graph records every transfer (T. Do et al., 2023), and the allowance graph has fewer than half of the transfer graph's edges (Section 4.2);

the two methods also restart differently. PageRank places stationary mass according to the edges it is given (Page et al., 1999), so a contract that receives many or large transfers from active senders but few approvals ranks high under AWP and modestly under EndorseRank, and the reverse holds for a contract that many owners approve without being a transfer hub. H1 is supported, largely by construction, since two walks over different graphs with different restarts were not expected to agree. In practice one score cannot stand in for the other: an operator who adopts EndorseRank because its graph is smaller gets a different ranking and not a cheaper approximation of AWP.

5.1.2 RQ2: Same-window alignment

RQ2 asks how closely each score agrees with contemporaneous allowance and transfer checks. Each score follows the edges it walks on (Table 4.3). EndorseRank and the allowance layer walked alone agree most with the allowance family, AWP and the transfer layer walked alone most with the transfer family, and the paired contrasts between the two single-layer scores exclude zero: EndorseRank agrees more with the allowance family than with the transfer family, and more with the allowance family than AWP does, while AWP agrees more with the transfer family than EndorseRank does (Table 8.7). The C-PR mixtures lie between their two layers on every family and move far less between $\lambda = 0.25$ and $\lambda = 0.75$ than between the two layers. Every interval in Table 4.3 excludes zero except EndorseRank's on the Sybil-stability family, so by the criterion of O2 EndorseRank does not agree with that construct, and every other score agrees with each construct to some degree; what separates the scores is the family they agree with most.

Two caveats bound H2. The first is circularity. Every proxy in a family is computed from the same events as the matching graph, so the agreement is an intended-construct check in the sense of Cronbach and Meehl (1955), and part of it is built in: the allowance proxy with which EndorseRank agrees most is the in-approve degree, the count it smooths (Table 8.9), and AWP, whose edges weigh each transfer by its dollar value, follows the summed value of inbound transfers almost as closely as the number of its senders (Table 8.8). Ties do not distort the comparison, since EndorseRank leaves few pairs tied and the isolated-node rule changes no coefficient (Table 4.5). The second

caveat is that a higher coefficient in the same window does not mark a better score out of window. EndorseRank with AWP's activity restarts agrees less with the allowance family than EndorseRank does ($\tau = 0.426$, $[0.418, 0.435]$, against 0.585, $[0.578, 0.592]$; no paired contrast was computed, and the two intervals do not overlap), yet it ranks both spender labels better in both holdout windows, with paired intervals above zero, in comparisons made after the labels were known (Table 4.9 and Section 4.5.4). In the narrow sense of agreement with the checks of a score's own edges, H2 is supported, largely by construction. The paired differences between EndorseRank and AWP, which carry no claim, are in Appendix 8.5.

5.1.3 RQ3: Computational cost

RQ3 asks what each score costs to compute on the same sample. The cost follows the edge count. On the same contracts AWP's graph has 2.23 times the edges of EndorseRank's, and its solver call takes 2.17 times as long (Section 4.2; Table 4.1 gives $|V|$, $|E|$, peak memory and iterations of each graph). C-PR and S-PR also walk the transfer layer and take longer than AWP. Matrix assembly and iteration are both linear in $|E|$ (Langville & Meyer, 2006), so the time ratio is expected to track the edge ratio. Both scores run the same solver, so the lower cost is a property of the sparser graph and not of the algorithm. Each graph and setting was timed once under the protocol of five timed runs after one warm-up, and that measurement is reused wherever the graph appears.

The scaling stages repeat the comparison on smaller graphs. Each stage keeps the edges incident to a deterministic subsample of the cohort, so both edge counts grow from stage to stage (Table 4.2). The time ratio of the two solves follows the edge ratio from the stage of $10{,}000$ contracts on, but not at the three smaller stages, where the time ratio departs from the edge ratio in either direction (Section 4.2.1). The stages are subsets of one ego network, so they say nothing about either solver on a chain-wide graph. The damping and value-slope checks leave the order of the families as it was and move the family means little (Tables 4.6 and 4.7); with the slopes of the value weight divided or multiplied by ten, each score still agrees most with its own family and ranks the contracts almost as the main score does. H3 is supported on the matched cohort and its subsamples, again largely

by construction, since both scores run the same solver. H1–H3 are thus descriptive checks of the instruments; the tests whose outcome was open are RQ4, RQ5 and Section 5.1.6.

5.1.4 RQ4: Temporal holdout of future new approvals

RQ4 has two clauses. The first asks whether scores frozen at the end of March 2026 predict new owner–spender approval pairs and new transfer senders on the spender cohort over the following three months. Every score predicts both labels, and the edge type decides how well (Table 4.10). Among the scores of that table, the interior mixtures and the in-approve degree lead on new approval pairs, followed by EndorseRank and the allowance layer, while the scores built on transfers alone stay lower. On new senders AWP leads, ahead of the transfer in-degree and the transfer layer, and EndorseRank, the allowance layer and the in-approve degree rank lowest. The window W1, with scores frozen at the end of June and labels counted from July to September, repeats this pattern among the same scores, except that on new approval pairs the in-approve degree has a higher point estimate than the interior mixtures, a difference for which no paired contrast was computed (Tables 4.14 and 4.16). The evidence concerns spender contracts; Section 5.1.6 turns to the traders. The second clause asks whether any PageRank beats the raw degree it smooths, and the two walks answer it differently (Tables 4.11 and 4.16). AWP ranks later new senders better than the transfer in-degree in both windows, with intervals above zero, and the lead holds on the spenders that had received a transfer before the freeze, so the spenders that AWP scores zero do not create it (Table 4.12). On the transfer side, then, the walk with its time and value weights and activity restarts added information that the count lacks. EndorseRank, with the uniform restarts fixed in the analysis plan, ranks new approval pairs below the in-approve degree in both windows, and so does the allowance layer walked alone in W0. On the allowance side the count was the better predictor. Known results lead one to expect the count to be a hard baseline to beat. In growing networks new links attach preferentially to nodes that already have many (Barabási & Albert, 1999), which makes degree a standard link-prediction baseline (Liben-Nowell & Kleinberg, 2007), also on Ethereum transaction networks (D. Lin et al., 2022). Where degree correlations are weak, as on the Web, PageRank is

on average proportional to in-degree (Fortunato et al., 2008). For most spenders EndorseRank is an approval count in which each approval is weighted by its age and dollar amount and divided by the summed weight of all the approvals its owner grants (Section 5.3.2), which leaves it little room to differ from the count.

The variant of EndorseRank with AWP's activity restarts ranks new approval pairs above the in-approve degree in both windows (Section 4.5.4 and Table 4.16). Only the restart rule separates the two variants, so on the allowance side the restart rule appears to decide whether the PageRank beats its count; whether the same holds for AWP was not tested, since AWP was run only with its published restarts. The variant's contrasts were computed after the labels were known in both windows, so they state a hypothesis for a future registered test and not a finding. None of the degree contrasts was fixed in advance: on the revised scores every one of them was computed after the labels of its window were known. In the registered analysis the four degree contrasts of EndorseRank and AWP had the same signs, with intervals on the same side of zero (Table 8.13). They do not undo the first clause, since the counts are statistics of the same edges.

The labels lie outside the score window but belong to the same constructs as the scores, and neither contains any element of profit, loss or repayment. RQ4 therefore shows how far each construct carries forward in time, with the raw degree as the baseline that a PageRank must beat, and leaves the safety and the credit risk of a contract aside; the liquidation label of W1 (Section 3.7.2), on which Section 5.1.6 compares the edge types, comes closest to a risk outcome. The revocation counts and the drain heuristic computed on the same spenders rise with every score, and the revocations most with the in-approve degree (Section 4.6). They track the volume of approvals a contract holds, and they are not offered as evidence about its safety.

5.1.5 RQ5: Coupling the two edge types

Both halves of RQ5 get a negative answer, each under its own rule. The rules were fixed for the registered analysis and are applied here to the revised scores after the registered results were known, so the verdicts below are a re-analysis and not registered tests; the registered analysis reached the

same verdicts (Table 8.13). The second half asks whether one walk over both layers ranks contracts better than either layer walked alone. Under rule A, fixed in the analysis plan before any score existed, it does not. On W0, C-PR is no worse than the allowance layer on new approval pairs but falls behind the transfer layer on new senders (Table 4.11), and in the same window its point estimates lie outside the interval of the better layer on both the transfer and the allowance family and between the two layers on Sybil stability (Section 4.3.1), so the secondary part fails on all three families as well. The first half asks whether the walk improves on the transfer layer walked alone. An exploratory comparison on W0 said that it did on new approval pairs, but it was made after the labels were known. Rule B, registered before the July–September logs were extracted, was written for this question, and on the revised scores it is not met. C-PR again beat the transfer layer on new approval pairs, by a little more than in W0, but it fell behind on new senders by more than the margin of 0.01, and on the traders' liquidation-free close rate it could not show that it was no worse: the two point estimates are equal, but the interval of their difference reaches below -0.01 (Section 4.6.2). On that trader label every score has an interval above zero (Table 4.14). Against AWP, the comparator of the first sensitivity analysis, C-PR falls behind on new senders by a wider margin. Against EndorseRank, in comparisons not fixed in advance, it is slightly above on new approval pairs in both windows (Tables 4.11 and 4.16).

The ablation and the layer weights explain both halves. S-PR keeps the transfer walk but restarts it from the allowance walk. In contrasts computed after the labels were known, it stays close to the transfer layer on new senders and falls far below the allowance layer on new approval pairs (Table 4.11), since a teleportation vector derived from endorsements changes little when transfer edges decide where the walk goes. In C-PR the allowance edges carry part of the walk, so the allowance signal survives on the spender holdout: every interior mixture has a slightly higher coefficient on new approval pairs than the allowance layer alone, and at $\lambda = 0.5$ the paired interval lies above zero (Tables 4.10 and 4.11). The price is paid on the transfer side, where every interior mixture has a lower coefficient on new senders than the transfer layer. C-PR's layers carry dollar amounts without a bound, apart from the cap on allowances, and restart uniformly, whereas AWP, which bounds the

weight of each transfer through V and restarts at active senders, ranks new senders better than C-PR in both windows (after the labels in W0, under the first sensitivity analysis applied to the revised scores in W1). On these data allowance edges help a walk only when they carry it. Other couplings were not tested (Section 5.9).

Both halves count as results. Rule A was fixed before any coupled score existed, so λ could not have been tuned to it; the revision changed the tokens, the cohort and the amounts in its layers, adding the cap U to the allowance layer, after its registered verdict was known, but neither λ nor its thresholds. It limits an obvious next step: a convex mixture of the two edge types at the transition level gained less over the allowance layer on new approval pairs than it lost against the transfer layer on new senders, and in the same window it stayed outside the better layer's interval on both families (Tables 4.11 and 4.3). Rule B was registered before its labels were extracted, and on the revised scores, as in the registered analysis, it fails on two of its three conditions. W1 shows something narrower: adding allowance edges to a transfer walk gains information about authorization that holds on new data, at a loss on inflow larger than the margin fixed in the registration.

5.1.6 Allowance against transfer on outcomes built from neither edge type

Unlike the spender labels of RQ4 and RQ5, which favor the scores built on their own edge type, the GMX trader outcomes are built from neither edge type and were measured after the scores were frozen. The comparison of the two edge types on them (Section 3.7.3) adds no research question and is reported as Claim C6 (Section 4.6.3). It was registered together with rule B, before the July–September logs were extracted and before the W0 liquidation labels had been computed for any score, and the registration fixed the wording of its conclusions. Here its rules R1–R3 and their wording are applied to the revised scores, after the registered results were known, so the verdicts are a re-analysis and not registered tests; the registered analysis reached the same three verdicts (Table 8.13).

This comparison bears on counterparty risk, the use that motivated the thesis (Chapter 1). For a protocol that extends leverage or credit to a contract account, a counterparty that is often force-

liquidated is a riskier one, and forced liquidation is the nearest outcome to default in these data. In the primary window W1 the paired difference between the approval count at the freeze and the transfer count on the liquidation-free close rate is $\Delta\tau = +0.084$ (95% interval $[-0.013, 0.176]$, Bonferroni 97.5% interval $[-0.026, 0.188]$), and stratified by the number of closes it is $+0.101$ ($[-0.024, 0.231]$); both intervals include zero. In W0 the same contrast was $+0.182$ ($[0.052, 0.314]$), but W0 is the secondary window. The like-for-like PageRank pair, EndorseRank with AWP's restarts against AWP, is negative in both windows with intervals that include zero (Table 4.18). Under the wording fixed in the registration, the result therefore reads: no count-level lead of allowance over transfer on later liquidation avoidance; the PageRank versions are not resolved; and, since the approval count ranks traders lower than the transfer count on the profitable share and on the non-loss share, the signal concerns forced-liquidation avoidance, not trading success.

What remains is descriptive. Every trader that had received an approval by the freeze avoided liquidation in both windows, 37 of 37 in W1 and 40 of 40 in W0, against about three in four of the other traders (Table 4.19). Among the traders the in-approve degree is at most one in W1 and three in W0, so the approval count works here as a flag for having been approved at all, and the transfer count also ranks the label in W1 (Table 4.14).

Three limits bound this reading. First, the samples are small. Few contract accounts close positions on GMX V2 in these months, so the trader cohorts hold 70 and 88 contracts and every interval is wide. Second, account type cannot explain the pattern, because every trader in both cohorts is a contract, but the role of a contract may: the recipients close more positions than the other traders and were authorized by other addresses to spend their tokens, a pattern that fits managed or pooled trading contracts, which the data do not identify. Third, the recipients have lower profitable and non-loss shares, so what they avoid is forced liquidation, not loss; the lower non-loss share is partly built in, because the label counts a liquidated close as not a loss. Within these limits, the comparison did not separate allowance-edge from transfer-edge information on the outcome that a margin or lending protocol screens for.

5.2 Achievement of the research objectives

Chapter 1 attached a measurement and a criterion to each of its five objectives. Table 5.1 checks them against the evidence of the revised analysis. Where a criterion is a rule fixed in the plan or the registration, the outcome is that of the rule applied to the revised scores after the registered results were known; the registered outcomes are in Table 8.13.

Table 5.1: Research objectives, the criterion stated in Chapter 1, and the outcome.

Obj.	Criterion	Evidence	Outcome
O1	95% interval of inter-method τ contains neither 0 nor 1	Table 4.5, last row: $\tau = 0.174$, interval $[0.163, 0.185]$, the same under the isolated-node rule	Met (largely by construction)
O2	A score agrees with a construct when its interval on that family excludes 0; agreement with the construct of a score's own edges is an intended-construct check	Table 4.3: every interval excludes 0 except EndorseRank's on Sybil stability (0.007, $[-0.001, 0.015]$); each single-layer score agrees most with its own family (EndorseRank allowance 0.585, $[0.578, 0.592]$; AWP transfer 0.572, $[0.564, 0.580]$), and the paired contrasts of Table 8.7 exclude 0; the proxies come from the same events as the graphs	Met (largely by construction)
O3	Time ratio reported only together with the ratio of $\lvert E\rvert$, under one fixed protocol	Section 4.2 and Table 4.1: five timed runs after one warm-up, with $\lvert V\rvert$ and $\lvert E\rvert$ of each solver graph; AWP's time ratio to EndorseRank 2.17 beside an edge ratio of 2.23; one solver, so time follows $\lvert E\rvert$	Met (largely by construction)
O4	A score predicts a label when its interval excludes 0; a PageRank improves on its degree only when the interval of the paired difference lies above 0	Tables 4.10, 4.14 and 4.16: every score predicts both labels in W0 and W1. Tables 4.11 and 4.16: AWP minus transfer in-degree $+0.043$ $[0.038, 0.048]$ (W0) and $+0.022$ $[0.017, 0.027]$ (W1); EndorseRank minus in-approve degree -0.019 $[-0.026, -0.011]$ (W0) and -0.034 $[-0.038, -0.029]$ (W1); C-PR ($\lambda = 1$) minus in-approve degree -0.027 $[-0.034, -0.019]$ (W0); all computed after the labels were known	Partly met (first clause met; second clause met for AWP, not for EndorseRank)

Table 5.1: (continued)

Obj.	Criterion	Evidence	Outcome
O5	(a) Rule A (analysis plan, 9 October 2026): on W0, C-PR does no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new senders; in the same window, its point estimate lies within the better layer's interval on the transfer and allowance families, and it exceeds both layers on Sybil stability. (b) Rule B (registered at 16:49 UTC on 9 October 2026, before the July–September logs were extracted): on W1, C-PR beats C-PR ($\lambda = 0$) on new approval pairs (F1) and is not worse by more than $\delta = 0.01$ on new senders (F2) and on the traders' liquidation-free close rate (F3)	(a) Table 4.11: no worse holds ($+0.029$, $[0.022, 0.036]$), better fails (-0.034, $[-0.040, -0.029]$). Same window (Table 4.3): transfer 0.365 $[0.355, 0.375]$ outside $[0.653, 0.666]$; allowance 0.477 $[0.469, 0.485]$ outside $[0.590, 0.604]$; Sybil stability 0.142 $[0.134, 0.151]$, between the layers. (b) Table 4.15: F1 $+0.110$ $[0.098, 0.122]$ holds; F2 -0.028 $[-0.033, -0.023]$ and F3 $+0.000$ $[-0.014, 0.010]$ fail	(a) Not met; (b) not met; both on the revised scores, as in the registered analysis

Six findings outside the objectives also shape the conclusions. Three limit the claims. The sample-definition stages move the same-window coefficients enough that they cannot serve as robustness evidence (Table 4.8); the value-slope check bounds the effect of the slopes chosen in the revision only in the same window, where dividing or multiplying them by ten keeps each score closest to its own family and its ranking close to the main one (Table 4.7), and no other slope was tried on the holdout labels; and on the trader cohorts no score ranks the share of profitable closes with an interval above zero in either window (Tables 4.13 and 4.17). Three add to the claims. EndorseRank with AWP's activity restarts ranked new approval pairs above the in-approve degree in both windows, in comparisons made after the labels and offered as a hypothesis (Section 5.1.4). Every trader contract that had received an approval avoided liquidation in both windows, in small samples and without a lead over the transfer count under rule R1 (Section 5.1.6). And EndorseRank matches a closed form to within 1% for every depth-one spender, while the graph also has deeper paths along which propagation changes the score (Section 5.3.2). That AWP beats its count and EndorseRank does

not is not among them, because it is the second clause of O4.

5.3 Theoretical implications

The results bear on how propagation works on financial graphs, on what EndorseRank computes for spenders whose owners are not themselves approved, on the construct alignment of on-chain scores and on trust under cheap pseudonyms.

5.3.1 Propagation semantics beyond transfers

PageRank was first formulated for hyperlinks, read as a citation network (Page et al., 1999). H.-D. Do and Do (2023) carried this reading over to Ethereum transactions, and AWP (T. Do et al., 2023) refined it with value and age weights and restarts at active senders. EndorseRank applies it to ERC-20 allowances, where an owner who lets a spender contract act for it issues a directed endorsement of spending authority (Vogelsteller & Buterin, 2015). The RQ1 divergence shows that changing what the edges mean, together with the restart rule that goes with each graph, changes the ranking even on the same nodes, so graph-based reputation is a family of methods, each tied to the relationships its edges encode. Peer-to-peer and web trust systems reached the same conclusion from the other side: in EigenTrust (Kamvar et al., 2003) and TrustRank (Gyöngyi et al., 2004) the seed set and the edge definitions largely decide the outcome. EndorseRank's edge is native to a token standard, where theirs belonged to a message-passing overlay or a hyperlink graph.

The holdouts add a qualification that the propagation metaphor hides. Changing the edge type changed what the score could anticipate: on new approval pairs the scores built on allowance edges led and those built on transfers alone stayed lower, and on new senders the reverse held, in both windows. Whether propagating along the edges added to the count depended on the side. On transfers AWP beat its count; on allowances EndorseRank did not, while its variant with activity restarts did. None of these comparisons was fixed in advance, and on the revised scores all of them were made after the labels were known (Section 5.1.4). On these data the edge type sets what

a score anticipates, and the restart rule, more than the walk itself, appears to decide whether the score improves on the count. When the two layers share one walk, the allowance signal survives where allowance edges carry the walk and is lost where they only seed its restarts (Section 5.1.5).

5.3.2 EndorseRank on depth-one spenders

The allowance graph keeps the approvals of the five tokens that a cohort contract grants or receives (Section 3.6). Most of its nodes are pure owners, with out-edges only, or pure spenders, with in-edges only. A cohort contract can be both, since it can receive approvals from owners and approve other contracts in turn, so the graph has paths of more than one step. For a spender whose owners receive no approvals, EndorseRank has a closed form.

Proposition. Let PageRank be computed by Equation (3.1) with damping d and uniform restarts, so that dangling mass is also spread uniformly. Then every node without in-edges has the same score c, and every node v whose in-neighbors all lack in-edges has

$$r_v = c\,\big(1 + d\,\delta(v)\big), \qquad \delta(v) = \sum_u \frac{w(u,v)}{o(u)}, \tag{5.1}$$

where $\delta(v)$ is the owner-normalized in-strength of v: each owner's endorsement is divided over all the spenders it approves.

Derivation. At the fixed point, $r_v = (1 - d + d\,r_D)/|V| + d\sum_u r_u\,w(u,v)/o(u)$, where r_D is the mass held by the dangling nodes. A node without in-edges receives only the first term, so every such node has the score $c = (1 - d + d\,r_D)/|V|$, the constant that Section 5.6.3 also writes c. If every in-neighbor u of v is such a node, substituting $r_u = c$ gives Equation (5.1).

On the graph at T_{obs} the nodes without in-edges share one score, $c = 7.64 \times 10^{-8}$, to numerical precision. Of the 14,406 cohort spenders in that graph, 12,039 have depth one, and EndorseRank as computed in this thesis lies within 1% of Equation (5.1) for every one of them. Of the other 2,367, which have an owner that itself receives approvals, 26.4% lie within 1%. Over all spenders the share is 87.9%. Of the 17,032 W0 spenders in the graph at t_1, 11,326 have depth one, every

one of them again within 1%, and the share over all W0 spenders is 73.7%. The share of deeper spenders is larger there partly because the W0 cohort also contains contracts outside the matched cohort, which enter the extracted data only through approvals from cohort contracts, and every cohort contract receives approvals of its own in the observation window (Section 4.1.1). The check is computed by `scripts/run_same_window.py` and, for W0, `scripts/run_holdout.py` of the experiment code, run with the switch that selects the revised analysis (Section 3.13).

Two consequences follow. First, for a depth-one spender EndorseRank only reweights the approval count: each owner's latest approval of a token counts by its age and its dollar amount, never for more than one, and an owner that approves many spenders counts for less at each. These edge weights and that normalization are all that separate it from the count, and for these spenders there is no propagated information for the walk to add. Second, the graph does have deeper paths: about a sixth of the cohort spenders at T_{obs} and a third of the W0 spenders have an owner that is itself endorsed, and for most of them the score departs from the closed form. Multi-hop propagation is therefore present in this graph, but it did not make EndorseRank with uniform restarts beat the raw count out of window (Section 5.1.4). Whether it helps on a deeper sample, such as the two-hop neighborhood of the cohort or a chain-wide allowance graph, or with restarts seeded at vetted contracts, is not tested here.

5.3.3 Multi-construct domain alignment

The checks were designed to separate transfer centrality from allowance delegation, and each score aligns with the construct of the edges it walks on, the mixtures falling between their layers. Classical construct-validity theory would predict this pattern for instruments that measure different constructs (Cronbach & Meehl, 1955), and Messick (1995) goes further, counting the consequences of using a score as part of its validity. This thesis applies that logic to ordinal reputation scores with Spearman's ρ (Spearman, 1904) and Kendall's τ (Kendall, 1938), which need no classification threshold, and the paired bootstrap (Efron, 1979) makes the pattern testable. The same theory says what such a pattern cannot settle. When two instruments measure different constructs, the one that comes out

ahead on a check is the one built for that check, so ordering the instruments by their coefficients does not show which is the better measure of reputation. The two EndorseRank variants add a sharper point: the variant that agrees less with its own construct in the same window anticipates that construct's later events better, in comparisons made after the labels were known (Section 5.1.2), so same-window agreement does not rank instruments even within one construct. Outcomes built from neither edge type are the exception, since neither instrument was built for them, which is why the thesis compares the edge types there (Section 5.1.6); in the primary window that comparison, its registered rules applied to the revised scores, did not separate them. Such comparisons are possible here because no score leaves many pairs tied; with a heavily tied score, two rank correlations could be compared only when they share the same attainable maximum.

5.3.4 Reputation under cheap pseudonyms

Where identities cost little to create, one entity can present many (Douceur, 2002), and under Ethereum's account model a new address needs no registration step at all (Wood, 2025). Any reputation system that rewards inbound transfers or approvals is therefore exposed to Sybil strategies, in which an attacker creates edges among addresses it controls. For a contract the strategy is direct, since whoever deploys it can create owner addresses and have each of them approve it. The empirical findings do not rule this out. Section 5.6 formalizes the cost of attacking an allowance graph, where a closed farm of addresses that approve one another needs, unlike transfer wash trading, no token balance and pays little beyond gas. On the graph studied here such a farm buys a place in the upper part of the ranking for a few dozen approvals per window but would need more than a hundred thousand addresses to reach its top, because endorsement is concentrated: the highest-ranked cohort contract holds about 3% of all stationary mass (Section 5.6.3).

5.4 Practical implications and deployment scenarios

Wallet providers, protocols and security tools that have to judge contracts can read either the flow into a contract or the authorizations it holds. The measured results support choosing the score by scenario over recommending a single one.

5.4.1 Transfer-hub and flow monitoring

A protocol whose priority is describing current flow into contracts (for market-maker identification, routing analysis or transfer-based compliance heuristics) can read that flow from in-degree and in-value directly. AWP (T. Do et al., 2023) adds propagation, time and value weights and restarts at active senders. It agrees with the transfer family in the same window (Table 4.3), and out of window it has the highest coefficient of any score on later new senders, with a paired difference to the transfer in-degree above zero in both windows, although neither contrast was fixed in advance (Tables 4.11 and 4.16). A protocol that wants to anticipate flow into a contract has reason to use AWP rather than the count.

5.4.2 Authorization-aware and low-latency refresh

Where explicit spending authorization matters most (token vaults, allowance-gated integrations, or risk engines that treat approvals as acts of trust), protocols have reason to use allowance edges. The cheapest form is a spender's in-approve degree, which ranked new approval pairs better than EndorseRank in both windows, in contrasts not fixed in advance (Tables 4.11 and 4.16), and needs no solver at all. EndorseRank is its propagated version. A refresh costs one solve over the allowance graph (Table 4.1); the time comes from the size of that graph and would grow with any denser one. Its allowance alignment (Table 4.3) and its holdout results (Table 4.10) make it usable where the operator wants a ranking that also reflects who the endorsing owners are. For most spenders, however, it reflects them only through how many spenders each owner approves and how recent and large each approval is (Section 5.3.2), weights not shown to improve any label over the count,

and under uniform teleportation the propagated score is open to cheap inflation by closed farms (Section 5.6). The variant with activity restarts did rank new approval pairs above the count, in comparisons made after the labels; an operator who wants a propagated allowance score has more reason to test that variant than the uniform one. In return for authorization semantics, operators accept a score that carries little transfer information.

5.4.3 Evaluation design and combined use

Whether to combine the two signals is a question about deployment and about method, and the evidence answers the two differently. An operator can run an allowance score for authorization-aware monitoring and AWP for flow, at the cost of both solves. Folding the two edge types into one walk did not pay on these data. C-PR walks both layers, so its solve costs more than AWP's (Section 4.2), and although it ranks new approval pairs slightly better than EndorseRank in both windows, it trails AWP on new senders by a wider margin (Section 5.1.5), so an operator free to choose has reason to keep the two scores apart. For an operator tied to a single transfer walk weighted by dollar amounts, rule B, applied to the revised scores, shows that adding allowance edges bought authorization information that held on new data, at a loss on inflow larger than the margin fixed in the registration. Because the pipeline is open, the analysis can be replicated on any ERC-20 chain that has allowance logs and a daily price series for its main tokens (Appendix 8.1); integration with live risk oracles is left for future work (Chapter 6).

5.4.4 Industry adoption scenarios

The scenarios below say for whom each score is useful and for whom it is not.

- Wallet software and approval prompts. An owner usually decides whether to trust a spender contract when its wallet shows the approval for signature, and weaknesses in how wallets present approvals make approval frauds easier (Adamczyk, 2025). A prompt could show, beside the amount requested, how many distinct owners hold a positive allowance to the spender, or its EndorseRank. That tells the owner how widely the contract is authorized. It tells noth-

ing about what the contract will do with the allowance: a contract that collects approvals by phishing gains endorsement edges as an honest router does (Section 2.17), and a new contract scores low whatever its code. A low score is a reason to look closer, and a high score is no reason to sign.

- Protocol integrations. A protocol that routes user funds through external contracts, such as an aggregator choosing routers or a vault choosing strategies, has to decide which contracts to trust. The in-approve degree and EndorseRank rank how widely owners already authorize each candidate, and out of window they rank the contracts that go on to gain new approvals; AWP ranks those that go on to gain new senders (Table 4.10). Either is one input beside code review and audits, and neither says that a contract is safe to integrate.
- Security tooling. Tools that list a user's open allowances and help revoke them could sort spenders by standing, so that the least endorsed are reviewed first. The revocation counts and the drain heuristic rise with every score on the spender holdout and track the volume of approvals a contract holds (Section 4.6), so no score here is a drainer or phishing detector. No score is a security audit either: a contract can contain malicious functions whatever its standing among owners (Section 2.8).
- Lending and margin protocols. Lending markets want to put more of their capital to use without cutting collateral (Bartoletti et al., 2021), and the stress of 2020 showed the cost of updating risk parameters late (Gudgeon, Perez, et al., 2020). When the borrower or trader is itself a contract, the screening question is whether it will be force-liquidated. On that outcome the comparison, its registered rules applied to the revised scores, found no count-level lead of allowance over transfer, although every trader contract that had received an approval avoided liquidation in both windows, in samples of 70 and 88 (Section 5.1.6). A screen built on that pattern would need a registered test on a larger sample. The scores are not tools for predicting default.
- Registries and incentive programs. Airdrop farming shows how readily on-chain rewards

attract many-address strategies (Luo et al., 2025), and a program that lists or rewards contracts by their endorsements would face the same pressure. EndorseRank as defined here offers little protection below the top of its ranking, because a closed farm collects about three times its population share and can concentrate it on one target (Section 5.6). Programs can require diverse incoming endorsements from owners with independent histories, discount mass that circulates inside closed clusters, or send teleportation and dangling mass only to a vetted seed set; none of these stops an attacker who buys approvals from honest owners.

- Liquidity providers and market makers. To control inventory and spreads, these participants need to find the contracts that act as transfer hubs and see how flow is structured. Transfer counts and AWP (T. Do et al., 2023) are the suitable tools for that goal, and their benefit is lower screening cost, not higher profit and loss.
- Analytics and compliance providers. Exchanges, funds and compliance teams need contract rankings that outsiders can recompute. EndorseRank and AWP can be published for defined windows and parameters and audited, unlike proprietary machine learning models (Packin & Lev-Aretz, 2024), provided that their Sybil exposure is disclosed and they are not presented as safety ratings or profit predictions.

5.4.5 Standardized scores as shared infrastructure

A score shared by several parties is useful only if each of them can recompute it and see how it might be manipulated. EndorseRank and AWP can be recomputed from public logs, a versioned token list and a public daily price series with the open pipeline (Appendix 8.1), and Sections 5.6 and 5.7 document how they can be manipulated.

5.5 Societal and practical impact

In the DeFi market, vetting a counterparty is expensive, and the capacity to do it is unevenly spread. Wealthy actors may be able to pay for code audits and proprietary risk tools, while retail participants

work from anecdotal evidence or skip vetting altogether. A score built from the permissions owners have already granted, and recomputable from public logs and public prices at a cost set by the size of the allowance graph (Section 4.2), lowers the barrier to vetting the contracts that ask for those permissions. An approval that a user granted earlier is one route by which tokens are drained (Section 2.17). A score centered on the spender contract therefore shows where that exposure lies, which transfer volume does not, although it does not say which spender will misuse an allowance (Section 4.6). The same score cuts the other way for developers: a new contract, however sound its code, starts with no endorsements, so a score used as a gate would favor contracts that are already widely approved.

For protocol developers, the results argue for replacing a single "reputation" construct with specific scores, each fitted to the decision it informs, which reduces the risk of using a flow score to decide authorization.

Where the decentralized credit scoring systems discussed by Packin and Lev-Aretz (2024) are opaque, a score built from public events and public prices, with a versioned token list and versioned parameters, supplies the explanatory information that regulators suggest should be required when DeFi protocols are analyzed (Zetzsche et al., 2020). It also brings a duty, described in Section 5.7, to ensure that an ordinal, graph-based statistic is not mistaken for a security audit, a credit history, or an endorsement by whoever publishes it.

Any reputation metric that affects how value is allocated is open to exploitation; the next section models what it costs to manipulate EndorseRank.

5.6 Sybil attack cost model on allowance graphs

A Sybil attack exploits cheap pseudonyms to create a false impression of trustworthiness (Douceur, 2002). On a transfer graph, a malicious actor can inflate in-degree and in-flow artificially by moving assets among accounts it controls, paying for this in transaction fees and possible inventory loss (Cong et al., 2023). Allowance graphs expose a different attack surface, since each edge records

an authorization, not an actual transfer, and for a contract the attack is direct: whoever deploys it can create owner addresses and have each of them approve it. In this section we formulate a simple cost model for inflating EndorseRank with fake allowances. It builds on the optimal link farms of Gyöngyi and Garcia-Molina (2005) and on the result of Cheng and Friedman (2005) that symmetric functions such as PageRank are not Sybil-proof (Section 2.6), and adds the cost of an allowance edge and the size of the gain on this study's graph.

5.6.1 Attack setup

Let the attacker control a set of addresses $\mathcal{S}$, and write $\mathcal{H} = V \setminus \mathcal{S}$ for the honest addresses in the endorsement graph $G_{\text{approve}} = (V, E, w)$. An edge (u, v) of weight $w_{uv} > 0$ records that u has authorized v to spend its tokens. The weight is that of Equation (3.7): for each of the five tokens, the time weight of the latest in-window approval times the bounded value weight of its US-dollar amount, summed over the tokens. With damping parameter $d \in (0, 1)$, EndorseRank finds the PageRank vector $\mathbf{r}$ on G_{approve} as the solution of (Page et al., 1999):

$$\mathbf{r} = d\, P^{\top}\mathbf{r} + \frac{1-d}{|V|}\mathbf{1}, \tag{5.2}$$

where P is the transition matrix of Equation (2.1). A node with out-edges splits its row in proportion to its outgoing allowance weights. A dangling node, one without out-edges, has every entry of its row equal to $1/|V|$, so its mass is spread uniformly over all nodes (Langville & Meyer, 2006). The fixed point is the same as that of the iteration in Equation (3.1), which the implementation runs and which feeds dangling mass back through the uniform teleportation vector. The attacker's objective is to maximize the mean EndorseRank of a target set $\mathcal{G} \subseteq \mathcal{S}$,

$$J(\mathcal{G}) = \frac{1}{|\mathcal{G}|}\sum_{i \in \mathcal{G}} r_i, \tag{5.3}$$

subject to a budget constraint on the cost of the attack, which is defined next.

5.6.2 Cost components

The attacker creates a set of edges $E_{\mathcal{S}}$, all with sources in $\mathcal{S}$, and may also buy a set $E_{\mathcal{H}\to\mathcal{S}}$ of edges from honest owners into $\mathcal{S}$. Costs are counted over q observation windows, the number of score refreshes for which the targets are meant to hold their rank. Gas is the only cost that every edge incurs. The other components arise only for particular edges or strategies.

1. Gas cost. Each `approve` or `permit` call consumes gas; write c_{gas} for the expected fee of one allowance update. EndorseRank reads only the `Approval` events inside its observation window, so an edge enters a window's graph only if its approval is emitted in that window. To keep $E_{\mathcal{S}}$ in place for q windows, the attacker has to emit every approval again in each window, which also keeps its time weight near the maximum $\sigma(0) = 0.86$, and the gas is paid q times:

$$C_{\text{gas}} = q\,|E_{\mathcal{S}}|\,c_{\text{gas}}.$$

2. Exposure cost. An approval locks no tokens, and the owner need not hold any balance of the token when it approves. The amount sets the weight but costs nothing: the value weight V grows with the dollar amount, and an unlimited allowance takes its full value, so an owner that holds none of the token gives its edge the largest weight without exposing anything. After row normalization, moreover, a single out-edge gives $P_{uv} = 1$ whatever its weight. The token list does not raise this cost, since approving one of the five tokens costs an attacker no more than approving any other token; what the list removes are events that a token contract can fabricate in the name of owners that never approved anything (Section 3.5.1), not approvals that a farm emits itself. An edge inside $\mathcal{S}$ therefore exposes nothing and costs nothing beyond gas. Exposure arises only when an attacker address that holds tokens approves a spender $v \notin \mathcal{S}$, for instance to look like an ordinary user of known contracts, since v can then move those tokens. Such an edge also lets score mass leave $\mathcal{S}$. With ℓ_{uv} the expected cost to the

attacker of one such exposure over one window,

$$C_{\text{exp}} = q \sum_{(u,v)\in E_{\mathcal{S}},\ v\notin\mathcal{S}} \ell_{uv},$$

which is zero for a closed farm.

3. Revocation and detection cost. Legitimate users revoke old allowances from time to time, and a farm whose approvals reappear unchanged in every window might be identifiable. Let c_{rev} be the expected cost per window of detection and of the revocations the attacker issues to avoid it. Over q windows,

$$C_{\text{rev}} = q\, c_{\text{rev}}.$$

4. Bought honest edges. The attacker can also pay honest owners to approve its addresses, for instance through an app that asks users for permission. A bought edge (h, a), with $h \in \mathcal{H}$ and $a \in \mathcal{S}$, falls under the same window rule and costs, per window,

$$c_{\text{in}}(h, a) = c_{\text{gas}} + p_h + \rho_h, \tag{5.4}$$

where p_h is the payment to the honest owner and ρ_h is a risk premium the owner asks against the allowance being spent by the attacker.

Over the q windows, the total cost of the attack is

$$C = C_{\text{gas}} + C_{\text{exp}} + C_{\text{rev}} + q \sum_{(h,a)\in E_{\mathcal{H}\to\mathcal{S}}} c_{\text{in}}(h, a). \tag{5.5}$$

For a closed farm, $C_{\text{exp}} = 0$ and $E_{\mathcal{H}\to\mathcal{S}}$ is empty, so $C = q\,(|E_{\mathcal{S}}|\, c_{\text{gas}} + c_{\text{rev}})$.

5.6.3 Influence and a cost-effectiveness bound

By the usual sensitivity results for PageRank, an edge moves **r** in proportion to the stationary mass at its source node and to its normalized outgoing weight (Langville & Meyer, 2006). Consider first

a closed farm: every node of $\mathcal{S}$ has at least one out-edge, all of these edges end in $\mathcal{S}$, and no honest owner approves an address in $\mathcal{S}$. Let D be the set of dangling nodes, all of which then lie in $\mathcal{H}$, and let $r_D = \sum_{j \in D} r_j$ be their stationary mass. By Equation (5.2), a node $i \in \mathcal{S}$ receives $d \sum_{j \in \mathcal{S}} r_j P_{ji}$ from the farm, $(1-d)/|V|$ from teleportation and $d\, r_D/|V|$ from the dangling rows. The rows of the farm's nodes sum to one inside $\mathcal{S}$, so summing over $\mathcal{S}$ gives $\sum_{i \in \mathcal{S}} r_i = d \sum_{i \in \mathcal{S}} r_i + |\mathcal{S}|\,(1 - d + d\, r_D)/|V|$, that is,

$$\sum_{i \in \mathcal{S}} r_i = \frac{|\mathcal{S}|}{|V|} \cdot \frac{1 - d + d\, r_D}{1 - d}. \tag{5.6}$$

The first factor is the farm's population share, and the second says how many times that share the farm collects. The second factor equals one only when $r_D = 0$, that is, on a graph without dangling nodes; otherwise the dangling rows pass part of the honest mass to the farm. Since the dangling nodes lie outside $\mathcal{S}$, $r_D \le 1 - \sum_{i \in \mathcal{S}} r_i$, and substituting this into Equation (5.6) gives a bound that holds on any graph:

$$\sum_{i \in \mathcal{S}} r_i \le \frac{|\mathcal{S}|}{(1-d)\,|V| + d\,|\mathcal{S}|}. \tag{5.7}$$

For a farm that is small against $|V|$, the bound is close to $1/(1-d)$ times the population share. On the EndorseRank graph at T_{obs}, with $6{,}219{,}063$ nodes and $11{,}210{,}114$ edges, the pipeline's own PageRank code gives $r_D = 0.355$, 0.382 and 0.404 at $d = 0.75$, 0.85 and 0.95, so a closed farm on this graph collects 2.06, 3.17 and 8.68 times its population share. A direct check with the same code agrees: a ring of 20 addresses, each approving the next, added to the graph collects 3.17 times its population share at $d = 0.85$.

Equation (5.7) limits the farm's total mass, not the score of one target, and the attacker can concentrate that mass. Take a star in which f farm addresses each approve the target t and t approves each of them back. Every farm address a has t as its only out-neighbor, so $P_{at} = 1$, and the row of t sums to one over the farm. Write $c = (1 - d + d\, r_D)/|V|$ for the mass that each node receives from teleportation and the dangling rows. The balance equations $r_a = c + d\, r_t P_{ta}$ and $r_t = c + d \sum_a r_a$

then give

$$r_t = \frac{c\,(1 + df)}{1 - d^2}.$$

At $d = 0.85$, $c = 7.64 \times 10^{-8}$, the common score of every address that receives no approval (the constant c of Section 5.3.2). With $f = 10$ (eleven addresses and twenty approvals per window), the target reaches $r_t = 2.62 \times 10^{-6}$ on the graph with the star added, the value of the closed form. That is above the median EndorseRank of the cohort contracts (1.88×10^{-7}) and about half of their 90th percentile (5.28×10^{-6}), but about one twelve-thousandth of the highest cohort score (0.0318). The score grows linearly in f: each further farm address adds $d\,c/(1 - d^2)$ to r_t and costs two approvals per window. Solved for f, the formula needs no farm at all for the median, since two addresses that approve each other already score $c/(1 - d)$, above it; 22 farm addresses pass the 90th percentile, 2,074 the 99th (4.86×10^{-4}), and about 135,895 reach the highest cohort score, before counting that a farm of that size would itself enlarge $|V|$ and lower c. Passing the median takes one pair of addresses and passing the 90th percentile a few dozen approvals per window, whereas reaching the top would take more than a quarter of a million approvals per window. The reason is concentration: the highest-ranked cohort contract holds about 3% of the graph's stationary mass, more than four hundred thousand times c.

Honest in-edges are therefore not needed to reach the upper part of the ranking; without them, reaching its top takes a farm of the size just given, and bought honest edges add to the gain. When the farm has no out-edges to honest addresses, an edge (h, a) bought from an honest owner h adds, to first order,

$$\Delta \sum_{i \in \mathcal{S}} r_i \approx \frac{d\,r_h\,P_{ha}}{1 - d}$$

to the farm's mass, where P_{ha} is the share of h's row that the new edge takes. The gain depends on this share, and through V on the dollar amount approved. A fresh approval of z dollars bought from h takes $\sigma(0)V(z)/(o(h) + \sigma(0)V(z))$ of h's row, and an approval of a few hundred dollars or more, or an unlimited one, has $V(z)$ close to one and takes $\sigma(0)/(o(h) + \sigma(0))$. For the median owner in the graph that share is 0.957, since the latest approvals of most owners carry small weights, through

their age, their amount or both. The edge an attacker would buy is therefore a large or unlimited approval from an owner whose own approvals are old, small or few, preferably one with a high r_h. Such an approval puts the owner's balance of the token, up to the approved amount, at the attacker's disposal, so the risk premium ρ_h in Equation (5.4) grows with what the owner holds; it can still be close to zero for an owner that holds none of the token. The allowance layer of C-PR weighs each allowance by its dollar amount up to the cap U, so there a bought approval takes a large share of an owner's row only if it is large against the owner's other allowances, and an approval that reaches the cap authorizes the spending of at least U dollars of the owner's balance (Section 3.6.3). The attack's cost-effectiveness is

$$\eta = \frac{\Delta J(\mathcal{G})}{C}, \tag{5.8}$$

with $\Delta J(\mathcal{G})$ the gain in the objective of Equation (5.3) that the attack edges produce, held over the q windows that C in Equation (5.5) pays for.

Equations (5.6)–(5.8) show that EndorseRank, as defined here, is open to cheap inflation by closed farms. Under uniform teleportation and uniform redistribution of dangling mass, a closed cluster is not held to its population share. On this graph the inflation is cheap in the middle and upper part of the ranking and expensive at its top. The control that does bound closed clusters changes the walk itself: if teleportation and dangling redistribution go only to a vetted seed set, as in TrustRank (Gyöngyi et al., 2004), a cluster outside that set gains mass only through edges from honest owners, which the attacker has to buy at the price of Equation (5.4). AWP's restart rule gives the transfer side no protection either: a farm whose addresses send small amounts to one another raises their activeness and so draws restart mass to itself. The variant of EndorseRank with activity restarts, which ranked new approval pairs above the count (Section 5.1.4), inherits the same exposure, since every farm address that approves raises its own approving activity. The Sybil-adjusted stability proxies (Section 3.9.3) flag repeated transfers from a few addresses but not a farm of fresh addresses, so they cannot replace an adversarial assessment, which we leave to future work.

5.7 Regulatory and ethical considerations

On-chain reputation metrics raise questions of transparency as well as equity. Legal work has shown that decentralized finance markets remain bound by disclosure and consumer protection rules even though settlement happens on-chain (Zetzsche et al., 2020). Packin and Lev-Aretz (2024) add a warning of their own: decentralized credit scoring may perpetuate black-box harms when models cannot be interpreted or contested, or when they are linked to identity databases without due process. Both EndorseRank and AWP are computed deterministically from on-chain events and a public daily price series, with open-source code and a versioned token list, so anyone can recompute them (Appendix 8.1). That openness partly answers the black-box criticism. It leaves the distributional concern untouched, because contracts with few approvals, every new contract among them, score lower whatever their code.

Ethically, these scores should never be presented as security ratings, as audits or as guarantees that a contract is safe to approve, nor as creditworthiness measures in the traditional sense. They are rank-based indicators of a contract's position in the defined graph. If a wallet, protocol or registry acts on score thresholds, for instance by warning users or by declining an integration, it should document the graph definitions behind the scores, the window and the limitations, above all the exposure to Sybil attacks and the fact that a phishing contract collects approvals as an honest one does. A score that wrongly marks a contract as little endorsed can harm its developers, and one that wrongly marks it as widely endorsed can harm the owners who trust it. A regulatory framework centered on interpretability would prefer this kind of ranking to proprietary machine learning models like those reviewed by Kotb (2023), as long as providers make no exaggerated claims about what it predicts (Packin & Lev-Aretz, 2024).

Privacy on an open blockchain is limited by the ledger's very nature (Narayanan et al., 2016), and computing the scores draws on nothing beyond public data. The scored objects are contracts. The externally owned accounts that approve, pay or are paid by them enter the graphs only as owners, senders and recipients; none of them is scored, ranked or reported, and no attempt is made to link

any address to a real-world person (Section 3.16). Owners' approvals still enter the computation, and the score of a contract that one person controls could say something about that person, so the scores should not be treated as permanent, fingerprint-like identifiers.

5.8 Threats to validity

The findings reported here should be read against the limitations below, which are grouped under construct, internal, external and adversarial validity.

5.8.1 Construct validity

Seven points concern what the scores and their checks measure.

- These scores are not credit ratings or security ratings: no default labels from uncollateralized lending and no exploit or audit labels enter the analysis.
- Same-window checks rest on local allowance and transfer statistics of the same contracts.
- Holdout construct. A future new approval is still an endorsement event and a future new sender a flow event: both lie outside the score window, yet neither is an independent measure of trust or safety. Being counts of arrivals, they also favor any score that tracks the current count of counterparties (Section 5.1.4).
- Unit of analysis. EndorseRank scores the receiving side of an approval, and every scored address here is a contract: the matched cohort holds the contracts that received approvals of one of the five tokens from at least three owners, the spender cohorts the contracts that held a positive allowance of one of them at the freeze, and the trader cohorts the GMX V2 contract accounts with at least three closes (Section 4.1.1). The same-window results say nothing about contracts that fewer owners approve or that owners approve only on other tokens, and no result says how the scores would order externally owned accounts. Claims about trading outcomes rest on the two small trader cohorts (Section 5.1.6).
- Account type. An address counts as a contract when it holds code at one block read on 9 Oc-

tober 2026, after both label windows (Section 4.1.1). A contract could once remove its own code with SELFDESTRUCT (Wood, 2025); since EIP-6780 (Ballet et al., 2023), only a contract created and destroyed in one transaction loses its code. A contract that destroyed itself before that change has no code at the block read, counts as an externally owned account and is left out of every cohort, so if such contracts differ from the ones that survive, for instance because some of them were scams, the cohorts lean toward contracts that still exist. In the other direction, an address that received approvals before code was deployed at it counts as a contract for the whole window, and an account with an EIP-7702 delegation (Buterin et al., 2024) counts as an externally owned account although it runs code.

- Own-construct agreement is partly mechanical. EndorseRank smooths spender in-approve degree over the whole graph, and AWP does the same for incoming transfers weighted by their value, so each score is expected by construction to agree with its own family. The contrasts of Table 8.7 were computed for this reason; with few ties in either score they compare coefficients with similar attainable maxima, but they remain checks of the instruments.
- Value weights. EndorseRank and AWP pass each US-dollar amount through $V(z)$ with the slope $b = \ln 3/m$, where m is the median dollar amount of the transfers or of the finite latest allowances up to t_1, rounded to two significant digits (Section 3.5.1). An amount at the median weighs one half and one at twice the median 0.8, so the weights record the value of the approvals and transfers an edge carries as well as their number and age (Table 8.8). The slopes are choices of the revision (Section 5.8.2). In the registered analysis, with $b = 1$ on raw base units, V was within 10^{-4} of one for almost every amount, so the weights there recorded number and age and almost nothing about value (Appendix 8.6.10). The layers of C-PR add dollar amounts without the bound of V, capping each allowance at $U = 20{,}000$ dollars, so an unlimited allowance weighs as much as a large finite one and an owner that holds both kinds splits its row among them in proportion to their capped amounts (Section 3.6.3). On both spender holdouts EndorseRank has a slightly higher coefficient on both labels than the allowance layer walked alone (Tables 4.10, 4.14 and 4.16). No paired contrast between the two

was computed, so the gap suggests, without showing, that how approvals are weighted matters for this construct. The comparison mixes the value weight with the time weight, which only EndorseRank applies to approvals, and neither weighting is shown to be the right one. Such weighting choices propagate into $\mathbf{r}$ and into the rank correlations (Langville & Meyer, 2006).

5.8.2 Internal validity

Six points concern whether the comparisons isolate what they are read as showing.

- Shared observation window. Every method and proxy is computed over 1 October 2023 to 30 June 2026, with the time decay anchored at the end. Market regimes and protocol changes peculiar to those months may push alignment up or down. The checks on damping and on the value slopes measure sensitivity to these parameters, but none of them varies the calendar window, and the two holdout freezes, three months apart, vary it only a little.
- Cohort selection. The matched cohort is every contract that received non-zero approvals of one of the five tokens from at least three owners in the window. It is a subset of the plan's cohort, taken from that cohort's approval logs and code check, and was fixed in the revision before any revised score was computed; results may not carry over to contracts with fewer approvals or with approvals of other tokens only. Not every cohort contract is in both graphs (Section 4.1.1). Keeping the missing contracts as isolated nodes instead of scoring them zero changes no coefficient (Table 4.5), and AWP's zero for spenders without a transfer edge does not explain its holdout results (Table 4.12). The sample-definition stages show how strongly the coefficients react when the evaluation set changes (Table 4.8), which is why they are not read as robustness evidence.
- Baseline fidelity. AWP follows the published method in its edge weights and its restarts (Section 3.6.1). The paper gives no parameter values, so k, t_0 and b are this study's choices, b set in the revision (see value weights above). The paper scored Ethereum transactions; here AWP scores transfers of five ERC-20 tokens on Arbitrum One, valued in US dollars, within the ego network of the cohort. EndorseRank takes AWP's weights, with its own slope for allowances,

and restarts uniformly, as fixed in the analysis plan, and the variant with AWP's restarts is reported beside it (Table 4.9).

- Order of analysis. The cohorts, scores, labels, statistics and rule A were fixed in the analysis plan of 9 October 2026, before any log was extracted. Rule B with its margin, its further results F4–F6, the two sensitivity analyses and the comparison on trader outcomes were registered at 16:49 UTC on the same day, after the W0 results were known and before the July–September logs were extracted (Section 3.7.2). These dates order the registered analysis; on the revised scores every rule was applied after the registered results on the same labels were known (next point). Every other comparison was computed after the relevant labels were known: in W0, the degree contrasts of the allowance layer, the S-PR contrasts, the comparisons of C-PR with its transfer layer on new approval pairs and with AWP, and the restriction to spenders that had received a transfer; in both windows, on the spender labels, the degree contrasts of EndorseRank and AWP, the comparisons of C-PR with EndorseRank, the comparisons of EndorseRank with AWP and every contrast of the activity-restart variant. They are reported with intervals, and none of them decides a question.
- Revision of the amounts. The amounts were revised on 10 October 2026, after the registered results of the same window, W0 and W1 were known (Section 3.5.1), so the revised analysis is post hoc: its choices, the five tokens, the price source, the slopes b and the cap U, were made with those results in view, and a revision made after the results can favor one outcome over another. Three facts limit that risk without removing it. The revision was committed to the public repository before any revised score was computed; it derives the slopes and the cap from quantiles of the events up to t_1; and the registered analysis reached the same verdicts under rules A, B and R1–R3 (Appendix 8.6.10, Table 8.13). The slopes and the cap remain choices of the revision: dividing or multiplying the slopes by ten changes the same-window results little (Table 4.7), and no other cap was tried.
- Timing environment. All runtimes are wall-clock times from one machine (Section 4.2), and their standard deviations cover only repeated solves of one configuration. Variation between

machines is not measured, so the comparison is stated as a time ratio beside the edge ratio, with $|V|$ and $|E|$ of each graph.

5.8.3 External validity

Five points limit how far the findings carry beyond this sample.

- Single chain and window. The evidence comes from Arbitrum One alone, over 33 months of scored activity and two three-month label windows. Extending the findings to other rollups, Layer-1 chains or lending-centered venues would need the same evaluation on those venues, since DeFi venues differ in protocol design (Werner et al., 2022) and in market structure (Schär, 2021).
- Tokens and prices. The graphs hold the approvals and transfers of five tokens only, WETH, USDC, USDT0, ARB and WBTC (Table 8.4). Every other token is left out, legitimate ones included, so a contract that owners approve only on such tokens is outside the cohort, and native ether, which moves without `Transfer` logs, enters the graphs only as WETH. Every price comes from one public source, the daily series of the DefiLlama API (DefiLlama, n.d.). On the dates sampled for the check it agreed closely with the Chainlink feeds read on the chain (Chainlink, n.d.), but an error in the series on another date would pass into every amount of that day (Appendix 8.4.2).
- Ego network. Each graph keeps only the approvals and transfers of the five tokens in which a cohort contract takes part (Section 4.1.1). An owner's approvals of contracts outside the cohort, and transfers between two addresses outside it, are missing, so owners' out-strengths, senders' activeness and the paths the walks can take are those of the ego network and not of the chain. Paths of more than one step exist only through cohort contracts that both approve and are approved (Section 5.3.2), and the findings about propagation do not carry over to graphs with longer endorsement paths.
- Scaling stages. The scaling stages are subsets of the cohort and its ego network, so they show

how the solve time follows $|E|$ up to the size of that graph and not beyond (Section 4.2). Same-window alignment remains anchored to the matched cohort and the holdout to spenders.

- GMX contract traders. Few contract accounts close positions on GMX V2 in the label windows, and the trader cohorts hold 70 contracts in W1 and 88 in W0, so every trader result has wide intervals. The comparison on outcomes built from neither edge type says nothing about the externally owned accounts that also trade on GMX V2, which this study does not score.

5.8.4 Adversarial validity

Three points concern deliberate manipulation and deception.

- Sybil and wash trading. Rankings may be manipulated through adversarial allowance or transfer patterns (Douceur, 2002); the Sybil-adjusted proxies only partly address this, and the cost model of Section 5.6 is analytical, with one ring and one star added to the observed graph, not an empirical red-team evaluation.
- Phishing. A contract that obtains approvals by deception gains endorsement edges as an honest one does (Section 2.17), and nothing in the scores separates the two.
- Strategic approval behavior. Once a score is live, the developers of a contract may arrange approvals to it to farm rank. An observational τ estimated before deployment may overstate how well the ranking resists such behavior.

5.9 Limitations

On this sample, EndorseRank's strengths are agreement with the allowance checks, a ranking distinct from AWP's and, on spender contracts, a ranking of new approval pairs in both windows that the scores built on transfers alone do not match. Its lower cost follows from a graph with fewer than half of the transfer graph's edges and is a property of that graph, not of the method. Its weaknesses are a ranking that carries little transfer information, a holdout coefficient below that of the raw in-approve degree it smooths in both windows, exposure to closed farms below the top of its ranking,

and nothing to say about contracts that few owners approve or that owners approve only on other tokens. These strengths and weaknesses are those of the revised analysis, whose tokens, prices, slopes and cap were chosen after the registered results were known (Section 5.8.2); the registered analysis reached the same verdicts on every rule.

The coupled operator was tried at three fixed interior values of λ, with uniform teleportation and a per-node convex mixture at the transition level, on two layers weighted by dollar amounts, with each allowance capped at U. Layers weighted as in EndorseRank and AWP, node-dependent or learned weights, mixtures of the stationary vectors instead of the transition rows, and other Markov chains went untested, so both answers to RQ5 apply only to that family of methods on this dataset. Deployment remains bound by the considerations of Section 5.7.

EndorseRank draws on no off-chain evidence about who controls an address, and neither does any other score here; the only off-chain input is the daily price series that values the amounts. Attestation tooling such as Human Passport uses identity checks, biometrics and vouching to decide whether an address belongs to a distinct person (Human Passport, n.d.). That question has no direct counterpart for a contract, and EndorseRank gives no guarantee that the owners who approve a contract are controlled by different actors, so it should be read as an ordering of contracts by the approvals they hold and not as a Sybil defense. Counting only the approvals of owners that pass such a gate before computing EndorseRank is a plausible combination, but it is untested here: the gate's own weaknesses, such as credential farming (Bordeianu & Popescu, 2026), would carry over to it, and evaluating the combination needs labeled Sybil data that this study did not collect.

These limitations narrow what can be concluded. For contracts that at least three owners approved on WETH, USDC, USDT0, ARB or WBTC on Arbitrum One in these months, EndorseRank is an allowance-specific reputation score for spender contracts. It is not a security audit or a credit score, it does not resist Sybil farms, and it does not replace protocol-level collateralization (Gudgeon, Perez, et al., 2020). The allowance edge it rests on is worth adding to on-chain reputation, and the approval count is the cheapest way to use it. Chapter 6 draws the conclusions and names the work that would extend these bounds.

6 Conclusion and Future Work

Chapter 5 interpreted the findings and judged whether the study met its objectives. The tables of Chapter 4 form the record of the revised analysis, and Table 8.13 in Appendix 8.6.10 keeps the decision contrasts and verdicts of the registered analysis.

6.1 Answers to the research questions

The thesis proposed EndorseRank, PageRank with the edge weights of Adaptive Weighted PageRank (AWP) on a latest-allowance endorsement graph, and evaluated it beside AWP itself (T. Do et al., 2023), a coupled walk over both edge types and the raw degrees these walks smooth. Every scored address was a smart contract on Arbitrum One. Both graphs kept the approvals and transfers of WETH, USDC, USDT0, ARB and WBTC, with every amount valued in US dollars. In the same window the scores ranked the matched cohort, the contracts approved on these tokens by at least three distinct owners between October 2023 and June 2026 (Section 4.1.1). Out of window they ranked the contract spenders that held a positive allowance of these tokens at a freeze date, with labels counted from April to June 2026 (W0) and from July to September 2026 (W1); the rule for W1 was registered before its logs were extracted (Sections 3.7 and 3.7.2).

The amounts were revised on 10 October 2026, after the results of the registered analysis were known (Section 3.5.1). The analysis plan had fixed amounts in raw token base units, every ERC-20 token and a value weight that was close to one for almost any amount. The revision values every amount in US dollars, keeps the five tokens with the most rows among those that pass a price-feed rule and have a daily price, sets the slope of the value weight so that the median amount weighs one half and caps the allowances of the coupled walk; it was committed before any revised score was computed. The answers below come from the revised analysis. Rules A and B and the

comparison on traders keep the thresholds and the wording fixed for the registered analysis, but they were applied to the revised scores after the registered results were known, so their verdicts here are a re-analysis and not registered tests. The registered analysis reached the same verdicts (Appendix 8.6.10; Table 8.13).

RQ1 asked whether EndorseRank and AWP order the contracts of one cohort differently. They do, and H1 is supported, largely by construction (Section 4.4; Claim C3). Every cohort contract received approvals of one of the five tokens from at least three owners, so EndorseRank leaves few pairs of contracts tied, and keeping the contracts that a graph misses as isolated nodes leaves the coefficient between the two scores unchanged (Table 4.5). The divergence therefore comes from the two orderings themselves. Whether they reflect different kinds of reputation is a question for the checks of RQ2 and the holdouts of RQ4, not for the divergence alone.

RQ2 asked how closely each score agrees with allowance and transfer checks in the same window. Each score agrees most with the proxies of the edge type it walks on, EndorseRank with the allowance family and AWP with the transfer family, and the paired contrasts between families and between the two scores, none of them fixed in advance, have intervals that exclude zero; the C-PR mixtures lie between their two layers on every family (Tables 4.3 and 8.7; Claim C2). With amounts in US dollars, AWP follows the summed value of a contract's inbound transfers almost as closely as the number of its senders. In the narrow sense of agreement with the checks of a score's own edges, H2 is supported, largely by construction. The proxies are built from the same events and amounts as the graphs, so the agreement is an intended-construct check, and part of it is built in.

RQ3 asked what each score costs to compute on the same contracts. On the full graphs of the matched cohort, the time of a solver call follows the edge count of its graph. The allowance graph has less than half as many edges as the transfer graph, EndorseRank's call takes correspondingly less time than AWP's, and C-PR and S-PR, which also walk the transfer layer, take longer than AWP (Section 4.2; Claim C1). H3 is supported, largely by construction. The same solver runs on every graph, so the speed belongs to the sparser graph and not to the algorithm.

RQ4 asked whether scores frozen at a given date predict the new approval pairs and new transfer

senders that spender contracts gain afterwards, and whether any PageRank predicts them better than the raw degree it smooths. The first part holds in both windows: the interval of every score excludes zero on both labels, and in contrasts computed after the labels EndorseRank ranks later approval pairs above AWP, while AWP ranks later senders above EndorseRank (Tables 4.10, 4.14, 4.16 and 8.11). The second part has a different answer for each walk (Claim C4). AWP ranks later new senders above the raw transfer in-degree in both windows, so on the transfer side the walk, with its time and value weights and activity restarts, added information that the count lacks. EndorseRank with uniform restarts ranks later approval pairs below the raw in-approve degree in both windows (Tables 4.11 and 4.16). Leaving out the spenders that had received no transfer, which AWP scores zero, keeps both degree comparisons of W0 in place (Table 4.12). These degree contrasts were computed after their labels were known, but each points the same way in both windows, and the registered analysis gave the same directions (Table 8.13). Multi-hop propagation is present in these graphs: some spenders are approved by owners that are themselves approved, and for most of them EndorseRank departs from its depth-one closed form. It still did not lift EndorseRank above the count (Section 5.3.2). With AWP's activity restarts, EndorseRank ranks later approval pairs above the degree in both windows (Section 4.6; Tables 4.9 and 4.16), but that comparison lies outside every rule, in the registered analysis as in the revised one, and is a hypothesis for a registered test, not a finding.

RQ5 asked whether a coupled PageRank over both edge types improves on a walk over the transfers alone, and whether it ranks contracts better than a walk over either layer alone. On the revised scores both answers are no, as they were in the registered analysis (Claim C5). Rule A, written in the analysis plan before any log was extracted, asked C-PR to do no worse than the allowance walk on new approval pairs and better than the transfer walk on new transfer senders in W0. The first condition holds and the second fails, and the same-window part of the rule fails as well (Section 4.3.1; Tables 4.11 and 4.4). Rule B, registered before the logs of W1 were extracted, asked C-PR to beat the transfer walk on new approval pairs and to be no worse than it, within a fixed margin, on new transfer senders and on the traders' liquidation-free close rate. The first condition

holds; it also held in the registered analysis, where it tested whether the gain seen in W0 on new approval pairs would replicate. The loss on new senders exceeds the margin. On the trader label the two scores have equal point estimates, but the interval of their difference reaches further below zero than the margin allows, so that condition fails narrowly; with AWP in place of the transfer walk, the first sensitivity analysis of the registration, the loss on new senders is larger still (Table 4.15). In contrasts computed after the labels, the seeded ablation S-PR ranks new senders almost as the transfer walk does and ranks new approval pairs below the allowance walk (Table 4.11). Allowance edges help when they carry the walk, not when they only decide where it restarts.

Beside the research questions, the registration of W1 set allowance-side against transfer-side scores on an outcome built from neither edge type, the liquidation-free close rate of GMX V2 contract traders (Section 4.6.3; Claim C6). On the revised scores, and in the wording fixed at registration, there is no count-level lead of allowance over transfer on later liquidation avoidance: the approval count led the transfer count in W0, but in W1, the primary window, the interval of the contrast includes zero. The PageRank versions are not resolved. In W1 the approval count ranks traders below the transfer count on the profitable share and on the non-loss share, so whatever signal it carries concerns forced-liquidation avoidance, not trading success (Table 8.12). Every trader that had received an approval of the five tokens avoided liquidation in both windows, and some of the other traders did not (Table 4.19); the transfer count, however, also ranks this label in W1 (Table 4.14). Contract accounts traded little on GMX V2 in 2026, so these samples are small and every trader interval is wide.

Claims C1–C6 (Section 4.7) state these findings formally.

The main research question asked what ERC-20 allowance edges add to PageRank as a reputation score for smart contracts on Arbitrum One, compared with the transfer walk and with the raw degrees that both walks smooth. Compared with the transfer walk, the allowance edges of the five tokens, valued in US dollars, add a second construct. The allowance walk orders contracts by the spending authority that owners delegate to them, each allowance weighted by its dollar value through a bounded weight; this ordering differs from the ordering by transfers, agrees with the

allowance checks and is computed on a sparser graph. Allowance-based scores also rank later approval pairs above the transfer walk: C-PR beat its transfer layer on that label in W1 under the first condition of rule B, and EndorseRank beat AWP in both windows in contrasts computed after the labels (Tables 4.15 and 8.11). Compared with the raw degrees, the allowance walk with uniform restarts did worse than the count it smooths. In contrasts computed after the labels, EndorseRank ranked later approval pairs below the count of approving owners in both windows, whereas AWP improved on its own count. Mixed into one walk with the transfers, allowance edges lift the ranking of later approval pairs above that of either layer walked alone, by a small margin over the allowance layer, but they lower the ranking of later senders below that of the transfer walk, so the mixture does not improve on the better single layer for both labels at once. On the one outcome built from neither edge type, the rules fixed at registration found no lead for the allowance side, on the revised scores as in the registered analysis.

6.2 Original contribution to knowledge

In the terms of the literature reviewed in Chapter 2, the five contributions of Section 1.9 establish the following.

Before this work, graph reputation on public ledgers ranked accounts by their transactions. H.-D. Do and Do (2023) applied PageRank and HodgeRank to Ethereum transaction graphs, AWP weighted transfer edges by value and age (T. Do et al., 2023), and RiskProp propagated risk through the transaction network (D. Lin et al., 2025). Blockchain reputation systems mostly took explicit ratings or transfer volume as their trust signal (Bellini et al., 2020), the attestation tooling deployed in DeFi scores the holder of an address from credentials (Human Passport, n.d.), and the risk of a contract has been judged from its code (Z. Lin et al., 2024). None of this work used the ERC-20 allowance, the on-chain record of a deliberate delegation of spending power (Vogelsteller & Buterin, 2015), as the edge of a reputation ranking, and none ranked contracts by the approvals they receive. Approvals had been used to cluster the addresses of one entity (Victor, 2020) and measured as a

security risk (Wang et al., 2022). Nor had on-chain reputation scores been evaluated on one cohort against the checks of each construct and against the raw degrees they smooth, with an interval for every coefficient, paired contrasts and temporal holdouts judged by rules committed before the data were extracted. This thesis supplies both, and its main results apply those rules to amounts revised after the registered results were known (Section 3.5.1). PageRank (Page et al., 1999) and AWP's weighting are taken from the literature, with the slope of the value weight set here from the median dollar amount; EndorseRank changes only the edge, and the coupled operator changes the walk.

The new knowledge has five parts. First, what an on-chain reputation score of a contract measures depends on the edges it walks, and this can be checked on one cohort with quantified uncertainty: the allowance walk and the transfer walk order the same contracts differently, and each agrees most with the checks of its own edges (Claims C2 and C3). The allowance graph is also the sparser of the two, so its walk costs less; that follows from its edge count and is not a property of the method (Claim C1). Second, out of window the two walks relate differently to the counts they smooth, in contrasts computed after the labels were known. The transfer walk, AWP, ranked later new senders above the transfer in-degree in both windows. The allowance walk, EndorseRank with uniform restarts, ranked later approval pairs below the in-approve degree in both windows (Claim C4), even though the allowance-based scores ranked those pairs above the transfer walk (Tables 4.15 and 8.11). Third, the closed form accounts for part of this result. For a spender whose owners receive no approvals, EndorseRank is a linear function of the owner-normalized in-strength, which divides each owner's endorsement over all the spenders it approves. For the other spenders the walk propagates over more than one step, so multi-hop propagation is present in these graphs; over all spenders EndorseRank still did not beat the count, and whether propagation adds anything where it occurs was not tested separately (Section 5.3.2). A propagated allowance score therefore has yet to earn its place against the count, and the variant with activity restarts is the candidate for that test. Fourth, a weighted combination of the two edge types does not improve on the better single layer for both spender labels at once: it gains a little over the allowance walk on later approval pairs and loses against the transfer walk on later senders. Under rule B it beats the transfer walk

on later approval pairs, but its loss on later senders exceeds the margin and it cannot show that it is no worse on the traders' label, so the rule is not met (Claim C5). Fifth, on an outcome that neither graph records, the comparison registered for W1 found no count-level lead of allowance over transfer and left the PageRank pair unresolved (Claim C6). On the few contract traders available, the comparison did not separate allowance-edge from transfer-edge information, and this is not evidence that allowance edges carry no such information. The fourth and fifth parts apply rules fixed at registration to the revised scores after the registered results were known, so their verdicts are a re-analysis; the registered analysis reached the same verdicts (Appendix 8.6.10).

The construct-validity framing (Cronbach & Meehl, 1955) that makes these claims testable is part of the contribution. So is the design that fixed the cohort, the scores and the decision rules before the logs they were judged on were extracted (Section 3.7). The registered analysis followed it; the revision of the amounts departed from it after the registered results were known, but was committed before any revised score was computed and is reported beside the registered verdicts (Appendix 8.6.10). The open pipeline publishes the summaries of both analyses, so others can check every table (Section 3.13), since reputation systems for pseudonymous users must remain auditable (Packin & Lev-Aretz, 2024).

The scope is deliberately narrow. DeFi lending rests on collateral (Schär, 2021), and the thesis does not show that on-chain reputation can replace it. It does not show that the results extend beyond contracts approved on WETH, USDC, USDT0, ARB or WBTC by at least three owners on Arbitrum One in the months studied, or to the approvals and transfers of other tokens, and it does not show that any score measures the safety of a contract, the creditworthiness of an account or trading skill (Section 2.8). Nor does it claim that EndorseRank resists Sybils. Under uniform teleportation a small closed farm of owner addresses makes a contract look ordinary or better at little cost, although reaching the top of the cohort would take a farm larger by orders of magnitude. Keeping only the five tokens does not raise that cost, since approving a token needs no balance of it (Section 5.6).

6.3 Future work

Each suggestion below answers a limitation discussed in Chapter 5. Any new score, weighting or label it leads to should be fixed and registered before the logs it is judged on are extracted, as the decision rules of the registered analysis were and the revised amounts were not, because registration keeps a confirmatory test apart from exploration (Nosek et al., 2018). It should also be checked out of window against the raw degrees.

- A registered test of the revised analysis. The US-dollar amounts, the five tokens, the value slopes and the allowance cap of the main analysis were fixed after the registered results were known (Section 3.5.1), so its verdicts are a re-analysis even where they agree with the registered ones (Table 8.13). A registration committed before the logs of a later window are extracted, such as October to December 2026, would fix these choices together with the cohort rule, rules A and B, the margin and the comparison on traders, and would test whether the verdicts of the revised analysis hold on logs extracted after the registration. The value-slope check moved the same-window family means little (Table 4.7), but it did not cover the holdout labels, so the slopes and the cap should be fixed in the registration and not tuned on the new window.
- A registered test of EndorseRank with activity restarts against the in-approve degree. With AWP's restarts, which weight each owner by how actively it approves, EndorseRank ranked later approval pairs above the raw degree in both windows, but the comparison was made after the labels were known (Section 4.6; Table 4.16). A test on a later window whose logs have not been extracted, which could form part of the registration above, would fix the contrast, its margin and the label in advance and keep the uniform-restart version as a second comparator. It should report the contrast separately for spenders whose owners receive no approvals, where the walk reduces to a weighted count of approving owners, and for the other spenders, where it propagates over more than one step (Section 5.3.2). Only the second group can show whether propagation, and not the restart weights alone, adds to the count.

- Seed-restricted teleportation against farms. Under uniform teleportation every address, a fresh one included, receives the same restart mass, so a closed farm of owner addresses collects more than its share of the population and a small farm lifts the contract it approves (Section 5.6). Restricting teleportation and the redistribution of dangling mass to a vetted seed set, as TrustRank does (Gyöngyi et al., 2004), would remove that inflow, and some asymmetric reputation functions based on flows from a trusted node resist Sybil attacks where PageRank does not (Cheng & Friedman, 2005). Such a variant should be evaluated under the attacks of the cost model and on the same holdout labels, with attention to which contracts the choice of seeds favours. The cost model is analytical and no attack was carried out on chain, so an empirical study with wash approvals and coordinated owner farms is needed to measure what an attack costs in practice.
- Combination with personhood credentials. EndorseRank uses no off-chain information, so it cannot tell whether the owners that approve a contract are distinct people. Proof of personhood does not apply to a contract, but it does apply to the externally owned accounts among its owners. Counting only the approvals of owners that pass a Passport-style check (Human Passport, n.d.) would make every farm address cost a credential as well as gas. The weaknesses of the gate, such as credential farming (Bordeianu & Popescu, 2026), would carry over, and personhood protocols trade Sybil resistance against privacy (Siddarth et al., 2020). Evaluating the combination needs labelled Sybil data, which this study did not collect, and a rule for owners that are themselves contracts.
- Labelled contract-risk outcomes. The out-of-window labels of this thesis count later approval pairs and senders, and the trader label covers few contracts; none of them says whether a contract is safe. The revocation and drain counts computed on the holdout rose with every score and with the raw degrees, so they track volume and not risk (Section 4.6). A test against labelled outcomes, such as contracts later tied to drainer phishing operations (He et al., 2025) or flagged for contract-related rug-pull risk (Z. Lin et al., 2024), with scores frozen before the label window and compared within strata of activity, would show whether the standing of

a contract among its owners says anything about its risk beyond its volume. Such labels are rare, so the window and the cohort would need to be larger than here.

- Other chains and later windows. The results hold for contracts approved on one of the five tokens by at least three owners on Arbitrum One in the months studied. Contracts on Arbitrum face security risks that differ from those on Ethereum mainnet (Tang & Shi, 2024), and DeFi protocols differ widely in design (Werner et al., 2022), so the same analysis is needed on other rollups, on Layer-1 networks and over later windows before any result is read as general. A chain or protocol on which more contract accounts trade would also give the comparison on outcomes built from neither edge type the sample it lacked here, since contract activity on GMX V2 fell after 2025.
- Coupling beyond a fixed transition-level mixture. Only fixed values of λ with uniform teleportation on layers weighted by US-dollar amounts, with each allowance capped, were tried (Section 3.6.3). Node-dependent or learned weights, per-layer damping, mixtures of the two stationary vectors and layers weighted as in EndorseRank and AWP remain untested, and ranking methods for multilayer networks offer designs for several of them (De Domenico et al., 2015). Any such design should keep the auditability that separates graph propagation from learned credit models, which give up interpretability to predict labelled defaults (Kotb, 2023).
- A wider token list under the same rule. The revised analysis keeps the five tokens with the most rows among those that pass the feed rule and have a daily price; they hold most of the rows, and the approvals and transfers of every other token are left out (Section 3.5.1; Table 8.4). Widening the list to every token that passes the same rule, fixed before any score is computed, would show whether the results depend on these five tokens, and it would bring back into the cohort contracts that are approved mainly on other tokens. Tokens without a qualifying feed would need another check of their prices before they could enter. Native ether moves without `Transfer` logs and enters the graphs only through WETH; adding it would need the values of transactions and internal calls, which this study did not extract.

6.4 Closing reflection

On a public ledger, an owner's decision to let a contract spend its tokens is recorded as plainly as a payment, and the two records describe different relations. A transfer records economic flow, and an allowance records delegated authority. PageRank will rank either graph, yet one ranking does not serve both constructs, and a walk over both graphs gained little over the better layer on one label and lost on the other. For a protocol or researcher that needs an auditable signal of which contracts owners entrust with their tokens, the allowance edge is the one to use on data like those studied here, the allowances of widely traded tokens valued in US dollars. On these data the count of approving owners ranked later approvals better than the walk with uniform restarts, so a propagated version has yet to show, in a registered test, that it improves on that count. For describing transfer hubs and anticipating their inflow, the transfer walk did improve on its count. None of these orderings says that a contract is safe, and a closed farm of owner addresses can raise a contract's standing at little cost (Section 5.6). Whether the authority that owners delegate to a contract tells anything about outcomes that neither graph records, such as a trader's liquidations or a contract's later misuse, remains open. The comparison registered for the few contract traders available found no lead in the registered analysis, nor when its rules were applied to the revised scores as a re-analysis, and settling the question will take more traders or labels of contract risk.

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8 Appendix

8.1 Evaluation Pipeline and Reproducibility

This appendix documents the evaluation pipeline behind Chapter 4: its stages, the extraction design, the scaling tables, the account-type classification, the log of every timed run, the default parameters, the proxies and a glossary. The pipeline computes two analyses from the same extracted logs: the registered analysis of the plan and the registration (Appendix 8.6.10), and the revised analysis of Section 3.5.1, which Chapter 4 reports. Paths are given relative to the folder `1-Dissertation-Works/ERC20-Allowance-PageRank-Wallet-Reputation/experiments` of the public repository https://github.com/abitocodes/phd_works.

8.1.1 Pipeline stages

The evaluation runs in seven stages. Stages 1 to 6 produce the registered analysis, and stage 7 the revised analysis. Stages 2 to 4 read the logs of the observation window, 1 October 2023 to 30 June 2026 (UTC, inclusive at both ends); stage 5 adds July to September 2026; stages 6 and 7 read events only from the project tables, stage 6 those of the observation window from stage 3 and stage 7 those of both windows from stages 2 to 5, and scan no public log; stage 1 reads no logs.

1. Plan and configuration: The analysis plan (`docs/analysis_plan.md`), the configuration (`config/contract_reputation.yaml`), the materialization query (`sql/01_materialize_logs.sql`) and the BigQuery runner (`scripts/run_bq.py`) were committed together in `43d8045`, before any log of the study was extracted. The plan fixes the cohorts, the scores, the proxies and labels, the statistics and rule A for the coupled operator. Stage 7 departs from it in the tokens, the amounts and with them the cohort, and in the robustness check, where the value-slope check replaces the top-token subgraphs.

2. Scan A and the cohort: `scripts/run_bq.py` dry-runs every query, checks the estimate against the budget in the configuration, runs the query and writes the billed bytes to `data/bq_ledger.json`. Scan A materializes the observation window (Appendix 8.1.2). `sql/02_spender_candidates.sql` lists the spenders that received a non-zero approval from at least three distinct owners in the window, `scripts/check_account_code.py` reads the account type of each (Appendix 8.1.4), and `scripts/build_cohort.py` keeps the contracts as the plan's cohort and loads their addresses into BigQuery. The revised cohort of stage 7 is a subset of it.
3. Graph tables, proxies and labels: `sql/03_ego_events.sql` extracts the ego network of the cohort, and `sql/04_anchor_pairs.sql` aggregates it into the pair tables of both graphs at T_{obs} and at t_1. `sql/05_proxies_same_window.sql` computes the transfer and Sybil-stability proxies and `sql/06_holdout_labels.sql` the labels of W0. `sql/07_gmx_accounts.sql` and `sql/09_gmx_contract_closes.sql` select the GMX V2 closes of contract accounts, which `scripts/decode_gmx_events.py` decodes with the event ABI kept in `data/0-gmx-abi-for-decoding-logs/`. `sql/08_export_ids.sql` replaces addresses by integer node ids, `scripts/download_graph_tables.sh` downloads the tables in parquet parts, and `sql/10_describe.sql` counts the events and allowance amounts of the plan's ego tables.
4. Same window and W0: `scripts/run_same_window.py` scores every method at T_{obs} and computes the same-window checks, the contrasts and the secondary part of rule A. `scripts/run_holdout.py` scores every method at t_1 and evaluates the spender and trader cohorts of W0 together with the primary part of rule A.
5. Registration and scan B: The registration of W1 (`docs/registration_w1.md`) was committed in `0d3a057` with the configuration status set to `registered`. `scripts/extract_w1.py` refuses to run unless both files are committed without local changes and the status is `registered`; it writes their SHA-256 hashes and the commit to `registered-w1/extraction_manifest.json` and then runs scan B, the ego extraction of July–September and the

GMX account query of that window. The W1 labels and closes follow from `sql/06` and `sql/09`, and `scripts/run_registered.py` evaluates rule B, its two sensitivity analyses and the comparison on outcomes built from neither edge type, in the order of the registration.

6. After the labels: `scripts/run_supplementary.py` computes the spender contrasts made after the labels were known, `scripts/run_robustness.py` the top-token subgraphs (with `sql/11_top_tokens.sql`) and the sample-definition stages, `scripts/sybil_model.py` the Sybil cost model, and `scripts/run_benchmark.py` the timed runs. `scripts/publish_results.py` copies the summaries to `data/3-published-results-for-thesis/` with a digest that names the source of every number and a list of checksums.
7. Revision: After the results of stages 4 to 6 were known, the amounts were revised (Section 3.5.1; `docs/revision_price_weighting.md`). `scripts/usd_inputs.py` applies the token rule to the Chainlink directory, reads the decimals of the listed tokens with `eth_call` at one block and builds the daily price table, and `scripts/chainlink_check.py` compares the prices with the Chainlink feeds; both write to `data/0-usd-token-list-and-prices/`. `scripts/run_usd_pipeline.py` loads the five tokens and the price table into BigQuery (`sql/19_usd_load.sql`) and runs `sql/12` to `sql/18` in order: the revised cohort and its ego tables, the value slopes and the cap from the events up to t_1, the pair tables at both anchors, the proxies, the labels of W0 and W1, the node ids and the counts described in Appendix 8.4. With `CONTRACT_REP_VARIANT=usd` the scripts of stages 4 to 6 then compute the revised analysis from the downloaded tables, in the order of the registered one (`scripts/run_usd_local.ps1`). `scripts/run_registered.py` applies rule B and the comparison on outcomes built from neither edge type to the revised scores, a re-analysis and not a registered test, and `scripts/run_robustness.py` runs the value-slope check in place of the top-token subgraphs. `scripts/publish_results.py` publishes the revised summaries to `data/3-published-results-for-thesis/revised-usd/` with their own digest, and `scripts/export_latex.py` writes the tables of `results/tables/` and the figures of `Figures/generated/` from them; Table 8.13 also reads the registered summaries.

The order of these steps is recorded in the commit history, in the cost ledger and in the extraction manifest of W1. The plan was committed at 13:41 UTC on 9 October 2026; the ledger, which records each job when it completes, lists the first job of scan A at 13:43 UTC. The other queries and scripts of stages 2 to 4 entered the repository with the registration in `0d3a057`, after they had run, together with the summaries of the same window and of W0 and with the scripts of stage 5, which had not yet run. That commit was made at 16:49:32 UTC on 9 October 2026, which the commit records in Korea Standard Time as 01:49:32 on 10 October. `scripts/extract_w1.py` wrote its manifest one second later, before its first query, and the manifest records that no override was used. The digest of the registered analysis, `results_digest.md`, records the code revision at which its summaries were published, `8d8b568`, and names the scripts that differed from that revision when the digest was written. Commit `8d8b568` added or extended the scripts of stage 6 and changed `scripts/run_holdout.py` only in how it groups the contrasts of W0, which the summary of W0 records as leaving every value unchanged. The benchmark of the registered analysis ran with `scripts/run_benchmark.py` as committed in `8d8b568`, the revision its digest records.

The revision was committed on 10 October 2026, after the registered summaries had been published: the revision document at 07:54 UTC (`a897475`), the choice of the five tokens at 08:12 UTC (`09dfc32`) and the amended cap at 11:24 UTC (`2705b28`). In the ledger, the query that built the revised cohort follows the second of these commits, and the queries of the proxies and labels follow the third; the scores were computed locally after that. The digest of the revised analysis records the code revision `2705b28` and names five files that differed from it. Of these, `scripts/run_holdout.py` differed in selecting the two degree contrasts of the spenders that had received a transfer by name instead of by position, and `scripts/chainlink_check.py` in restricting the price comparison to the five tokens; all five were committed with the revised summaries in `77e71f5`. The token inputs were also written again after `2705b28`, with the same five tokens, decimals and prices: the decimals loaded into BigQuery had been read at block $513{,}457{,}143$ and were read a second time at block $513{,}498{,}909$, which the published `usd_inputs.json` records, and the evidence columns of the token list and of the candidate lists were updated (the verification of each listing, and the relation

of USDT0 and WBTC to their feeds, now a canonical bridge); these files were also committed in `77e71f5`. The benchmark of the revised analysis ran with `scripts/run_benchmark.py` as it stands at `2705b28`, which differs from the version of `8d8b568` only in also recording the NumPy and SciPy versions. The timed runs of both analyses were made on an otherwise idle machine with an Intel Core Ultra 9 185H processor and 64 GB of memory, under Windows 11 (build 26200), with Python 3.11.9. `benchmark/benchmark.json` records the platform (`Windows-10-10.0.26200-SP0`, the string that Python reports for this build), the processor (Intel64 Family 6 Model 170 Stepping 4) and the Python version, and in the revised analysis also NumPy 2.3.2 and SciPy 1.16.1. In the registered analysis the processor name, the memory (recorded as 63.4 GB) and the same NumPy and SciPy versions were added to the file after the run, on the same machine.

The public repository https://github.com/abitocodes/phd_works contains the evaluation source in `1-Dissertation-Works/ERC20-Allowance-PageRank-Wallet-Reputation/experiments`: the queries (`sql/`), the scripts (`scripts/`), the tests (`tests/`), the versioned configuration (`config/contract_reputation.yaml`), the plan, the registration and the revision document (`docs/`), the inputs of the revision, the extracted tables, the summaries and the cost ledger. The inputs of the revision are in `data/0-usd-token-list-and-prices/`: the token list with decimals, feeds and the evidence for each listing (`listed_tokens.csv`), the daily prices (`daily_prices_usd.csv`) with the raw DefiLlama points, the constants (`constants.json`) and the comparison with Chainlink. Every extracted table is split into parquet parts below 90 MB so that it can be committed, and a JSON file beside each table records the BigQuery table it came from, its row count and the time of the download (for the account types, the endpoint, block and time of the read). The summaries behind the tables are in `data/3-published-results-for-thesis/revised-usd/`, and those of the registered analysis in `data/3-published-results-for-thesis/`. Each folder holds copies of the configuration, the plan and the registration, a digest `results_digest.md`, which gives the summary file and key of every number, and `SHA256SUMS.txt`; the revised folder adds the revision document, the token list, the constants and the price comparison. Any number in a table can thus be checked against its summary without re-running the pipeline.

The scoring and statistics code is checked offline: `tests/test_scoring.py` and `tests/test_stats.py` compare the solver, the edge weights and the paired bootstrap on small synthetic graphs with a reference implementation of the same definitions, and `tests/test_usd.py` checks the US-dollar conversion, the slope $b = \ln 3/m$, the value transform of the revised SQL against the scoring code and the price rules, all with no cloud access. Real extraction needs Google Cloud credentials and a billing project, and its scans read only the public Arbitrum One log table. The revision also read token decimals and Chainlink feeds from public RPC endpoints and daily prices from the DefiLlama API; these inputs are committed, so its BigQuery steps can be rerun without either source.

8.1.2 BigQuery extraction design

The public table `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs` is read once for each window. Scan A runs `sql/01_materialize_logs.sql` over 1 October 2023 to 30 June 2026 and keeps four kinds of log in one project table, `contract_rep.logs_obs`, partitioned by month: ERC-20 `Approval` and `Transfer` logs with exactly three topics, identified by their standard topic-0 hashes (Vogelsteller & Buterin, 2015); GMX V2 `PositionDecrease` logs of the protocol's event emitter; and the `Borrow`, `Repay` and `LiquidationCall` logs of the Aave V3 pool, which no analysis of this thesis uses. Scan B runs the same query over July to September 2026 into `contract_rep.logs_w1`. Every later query reads these project tables, so the cost of the chain-wide scan is paid once per window; the revision scanned no public log again. Appendix 8.2 shows the queries on which the graphs of both analyses rest.

The ego extraction (`sql/03_ego_events.sql`) keeps an approval when a contract of the plan's cohort is its owner or its spender, and a transfer when such a contract is its sender or its recipient. Each graph is therefore the ego network of the cohort, and the size of the extract follows the cohort and not the chain. These ego tables keep amounts in raw base units, the `uint256` value read as a double, for every token. Zero-value transfers are dropped, while zero-value approvals are kept because they delete an allowance. For the observation window they hold $389{,}866{,}882$ approval

logs and 1,538,404,067 transfer logs. The revision cuts its own ego tables from these (`sql/12_usd_ego.sql`). It joins the events with the table of the five tokens and divides each value by ten to the power of the token's decimals; a transfer is then multiplied by the token's price on the UTC day of its block, and an allowance is valued later, at the price of the anchor's day. It keeps the events of the five tokens with a contract of the revised cohort at one end, 224,954,202 approvals and 1,149,168,538 transfers in the observation window; Table 8.4 counts the rows it leaves out.

The pair tables are the edge lists of the two graphs at one anchor. They are built at T_{obs} and at t_1 from the events up to the anchor, with the time decay anchored there, and each pair carries the weights of every method (Appendix 8.2.3). In the revised analysis `sql/14_usd_anchor_pairs.sql` builds them in US dollars: at T_{obs} they hold 11,219,116 allowance pairs and 25,018,892 transfer pairs, and at t_1, 10,650,258 and 23,680,692. EndorseRank and AWP keep the pairs with a positive weight, so a pair whose amounts are a tiny fraction of a cent, and whose weight is zero in double precision, is not an edge of their graphs (Section 4.1 gives their sizes). The freeze of W1 coincides with T_{obs}, so W1 reuses the pair tables of T_{obs}. `sql/17_usd_export_ids.sql` numbers the addresses that occur in a revised pair table of either anchor or in the revised cohort; the JSON record of the node table gives their count. In the registered analysis `sql/04_anchor_pairs.sql` and `sql/08_export_ids.sql` did the same in raw base units, with 22,321,470 allowance pairs and 59,737,617 transfer pairs at T_{obs}. Only the id-coded pair tables, the node table, the proxies and the labels are downloaded; the event tables stay in BigQuery.

Every job is listed in `data/bq_ledger.json` with its query, parameters, estimated and billed bytes and cost. The ledger holds 32 jobs that billed 6,122,812,473,344 bytes (5.57 TiB), or USD 34.80 at the on-demand rate of USD 6.25 per TiB recorded in the ledger, against a budget of USD 50 fixed in the configuration. The 17 jobs of the registered analysis billed 5,226,856,710,144 bytes (4.75 TiB), or USD 29.71. Scan A alone billed 3.29 TiB (USD 20.55) and scan B 0.26 TiB (USD 1.64). When the registration was committed, the ledger stood at USD 26.20, and when the revision was committed at USD 29.71. The 15 jobs of the revision billed USD 5.09. Three of them, two reads of the table of constants and the query of the allowance and transfer distributions behind the amended cap,

ran outside `scripts/run_bq.py` and were entered in the ledger by hand.

8.1.3 Scaling tables

Table 4.2 in Section 4.2.1 reports the runtime of EndorseRank and AWP on deterministic SHA256 subsamples of the matched cohort, measured under the timing protocol of Section 3.11; this section describes how its stages were drawn. The contracts are ordered by the SHA256 hash of the seed string `contract-benchmark-v1` joined to the address, and the first n of them form a stage, for $n = 1{,}000$, $2{,}000$, $5{,}000$ and $10{,}000$; the last row is the full cohort, $n = 15{,}107$, whose measurement is the one in Table 4.1, reused. A stage keeps every extracted edge with an end point among its contracts, so $|V|$ and $|E|$ of both graphs grow with n, from $424{,}888$ allowance and $1{,}692{,}513$ transfer edges at $n = 1{,}000$ to the full graphs with $11{,}210{,}114$ and $25{,}001{,}216$, and the table gives both counts for every stage. The stages are the evaluation sets of Table 4.8, which repeats the runtimes of this table.

The ratio of transfer edges to allowance edges is not constant across the stages. It is highest at the smallest stage, nearly constant from $n = 2{,}000$ to $n = 10{,}000$ and lowest on the full cohort; the table gives it for every stage, and Section 4.2.1 discusses it. The table therefore shows whether the ratio of the two runtimes follows the ratio of the edge counts from stage to stage. No stage is larger than the cohort graph, so the table says nothing about graphs beyond it.

8.1.4 Account-type classification

Only contracts are scored and evaluated, so every address that could enter a cohort was classified by its code first. `scripts/check_account_code.py` calls `eth_getCode` for each address at one block, fetched once at the start of a run and written to the output with the time of the read. An address with non-empty code is a contract unless the code is an EIP-7702 delegation designator, the bytes `0x ef0100` followed by a 20-byte address (Buterin et al., 2024); such an address, like one without code, is an externally owned account (EOA). Table 8.1 lists the four runs, all made on 9 October 2026 for the address sets of the registered analysis, in which approvals of every token counted. The

candidates of the plan's cohort were read from the public endpoint https://arb1.arbitrum.io/rpc until it limited the request rate after $20{,}000$ addresses; the other $9{,}022$ were read at the same block from https://arbitrum-one.public.blastapi.io, which also served the three later runs. The revision made no new read, because each of its cohorts is a subset of a set classified here.

Table 8.1: Account types read with `eth_getCode` on Arbitrum One, one block per run. Contract: non-empty code that is not a delegation designator. No code and EIP-7702 (a delegation designator) are both externally owned accounts. The address sets are those of the registered analysis (approvals of any token); the cohorts of the revised analysis are subsets of them. The W0 run read the spenders that were not cohort candidates, and the W1 run the addresses that no earlier run had classified. Sources: the JSON files beside the parquet outputs in `data/1-raw-blockchain-extracts/cohort/` and `gmx/`.

Address set	Block	Read at (UTC)	Addresses	Contract	No code	EIP-7702
Cohort candidates: non-zero approvals from ≥ 3 owners, observation window	513,216,506	9 Oct 2026, 13:55	29,022	27,844	1,121	57
GMX V2 accounts with ≥ 3 closes in the observation window	513,219,999	9 Oct 2026, 14:10	40,307	7,633	28,075	4,599
W0 spenders with a positive allowance at t_1, not cohort candidates	513,233,527	9 Oct 2026, 15:08	8,224	7,792	413	19
W1 spenders and GMX V2 accounts with ≥ 3 closes in W1, not read before	513,257,476	9 Oct 2026, 16:52	1,153	350	568	235

The plan's cohort consists of the contracts among the candidates. The revised matched cohort is the part of it that at least three owners approved on the five tokens (Section 3.5.1), so each of its contracts was classified in the first run. The W0 spender cohort draws on the first and third runs: of the spenders with a positive latest allowance of one of the five tokens at t_1, $17{,}050$ are contracts, 104 have no code and 13 carry a delegation designator. Such a spender also held that allowance in the registered ego data, since a latest allowance depends only on the events of its token, owner and spender, so it was read in one of these two runs. The W1 spender cohort and the two trader cohorts take their account types from all four runs. Every read refers to the chain on 9 October 2026, so an address is classified by the code it held on that day.

8.1.5 Benchmark run logs

Table 8.2 lists the wall-clock seconds of every timed solver call of the benchmark on the graphs of the revised analysis, in the order the runs were made, with their mean and standard deviation: the four scores on the full graphs at $d = 0.85$, EndorseRank and AWP at $d = 0.75$ and $d = 0.95$, and the two scores at every scaling stage below the full cohort. Each configuration had one untimed

warm-up before its five timed runs, and peak memory was traced on the warm-up alone. Each graph and setting was timed once, and that measurement is reused wherever the same graph and setting appear: the $d = 0.85$ rows of Table 4.6 and the full-cohort row of Table 4.2 repeat the first block of Table 8.2. Across all configurations the standard deviation of the five runs is between 0.7% and 21.5% of the mean; the largest share belongs to EndorseRank at $n = 1{,}000$, whose calls take about half a second. The benchmark of the registered analysis, on the graphs of the plan's cohort, is published with the registered summaries (Appendix 8.6.10).

8.1.6 Parameter summary

Table 8.3 collects the default parameters of the revised analysis. Most of them are keys of the configuration (Appendix 8.2.4) and were set with the plan before any log was extracted. The registration of W1 added the margin of rule B and the $2{,}000$ resamples of the trader comparisons, and the block of the code check was fixed when the check ran. The revision, made after the registered results were known, set the tokens, the prices, the value slopes and the cap of the C-PR allowance layer, and replaced the token check by the value-slope check (Section 3.5.1); where it changed a setting of the plan, the table gives the plan's setting in brackets. Rules A and B keep the conditions and the margin fixed for them, but on the revised scores they are applied after the registered results were known, as a re-analysis and not as registered tests.

8.1.7 Proxy operational definitions (reference)

The proxies are defined in Section 3.9; the list below names each one with its equation and states in one line what it counts for a cohort contract v. Every proxy counts events of the five tokens only.

- Transfer family: the in-degree (Equation (3.10)) counts the distinct senders of transfers to v, self-transfers included; the in-value (Equation (3.11)) sums their US-dollar amounts, each at the price of its day, across tokens.
- Allowance family: the in-approve degree (Equation (3.12)) counts the owners with a positive latest allowance to the spender v; the in-approve value (Equation (3.13)) sums those

Table 8.2: Wall-clock seconds of every timed solver call of the benchmark (matched cohort, $n = 15{,}107$ contracts), in the order the runs were made, with their mean and standard deviation. Each configuration had one untimed warm-up before its timed runs, and each graph and setting was timed once: the $d = 0.85$ rows of Table 4.6 and the full-cohort row of Table 4.2 are the first block, reused. The stages are the subsamples of Table 4.2.

Method	Run 1	Run 2	Run 3	Run 4	Run 5	Mean	SD
Full graphs, $d = 0.85$							
EndorseRank	9.570	11.063	10.497	9.811	10.530	10.294	0.601
AWP	21.139	20.446	24.122	24.492	21.637	22.367	1.825
C-PR ($\lambda = 0.5$)	33.813	39.576	35.439	35.793	36.187	36.162	2.112
S-PR	38.769	36.970	29.310	35.214	31.848	34.422	3.833
Full graphs, $d = 0.75$							
EndorseRank	6.045	5.963	6.003	5.984	5.927	5.984	0.044
AWP	13.208	16.968	16.667	15.870	13.469	15.236	1.781
Full graphs, $d = 0.95$							
EndorseRank	31.701	32.171	30.886	30.574	30.836	31.233	0.673
AWP	58.498	56.896	59.713	64.490	65.212	60.962	3.697
Stage $n = 1{,}000$							
EndorseRank	0.425	0.489	0.585	0.683	0.719	0.580	0.125
AWP	1.892	1.408	1.870	1.758	1.458	1.677	0.229
Stage $n = 2{,}000$							
EndorseRank	0.944	0.956	0.925	0.932	0.937	0.939	0.012
AWP	3.748	4.876	4.133	4.811	3.792	4.272	0.543
Stage $n = 5{,}000$							
EndorseRank	2.293	2.420	2.489	2.555	2.723	2.496	0.159
AWP	8.713	9.861	9.895	8.051	8.453	8.995	0.840
Stage $n = 10{,}000$							
EndorseRank	6.253	6.375	6.065	6.285	6.119	6.220	0.126
AWP	17.640	17.362	17.160	16.124	15.816	16.821	0.802

Table 8.3: Default parameters used in the evaluation (revised analysis; the plan's setting in brackets where the revision changed it).

Parameter	Value	Role
PageRank damping d	0.85; sweep 0.75, 0.95	Teleport mixture
Convergence tolerance ε	10^{-8} in L_1 norm	Power-iteration stop
Max iterations $I_{\max}$	300	Power-iteration cap
Logistic k	0.01	Decay slope (AWP, EndorseRank, transfer layer of C-PR)
Logistic t_0	180 days	Decay midpoint (AWP, EndorseRank, transfer layer of C-PR)
Value transform b	$\ln 3/m$ per dollar, $m_T = 200$, $m_A = 62$ (1 on raw base units)	Value weight $V(z)$ of AWP (m_T) and EndorseRank (m_A)
Allowance cap U	20,000 dollars (none)	C-PR allowance layer; in-approve value
Tokens	WETH, USDC, USDT0, ARB, WBTC (every ERC-20 token)	Events of graphs, proxies and spender labels (Table 8.4)
Prices	DefiLlama daily close; decimals at block 513,457,143 (none)	US-dollar amounts
Restart distribution	$\propto X_u$; uniform	AWP; EndorseRank, C-PR
Coupling weight λ	0.5 primary; 0.25, 0.75	C-PR allowance-layer weight, fixed in the plan
S-PR restart vector	C-PR ($\lambda = 1$) on the transfer-layer nodes, renormalized	Endorsement-seeded teleportation
Cohort rule	non-zero approvals of the five tokens from ≥ 3 owners; contract code (approvals of any token)	Matched cohort
Account type	`eth_getCode`; block 513,216,506 for the cohort	Contract or EOA
Observation window	2023-10-01 – 2026-06-30	UTC inclusive; T_{obs} anchors the decay
W0 freeze / labels	2026-03-31 / 2026-04-01 – 2026-06-30	Spender and trader cohorts; rule A (re-analysis)
W1 freeze / labels	2026-06-30 / 2026-07-01 – 2026-09-30	Registered window; rule B, margin $\delta = 0.01$ (re-analysis)
Trader rule	contract code, node at the freeze, ≥ 3 closes in the label window	GMX V2 trader cohorts
Bootstrap resamples / seed	400 / 42; 2,000 / 42	Paired contract bootstrap; trader labels of Table 8.12
Timed runs / warm-up	5 / 1	Runtime protocol
Scaling seed	`contract-benchmark-v1`	Deterministic SHA256 subsamples
Value-slope check	$b/10$ and $10b$ (graphs of the matched cohort at T_{obs})	Sensitivity of EndorseRank and AWP (Table 4.7)
BigQuery budget	USD 50	Checked before every query

allowances across owners and tokens in US dollars at the anchor's price, each capped at $U = 20{,}000$ dollars.

- Sybil-adjusted stability family: the inbound counterparty ratio (Equation (3.14)) divides the distinct other senders by the transfers received from them; the transfer tenure (Equation (3.15)) is the span in days of the transfers attributed to v; the active months (Equation (3.16)) count the calendar months with such transfers.

8.1.8 Glossary

The terms below are used throughout the thesis with the meanings given here.

Account type Contract or externally owned account, read with `eth_getCode` at one block (Appendix 8.1.4).

Activeness AWP's restart weight of an address: over the addresses it has sent to, the sum of the time weights of its latest transfer to each.

Allowance An ERC-20 authorization that lets a spender move tokens on an owner's behalf.

Alignment Rank agreement between a score and a proxy, measured by Spearman's ρ on mid-ranks and by Kendall's τ; every reported τ is the tie-corrected τ_b.

Construct overlap A ranking method and a proxy sharing a data source, which raises their agreement; such agreement is a check of the intended construct and not evidence from outside it.

C-PR end points C-PR at $\lambda = 1$ and at $\lambda = 0$, which walk the US-dollar allowance layer, with each allowance capped at U, or the US-dollar transfer layer alone with uniform restarts.

Ego network The events of the five tokens with a contract of the matched cohort at one end: the owner or spender of an approval, the sender or recipient of a transfer. In the registered analysis, the events of every token with a contract of the plan's cohort at one end.

Endorsement graph Directed graph of positive latest allowances of the five tokens from owner to spender, the input of EndorseRank and the allowance layer of C-PR.

Externally owned account (EOA) An address without code or with an EIP-7702 delegation designator. EOAs appear in the graphs as owners, senders and recipients and are never scored or evaluated.

Matched cohort The primary sample: the contracts that received a non-zero `Approval` of WETH, USDC, USDT0, ARB or WBTC from at least three distinct owners in the observation window (Sections 3.5.1 and 4.1.1). It was fixed from the approvals of the five tokens and the code check, without reference to transfers, after the registered results were known. The plan's cohort of the registered analysis applied the same rule to approvals of every token and was fixed before any transfer was aggregated. Membership does not depend on either graph at T_{obs}, so a cohort contract can lie outside one graph or both.

Paired bootstrap Resampling of contracts with replacement through a single index matrix shared

by every coefficient, so that differences between coefficients can be tested on the same resamples.

PageRank Spectral ranking of nodes by the stationary distribution of a damped random walk.

Proxy An observable statistic of a contract against which a ranking is checked for the construct it is built to measure.

Registered analysis The analysis fixed by the plan and the registration of W1, with amounts in raw base units and every ERC-20 token; its results are kept as they were obtained (Appendix 8.6.10).

Registered window (W1) Scores frozen at 30 June 2026 and labels counted from July to September 2026, with rule B and the comparison on outcomes built from neither edge type registered before the logs of those months were extracted (Section 3.7.2). On the revised scores these rules are applied after the registered results were known, as a re-analysis.

Revised analysis The main analysis of the thesis: the protocols and rules of the registered analysis with amounts in US dollars, the five tokens and the cohort rebuilt on them, decided after the registered results were known (Section 3.5.1).

Scaling stage A deterministic SHA256 subsample of the matched cohort together with the extracted edges incident to its contracts.

Spender cohort Every contract spender with at least one positive latest allowance of one of the five tokens at the freeze in the extracted approval data: at t_1 (31 March 2026, 23:59:59 UTC) for W0 and at 30 June 2026, 23:59:59 UTC, for W1.

Teleport The jump that PageRank makes with probability $1 - d$, to a node drawn from the restart distribution: uniform for EndorseRank and C-PR, weighted by sending activity for AWP.

Trader cohort The GMX V2 accounts that hold contract code, are nodes of the graph at the freeze and close at least three positions in the label window.

US-dollar amount The raw value of a transfer or an allowance divided by ten to the power of the token's decimals and multiplied by the token's DefiLlama daily close: of the transfer's UTC day, or of the anchor's day for a latest allowance (Section 3.5.1). The registered analysis used raw base units.

Value slope The slope $b = \ln 3/m$ of the value weight V, chosen so that an amount of m dollars weighs $1/2$ (V never exceeds one): $m_T = 200$ for transfers and $m_A = 62$ for allowances.

8.2 SQL Templates, Configuration, and Extended Notes

To aid reproduction, this appendix shows the SQL patterns on which the graphs of both analyses rest, the configuration keys, and notes on complexity and on the conventions of the result tables. The three queries of the plan build the graphs of the registered analysis; the revised analysis cuts its tables from their output and values the amounts in US dollars (Appendix 8.2.3). The listings are abridged; the full files are in `sql/` of the experiments folder (Appendix 8.1). Placeholders in double underscores, such as `__DEST__`, are replaced by `scripts/run_bq.py` before a query runs, and names that start with `@` are query parameters.

8.2.1 Log materialization pattern

Scan A and scan B run the same query, `sql/01_materialize_logs.sql`, over different windows. It reads the public log table once, tags each log by kind and keeps only the four kinds the study needs, in a project table partitioned by month and clustered by kind. With the GMX and Aave branches shortened, it reads:

```
CREATE TABLE `__DEST__`
PARTITION BY TIMESTAMP_TRUNC(block_timestamp, MONTH)
CLUSTER BY kind
AS
WITH src AS (
```

```
  SELECT block_timestamp, block_number, log_index,
         LOWER(address) AS address, topics, data
  FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
  WHERE block_timestamp >= TIMESTAMP(@start_ts)
    AND block_timestamp < TIMESTAMP(@end_ts)
),
tagged AS (
  SELECT *,
    CASE
      WHEN ARRAY_LENGTH(topics) = 3 AND topics[OFFSET(0)] =
        '0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925'
        THEN 'A'                                    -- ERC-20 Approval
      WHEN ARRAY_LENGTH(topics) = 3 AND topics[OFFSET(0)] =
        '0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'
        THEN 'T'                                    -- ERC-20 Transfer
      WHEN address = '0xc8ee91a54287db53897056e12d9819156d3822fb'
           AND ... THEN 'G'                         -- GMX V2 PositionDecrease
      WHEN address = '0x794a61358d6845594f94dc1db02a252b5b4814ad'
           AND ... THEN 'V'                         -- Aave V3 pool events
    END AS kind
  FROM src
)
SELECT
  block_timestamp, block_number, log_index, kind,
  FROM_HEX(SUBSTR(address, 3)) AS emitter,
  IF(kind IN ('A', 'T'), FROM_HEX(SUBSTR(topics[OFFSET(1)], 27)), NULL) AS a,
  IF(kind IN ('A', 'T'), FROM_HEX(SUBSTR(topics[OFFSET(2)], 27)), NULL) AS b,
  IF(kind IN ('A', 'T'), SAFE.FROM_HEX(SUBSTR(data, 3)), NULL) AS val,
```

```
  IF(kind IN ('G', 'V'), topics, NULL) AS raw_topics,
  IF(kind IN ('G', 'V'), data, NULL) AS raw_data
FROM tagged
WHERE kind IS NOT NULL
```

For an approval, `a` is the owner and `b` the spender; for a transfer, `a` is the sender and `b` the recipient. In both, `emitter` is the token contract and `val` the 32-byte amount. The two topic-0 hashes are the keccak256 hashes of `Approval(address,address,uint256)` and `Transfer(address,address,uint256)` (Vogelsteller & Buterin, 2015). Requiring exactly three topics keeps the ERC-20 logs and leaves out the ERC-721 logs, which share both hashes but index the token id as a fourth topic. Addresses are stored as 20-byte values, not as hexadecimal strings.

8.2.2 Ego-network extraction pattern

`sql/03_ego_events.sql` reads the materialized table of one window and keeps the approvals and transfers with a cohort contract at one end. The table `contract_rep.cohort` holds the 20-byte addresses of the plan's cohort, loaded by `scripts/build_cohort.py`. Abridged:

```
CREATE TEMP FUNCTION u256(v BYTES) AS (
  IFNULL((SELECT SUM(c * POW(256.0, LENGTH(v) - 1 - o))
          FROM UNNEST(TO_CODE_POINTS(v)) AS c WITH OFFSET o), 0.0)
);

CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.approvals___SUFFIX__`
PARTITION BY TIMESTAMP_TRUNC(block_timestamp, MONTH)
AS
SELECT
  l.block_timestamp, l.block_number, l.log_index,
  l.emitter AS token_address,
  l.a AS owner,
```

```
  l.b AS spender,
  u256(l.val) AS value
FROM `dissertation-bq.contract_rep.__SRC__` AS l
WHERE l.kind = 'A'
  AND l.block_timestamp >= TIMESTAMP(@start_ts)
  AND l.block_timestamp < TIMESTAMP(@end_ts)
  AND (l.a IN (SELECT address FROM `dissertation-bq.contract_rep.cohort`)
       OR l.b IN (SELECT address FROM `dissertation-bq.contract_rep.cohort`));

-- transfers___SUFFIX__: the same with kind = 'T', from_address = a and
-- to_address = b, and zero amounts dropped:
--   AND LTRIM(TO_HEX(l.val), '0') != ''
```

The function `u256` reads the 32-byte amount as a double in raw base units; the queries of the plan apply no token decimals. Approvals of zero are kept, because an approval of zero deletes an allowance, while zero-value transfers are dropped. The suffix is `ego` for the observation window and `w1ego` for July–September 2026. The revised analysis does not run this query again. It reads these tables of every token and the plan's cohort and keeps the part that concerns the five tokens and the revised cohort, which is a subset of the plan's: a contract that three owners approved on the five tokens was approved by three owners on some token.

8.2.3 Latest-allowance and pair-table pattern

In the registered analysis, `sql/04_anchor_pairs.sql` turns the ego events up to one anchor into the edge lists of both graphs. Abridged:

```
CREATE TEMP FUNCTION sig(age_days FLOAT64) AS
  (1.0 / (1.0 + EXP(0.01 * (age_days - 180.0))));
CREATE TEMP FUNCTION vt(x FLOAT64) AS
  (IF(x >= 40.0, 1.0, 2.0 / (1.0 + EXP(-x)) - 1.0));
```

```
CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.allow_triples___TAG__` AS
SELECT token_address, owner, spender, value, block_timestamp,
       TIMESTAMP_DIFF(TIMESTAMP(@anchor_ts), block_timestamp, MICROSECOND)
         / 8.64e10 AS age_days
FROM (
  SELECT *, ROW_NUMBER() OVER (
           PARTITION BY token_address, owner, spender
           ORDER BY block_number DESC, log_index DESC) AS rn
  FROM `dissertation-bq.contract_rep.approvals_ego`
  WHERE block_timestamp >= TIMESTAMP(@start_ts)
    AND block_timestamp <= TIMESTAMP(@anchor_ts)
)
WHERE rn = 1 AND value > 0;

CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.allow_pairs___TAG__` AS
SELECT owner, spender,
       COUNT(*) AS n_tokens,
       SUM(sig(age_days) * vt(value)) AS w_er,   -- EndorseRank
       SUM(value) AS w_raw,                      -- C-PR allowance layer
       MAX(sig(age_days)) AS max_sig,
       MAX(block_timestamp) AS last_ts
FROM `dissertation-bq.contract_rep.allow_triples___TAG__`
GROUP BY owner, spender;

-- transfer_pairs___TAG__: per (from_address, to_address) over transfers_ego
-- up to the anchor, with
--   SUM(sig(age_days) * vt(value)) AS w_awp    -- AWP
```

```
--   SUM(value * sig(age_days))     AS w_raw    -- C-PR transfer layer
--   MAX(sig(age_days))             AS max_sig  -- AWP activeness
```

The function `sig` is the logistic decay of Equation (3.3) with $k = 0.01$ and $t_0 = 180$ days, and `vt` the value transform of Equation (3.4) with the plan's slope $b = 1$ on raw base units, set to one from $z = 40$ on, where it equals one in double precision. Among the approvals of one token, owner and spender up to the anchor, the latest is the one with the largest block number and, within the block, the largest log index. It is kept only when its value is positive, so a revocation removes that token from the pair. Each pair row carries the weights of every method, so all scores at one anchor come from the same rows. The tag is `tobs` for T_{obs} and `t1` for the freeze of W0.

The revised analysis adds a US-dollar valuation to these patterns (Section 3.5.1). `scripts/run_usd_pipeline.py` loads the five tokens with their decimals and the daily price table from `data/0-usd-token-list-and-prices/` into the tables `u_tokens` and `u_prices`, which `sql/19_usd_load.sql` keys by 20-byte address. `sql/12_usd_ego.sql` then forms the revised cohort and joins the ego events with these tables, and `sql/14_usd_anchor_pairs.sql` builds the pair tables with the slope and the cap as query parameters. Abridged:

```
-- sql/12_usd_ego.sql
CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.u_cohort` AS
SELECT spender AS address, COUNT(DISTINCT owner) AS owners_nonzero
FROM `dissertation-bq.contract_rep.approvals_ego`
WHERE block_timestamp >= TIMESTAMP(@start_ts)
  AND block_timestamp < TIMESTAMP(@end_ts)
  AND value > 0
  AND token_address IN
        (SELECT token FROM `dissertation-bq.contract_rep.u_tokens`)
  AND spender IN (SELECT address FROM `dissertation-bq.contract_rep.cohort`)
GROUP BY spender
HAVING COUNT(DISTINCT owner) >= 3;
```

```
CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.u_transfers` AS
SELECT x.block_timestamp, x.token_address, x.from_address, x.to_address,
       x.value / POW(10.0, t.decimals) * p.price_usd AS usd
FROM `dissertation-bq.contract_rep.transfers_ego` AS x
JOIN `dissertation-bq.contract_rep.u_tokens` AS t
  ON t.token = x.token_address
LEFT JOIN `dissertation-bq.contract_rep.u_prices` AS p
  ON p.token = x.token_address AND p.day = DATE(x.block_timestamp)
WHERE x.from_address IN
        (SELECT address FROM `dissertation-bq.contract_rep.u_cohort`)
   OR x.to_address IN
        (SELECT address FROM `dissertation-bq.contract_rep.u_cohort`);
-- u_approvals: the same filter on approvals_ego, amount = value / 10^decimals

-- sql/14_usd_anchor_pairs.sql
CREATE TEMP FUNCTION vt(z FLOAT64, b FLOAT64) AS
  (IF(b * z >= 40.0, 1.0, 2.0 / (1.0 + EXP(-b * z)) - 1.0));
-- u_allow_triples___TAG__: the latest approvals of 04_anchor_pairs.sql, taken
-- from u_approvals and valued at the price of the anchor's day:
--   r.amount * p.price_usd AS usd ... JOIN u_prices AS p
--   ON p.token = r.token_address AND p.day = DATE(TIMESTAMP(@anchor_ts))
CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.u_allow_pairs___TAG__` AS
SELECT owner, spender,
       COUNT(*) AS n_tokens,
       SUM(sig(age_days) * vt(usd, @b_allowance)) AS w_er,  -- EndorseRank
       SUM(LEAST(usd, @cap_allowance)) AS w_raw,   -- C-PR allowance layer
       MAX(sig(age_days)) AS max_sig,
```

```
       MAX(block_timestamp) AS last_ts
FROM `dissertation-bq.contract_rep.u_allow_triples___TAG__`
GROUP BY owner, spender;
-- u_transfer_pairs___TAG__: per (from_address, to_address) over u_transfers
-- up to the anchor, rows with usd IS NULL left out, with
--   SUM(sig(age_days) * vt(usd, @b_transfer)) AS w_awp   -- AWP
--   SUM(usd * sig(age_days))                  AS w_raw   -- C-PR transfer layer
--   MAX(sig(age_days))                        AS max_sig -- AWP activeness
```

A transfer is valued at the price of the UTC day of its block, and a latest allowance at the price of the anchor's day, since the edge stands for the permission in force at the anchor. The parameters come from `constants.json`: $b_A = \ln 3/62$ and $b_T = \ln 3/200$ per dollar and $U = 20{,}000$ dollars, computed from the events up to t_1 (`sql/13_usd_constants.sql`). An unlimited allowance reaches $bz \geq 40$, so it enters EndorseRank with $V = 1$ and the allowance layer of C-PR with U. A transfer on a day before its token's first price would keep a null value and be left out; no kept transfer lacks a price (Table 8.4). The same query writes the EndorseRank and AWP weights with the slope multiplied and divided by ten, from which the value-slope check of Table 4.7 is computed. `sql/15` to `sql/18` compute the proxies, the labels, the node ids and the description from the revised tables as `sql/05`, `sql/06`, `sql/08` and `sql/10` do from the registered ones, with the in-value summed in US dollars.

8.2.4 Configuration keys

The pipeline's settings are kept in one versioned file, `config/contract_reputation.yaml`, which the scripts read at start-up. The environment variable `CONTRACT_REP_VARIANT` selects the analysis. Unset, the scripts compute the registered analysis from the keys below; set to `usd`, they read the revised tables in `data/1-raw-blockchain-extracts/graph-tables-usd/` and write every result under `revised-usd/`. The revised analysis takes its tokens, decimals, prices, value slopes and cap from `data/0-usd-token-list-and-prices/` (`listed_tokens.csv`, `dai`

ly_prices_usd.csv and constants.json) and not from the configuration. The groups of keys of the configuration are:

- chain_id, chain_name and bigquery: chain 42161 (Arbitrum One); the project, dataset, location and public log table; the budget in USD; and the windows of scan A and scan B;
- period, holdout and registered_window: the observation window with its anchor T_{obs}, the freeze and label window of W0, and the freeze, label window, status and margin δ of W1;
- events, gmx_arbitrum and aave_arbitrum: the topic-0 hashes of Approval and Transfer; the GMX V2 event emitter, the EventLog1 topic, the PositionDecrease hash, the liquidation handler, the order type of a liquidation and the minimum of three closes; and the Aave V3 pool with its event topics;
- accounts and cohort: the RPC endpoint and the delegation prefix 0xef0100 of the code check, and the minimum of three distinct owners;
- reputation: damping, tolerance, iteration cap, decay k and t_0, the value parameter $b = 1$ of the registered analysis, the primary and sensitivity values of λ, and the two parts of rule A;
- alignment, benchmark and robustness: the bootstrap resamples and seed; the warm-up and timed runs and the scaling seed; the damping values and the number of top tokens of the registered token check;
- storage: the largest size of a data file part, 90 MB.

Changing the damping factor, the decay, the value slopes, the cap or the tokens requires running the evaluation again, the damping, value-slope and sample-definition checks included, and exporting the tables and figures again from the same summaries, so that every reported number matches its machine-readable source. The revised analysis is such a rerun: scripts/run_usd_pipeline.py ran its BigQuery steps in order, and every local script ran again with CONTRACT_REP_VARIANT=usd (Appendix 8.1.1). scripts/extract_w1.py records the SHA-256 hash of this file at the start of scan B (Appendix 8.6.8).

8.2.5 Complexity notes

Write $m = |E|$. In BigQuery the latest allowances come from a window function over the R_a approval rows, which sorts within each token, owner and spender, in $O(R_a \log R_a)$ time at most; the transfer pairs are one aggregation over the R_t transfer rows, in $O(R_t)$. The joins of the revised analysis with the token and price tables match each row with one token and one day, so they add work linear in R_a and R_t. Locally, PageRank over I iterations costs $O(m + mI)$, the first term for building the sparse operator and the second for one sparse product per iteration (Appendix 8.3.3). C-PR walks the union of both layers, so one of its iterations costs as much as one on each layer, and S-PR solves the allowance layer first and then the transfer layer. The edge lists and score vectors take most of the memory, so the peak memory of a solve grows with m; in Table 4.1 the peak-memory ratio of AWP to EndorseRank is 1.95 ($1{,}712.6$ against 879.7 MB), at an edge ratio of 2.23.

8.2.6 Conventions for the result tables

The tables of Chapter 4 and of this appendix follow these conventions:

1. Every τ is Kendall's τ_b on raw scores and proxies, reported with its 95% paired-bootstrap interval: 400 resamples over contracts with seed 42, and $2{,}000$ for the trader comparisons on outcomes built from neither edge type, whose decision contrasts P and Q also carry 97.5% intervals. Differences between coefficients come from the paired contrast tables, not from subtracting point estimates.
2. A cohort contract missing from a method's graph scores zero under that method. Keeping it as an isolated node instead is the sensitivity analysis of Table 4.5.
3. Each contrast table says which of its contrasts decide a rule: rule A, fixed by the plan, or rule B and the comparison on outcomes built from neither edge type, fixed by the registration of W1. In the tables of the revised analysis these rules are applied to scores defined after the registered results were known, so none of their contrasts is a registered test; the registered decision contrasts are in Table 8.13. Every other contrast was computed after the labels were

known.

4. A runtime is the mean wall-clock time of five timed solver calls after one untimed warm-up. A call turns edge lists built in advance into scores: node indexing, sparse-matrix assembly, row normalization, restart vector and power iteration, without extraction or preprocessing. Every runtime appears with $|V|$ and $|E|$ of its solver graph, and the same graph under the same settings shows the same measurement in every table (Section 3.11).
5. The sample-definition stages (Table 4.8) show how the statistic moves with the evaluation set; they are not robustness evidence, and their values are not compared with the full-cohort tables.

8.2.7 Limitations

The queries above were run on Arbitrum One only. Their results depend on the public log table as it stood on 9 October 2026, when scan A and scan B ran. The revision read no public log again, but its results also depend on the Chainlink directory as read on 10 October 2026, on the token decimals read at block $513{,}457{,}143$ (read again at block $513{,}498{,}909$ with the same values) and on the DefiLlama daily prices as fetched for it. The ego filter keeps only events with a cohort contract at one end, so the graphs contain no edge between two addresses outside the cohort. The revised graphs keep the five tokens only: the rows of other tokens are counted and left out (Table 8.4), and native ether, which moves without `Transfer` logs, enters only as WETH. The numbers of Chapter 4 come from the machine-readable summaries of the revised analysis, for the matched cohort, the spender and trader cohorts of W0 and W1, the scaling stages and the Sybil model; the July–September results among them apply the registered rules to the revised scores after the registered results were known, as a re-analysis (Section 4.6.2). The registered results come from the summaries of the registered analysis (Appendix 8.6.10).

8.3 Elementary Properties of PageRank and Rank Correlation

Chapters 2–4 rely on the elementary properties summarized here. The material is standard: the PageRank properties follow Page et al. (1999) and Langville and Meyer (2006), and the rank correlations Spearman (1904) and Kendall (1938).

8.3.1 Existence and uniqueness of PageRank

Take P to be a row-stochastic matrix on $|V|$ nodes, possibly after the dangling-node repair of Equation (2.1). For $d \in (0, 1)$ the Google matrix of Equation (2.2),

$$G_{\text{PR}} = dP^{\top} + \frac{1-d}{|V|}\mathbf{1}\mathbf{1}^{\top},$$

is positive and column-stochastic. By the Perron–Frobenius theorem for positive matrices, G_{PR} has a single eigenvalue of modulus one, and the corresponding eigenvector $\mathbf{r}$ is strictly positive; we can scale it so that $\sum_i r_i = 1$. It follows that, once the damping and the dangling-node rules are fixed, every directed graph has unique PageRank scores (Langville & Meyer, 2006).

Two restart vectors in this thesis are not uniform. AWP's, proportional to activeness, is zero at every address that has sent nothing, and S-PR's, the normalized score of the allowance walk, is zero at every node of the transfer layer that has no allowance score, so neither Google matrix is positive. The fixed point is still unique: for two probability vectors the restart term cancels in their difference, and the operator multiplies their $\|\cdot\|_1$ distance by at most d. In general it is not strictly positive, however, because a node with no restart mass that no path from a restart node reaches keeps a score of zero. This is why AWP scores zero every cohort contract outside the transfer graph under both rules of Table 4.5.

Uniqueness guarantees a well-defined comparison between EndorseRank and AWP. Once its parameters are fixed, each method assigns exactly one score vector to a given edge set, so any difference between $\mathbf{s}_{\text{ER}}$ and $\mathbf{s}_{\text{AWP}}$ must arise from their edges and restart distributions and cannot be traced to

the solver.

8.3.2 Continuity in edge weights

For fixed $d \in (0, 1)$, PageRank depends continuously on the entries of P. If $\tilde{P}$ is a second row-stochastic matrix on the same nodes, with PageRank vector $\tilde{\mathbf{r}}$, then

$$\|\mathbf{r} - \tilde{\mathbf{r}}\|_1 \leq \frac{d}{1-d} \max_i \left\| P_{i\cdot} - \tilde{P}_{i\cdot} \right\|_1,$$

a bound that does not depend on $|V|$ (Langville & Meyer, 2006).

Continuity in P is not the same as continuity in the edge weights. A row $P_{i\cdot} = w_{i\cdot}/o_i$ moves continuously with the weights only while the out-strength o_i stays positive. When a node gains its first out-edge or loses its last one, its row jumps between the uniform dangling row and a row concentrated on its out-neighbors, however small the weight involved. In the worked example of Section 2.13, adding an edge $D \to A$ of weight 10^{-12} moves $\mathbf{r}$ by 0.37 in the $\|\cdot\|_1$ norm and raises r_A from 0.1375 to 0.3202.

The damping sweep is covered by continuity: it keeps P fixed and changes d smoothly, and for $d < 1$ the PageRank vector varies continuously with d. The value-slope check is covered only in part. Changing the slope b of V changes the weight of every finite amount, while an unlimited allowance keeps $V = 1$, and in exact arithmetic it adds or removes no edge, since $V(z) > 0$ for every $z > 0$, so P moves continuously with b. In floating point an edge that carries only tiny amounts can round to zero weight (Section 4.4), and a change of b can move such an edge to or from zero. A tenfold change of b is not small, however, and the bound above says little about it, so the stability of that check rests on the data (Table 4.7). Empirically, the family means of EndorseRank and AWP change little across $d \in \{0.75, 0.85, 0.95\}$, AWP's transfer mean the most (Section 4.5.1; Table 4.6), although the number of iterations grows with d.

8.3.3 Sparsity and per-iteration cost

With $m = |E|$ nonzero edges, the sparse product $P^{\top}\mathbf{x}$ costs $O(m)$ arithmetic operations, and building the sparse operator from the edge list is also $O(m)$, so power iteration over I iterations is $O(m + mI)$. On the matched cohort the transfer graph has 2.23 times as many edges as the allowance graph (Section 4.2), so at equal iteration counts the transfer method would take about 2.23 times as long. Both scores run the same solver, so any further difference in time comes from the iteration counts and from the vector operations over $|V|$, not from the algorithm. In the measured calls the time ratio of AWP to EndorseRank is 2.17, with 92 and 69 iterations (Section 4.2), so the transfer method's time per edge and iteration is lower than EndorseRank's on these graphs.

8.3.4 Properties of Spearman's ρ

Spearman's ρ is Pearson's correlation computed on ranks. It lies in $[-1, 1]$ and does not change under strictly increasing transformations of either variable. A value of $\rho = 1$ means perfect monotone agreement, and $\rho = -1$ perfect reverse agreement. Tied values receive mid-ranks, and the evaluation pipeline relies on standard tie-aware implementations. Because reputation scores and proxies are continuous and heavy-tailed, rank-based association suits them better than Pearson correlation on the raw values. The AWP baseline paper likewise evaluates its ranking with rank statistics (T. Do et al., 2023).

8.3.5 Properties of Kendall's τ

Kendall's τ compares the numbers of concordant and discordant pairs, n_c and n_d. Without tie correction, $\tau_a = (n_c - n_d)/n_0$ with $n_0 = n(n-1)/2$ is the probability that two randomly drawn contracts are ordered the same way by both rankings minus the probability that they are ordered oppositely (Kendall, 1938). Every reported τ is the tie-corrected τ_b of Equation (2.6), whose denominator removes the pairs tied on either ranking from n_0. The two coincide without ties, and among the scores of this thesis ties are rare: on the matched cohort no score ties more than 3.2%

of the contract pairs, and Section 4.4 gives the shares of EndorseRank and AWP. For EndorseRank against AWP the denominator of τ_b therefore differs from n_0 by less than two percent, and τ_a, computed from the pair counts of Section 4.4, lies within two percent of the reported τ_b (Table 4.5). Ties in a proxy cap τ_b as well. The integer proxies, such as the in-degree and the in-approve degree, tie many pairs, and the largest τ_b any score can reach against such a proxy falls with the share of tied pairs. Because no score here ties more than 3.2% of the pairs, this cap differs little between the scores compared on one proxy, so paired contrasts on one proxy are not distorted by it. Contrasts between proxies can be, to the extent that the proxies' ties differ.

When n is large, τ and ρ usually agree in sign, but they differ in magnitude. Across the fourteen pairs of EndorseRank or AWP with a proxy (Tables 8.8 to 8.10), every $|\tau|$ exceeds 0.07, the ratio ρ/τ lies between about 1.2 and 1.8, and $|\rho|$ is always the larger of the two. No pair has coefficients of opposite sign. We report both, as the AWP baseline does (T. Do et al., 2023). Our main summary statistic is the family-mean τ with a paired-bootstrap interval (Efron, 1979).

8.3.6 Construct validity note

Cronbach and Meehl (1955) distinguish criterion-related validity from construct validity. Since no definitive gold-standard label of contract reputation exists, this research takes a construct-oriented approach. Each score is checked against statistics that follow from its conceptual definition, such as receiving allowances for EndorseRank and receiving inbound transfers for AWP. Agreement with statistics built from a score's own edges is a check of the intended construct and is partly built in. Each score is also set against later arrivals of the same constructs, new approval pairs and new transfer senders, against the raw degree it smooths, and in the windows W1 and W0 against a trader outcome built from neither edge type. Agreement within a construct supports the intended reading of a score, subject to that overlap. Low agreement across constructs does not by itself signal failure, since a score built for one construct need not track the other.

8.3.7 Where each primary quantity is reported

For each primary quantity of the revised analysis, the index below names the table or section where it is established; its last rows point to the revision of the amounts and to the registered analysis.

Quantity	Reported in
Runtime, SD, memory, iterations, $\|V\|$, $\|E\|$ (four scores, full graphs)	Table 4.1
Scaling stages (subsamples of the matched cohort)	Table 4.2
Every timed run of the benchmark	Table 8.2
Family-mean τ with 95% CI, every PageRank	Tables 4.3 and 4.9
Paired $\Delta\tau$ contrasts of C-PR and S-PR	Table 4.4
Inter-method τ; contracts outside a graph as isolated nodes	Table 4.5
Damping, value-slope and sample-definition checks	Tables 4.6, 4.7, 4.8
EndorseRank with uniform and with activity restarts	Table 4.9
Spender holdout W0	Tables 4.10, 4.11 and 4.12
Trader run W0 (profit labels)	Table 4.13
Window W1, revised scores	Tables 4.14, 4.15, 4.17 and 4.16
Outcomes built from neither edge type	Tables 4.18, 4.19 and 8.12
Sybil cost model	Section 5.6
Comparisons between EndorseRank and AWP, per-proxy results	Appendix 8.5
Account types of the cohorts	Table 8.1
Extracted events, tokens, GMX V2 activity by quarter	Appendix 8.4
Five tokens, rows kept and left out, prices	Table 8.4
Observation window	2023-10-01 – 2026-06-30 (Table 8.3)
Revision of the amounts (US dollars, five tokens, slopes, cap)	Section 3.5.1
Registered analysis: decision contrasts and verdicts	Appendix 8.6.10, Table 8.13

Each quantity listed appears in a machine-readable summary and in the table or section named beside it; the registered results are in a summary of their own.

8.4 Data Description

This appendix describes the extracted data behind Chapter 4: the volume of the ego extraction, the rows that the revised analysis keeps and the size of its two graphs, the five tokens and their prices, the US-dollar amounts of transfers and allowances and how often owners grant unlimited allowances, and the activity of GMX V2 contract accounts by quarter. The account types of the cohorts are in Table 8.1, the extraction itself is described in Appendix 8.1.2, and the revision of the amounts in Section 3.5.1.

8.4.1 Extracted events and graphs

The ego extraction of the observation window was made for the plan's cohort of $27{,}844$ contracts. It holds $389{,}866{,}882$ approval logs, zero-value ones included, and $1{,}538{,}404{,}067$ transfer logs of non-zero value of every ERC-20 token, each with a contract of that cohort at one end. The revised analysis keeps the $224{,}954{,}202$ approvals and $1{,}149{,}168{,}538$ transfers that are of the five tokens and have a contract of the revised cohort at one end, 57.7% and 74.7% of the extracted logs; of the July–September logs of W1 it keeps $12{,}209{,}894$ approvals and $60{,}182{,}192$ transfers. Table 8.4 gives the rows of each token and those left out. From the kept events the allowance graph at T_{obs} has $6{,}221{,}665$ nodes and $11{,}210{,}114$ edges, and the transfer graph $7{,}884{,}602$ nodes and $25{,}001{,}216$ edges. These edge counts are those of the pairs with a positive EndorseRank or AWP weight: the $9{,}002$ allowance pairs and $17{,}676$ transfer pairs whose amounts give a value weight of zero in double precision are not counted, while the node counts include the $2{,}602$ and $2{,}246$ addresses that only such pairs touch. At the freeze of W0 the pair tables of the two graphs hold $10{,}650{,}258$ and $23{,}680{,}692$ pairs, counted without that filter. The allowance graph has fewer than half as many edges as the transfer graph. It keeps one latest allowance per token, owner and spender and drops the revoked ones, whereas the transfer graph is built from every sender–recipient pair that exchanged a transfer of the five tokens in the window.

Membership in the cohort does not depend on either graph at T_{obs}. Every cohort contract received

non-zero approvals of the five tokens from at least three owners in the window, but 523 of the $15{,}107$ neither hold nor grant a positive latest allowance at T_{obs}, so they lie outside the allowance graph; $2{,}670$ lie outside the transfer graph, and 80 outside both. Chapter 4 scores such a contract zero under the method whose graph it misses, and Table 4.5 reports the rule that keeps it as an isolated node instead.

8.4.2 Tokens and allowance amounts

The revised analysis keeps the approvals and transfers of five tokens (Section 3.5.1). A token qualified when Chainlink's directory of data feeds on Arbitrum One has a market-price feed rated low or medium market risk that prices the token itself, the asset it wraps one-to-one, or the Ethereum token of which it is the canonical-bridge copy (Chainlink, n.d.), and the token occurs in the observation-window data. Of the 133 Arbitrum One tokens that the directory maps, 64 qualified and occur in the data of the observation window, and 46 of these have a daily price from DefiLlama (DefiLlama, n.d.). Among the 46 the analysis keeps the five with the most rows, counting the transfers up to T_{obs} and the positive latest allowances at T_{obs} in the ego tables of the plan's cohort: WETH, the native USDC, USDT0, ARB and WBTC. Together they hold 75.75% of the rows of all tokens, and all five are priced by feeds rated low risk. Table 8.4 gives for each of them the address, the decimals and the feed, the approval and transfer logs, counted together, that the ego tables of the plan's cohort hold and those that the revised analysis keeps, the US-dollar volume of the kept transfers and the data behind their daily prices.

The price of a token on a UTC day is DefiLlama's daily close (Section 3.5.1). DefiLlama's series under the Arbitrum One address begins in August 2024 for WETH and WBTC and in January 2025 for USDT0, so 319, 308 and 481 of their $1{,}096$ days take the price of the same asset under its CoinGecko identifier, the second key, which differs from the Arbitrum series by less than 0.1% in median on the days both cover. USDC and ARB need no second key. None of the five tokens has a day filled from the previous day or a point replaced as a data error, and no kept transfer precedes its token's first price. Against the Chainlink feeds read on the chain on 38 dates per token, the

Table 8.4: The five tokens of the revised analysis, with their Chainlink feed (market-risk category), the rows of the observation window that the ego tables of the plan's cohort hold for them (approvals and transfers) and the rows that the revised analysis keeps, the US-dollar volume of the kept transfers, the days of the price series filled from the second DefiLlama key, filled from the previous day and replaced as one-day errors (of 1,096 days), and the median and largest absolute difference, in percent, between the DefiLlama price and the Chainlink feed read on the chain on 38 dates. The ego tables of the plan's cohort hold 389,866,882 approvals and 1,538,404,067 transfers in the observation window. Of these, 164,202,859 approvals and 364,468,917 transfers belong to other tokens (of the 142,958 other tokens in the ego tables of the observation window and W1) and are left out; 709,821 approvals and 24,766,612 transfers of the five tokens are left out because neither end point is a contract of the revised cohort; and 0 kept transfers precede their token's first price. At T_{obs}, 5,763,757 of the 12,590,613 positive latest allowances are unlimited and 6,776,289 reach the cap $U = 20{,}000$ dollars of the allowance layer of C-PR.

Token	Address	Dec.	Chainlink feed	Rows, plan	Rows, kept	Transfer USD	2nd key	Filled	Errors	CL diff. (%)
WETH	`0x82af…bab1`	18	ETH / USD (low)	586,587,366	569,011,073	8.4×10^{11}	319	0	0	0.06 / 0.72
USDC	`0xaf88…5831`	6	USDC / USD (low)	400,716,805	396,280,336	9.9×10^{11}	0	0	0	0.01 / 0.26
USDT0	`0xfd08…cbb9`	6	USDT / USD (low)	233,074,004	231,428,275	2.7×10^{11}	481	0	0	0.03 / 0.27
ARB	`0x912c…6548`	18	ARB / USD (low)	90,752,065	89,229,800	8.5×10^{10}	0	0	0	0.08 / 0.32
WBTC	`0x2f2a…5b0f`	8	WBTC / USD (low)	88,468,933	88,173,256	2.2×10^{11}	308	0	0	0.06 / 0.24

DefiLlama prices differ by a median of 0.01% to 0.08% per token and by at most 0.72%, for WETH on 15 January 2024.

Table 8.5 lists 27 tokens: the twenty with the most rows and seven more from the list by amount of the registered token check. The revised analysis keeps the first five; the sixth, the bridged USDC, qualifies and has a price but is left out. Seven of the twenty, among them NPC, UXUY and AIUS, do not qualify under the feed rule. The last two columns mark the two lists of the token check of the registered analysis, which rebuilt both graphs from twenty tokens chosen once by the number of rows and once by the summed raw amount (`sql/11_top_tokens.sql`) and which the revision replaced by the value-slope check (Section 3.12.3); the two lists share 13 tokens. In that analysis raw amounts were summed without regard to decimals, and an unlimited allowance counted as $2^{256} - 1$ base units, so the list by amount took in seven tokens with comparatively few rows, among them aArbWETH, the Aave deposit receipt for WETH, and two Fiat24 tokens with two decimals. In US dollars the kept transfers of the observation window have a median of 193.57 dollars. A tenth of them are worth less than 0.14 dollars and a quarter less than 12.19, while three quarters are

Table 8.5: Tokens with the most rows in the ego tables of the plan's cohort and tokens of the two top-20 lists of the token check of the registered analysis, ordered by rows: transfers in the ego tables up to T_{obs} plus positive latest allowances at T_{obs}. The revised analysis keeps the first five (Table 8.4). By amount: among the 20 tokens with the largest summed raw amount of the same rows. By rows: among the 20 tokens with the most rows. Symbols and names are those the token contracts returned to `symbol()` and `name()` calls on 10 October 2026, except that USDT0 is written with a T where its contract returns the tugrik sign; the extraction keeps only addresses, which are given in full in `data/1-raw-blockchain-extracts/graph-tables/top_tokens.csv`.

Symbol	Name	Address	Rows	By amount	By rows
WETH	Wrapped Ether	`0x82af49…3fbab1`	509,394,850	✓	✓
USDC	USD Coin	`0xaf88d0…8e5831`	366,294,721	✓	✓
USDT0	USDT0	`0xfd086b…9fcbb9`	163,462,893	✓	✓
ARB	Arbitrum	`0x912ce5…9e6548`	75,552,592	✓	✓
WBTC	Wrapped BTC	`0x2f2a25…fc5b0f`	71,824,604	✓	✓
USDC	USD Coin (Arb1)	`0xff970a…db5cc8`	71,329,831	✓	✓
PENDLE	Pendle	`0x0c880f…a8c9e8`	19,516,891	✓	✓
WBTC_ETH_FS_LT	WBTC ETH Futureswap Liquidity Token	`0xa522a5…0d8d98`	17,735,463		✓
NPC	NPC	`0xe1a498…0bb8b0`	12,163,764		✓
UXUY	UXUY Token	`0xe2035f…7c79f9`	9,761,886	✓	✓
AIUS	Arbius	`0x4a24b1…0cabc3`	8,804,452		✓
DAI	Dai Stablecoin	`0xda1000…000da1`	8,677,480	✓	✓
LINK	ChainLink Token	`0xf97f4d…539fb4`	8,675,046		✓
xUSD	xUSD	`0xe80772…afca65`	6,437,746		✓
ZRO	LayerZero	`0x698588…3271cd`	6,318,171	✓	✓
UXLINK	UXLINK Token	`0x1a6b3a…c8cff1`	6,187,243		✓
VSN	Vision	`0x6fbbbd…7db74b`	6,052,221	✓	✓
wstETH	Wrapped liquid staked Ether 2.0	`0x5979d7…800529`	5,553,387		✓
GRT	Graph Token	`0x962306…7e88c7`	4,761,427	✓	✓
GMX	GMX	`0xfc5a1a…35ad0a`	4,378,643	✓	✓
BEBE	BEBE	`0x406d59…64f97b`	2,720,903	✓	
STG	StargateToken	`0x669434…ce1eb6`	1,647,816	✓	
aArbWETH	Aave Arbitrum WETH	`0xe50fa9…8128c8`	1,343,622	✓	
OTMT	OTMAirdrop	`0x6b708d…37f770`	493,833	✓	
BPET	BPET	`0x6daf58…862ac5`	389,567	✓	
USD24	Fiat24 USD	`0xbe00f3…cc0d62`	207,237	✓	
CNH24	Fiat24 CNH	`0x7288ac…0144fd`	102,957	✓	

worth less than 891.94 dollars and the 90th and 99th percentiles are $2{,}967.6$ and $19{,}999.9$ dollars. The finite positive latest allowances at T_{obs}, valued at the price of 30 June 2026, have a median of 60.58 dollars, with a tenth below 0.33, a quarter below 5.44 and three quarters below $1{,}198.4$ dollars; their 90th percentile is about 2.1×10^{9} dollars and their 99th about 5.9×10^{45}, because many allowances below $2^{256} - 1$ base units, such as 2^{255} or 2^{128}, are unlimited in effect. The constants of Section 3.5.1 come from the same quantities up to t_1: medians of 203.71 dollars for transfers and 61.51 dollars for finite allowances give $m_T = 200$ and $m_A = 62$ dollars, and the 99th percentile of single transfers, $20{,}266.4$ dollars, gives $U = 20{,}000$ dollars. At T_{obs} both medians lie just below m_T and m_A, so more than half of the transfers and of the finite allowances enter AWP and EndorseRank with a value weight below one half, and between the lower and the upper quartile of either amount the weight rises from near zero to near one.

Unlimited allowances are common. Of the $12{,}590{,}613$ positive latest allowances at T_{obs}, $5{,}763{,}757$ (45.8%) are of 1.15×10^{77} base units or more, close to the largest `uint256` value $2^{256} - 1 \approx 1.16 \times 10^{77}$, and $6{,}776{,}289$ (53.8%), the unlimited ones included, are worth at least the cap $U = 20{,}000$ dollars. The $12{,}590{,}613$ allowances belong to $6{,}208{,}769$ owners, of whom $3{,}294{,}370$ hold at least one unlimited allowance and $797{,}852$ hold both an unlimited and a finite one (`sql/18_usd_describe.sql`). EndorseRank gives an allowance of $m_A = 62$ dollars the value weight one half, one of 124 dollars 0.8, and an unlimited allowance one, the most an approval can weigh. In the allowance layer of C-PR every allowance at or above U counts U, so in the row of an owner with both kinds an unlimited allowance weighs as much as a finite allowance of $20{,}000$ dollars, and the smaller allowances keep a share of the row in proportion to their amounts (Section 3.6.3). In the registered analysis, which kept every token in raw base units with $b = 1$, 99.52% of the ego transfers and 99.78% of the $28{,}069{,}418$ latest allowances at T_{obs} were of ten base units or more, where V is within 10^{-4} of one, so the amount played no part in EndorseRank or AWP. In that analysis $1{,}532{,}301$ of the $13{,}029{,}306$ owners with a positive latest allowance held both an unlimited and a smaller allowance, and the raw allowance layer of C-PR let the unlimited ones take nearly all of such an owner's row (Appendix 8.6.10).

8.4.3 GMX V2 contract traders by quarter

The trader cohorts are GMX V2 accounts that hold contract code (Section 4.1.1). Table 8.6 counts their position closes, the `PositionDecrease` events of the protocol's event emitter (GMX, n.d.), by calendar quarter, together with the liquidations among them and the number of contract accounts that closed at least one and at least three positions in the quarter.

Table 8.6: GMX V2 closes of contract accounts by quarter. October 2023 to June 2026: closes by the 7,633 contracts among the 40,307 GMX V2 accounts with at least three closes in the observation window (Table 8.1). July to September 2026 (the label window of W1): closes by the contract accounts of the observation window and of the W1 run. Liquidation: a close with the liquidation order type or emitted through the liquidation handler. Source: the decoded closes in `data/2-processed-tables-and-evaluations/gmx/`.

Quarter	Closes	Liquidations	Accounts with a close	Accounts with ≥ 3 closes
2023 Q4	1,119	104	180	105
2024 Q1	7,956	692	1,247	961
2024 Q2	11,509	589	2,120	1,165
2024 Q3	7,947	456	1,807	724
2024 Q4	7,343	904	1,321	698
2025 Q1	6,929	498	497	273
2025 Q2	12,080	379	368	250
2025 Q3	28,753	897	1,750	1,567
2025 Q4	40,240	1,596	1,915	1,744
2026 Q1	6,026	233	269	171
2026 Q2	3,783	125	169	127
2026 Q3 (W1)	6,805	182	165	131
October 2023 – June 2026	133,685	6,473		

Contract accounts closed the most positions in the second half of 2025, 28,753 and 40,240 in the third and fourth quarters, when 1,567 and 1,744 of them closed at least three. In the first half of 2026 their closes fell to 6,026 and 3,783, and the accounts with at least three closes to 171 and 127. A trader cohort further requires the account to be a node of the graph at the freeze: of the 127 contract accounts with at least three closes in April–June 2026, 88 form the trader cohort of W0, and of the 131 in July–September, 70 form that of W1. These small cohorts are why every trader result of Chapter 4 has wide intervals.

8.5 EndorseRank and AWP Side by Side

Chapter 4 reports every score of the thesis together and reads each PageRank against the raw degree it smooths. This appendix collects the direct comparisons between the two single-layer scores, EndorseRank and AWP: their family means and paired differences in the same window, their per-proxy coefficients, their paired differences out of window, and their contrasts on every trader label. Every number here comes from the revised analysis, with amounts in US dollars over five tokens (Section 3.5.1); the registered analysis is in Appendix 8.6.10. None of the comparisons on the matched cohort or on the spender cohorts was fixed in advance, and none of them carries a claim of the thesis; the trader pairs were registered with W1 for the registered analysis and are recomputed here on the revised scores (Section 8.5.4). The two scores walk on different edges and are built for different constructs, so on each check over a full cohort the score ahead is the one whose edges the check counts. Outcomes built from neither edge type are the exception, since neither score was built for them; Section 4.6.3 reports them.

Coverage differs between the two. Of the matched cohort, 523 contracts lie outside the allowance graph and $2{,}670$ outside the transfer graph, and $2{,}891$ of the $17{,}050$ spenders of W0 had no transfer edge before the freeze, so AWP scores them zero. EndorseRank, by contrast, leaves almost no pair of cohort contracts tied (Section 4.4). Out of window the two scores stand differently against the raw degrees they smooth: AWP ranks later new senders above the transfer in-degree in both windows, while EndorseRank ranks later new approval pairs below the in-approve degree in both (Section 4.6). A lead of EndorseRank over AWP therefore does not show that propagation over approvals adds anything to the count of approving owners.

8.5.1 Same window

Table 4.3 gives the family means of the two scores with their intervals, and the last row of Table 4.5 their agreement. The two patterns cross. AWP tracks the transfer family most closely (0.572, interval $[0.564, 0.580]$) and the allowance family least (0.193, $[0.182, 0.203]$), while EndorseRank tracks

the allowance family most closely (0.585, $[0.578, 0.592]$) and stands at 0.158 ($[0.147, 0.169]$) on the transfer family. Table 8.7 tests five paired differences on the same contract resamples (Section 3.10), numbered i to v in the order of its rows.

Table 8.7: Paired bootstrap tests of differences between Kendall τ coefficients of EndorseRank and AWP in the same window (matched cohort, $n = 15{,}107$ contracts). Family entries use the family-mean τ; the same contract resamples are used for a and b, so the interval is on the paired difference. An interval excluding 0 indicates a difference not attributable to sampling variation at the 5% level. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
EndorseRank allowance minus EndorseRank transfer	0.585	0.158	+0.427	[0.416, 0.440]	1.000
AWP transfer minus EndorseRank transfer	0.572	0.158	+0.414	[0.401, 0.427]	1.000
EndorseRank allowance minus AWP allowance	0.585	0.193	+0.392	[0.380, 0.405]	1.000
AWP sybil-stability minus EndorseRank sybil-stability	0.395	0.007	+0.388	[0.376, 0.401]	1.000
EndorseRank in-approve degree minus EndorseRank in-degree	0.655	0.182	+0.473	[0.461, 0.486]	1.000

All five intervals lie above zero. EndorseRank's allowance coefficient exceeds its transfer coefficient (contrast i, $+0.427$, interval $[0.416, 0.440]$); AWP's transfer coefficient exceeds EndorseRank's (contrast ii, $+0.414$, $[0.401, 0.427]$); EndorseRank's allowance coefficient exceeds AWP's (contrast iii, $+0.392$, $[0.380, 0.405]$); AWP leads on the Sybil-stability family (contrast iv, $+0.388$, $[0.376, 0.401]$); and EndorseRank follows the in-approve degree more closely than the in-degree (contrast v, $+0.473$, $[0.461, 0.486]$). Contrasts i and v compare EndorseRank with itself across proxies. EndorseRank ties only 0.17% of the contract pairs, so its own ties cap none of these coefficients, and the ties of the proxies do not explain gaps of this size: on the in-degree AWP reaches 0.578 ($[0.569, 0.587]$), where EndorseRank reaches 0.182 ($[0.169, 0.194]$). Keeping the contracts outside the allowance graph as isolated nodes changes none of the family means (Table 4.5).

The opposite signs of contrasts ii and iii follow from the constructs. Each score is ahead on the family built from its own edges, and each of these families includes the in-degree that the corresponding PageRank smooths. The agreement is partly built in and checks the intended construct of each score; it is not evidence from outside the score. Every contract of the cohort received approvals of WETH, USDC, USDT0, ARB or WBTC from at least three owners, and AWP's allowance co-

efficients lie above zero (Table 8.9): contracts that many owners approve also tend to be central in the transfer graph. Neither difference says which score is the better measure of reputation.
On the scaling stages of Table 4.8, whose values are not comparable with those of the full cohort, the order of the two scores within each family is the same at every stage: AWP is ahead on the transfer and Sybil-stability families and EndorseRank on the allowance family.

8.5.2 Proxy detail

Each family mean condenses several per-proxy coefficients. Tables 8.8 to 8.10 report them one proxy at a time for EndorseRank and AWP, as Spearman ρ, Kendall τ and the 95% interval of τ. Every proxy counts the events of the five tokens, with amounts in US dollars (Section 3.9). Before correlation, every proxy is oriented so that a higher value means higher expected reputation, and the transfer proxies are the ones used in the evaluation of AWP by its authors (T. Do et al., 2023).

8.5.2.1 Transfer family

Table 8.8 gives the coefficients of the two scores with the in-degree and the in-value.

Table 8.8: Domain alignment of EndorseRank and AWP with the transfer proxies (in-degree and in-value of each contract), same window (matched cohort, $n = 15{,}107$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
In-degree	0.261	0.182	[0.169, 0.194]
In-value (sum inbound)	0.205	0.134	[0.123, 0.145]
AWP			
In-degree	0.744	0.578	[0.569, 0.587]
In-value (sum inbound)	0.741	0.566	[0.558, 0.575]

AWP agrees with the in-degree at 0.578 (interval $[0.569, 0.587]$) and with the in-value at 0.566 ($[0.558, 0.575]$). AWP's value weight is one half at the median transfer of 200 dollars and comes close to one only for amounts several times larger (Section 3.6.1), so larger transfers count for more,

and AWP follows the summed US-dollar value of a contract's inbound transfers almost as closely as their number. EndorseRank's transfer coefficients lie above zero, at 0.182 ($[0.169, 0.194]$) and 0.134 ($[0.123, 0.145]$), and their intervals lie far below AWP's: contracts that many owners approve receive transfers from more senders than the rest of the cohort. Out of window AWP ranks later new senders above the raw in-degree in both windows (Section 4.6), so on the transfer side the walk adds to the count it smooths.

8.5.2.2 Allowance family

Table 8.9 gives the coefficients of the two scores with the in-approve degree and the in-approve value.

Table 8.9: Domain alignment of EndorseRank and AWP with the allowance proxies (in-approve degree and value of each spender contract), same window (matched cohort, $n = 15{,}107$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
In-approve degree	0.811	0.655	[0.648, 0.662]
In-approve value (sum)	0.681	0.515	[0.506, 0.524]
AWP			
In-approve degree	0.286	0.206	[0.193, 0.217]
In-approve value (sum)	0.260	0.180	[0.169, 0.191]

EndorseRank agrees with the in-approve degree at 0.655 (interval $[0.648, 0.662]$) and with the in-approve value, the sum of the capped allowances, at 0.515 ($[0.506, 0.524]$). The value weight gives an allowance of the median finite amount, 62 dollars, one half, but more than half of the positive latest allowances are unlimited or reach 20,000 dollars (Table 8.4) and count fully, so EndorseRank follows the number of approving owners more closely than the summed amounts they approve. For a spender whose owners receive no approvals it equals the closed form of Section 5.3.2, a function of the owner-normalized in-strength. AWP agrees with the two proxies at 0.206 ($[0.193, 0.217]$) and 0.180 ($[0.169, 0.191]$). The comparison is partly built in, since EndorseRank is computed from the

same approvals as the proxies.

8.5.2.3 Sybil-adjusted stability family

Table 8.10 gives the coefficients of the two scores with the inbound counterparty ratio, the transfer tenure and the active months.

Table 8.10: Domain alignment of EndorseRank and AWP with the Sybil-adjusted stability proxies from transfer activity, same window (matched cohort, $n = 15{,}107$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
Inbound counterparty ratio	-0.231	-0.156	[-0.166, -0.147]
Transfer tenure (days)	0.124	0.080	[0.069, 0.091]
Active months	0.146	0.098	[0.087, 0.109]
AWP			
Inbound counterparty ratio	0.234	0.127	[0.114, 0.141]
Transfer tenure (days)	0.683	0.508	[0.499, 0.517]
Active months	0.704	0.549	[0.540, 0.558]

The table shows AWP ahead of EndorseRank on the transfer tenure (0.508, interval $[0.499, 0.517]$, against 0.080, $[0.069, 0.091]$) and on the active months (0.549, $[0.540, 0.558]$, against 0.098, $[0.087, 0.109]$), as one would expect from a transfer score that rewards activity spread over the window; the paired difference on the family mean excludes zero (Table 8.7, contrast iv). On the inbound counterparty ratio the two coefficients have opposite signs, -0.156 ($[-0.166, -0.147]$) for EndorseRank and 0.127 ($[0.114, 0.141]$) for AWP. The ratio divides the distinct other senders by the transfers received from them, so it falls when the same senders return; EndorseRank's negative coefficient says that contracts that many owners approve are, on this measure, contracts whose senders return. AWP's positive coefficient comes from coverage: the $2{,}670$ contracts outside the transfer graph have ratio zero and AWP score zero, so they fill the bottom of both orderings. Among the contracts that AWP scores above zero, its order also falls with the ratio, as EndorseRank's does;

this check was made after the results were known and has no interval. The family describes how much longitudinal transfer information each score carries.

8.5.3 Out of window

Table 8.11 collects the paired differences between the two scores in both holdout windows. In every row a is EndorseRank and b AWP.

Table 8.11: Paired bootstrap differences between EndorseRank and AWP out of window (W0 spenders $n = 17{,}050$; W0 traders $n = 88$; W1 spenders $n = 18{,}152$; W1 traders $n = 70$). None of them enters rule A or rule B: the spender rows were computed after the labels were known, and the trader rows are the EndorseRank-minus-AWP pair of the comparison on an outcome built from neither edge type (Tables 4.18 and 8.12). The W1 spender rows were computed from the saved W1 scores after the W1 evaluation, with the resamples of Table 4.14. Within each block a and b share the contract resamples (400 for the spenders and 2,000 for the traders; seed 42).

Contrast (a minus b)	Label	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
W0, spenders ($n = 17{,}050$)						
EndorseRank minus AWP	new approval pairs	0.231	0.196	+0.035	[0.019, 0.051]	1.000
EndorseRank minus AWP	new transfer senders	0.200	0.346	-0.146	[-0.156, -0.136]	0.000
W0, traders with at least three closes ($n = 88$)						
EndorseRank minus AWP	liquidation-free close rate	0.224	0.240	-0.017	[-0.156, 0.128]	0.394
W1, spenders ($n = 18{,}152$)						
EndorseRank minus AWP	new approval pairs	0.234	0.169	+0.065	[0.051, 0.081]	1.000
EndorseRank minus AWP	new transfer senders	0.194	0.330	-0.136	[-0.146, -0.127]	0.000
W1, traders with at least three closes ($n = 70$)						
EndorseRank minus AWP	liquidation-free close rate	0.297	0.329	-0.032	[-0.133, 0.066]	0.279

On the spenders of W0, EndorseRank ranks new approval pairs at $\tau = 0.231$ (interval $[0.220, 0.242]$) against 0.196 ($[0.181, 0.210]$) for AWP, a paired difference of $+0.035$ ($[0.019, 0.051]$). On new transfer senders the order reverses: AWP reaches 0.346 ($[0.336, 0.356]$) against 0.200 ($[0.189, 0.210]$) for EndorseRank, a difference of -0.146 ($[-0.156, -0.136]$). Every label falls after all scored events, so on this cohort each score ranks the later arrivals of its own edge type better than the other score does. Coverage accounts for part of the first gap. AWP scores zero for the $2{,}891$ spenders without a transfer edge before the freeze, and 13.3% of them gained a new approval pair, against 11.9% of the

rest. On the $14{,}094$ spenders that had received a transfer from another address, the order on new approval pairs reverses: AWP reaches 0.298 ($[0.287, 0.308]$) and EndorseRank 0.236 ($[0.224, 0.249]$), without a paired test (Table 4.12). On new senders AWP stays ahead there, at 0.357 ($[0.347, 0.367]$) against 0.231 ($[0.219, 0.244]$).

In July–September, with the contrasts computed from the revised W1 scores after all labels were known, the pattern on the full spender cohort repeats. EndorseRank ranks new approval pairs at 0.234 ($[0.222, 0.244]$) against 0.169 ($[0.157, 0.183]$) for AWP ($+0.065$, $[0.051, 0.081]$), and AWP ranks new senders at 0.330 ($[0.321, 0.340]$) against 0.194 ($[0.184, 0.205]$) (-0.136, $[-0.146, -0.127]$). On the liquidation-free close rate of the traders, a pair that the registration of W1 named for the registered analysis, EndorseRank reaches 0.224 ($[0.105, 0.339]$) in W0 and 0.297 ($[0.171, 0.410]$) in W1, and AWP 0.240 ($[0.073, 0.380]$) and 0.329 ($[0.207, 0.440]$). Neither difference can be told from zero (-0.017, $[-0.156, 0.128]$, and -0.032, $[-0.133, 0.066]$); the trader cohorts hold 88 and 70 contracts, so the intervals are wide.

On the W0 traders' profit labels (Table 4.13), EndorseRank's intervals on profitable closes and realized gain include zero, at 0.150 ($[-0.019, 0.295]$) and 0.078 ($[-0.094, 0.226]$), while AWP's lie above zero, at 0.155 ($[0.005, 0.297]$) and 0.196 ($[0.048, 0.351]$); both labels grow with the number of closes. On the two shares, which do not, both scores are negative, and for both the interval lies below zero on the non-loss share only: AWP at -0.254 ($[-0.404, -0.104]$) against -0.124 ($[-0.265, 0.015]$) on the profitable share, and EndorseRank at -0.178 ($[-0.298, -0.042]$) against -0.055 ($[-0.196, 0.084]$). The transfer in-degree is negative on both shares as well, at -0.191 ($[-0.329, -0.041]$) and -0.197 ($[-0.350, -0.035]$). Neither score ranks these contract traders by the share of successful closes.

8.5.4 All trader labels in both windows

Table 8.12 sets the allowance side against the transfer side on all six trader labels in both windows. Its four pairs and six labels were registered with W1 for the registered analysis, before the logs of July–September were extracted (Section 3.7.3); by then that analysis had scored the W0 traders on

the profit labels but not on their liquidation labels. The table gives the same pairs on the revised scores, which were defined after every registered result, the liquidation labels included, was known, so its entries are a re-analysis and not registered tests (Appendix 8.6.10). P and Q on the liquidation-free close rate, in W1 and with their signs in W0, enter rules R1 and R2, and P on the two shares in W1 enters rule R3, all applied to the revised scores (Claim C6, Section 4.6.3); the other entries, the column for EndorseRank minus AWP included, are reported beside them.

In W1 no pair separates the two sides on either liquidation label: every interval includes zero, P on the liquidation-free close rate at $+0.084$ ($[-0.013, 0.176]$) and the EndorseRank pair at -0.032 ($[-0.133, 0.066]$). In W0 the counts favor the allowance side on both liquidation labels, P at $+0.182$ ($[0.052, 0.314]$) on the liquidation-free close rate and $+0.197$ ($[0.059, 0.333]$) on the no-liquidation flag, while the three PageRank pairs lie below zero with intervals that include zero. On the four profit labels of W1 every pair is negative. The intervals of Q and of the EndorseRank pair lie below zero on all four, those of P on the two shares, and those of the C-PR layers on realized gain and on both shares. In W0 every interval on the profit labels includes zero except that of the C-PR layers on realized gain, -0.248 ($[-0.433, -0.067]$), which thus lies below zero in both windows on a label that grows with the number of closes; on the two shares Q, the C-PR layers and the EndorseRank pair lean to the allowance side with intervals that include zero, so the shares show no direction that holds in both windows. With 70 and 88 traders, none of these differences is precise.

8.6 Claim-to-Evidence Map

This appendix maps each hypothesis, research question and empirical claim to the relevant evidence in Chapter 4, and lists what the evidence does not show. The evidence is that of the revised analysis, whose amounts in US dollars on five tokens were set after the registered results were known (Section 3.5.1); the registered results are kept in Appendix 8.6.10.

Table 8.12: Allowance-side minus transfer-side scores, $\Delta\tau$ with 95% paired bootstrap interval, on all six trader labels of the window W1 ($n = 70$ GMX V2 contract traders) and of W0 ($n = 88$). P: in-approve degree minus transfer in-degree. Q: EndorseRank with activity restarts minus AWP (same edge weights and restart rule). C-PR: $\lambda = 1$ minus $\lambda = 0$ (same USD build). ER: EndorseRank minus AWP. Profitable closes and realized gain grow with the number of closes; the other four labels do not. CI = 95% percentile bootstrap over contracts (2,000 paired resamples, seed 42).

Label	P (counts)	Q (PageRanks)	C-PR layers	ER minus AWP
Window W1, July–September 2026 ($n = 70$ traders)				
Liquidation-free close rate	0.084 [-0.013, 0.176]	-0.041 [-0.135, 0.050]	-0.049 [-0.169, 0.060]	-0.032 [-0.133, 0.066]
No liquidation	0.078 [-0.012, 0.166]	-0.039 [-0.138, 0.062]	-0.051 [-0.170, 0.064]	-0.029 [-0.136, 0.080]
Profitable closes	-0.143 [-0.318, 0.013]	-0.231 [-0.429, -0.027]	-0.190 [-0.411, 0.037]	-0.214 [-0.423, -0.012]
Realized gain	-0.135 [-0.308, 0.025]	-0.350 [-0.536, -0.157]	-0.404 [-0.624, -0.182]	-0.346 [-0.547, -0.144]
Profitable share	-0.196 [-0.364, -0.046]	-0.336 [-0.523, -0.141]	-0.344 [-0.551, -0.120]	-0.316 [-0.517, -0.112]
Non-loss share	-0.252 [-0.391, -0.120]	-0.350 [-0.526, -0.175]	-0.324 [-0.531, -0.109]	-0.346 [-0.526, -0.161]
Window W0, April–June 2026 ($n = 88$ traders)				
Liquidation-free close rate	0.182 [0.052, 0.314]	-0.014 [-0.147, 0.130]	-0.041 [-0.187, 0.105]	-0.017 [-0.156, 0.128]
No liquidation	0.197 [0.059, 0.333]	-0.010 [-0.152, 0.146]	-0.051 [-0.198, 0.102]	-0.012 [-0.154, 0.137]
Profitable closes	0.075 [-0.080, 0.222]	-0.016 [-0.150, 0.118]	-0.067 [-0.222, 0.087]	-0.005 [-0.163, 0.155]
Realized gain	0.017 [-0.150, 0.176]	-0.107 [-0.252, 0.044]	-0.248 [-0.433, -0.067]	-0.118 [-0.286, 0.046]
Profitable share	0.045 [-0.113, 0.184]	0.046 [-0.071, 0.172]	0.027 [-0.112, 0.181]	0.069 [-0.060, 0.208]
Non-loss share	-0.080 [-0.228, 0.058]	0.055 [-0.070, 0.192]	0.037 [-0.111, 0.189]	0.076 [-0.063, 0.223]

8.6.1 H1 / RQ1: Rank divergence

RQ1 asks whether EndorseRank and AWP order the same contracts differently.

- Statement: on the matched cohort, EndorseRank and AWP rank contracts differently.
- Evidence: Section 4.4; Table 4.5, whose last row gives the inter-method τ_b with its interval, also with the contracts outside each graph kept as isolated nodes; Figures 4.3 and 4.2.
- Interpretation: the rankings are not equivalent. A large share of the pairs that both scores order are ordered oppositely, and the rule for contracts outside a graph does not change their agreement. The O1 criterion is met (Table 5.1), as it was in the registered analysis (Table 8.13).

8.6.2 H2 / RQ2: Same-window alignment

RQ2 asks how closely each score agrees with the allowance checks and with the transfer checks of the same window.

- Statement: in the same window, each score agrees most with the proxies of the edge type it walks on, EndorseRank with the allowance family and AWP with the transfer family, and the C-PR mixtures lie between their two layers on every family. With amounts in US dollars, AWP follows the summed value of a contract's inbound transfers almost as closely as their number of senders, the in-degree.
- Evidence: the family means of every score with their intervals (Table 4.3); the transfer proxies one by one (Table 8.8); the paired contrasts of C-PR and S-PR (Table 4.4); the same-window coefficients with the contracts outside a graph kept as isolated nodes (Table 4.5).
- Note: this is a check of the intended construct and is partly built in, since the proxies come from the same events as the graphs. EndorseRank leaves few ties among the cohort, and the rule for contracts outside a graph changes no coefficient. Only the same-window part of rule A, which concerns C-PR (Section 4.3.1), was fixed in the plan, and here it is applied to scores defined after the registered results were known, so none of these comparisons was fixed before its data. The paired differences between EndorseRank and AWP and their per-proxy results

are in Appendix 8.5.

8.6.3 H3 / RQ3: Computational cost

RQ3 asks what each score costs to compute on the same set of contracts.

- Statement: on the same contracts, the time of a solver call follows the edge count of its graph times its iterations, so EndorseRank, on the sparser graph, takes the least time, while C-PR, on the larger union of both layers, and S-PR, which adds a solve of the allowance layer for its restarts, take longer than AWP.
- Evidence: Table 4.1 (runtime, SD, memory, iterations, $|V|$, $|E|$); Table 4.2; every timed run in Table 8.2.
- Note: all scores run the same solver, so the speed is a property of $|E|$ and not an advantage of an algorithm (Appendix 8.3.3). The scaling stages are subsamples of the cohort graph on the five tokens, so the claim is confined to graphs up to its size.

8.6.4 RQ4: Temporal holdout

RQ4 asks whether scores frozen at t_1 rank the later arrivals of spender contracts, and whether any PageRank ranks them better than the raw degree it smooths.

- Statement: with scores frozen at the start of each label window (t_1 for W0, T_{obs} for W1), each score ranks best the later arrivals of its own edge type on the spender cohort. Against the raw degree at the freeze the two walks differ: AWP ranks later new transfer senders above the transfer in-degree in both windows, while EndorseRank, which keeps the uniform restarts fixed in the plan, ranks later new approval pairs below the in-approve degree in both windows. The registered analysis gave these contrasts the same signs (Table 8.13).
- Evidence: Tables 4.10 and 4.11 (Section 4.6); the window W1, Table 4.16; the spenders that had received a transfer before the freeze, Table 4.12; EndorseRank with activity restarts, Table 4.9.

- Qualification: both labels are the constructs as observed in a later period, counts of new approval pairs and new senders on the five tokens, not credit or trading outcomes. In the registered analysis the two contrasts of rule A were fixed in the plan and those of rule B, with F4–F6, in the W1 registration, each before its data; on the revised scores, defined after the registered results were known (Section 3.5.1), none was. The degree contrasts of EndorseRank and AWP, the increment of C-PR ($\lambda = 1$) over the in-approve degree, the contrasts of S-PR and the differences between EndorseRank and AWP (Appendix 8.5) were computed after the labels were known. EndorseRank with AWP's activity restarts ranks new approval pairs above the in-approve degree in both windows, but that comparison too was made after the labels were known, so it is a hypothesis for a registered test and not a finding.

8.6.5 RQ5: Coupled operator

RQ5 asks whether C-PR improves on a walk over the transfers alone, and whether it ranks contracts better than a walk over either layer alone.

- Statement: on the revised scores, one walk over both edge types does not beat its layers walked alone. Rule A fails: C-PR is no worse than the allowance walk on new approval pairs but worse than the transfer walk on new senders, and its same-window part fails as well. Rule B is not met: C-PR improves on the transfer walk on new approval pairs (F1), so the gain seen in W0 recurs in W1, but it fails the margin on new senders (F2) and, narrowly, on the liquidation-free close rate of the traders (F3), where the two point estimates are equal but the lower end of the interval lies below $-\delta$. With AWP as the comparator, the first sensitivity analysis, F2 fails clearly. In W0 contrasts computed after the labels were known, S-PR stays level with the transfer walk on new senders and below the allowance walk on new approval pairs: allowance edges help only when they carry the walk. The registered analysis reached the same verdicts (Appendix 8.6.10).
- Evidence, rule A: Table 4.11 for the primary part, and Tables 4.3 and 4.4 for the secondary part (Section 4.3.1).

- Evidence, rule B: Table 4.15, with F1–F3, F4–F6 reported beside them and both sensitivity analyses (Section 4.6.2); the exploratory contrast of W0 in Table 4.11; the registered verdicts beside the revised ones in Table 8.13.
- Evidence, cost: the rows of C-PR and S-PR in Table 4.1.
- Qualification: tested with fixed $\lambda \in \{0.25, 0.5, 0.75\}$, layers in US dollars on five tokens with each allowance capped at U, uniform teleportation, transition-level mixing and one chain. Rule A and the values of λ were fixed in the plan and configuration committed in `43d8045` on 9 October 2026, before any log was extracted (Section 3.7). Rule B, its margin and its two sensitivity analyses were written after the results of W0 were known and registered in `0d3a057` before scan B (Section 3.7.2). The US-dollar layers and the cap were set on 10 October 2026, after the registered results were known (Section 3.5.1), so the verdicts above apply these rules to the revised scores as a re-analysis and are not registered tests.

8.6.6 Claims C1–C5

Chapter 4 states Claims C1–C6. The first five rest on the evidence below; the evidence for C6 follows in Appendix 8.6.9.

- C1: the runtime evidence listed under H3/RQ3, with the edge ratio and the time ratio in Section 4.2 and Appendix 8.3.3. The speed belongs to the sparser graph, not to the algorithm, and C-PR and S-PR cost more than AWP.
- C2: the same-window agreement of every score with the proxies of its own edges (Table 4.3), a check of the intended construct that is partly built in, with AWP following the US-dollar value of inbound transfers almost as closely as their number of senders (Table 8.8); the ties and the rule for contracts outside a graph (Table 4.5). The damping and value-slope checks, which recompute the same-window family means of EndorseRank and AWP, are in Tables 4.6 and 4.7 and Figure 4.4; the sample-definition stages (Table 4.8) show how the statistic moves with the evaluation set and are not robustness evidence.

- C3: the rank divergence listed under H1/RQ1 (Table 4.5, Figure 4.3).
- C4: the comparison of each walk with the raw degree of its own edge type out of window (Tables 4.10, 4.11 and 4.16), with the coverage check (Table 4.12) and the variant with activity restarts (Table 4.9).
- C5: on the revised scores coupling does not beat its two layers walked alone, a bounded negative result, and rule B for the comparison with the transfer walk alone, applied to these scores, is not met (Table 4.15); the registered analysis reached the same verdicts (Table 8.13); see RQ5 above and Section 4.3.1.

8.6.7 Explicit non-claims

This research does *not* claim:

- that any of the scores is the better measure of reputation in general, or that AWP has become obsolete (the comparisons between EndorseRank and AWP in Appendix 8.5 follow the constructs and depend on coverage);
- that the agreement of a score with proxies built from its own edges is evidence from outside the score (it checks the intended construct and is partly built in);
- that EndorseRank with the uniform restarts of the plan ranks later new approval pairs better than the raw in-approve degree (it ranks them lower in both windows);
- that EndorseRank with activity restarts beats the in-approve degree (that comparison was made after the labels were known and awaits a registered test);
- that multi-hop propagation, which is present in the allowance graph, made EndorseRank beat the raw count (Section 5.3.2);
- that allowance edges lead transfer edges on liquidation avoidance, at the count level (R1) or at the PageRank level (R2), or that EndorseRank beats AWP there (neither rule is met on the revised scores or in the registered analysis);
- that any score measures creditworthiness, default risk or trading skill (in W1 the approval

count ranks traders below the transfer count on the profitable and non-loss shares, and the data contain no default label);

- that receiving approvals makes a contract trader avoid liquidation (the samples are small, the recipients close more positions, and the transfer count also ranks this label);
- that the value slopes $b = \ln 3/m$, the cap U of the allowance layer of C-PR or EndorseRank's uniform restart is the best choice (the slopes and the cap were set in the revision after the registered results were known, and the value-slope check of Table 4.7 varies the slopes but not the cap; the uniform restart was fixed in the plan, and the comparison with activity restarts was made after the labels were known);
- that the verdicts of rules A and B and of R1–R3 on the revised scores are registered tests (the rules were fixed for the registered analysis and are applied to scores defined after its results were known; the registered verdicts are in Appendix 8.6.10);
- that C-PR improves on a walk over the transfers alone (rule B fails on new senders and on the traders) or on AWP (it falls behind AWP on new senders in both windows);
- that no combination of the two edge types can beat the single-layer methods, only that the per-node convex mixture tested here, with fixed λ, does not;
- that EndorseRank's shorter solve time is an advantage of its algorithm (it follows from the smaller edge count), or anything about runtime on graphs larger than the ego network of the cohort;
- that EndorseRank resists Sybil farms (reaching an ordinary or above-median score is cheap, and an approval of one of the five tokens needs no balance, so the token list does not raise the cost of a farm; Section 5.6);
- that the results generalize beyond contracts approved by at least three owners on WETH, USDC, USDT0, ARB or WBTC on Arbitrum One in the months studied, or to the approvals and transfers of other tokens, which the revised analysis leaves out (Table 8.4);
- that on-chain scores substitute for collateral, audits or identity systems, or that they draw on

off-chain information, which this study does not use.

8.6.8 Replication artifacts

The paths below are relative to the experiments folder named in Appendix 8.1. The revised analysis, which the chapters report, has folders of its own; the folders of the registered analysis are kept as they were obtained.

1. The per-contract scores of every method, the cohort scores of the same window and the spender and trader scores of W0 and W1, in parquet files: for the revised analysis in `data/2-processed-tables-and-evaluations/revised-usd/` (`same-window/`, `holdout-w0/` and `window-w1/`), and for the registered analysis in `same-window/`, `holdout-w0/` and `registered-w1/` of `data/2-processed-tables-and-evaluations/`.
2. The machine-readable summaries of the same window (with the damping sweep), W0, W1, the robustness checks, the supplementary contrasts, the Sybil model, the benchmark and the data description, with every coefficient, interval, contrast and verdict: those of the revised analysis, with the value-slope check, in `data/3-published-results-for-thesis/revised-usd/`, and those of the registered analysis, with the top-token check, in `data/3-published-results-for-thesis/`, each folder with its own `results_digest.md` and `SHA256SUMS.txt`.
3. The analysis plan (`docs/analysis_plan.md`) and the configuration as committed in `43d8045` on 9 October 2026, before any log of the study was extracted.
4. The registration of W1 (`docs/registration_w1.md`) as committed in `0d3a057`, and the extraction manifest `data/2-processed-tables-and-evaluations/registered-w1/extraction_manifest.json`. The manifest records the commit, the times at which the extraction of W1 started and finished, that no override was used, and the SHA-256 hashes of the registration, `2126c8dc3fe3c0f9273aa4ecfc1e15bddc1f5a6f65e98548b9528f2b18ca2a6b`, and of the configuration, `d6a30adbcabcf511d6892332da73639efdbece25891b`

392e13e90faf09b7a833.

5. The revision of the amounts (docs/revision_price_weighting.md), committed in a897475 at 07:54 UTC on 10 October 2026, with the choice of the five tokens in 09dfc32 at 08:12 UTC and the amended allowance cap with the constants in 2705b28 at 11:24 UTC, all after the registered results were known and before any revised score was computed.
6. The token list and prices in data/0-usd-token-list-and-prices/: the snapshot of Chainlink's directory of feeds and the reviewed candidate tokens, the tokens that qualify, the daily US-dollar prices with the DefiLlama points they come from, the constants m_T, m_A and U, and the comparison of the prices with the Chainlink feeds read on the chain.
7. The cost ledger data/bq_ledger.json, with every BigQuery job, its parameters and its billed bytes.
8. The extracted tables in data/1-raw-blockchain-extracts/: the cohort candidates with their account types, the GMX V2 accounts and closes; in graph-tables/, the pair tables, the node table, the proxies, the labels and the top-token subgraphs of the registered analysis; and in graph-tables-usd/, the revised cohort with its pair tables, node table, proxies and labels on the five tokens. All are in parquet parts below 90 MB, each with a JSON record of its source table.
9. The exported table fragments (results/tables/) and vector figures (Figures/generated/) of the thesis folder, built from the revised summaries; Table 8.13 also reads the registered ones.

The tables and figures are rebuilt by running the evaluation scripts on the extracted tables and then exporting the tables and figures from the same summaries (Appendix 8.1.1). With CONTRACT_REP_VARIANT=usd the scripts read the tables of the five tokens, which scripts/run_usd_pipeline.py builds in BigQuery, and write the revised results; with the variable unset they reproduce the registered results.

8.6.9 C6: Outcomes built from neither edge type

C6 compares the two edge types on a trader outcome that neither graph contains.

- Statement: on the liquidation-free close rate of GMX V2 contract traders, with the rules registered with W1 applied to the revised scores, the count-level lead of the approval count over the transfer count (R1) is not supported in W1, the window on which the rules decide, although P is positive in W0; the PageRank lead (R2) is not supported either; and R3 applies. In the wording fixed at registration: there is no count-level lead of allowance over transfer on later liquidation avoidance, the PageRank versions are not resolved, and the signal concerns forced-liquidation avoidance, not trading success. These verdicts are a re-analysis, not registered tests, and the registered analysis reached the same ones (Table 8.13).
- Evidence: Tables 4.18 and 4.19 (Section 4.6.3); all six trader labels in both windows, Table 8.12; the protocol, Section 3.7.3; the registration, section 3 of `docs/registration_w1.md`, committed in `0d3a057` before scan B; its application to the revised scores, `docs/revision_price_weighting.md` (Section 3.5.1).
- Qualification: the trader cohorts are small, 70 contracts in W1 and 88 in W0, because contract accounts traded little on GMX V2 in 2026 (Table 8.6). Every trader that had received an approval of one of the five tokens avoided liquidation in both windows, against about three in four of the others, but the recipients closed more positions, and the transfer count also ranks this label. All traders are contracts, so account type is held fixed. Any use in screening would first need a registered test on a larger sample (Section 6.3).

8.6.10 The registered analysis

The analysis plan and the registration of W1 fixed amounts in raw token base units, every ERC-20 token, and the value transform with $b = 1$. The amounts were revised after the results of that analysis were known (Section 3.5.1). The chapters report the revised analysis, and the registered analysis is kept here as it was obtained. Table 8.13 sets its decision contrasts and verdicts beside

the same contrasts on the revised scores; its last four rows, the degree contrasts of both windows, were computed after the labels were known in both analyses and carry no verdict. Every verdict in the registered column is the outcome of a test fixed before its data; no verdict in the revised column is, because the rules were applied to scores defined after the registered results were known. The complete registered results, with every coefficient, interval and verdict, are published as they were computed in `data/3-published-results-for-thesis/` (`results_digest.md`); the revised results are in its subfolder `revised-usd/`.

8.7 Ethics Clearance Certificate

8.7.1 Certificate

This study uses public blockchain data only and involves no human participants (Section 3.16). The ethics clearance certificate issued by the Unisa School of Computing / College of Agriculture and Environmental Sciences research ethics review committee is reproduced below.

Ethics clearance certificate

Reference: [Unisa SoC/CAES ethics clearance ref: TBD]

Date of issue: [TBD]

(Scanned certificate to be inserted here.)

Table 8.13: Decision contrasts and verdicts of the registered analysis (amounts in raw base units, every ERC-20 token, $b = 1$), exactly as obtained under the analysis plan and the W1 registration, beside the same contrasts on the revised scores (amounts in USD, listed tokens only). The revision was decided after the registered results were known; the revised column applies the registered rules and thresholds to the revised scores and is not a registered test. The last block holds degree contrasts that no rule of either analysis decides; they are reported in both. $\Delta\tau$ with 95% paired bootstrap interval (400 and 2,000 resamples, seed 42); the rules and their thresholds are in Sections 3.7 and 3.7.2.

Quantity	Registered analysis	Revised analysis
Matched cohort (contracts)	27,844	15,107
W0 spender cohort	33,101	17,050
W1 spender cohort	34,587	18,152
W1 traders	70	70
EndorseRank vs. AWP, same window (τ)	0.217 [0.209, 0.225]	0.174 [0.163, 0.185]
A, W0: C-PR minus C-PR ($\lambda = 1$), new approval pairs	$+0.008$ [0.003, 0.013]	$+0.029$ [0.022, 0.036]
A, W0: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.030 [-0.034, -0.026]	-0.034 [-0.040, -0.029]
Rule A, primary part	not met	not met
Rule A, secondary part	not met	not met
B, F1: C-PR minus C-PR ($\lambda = 0$), new approval pairs	$+0.101$ [0.092, 0.111]	$+0.110$ [0.098, 0.122]
B, F2: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.025 [-0.029, -0.021]	-0.028 [-0.033, -0.023]
B, F3: C-PR minus C-PR ($\lambda = 0$), liquidation-free close rate	-0.044 [-0.100, -0.002]	$+0.000$ [-0.014, 0.010]
Rule B (F1, F2 and F3)	not met	not met
Rule B, sensitivity 1 (AWP as comparator)	not met	not met
Rule B, sensitivity 2 (isolated nodes)	not met	not met
F6 (rule A, primary part, on W1)	fails	fails
P, W1: in-approve degree minus transfer in-degree	$+0.063$ [-0.026, 0.148]	$+0.084$ [-0.013, 0.176]
Q, W1: EndorseRank (activity restarts) minus AWP	$+0.051$ [-0.085, 0.213]	-0.041 [-0.135, 0.050]
R1 (count-level lead)	not met	not met
R2 (PageRank-level lead)	not met	not met
R3 (specificity)	applies	applies
W0: EndorseRank minus in-approve degree, new pairs	-0.011 [-0.016, -0.006]	-0.019 [-0.026, -0.011]
W0: AWP minus transfer in-degree, new senders	$+0.038$ [0.035, 0.042]	$+0.043$ [0.038, 0.048]
W1: EndorseRank minus in-approve degree, new pairs	-0.029 [-0.032, -0.025]	-0.034 [-0.038, -0.029]
W1: AWP minus transfer in-degree, new senders	$+0.020$ [0.016, 0.023]	$+0.022$ [0.017, 0.027]